

ECA Table

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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8300 Hz - 9 kHz

METEOROLOGICAL AIDS (5.54A) 5.54B	METEOROLOGICAL AIDS (5.54A)	Lightning detection systems			
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9 kHz - 11.3 kHz

METEOROLOGICAL AIDS (5.54A) RADIONAVIGATION	METEOROLOGICAL AIDS (5.54A) RADIONAVIGATION	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Lightning detection systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

11.3 kHz - 14 kHz

RADIONAVIGATION	RADIONAVIGATION	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

14 kHz - 19.95 kHz

FIXED MARITIME MOBILE (5.57) 5.55 5.56	FIXED MARITIME MOBILE (5.57) 5.56 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

19.95 kHz - 20.05 kHz

STANDARD FREQUENCY AND TIME SIGNAL (20 KHZ)	STANDARD FREQUENCY AND TIME SIGNAL (20 KHZ)	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

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20.05 kHz - 70 kHz

FIXED MARITIME MOBILE (5.57) 5.56 5.58	FIXED MARITIME MOBILE (5.57) 5.56 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

70 kHz - 72 kHz

RADIONAVIGATION (5.60)	RADIONAVIGATION (5.60) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

72 kHz - 84 kHz

FIXED MARITIME MOBILE (5.57) RADIONAVIGATION (5.60) 5.56	FIXED MARITIME MOBILE (5.57) RADIONAVIGATION (5.60) 5.56 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		Standard frequency and time signal			77.5 kHz DCF time signal
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

84 kHz - 86 kHz

RADIONAVIGATION (5.60)	RADIONAVIGATION (5.60) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

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86 kHz - 90 kHz

FIXED MARITIME MOBILE (5.57) RADIONAVIGATION 5.56	FIXED MARITIME MOBILE (5.57) RADIONAVIGATION 5.56 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

90 kHz - 110 kHz

RADIONAVIGATION (5.62) Fixed 5.64	RADIONAVIGATION (5.62) Fixed 5.64 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

110 kHz - 112 kHz

FIXED MARITIME MOBILE RADIONAVIGATION 5.64	FIXED MARITIME MOBILE RADIONAVIGATION 5.64 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

112 kHz - 115 kHz

RADIONAVIGATION (5.60)	RADIONAVIGATION (5.60) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

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115 kHz - 117.6 kHz

RADIONAVIGATION (5.60) Fixed Maritime mobile 5.64 5.66	RADIONAVIGATION (5.60) Fixed Maritime mobile 5.64 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

117.6 kHz - 126 kHz

FIXED MARITIME MOBILE RADIONAVIGATION (5.60) 5.64	FIXED MARITIME MOBILE RADIONAVIGATION (5.60) 5.64 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

126 kHz - 129 kHz

RADIONAVIGATION (5.60)	RADIONAVIGATION (5.60) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

129 kHz - 130 kHz

FIXED MARITIME MOBILE RADIONAVIGATION (5.60) 5.64	FIXED MARITIME MOBILE RADIONAVIGATION (5.60) 5.64 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

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130 kHz - 135.7 kHz

FIXED MARITIME MOBILE 5.64 5.67	FIXED MARITIME MOBILE 5.64 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

135.7 kHz - 137.8 kHz

FIXED (5.64) MARITIME MOBILE Amateur (5.67A) 5.67B	FIXED (5.64) MARITIME MOBILE Amateur (5.67A) 5.67B ECA36	Amateur		EN 301 783	Within the band 135.7-137.8 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

137.8 kHz - 148.5 kHz

FIXED MARITIME MOBILE 5.64 5.67	FIXED MARITIME MOBILE 5.64 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330, EN 303 447, EN 303 454	Within the band 9-148.5 kHz
		Land military systems			
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

148.5 kHz - 255 kHz

BROADCASTING 5.68 5.69 5.70	BROADCASTING	Broadcasting		EN 302 017, EN 302 245	Frequency Assignment plan GE75. Digital systems to be introduced
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

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255 kHz - 283.5 kHz

AERONAUTICAL RADIONAVIGATION BROADCASTING 5.70	AERONAUTICAL RADIONAVIGATION BROADCASTING ECA36	Aeronautical military systems			
		Beacons (aeronautical)			Frequency Assignment plan GE85
		Broadcasting		EN 302 017, EN 302 245	Frequency Assignment plan GE75. Digital systems to be introduced
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

283.5 kHz - 315 kHz

AERONAUTICAL RADIONAVIGATION MARITIME RADIONAVIGATION (5.73) 5.74	AERONAUTICAL RADIONAVIGATION MARITIME RADIONAVIGATION (5.73) 5.74 ECA36	Aeronautical military systems			
		Beacons (aeronautical)			Frequency Assignment plan GE85
		Beacons (maritime)			Frequency Assignment plan GE85
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime military systems			
		ULP-AMI	ERC/REC 70-03	EN 302 195	Within the band 9-315 kHz

315 kHz - 325 kHz

AERONAUTICAL RADIONAVIGATION Maritime radionavigation (5.73) 5.75	AERONAUTICAL RADIONAVIGATION Maritime radionavigation (5.73) ECA36	Aeronautical military systems			
		Beacons (aeronautical)			Frequency Assignment plan GE85
		Beacons (maritime)			Frequency Assignment plan GE85. IALA - plan to allow differential GPS
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime military systems			
		ULP-AID	ERC/REC 70-03	EN 302 536	

325 kHz - 405 kHz

AERONAUTICAL RADIONAVIGATION	AERONAUTICAL RADIONAVIGATION ECA36	Aeronautical military systems			
		Beacons (aeronautical)			Frequency Assignment plan GE85
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz.
		RFID	ERC/REC 70-03	EN 300 330	within frequency range 400-600 kHz
		ULP-AID	ERC/REC 70-03	EN 302 536	

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405 kHz - 415 kHz

RADIONAVIGATION (5.76)	RADIONAVIGATION (5.76) ECA36	Aeronautical military systems			
		Beacons (aeronautical)			Frequency Assignment plan GE85
		Beacons (maritime)			Frequency Assignment plan GE85. IALA - plan to allow differential GPS
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz.
		Maritime military systems			
		RFID	ERC/REC 70-03	EN 300 330	within frequency range 400-600 kHz
		ULP-AID	ERC/REC 70-03	EN 302 536	

415 kHz - 435 kHz

AERONAUTICAL RADIONAVIGATION MARITIME MOBILE (5.79)	AERONAUTICAL RADIONAVIGATION MARITIME MOBILE (5.79) ECA36	Aeronautical military systems			
		Beacons (aeronautical)			Frequency Assignment plan GE85
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz.
		Maritime communications		EN 300 338	Frequency Assignment plan GE85
		Maritime military systems			
		RFID	ERC/REC 70-03	EN 300 330	within frequency range 400-600 kHz
		ULP-AID	ERC/REC 70-03	EN 302 536	

435 kHz - 472 kHz

MARITIME MOBILE (5.79) Aeronautical radionavigation (5.77) 5.82	MARITIME MOBILE (5.79) Aeronautical radionavigation 5.82 ECA36	Aeronautical military systems			
		Emergency detection	ERC/REC 70-03	EN 300 330, EN 300 718	Emergency detection is only with the band 456.9-457.1 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz.
		Maritime communications		EN 300 338	Frequency Assignment plan GE85
		Maritime military systems			
		RFID	ERC/REC 70-03	EN 300 330	within frequency range 400-600 kHz
		ULP-AID	ERC/REC 70-03	EN 302 536	

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472 kHz - 479 kHz

MARITIME MOBILE (5.79) Aeronautical radionavigation (5.77 5.80) Amateur (5.80A) 5.80B 5.82	MARITIME MOBILE (5.79) Aeronautical radionavigation Amateur (5.80A) 5.80B 5.82 ECA36	Aeronautical military systems			
		Amateur		EN 301 783	
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz.
		Maritime communications		EN 300 338	Frequency Assignment plan GE85
		Maritime military systems			
		RFID	ERC/REC 70-03	EN 300 330	within frequency range 400-600 kHz
		ULP-AID	ERC/REC 70-03	EN 302 536	

479 kHz - 495 kHz

MARITIME MOBILE (5.79 5.79A) Aeronautical radionavigation (5.77) 5.82	MARITIME MOBILE (5.79 5.79A) Aeronautical radionavigation 5.82 ECA36	Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz.
		Maritime communications		EN 300 338	Frequency Assignment plan GE85
		Maritime military systems			
		NAVTEX		EN 300 065	490 kHz: NAVTEX transmission in national language
		RFID	ERC/REC 70-03	EN 300 330	within frequency range 400-600 kHz
		ULP-AID	ERC/REC 70-03	EN 302 536	

495 kHz - 505 kHz

MARITIME MOBILE (5.82C 5.82D)	MOBILE ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz.
		Maritime military systems			
		RFID	ERC/REC 70-03	EN 300 330	within frequency range 400-600 kHz
		ULP-AID	ERC/REC 70-03	EN 302 536	

505 kHz - 526.5 kHz

AERONAUTICAL RADIONAVIGATION MARITIME MOBILE (5.79 5.79A 5.84)	AERONAUTICAL RADIONAVIGATION MARITIME MOBILE (5.79 5.79A 5.84) ECA36	Aeronautical military systems			
		Beacons (aeronautical)			Frequency Assignment plan GE85
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz.
		Maritime communications		EN 300 338	Frequency Assignment plan GE85
		Maritime military systems			
		NAVTEX		EN 300 065	518 kHz: NAVTEX transmission in national language
		ULP-AID	ERC/REC 70-03	EN 302 536	

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526.5 kHz - 1606.5 kHz

BROADCASTING 5.87 5.87A	BROADCASTING	Broadcasting		EN 302 017, EN 302 245	Frequency Assignment plan GE75. Digital systems to be introduced
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz.
		RFID	ERC/REC 70-03	EN 300 330	within frequency range 400-600 kHz
		ULP-AID	ERC/REC 70-03	EN 302 536	within frequency range 315-600 kHz

1606.5 kHz - 1625 kHz

FIXED LAND MOBILE MARITIME MOBILE (5.90) 5.92	FIXED LAND MOBILE MARITIME MOBILE (5.90) Radiolocation ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime communications		EN 303 402	Frequency Assignment plan GE85
		Maritime military systems			

1625 kHz - 1635 kHz

RADIOLOCATION 5.93	RADIOLOCATION 5.93 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Radiolocation (military)			

1635 kHz - 1800 kHz

FIXED LAND MOBILE MARITIME MOBILE (5.90) 5.92 5.96	FIXED LAND MOBILE MARITIME MOBILE (5.90) 5.96 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime communications		EN 303 402	Frequency Assignment plan GE85
		Maritime military systems			

1800 kHz - 1810 kHz

RADIOLOCATION 5.93	RADIOLOCATION 5.93 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Radiolocation (military)			

1810 kHz - 1850 kHz

AMATEUR 5.98 5.99 5.100	AMATEUR 5.98 5.100	Amateur		EN 301 783	Within the band 1810-2000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

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1850 kHz - 2000 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE 5.92 5.96 5.103	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE Amateur 5.96 5.103 ECA36	Amateur		EN 301 783	Within the band 1810-2000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime communications		EN 303 402	
		Maritime military systems			

2000 kHz - 2025 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.92 5.103	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.103 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime communications		EN 303 402	
		Maritime military systems			

2025 kHz - 2045 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) Meteorological aids (5.104) 5.92 5.103	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.103 5.104 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime communications		EN 303 402	
		Maritime military systems			
		Oceanographic buoys			Meteorological

2045 kHz - 2160 kHz

FIXED LAND MOBILE MARITIME MOBILE 5.92	FIXED LAND MOBILE MARITIME MOBILE 5.92 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime communications		EN 303 402	Frequency Assignment plan GE85
		Maritime military systems			

2160 kHz - 2170 kHz

RADIOLOCATION 5.93 5.107	RADIOLOCATION 5.93 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Radiolocation (military)			

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2170 kHz - 2173.5 kHz

MARITIME MOBILE	MARITIME MOBILE ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime communications		EN 303 402	
		Maritime military systems			

2173.5 kHz - 2190.5 kHz

MOBILE (DISTRESS AND CALLING) 5.108 5.109 5.110 5.111	MOBILE (DISTRESS AND CALLING) 5.108 5.109 5.110 5.111 ECA36	DSC		EN 302 885, EN 303 402	2187.5 kHz (DSC for distress and calling)
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime communications		EN 303 402	2182 kHz (Radiotelephony distress and calling). 2174.5 kHz (Telex distress traffic)

2190.5 kHz - 2194 kHz

MARITIME MOBILE	MARITIME MOBILE ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime communications		EN 303 402	
		Maritime military systems			

2194 kHz - 2300 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.92 5.103 5.112	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.103 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime communications		EN 303 402	
		Maritime military systems			

2300 kHz - 2498 kHz

BROADCASTING (5.113) FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.103	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.103 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime communications		EN 303 402	
		Maritime military systems			

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2498 kHz - 2501 kHz

STANDARD FREQUENCY AND TIME SIGNAL (2 500 KHZ)	STANDARD FREQUENCY AND TIME SIGNAL (2 500 KHZ)	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
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2501 kHz - 2502 kHz

STANDARD FREQUENCY AND TIME SIGNAL Space research	STANDARD FREQUENCY AND TIME SIGNAL Space research	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
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2502 kHz - 2625 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.92 5.103 5.114	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.92 5.103 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

2625 kHz - 2650 kHz

MARITIME MOBILE MARITIME RADIONAVIGATION 5.92	MARITIME MOBILE MARITIME RADIONAVIGATION 5.92 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime communications		EN 303 402	
		Maritime military systems			

2650 kHz - 2850 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.92 5.103	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.92 5.103 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

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2850 kHz - 3025 kHz

AERONAUTICAL MOBILE (R) 5.111 5.115	AERONAUTICAL MOBILE (OR) 5.111 5.115 ECA36	Aeronautical communications			Appendix 27 Allotment Plan
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		SAR (communications)		EN 303 402	3023 kHz (Aeronautical/Maritime radiotelephony SAR coordination)

3025 kHz - 3155 kHz

AERONAUTICAL MOBILE (OR)	AERONAUTICAL MOBILE (OR) ECA36	Aeronautical communications			Appendix 26 Allotment Plan
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

3155 kHz - 3200 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.116 5.117	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.116 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 3155-3400 kHz; and within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime communications		EN 303 402	
		Maritime military systems			

3200 kHz - 3230 kHz

BROADCASTING (5.113) FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.116	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.116 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 3155-3400 kHz; and within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime communications		EN 303 402	
		Maritime military systems			

3230 kHz - 3400 kHz

BROADCASTING (5.113) FIXED MOBILE EXCEPT AERONAUTICAL MOBILE 5.116 5.118	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE 5.116 ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 3155-3400 kHz; and within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime communications		EN 303 402	
		Maritime military systems			

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3400 kHz - 3500 kHz

AERONAUTICAL MOBILE (R)	AERONAUTICAL MOBILE (R) ECA36	Aeronautical communications			Appendix 27 Allotment Plan. Including HF Data Links
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

3500 kHz - 3800 kHz

AMATEUR FIXED MOBILE EXCEPT AERONAUTICAL MOBILE 5.92	AMATEUR FIXED MOBILE EXCEPT AERONAUTICAL MOBILE 5.92 ECA36	Amateur		EN 301 783	
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime communications		EN 303 402	
		Maritime military systems			

3800 kHz - 3900 kHz

AERONAUTICAL MOBILE (OR) FIXED LAND MOBILE	AERONAUTICAL MOBILE (OR) FIXED LAND MOBILE ECA36	Aeronautical communications			Appendix 26 Allotment Plan
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

3900 kHz - 3950 kHz

AERONAUTICAL MOBILE (OR) 5.123	AERONAUTICAL MOBILE (OR) ECA36	Aeronautical communications			Appendix 26 Allotment Plan
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

3950 kHz - 4000 kHz

BROADCASTING FIXED	BROADCASTING FIXED ECA36	Broadcasting		EN 302 017, EN 302 245	Digital systems to be introduced
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

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4000 kHz - 4063 kHz

FIXED MARITIME MOBILE (5.127) 5.126	FIXED MARITIME MOBILE (5.127) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime communications		EN 302 885, EN 303 402	Appendix 17 channelling plan. Appendix 25 allotment plan
		Maritime military systems			

4063 kHz - 4438 kHz

MARITIME MOBILE (5.79A 5.109 5.110 5.82D 5.130 5.131 5.132) 5.128	MARITIME MOBILE (5.109 5.110 5.130 5.131 5.132 5.79A) 5.128 ECA36	DSC		EN 302 885, EN 303 402	centre frequency 4207.5 kHz (DSC distress traffic). Ship stations centre frequencies 4208, 4208.5, 4209 kHz. Coast stations 4219.5, 4220, 4220.5 kHz (DSC calling)
		Eurobalise	ERC/REC 70-03	EN 302 608	centre frequency at 4234 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime communications		EN 303 402	Appendix 17 channelling plan. Appendix 25 allotment plan. centre frequency 4125 kHz (Radiotelephony distress and safety traffic. centre frequency 4177.5 kHz (Telex distress traffic). 4209.5 kHz (Meteorological and navigational warnings. centre frequency 4210 kHz (Safety Information)
		Maritime military systems			
		NAVTEX		EN 300 065	centre frequency 4209.5 kHz

4438 kHz - 4488 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) Radiolocation (5.132A) 5.132B	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) Radiolocation (5.132A) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			
		Radiolocation (military)			

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4488 kHz - 4650 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R)	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

4650 kHz - 4700 kHz

AERONAUTICAL MOBILE (R)	AERONAUTICAL MOBILE (R) ECA36	Aeronautical communications			Appendix 27 Allotment Plan. Including HF Data Links
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

4700 kHz - 4750 kHz

AERONAUTICAL MOBILE (OR)	AERONAUTICAL MOBILE (OR) ECA36	Aeronautical communications			Appendix 26 Allotment Plan
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

4750 kHz - 4850 kHz

AERONAUTICAL MOBILE (OR) BROADCASTING (5.113) FIXED LAND MOBILE	AERONAUTICAL MOBILE (OR) FIXED LAND MOBILE ECA36	Aeronautical communications			
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

4850 kHz - 4995 kHz

BROADCASTING (5.113) FIXED LAND MOBILE	FIXED LAND MOBILE ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

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4995 kHz - 5003 kHz

STANDARD FREQUENCY AND TIME SIGNAL (5 000 KHZ)	STANDARD FREQUENCY AND TIME SIGNAL (5 000 KHZ)	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
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5003 kHz - 5005 kHz

STANDARD FREQUENCY AND TIME SIGNAL Space research	STANDARD FREQUENCY AND TIME SIGNAL Space research	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
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5005 kHz - 5060 kHz

BROADCASTING (5.113) FIXED	FIXED	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
	ECA36	Land military systems			

5060 kHz - 5250 kHz

FIXED Mobile except aeronautical mobile 5.133	FIXED Mobile except aeronautical mobile ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

5250 kHz - 5275 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE Radiolocation (5.132A) 5.133A	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE Radiolocation (5.132A) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			
		Radiolocation (military)			

5275 kHz - 5351.5 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

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5351.5 kHz - 5366.5 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE Amateur (5.133B)	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE Amateur (5.133B) ECA36	Amateur		EN 301 783	
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

5366.5 kHz - 5450 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

5450 kHz - 5480 kHz

AERONAUTICAL MOBILE (OR) FIXED LAND MOBILE	AERONAUTICAL MOBILE (OR) FIXED LAND MOBILE ECA36	Aeronautical communications			
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

5480 kHz - 5680 kHz

AERONAUTICAL MOBILE (R) 5.111 5.115	AERONAUTICAL MOBILE (OR) 5.111 5.115 ECA36	Aeronautical communications			Appendix 27 Allotment Plan Including HF Data Links
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		SAR (communications)		EN 303 402	5680 kHz (Aeronautical/Maritime radiotelephony SAR coordination)

5680 kHz - 5730 kHz

AERONAUTICAL MOBILE (OR) 5.111 5.115	AERONAUTICAL MOBILE (OR) 5.111 5.115 ECA36	Aeronautical communications			Appendix 26 Allotment Plan
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		SAR (communications)		EN 303 402	5680 kHz (Aeronautical/Maritime radiotelephony SAR coordination)

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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5730 kHz - 5900 kHz

FIXED LAND MOBILE	FIXED LAND MOBILE ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

5900 kHz - 5950 kHz

BROADCASTING (5.134) 5.136	BROADCASTING (5.134) 5.136	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced.
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

5950 kHz - 6200 kHz

BROADCASTING	BROADCASTING	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

6200 kHz - 6525 kHz

MARITIME MOBILE (5.109 5.110 5.130 5.132 5.137A) 5.137	MARITIME MOBILE (5.109 5.110 5.130 5.132 5.137A) 5.137 ECA36	DSC		EN 302 885, EN 303 402	6312 kHz (DSC distress traffic). 6312.5, 6313, 6313.5, 6331, 6331.5, 6332 kHz (DSC calling)
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime communications		EN 303 402	Appendix 17 channelling plan. Appendix 25 allotment plan. 6215 kHz. (Radiotelephony distress and safety traffic). 6268 kHz (Telex distress traffic). 6314 kHz (Maritime Safety Information)
		Maritime military systems			

6525 kHz - 6685 kHz

AERONAUTICAL MOBILE (R)	AERONAUTICAL MOBILE (R) ECA36	Aeronautical communications			Appendix 27 Allotment Plan. Including HF Data Links
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

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6685 kHz - 6765 kHz

AERONAUTICAL MOBILE (OR)	AERONAUTICAL MOBILE (OR) ECA36	Aeronautical communications			Appendix 26 Allotment Plan
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

6765 kHz - 7000 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.138	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.138 ECA36	ISM			Within the band 6765-6795 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 6765-6795 kHz; and within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

7000 kHz - 7100 kHz

AMATEUR AMATEUR-SATELLITE 5.140 5.141 5.141A	AMATEUR AMATEUR-SATELLITE	Amateur		EN 301 783	Within the band 7000-7200 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

7100 kHz - 7200 kHz

AMATEUR 5.141A 5.141B	AMATEUR	Amateur		EN 301 783	Within the band 7000-7200 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

7200 kHz - 7300 kHz

BROADCASTING	BROADCASTING	Broadcasting		EN 302 017, EN 302 245	RR - Article 12 planning procedure. Digital systems to be introduced. Within the band 7200-7450 kHz.
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

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7300 kHz - 7400 kHz

BROADCASTING (5.134) 5.143 5.143A 5.143B 5.143C 5.143D	BROADCASTING (5.134) 5.143 5.143B	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced. Within the band 7200-7450 kHz.
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

7400 kHz - 7450 kHz

BROADCASTING 5.143B 5.143C	BROADCASTING 5.143B	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced. Within the band 7200-7450 kHz.
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 7400-8800 kHz; and within the band 148.5 kHz - 30 MHz

7450 kHz - 8100 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.144	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 7400-8800 kHz; and within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

8100 kHz - 8195 kHz

FIXED MARITIME MOBILE	FIXED MARITIME MOBILE ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 7400-8800 kHz; and within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime communications		EN 303 402	Appendix 17 channelling plan
		Maritime military systems			

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8195 kHz - 8815 kHz

MARITIME MOBILE (5.109 5.110 5.132 5.145 5.137A) 5.111	MARITIME MOBILE (5.109 5.110 5.132 5.137A 5.145) 5.111 ECA36	DSC		EN 302 885, EN 303 402	8414.5 kHz (DSC distress traffic). 8415, 8415.5, 8416, 8436.5, 8437, 8437.5 kHz (DSC calling)
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 7400-8800 kHz; and within the band 148.5 kHz - 30 MHz
		Maritime communications		EN 303 402	Appendix 17 channelling plan. Appendix 25 allotment plan. 8291 kHz (Radiotelephony distress and safety traffic).8376.5 kHz (Telex distress traffic). 8416.5 kHz (Maritime Safety Information)
		Maritime military systems			

8815 kHz - 8965 kHz

AERONAUTICAL MOBILE (R)	AERONAUTICAL MOBILE (R) ECA36	Aeronautical communications			Appendix 27 Allotment Plan1 Including HF Data Links
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

8965 kHz - 9040 kHz

AERONAUTICAL MOBILE (OR)	AERONAUTICAL MOBILE (OR) ECA36	Aeronautical communications			Appendix 26 Allotment Plan
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

9040 kHz - 9305 kHz

FIXED	FIXED ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

9305 kHz - 9355 kHz

FIXED Radiolocation (5.145A) 5.145B	FIXED Radiolocation (5.145A) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

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9355 kHz - 9400 kHz

FIXED	FIXED	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
	ECA36	Land military systems			

9400 kHz - 9500 kHz

BROADCASTING (5.134) 5.146	BROADCASTING (5.134) 5.146	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Within 9400-9900 kHz.
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

9500 kHz - 9900 kHz

BROADCASTING 5.147	BROADCASTING 5.147	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. . Within 9400-9900 kHz.
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

9900 kHz - 9995 kHz

FIXED	FIXED	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
	ECA36	Land military systems			

9995 kHz - 10003 kHz

STANDARD FREQUENCY AND TIME SIGNAL (10 000 KHZ) 5.111	STANDARD FREQUENCY AND TIME SIGNAL (10 000 KHZ) 5.111	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
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10003 kHz - 10005 kHz

STANDARD FREQUENCY AND TIME SIGNAL Space research 5.111	STANDARD FREQUENCY AND TIME SIGNAL Space research 5.111	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		SAR (communications)			10003 kHz (+/-3 kHz) concerning manned space vehicles

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10005 kHz - 10100 kHz

AERONAUTICAL MOBILE (R) 5.111	AERONAUTICAL MOBILE (R) 5.111 ECA36	Aeronautical communications			Appendix 27 Allotment Plan. Including HF Data Links
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

10100 kHz - 10150 kHz

FIXED Amateur	FIXED Amateur ECA36	Amateur		EN 301 783	
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

10150 kHz - 11175 kHz

FIXED Mobile except aeronautical mobile (R)	FIXED Mobile except aeronautical mobile (R) ECA36	Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 10200-11000 kHz; and within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

11175 kHz - 11275 kHz

AERONAUTICAL MOBILE (OR)	AERONAUTICAL MOBILE (OR) ECA36	Aeronautical communications			Appendix 26 Allotment Plan
		Aeronautical military systems			
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

11275 kHz - 11400 kHz

AERONAUTICAL MOBILE (R)	AERONAUTICAL MOBILE (R) ECA36	Aeronautical communications			Appendix 27 Allotment Plan. Including HF Data Links
		Aeronautical military systems			
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

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11400 kHz - 11600 kHz

FIXED	FIXED ECA36	Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

11600 kHz - 11650 kHz

BROADCASTING (5.134) 5.146	BROADCASTING (5.134) 5.146	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced. Within frequency range 11600-12100 kHz
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

11650 kHz - 12050 kHz

BROADCASTING 5.147	BROADCASTING 5.147	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced. Within frequency range 11600-12100 kHz
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz

12050 kHz - 12100 kHz

BROADCASTING (5.134) 5.146	BROADCASTING (5.134) 5.146	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced. Within frequency range 11600-12100 kHz
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz

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12100 kHz - 12230 kHz

FIXED	FIXED ECA36	Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz

12230 kHz - 13200 kHz

MARITIME MOBILE (5.109 5.110 5.132 5.145 5.137A)	MARITIME MOBILE (5.109 5.110 5.132 5.137A 5.145) ECA36	DSC		EN 302 885, EN 303 402	Centre frequency 12577 kHz (DSC distress traffic). Centre frequencies 12577.5, 12578, 12578.5, 12657, 12657.5, 12658 kHz (DSC calling)
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime communications		EN 303 402	Appendix 17 channelling plan. Appendix 25 allotment plan. Centre frequency 12290 kHz (Radiotelephony distress and safety traffic). centre frequency 12520 kHz (Telex distress traffic). 12579 kHz (Maritime Safety Information)
		Maritime military systems			
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

13200 kHz - 13260 kHz

AERONAUTICAL MOBILE (OR)	AERONAUTICAL MOBILE (OR) ECA36	Aeronautical communications			Appendix 26 Allotment Plan
		Aeronautical military systems			
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

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13260 kHz - 13360 kHz

AERONAUTICAL MOBILE (R)	AERONAUTICAL MOBILE (R) ECA36	Aeronautical communications			Appendix 27 Allotment Plan. Including HF Data Links
		Aeronautical military systems			
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		RFID		EN 300 330	within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

13360 kHz - 13410 kHz

FIXED RADIO ASTRONOMY 5.149	FIXED RADIO ASTRONOMY 5.149 ECA36	Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		Radio astronomy			Continuum observations
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

13410 kHz - 13450 kHz

FIXED Mobile except aeronautical mobile (R)	FIXED Mobile except aeronautical mobile (R) ECA36	Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

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13450 kHz - 13550 kHz

FIXED Mobile except aeronautical mobile (R) Radiolocation (5.132A) 5.149A	FIXED Mobile except aeronautical mobile (R) Radiolocation (5.132A) ECA36	Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz. Centre frequency 13.547 MHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

13550 kHz - 13570 kHz

FIXED Mobile except aeronautical mobile (R) 5.150	FIXED Mobile except aeronautical mobile (R) 5.150 ECA36	Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		ISM			Within the band 13553-13567 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 13553-13567 kHz; and within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			
		Non-specific SRDs	ERC/REC 70-03	EN 300 330	Within the band 13553-13567 kHz
		RFID		EN 300 330	Centre frequency is 13.56 MHz. Within frequency range 11.810-15.310 MHz.
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

13570 kHz - 13600 kHz

BROADCASTING (5.134) 5.151	BROADCASTING (5.134) 5.151	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

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13600 kHz - 13800 kHz

BROADCASTING	BROADCASTING	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

13800 kHz - 13870 kHz

BROADCASTING (5.134) 5.151	BROADCASTING (5.134) 5.151	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

13870 kHz - 14000 kHz

FIXED Mobile except aeronautical mobile (R)	FIXED Mobile except aeronautical mobile (R) ECA36	Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

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14000 kHz - 14250 kHz

AMATEUR AMATEUR-SATELLITE	AMATEUR AMATEUR-SATELLITE	Amateur		EN 301 783	Within the band 14000-14350 kHz
		Amateur-satellite			
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

14250 kHz - 14350 kHz

AMATEUR 5.152	AMATEUR	Amateur		EN 301 783	Within the band 14000-14350 kHz
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

14350 kHz - 14990 kHz

FIXED Mobile except aeronautical mobile (R)	FIXED Mobile except aeronautical mobile (R) ECA36	Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

14990 kHz - 15005 kHz

STANDARD FREQUENCY AND TIME SIGNAL (15 000 KHZ) 5.111	STANDARD FREQUENCY AND TIME SIGNAL (15 000 KHZ) 5.111	Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		SAR (communications)			14993 kHz (+/-3 kHz) concerning manned space vehicles
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

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15005 kHz - 15010 kHz

STANDARD FREQUENCY AND TIME SIGNAL Space research	STANDARD FREQUENCY AND TIME SIGNAL Space research	Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

15010 kHz - 15100 kHz

AERONAUTICAL MOBILE (OR)	AERONAUTICAL MOBILE (OR) ECA36	Aeronautical communications			Appendix 26 Allotment Plan
		Aeronautical military systems			
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		RFID		EN 300 330	Within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

15100 kHz - 15600 kHz

BROADCASTING	BROADCASTING	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		RFID		EN 300 330	within frequency range 11.810-15.310 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

15600 kHz - 15800 kHz

BROADCASTING (5.134) 5.146	BROADCASTING (5.134) 5.146	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced
		Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

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15800 kHz - 16100 kHz

FIXED	FIXED ECA36	Euroloop	ERC/REC 70-03	EN 302 609	Mainly within the band 11100-16000 kHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

16100 kHz - 16200 kHz

FIXED Radiolocation (5.145A) 5.145B	FIXED Radiolocation (5.145A) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

16200 kHz - 16360 kHz

FIXED	FIXED ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		ULP-AID		EN 300 330	Within frequency range 12500-20000 kHz

16360 kHz - 17410 kHz

MARITIME MOBILE (5.109 5.110 5.132 5.145 5.137A)	MARITIME MOBILE (5.109 5.110 5.132 5.137A 5.145) ECA36	DSC		EN 302 885, EN 303 402	16804.5 kHz (DSC distress traffic).16805, 16805.5, 16806, 16903, 16903.5, 16904 kHz (DSC calling)
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime communications		EN 303 402	Appendix 17 channelling plan. Appendix 25 allotment plan.16420 kHz (Radiotelephony distress and safety traffic).16695 kHz (Telex distress traffic).16806.5 kHz (Maritime Safety Information)
		Maritime military systems			
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

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17410 kHz - 17480 kHz

FIXED	FIXED ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		ULP-AID		EN 300 330	Within frequency range 12500-20000 kHz

17480 kHz - 17550 kHz

BROADCASTING (5.134) 5.146	BROADCASTING (5.134) 5.146	Broadcasting		EN 302 017, EN 302 245	Digital systems to be introduced
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

17550 kHz - 17900 kHz

BROADCASTING	BROADCASTING	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

17900 kHz - 17970 kHz

AERONAUTICAL MOBILE (R)	AERONAUTICAL MOBILE (R) ECA36	Aeronautical communications			Appendix 27 Allotment Plan. Including HF Data Links
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

17970 kHz - 18030 kHz

AERONAUTICAL MOBILE (OR)	AERONAUTICAL MOBILE (OR) ECA36	Aeronautical communications			Appendix 26 Allotment Plan
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

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18030 kHz - 18052 kHz

FIXED	FIXED ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

18052 kHz - 18068 kHz

FIXED Space research	FIXED Space research ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

18068 kHz - 18168 kHz

AMATEUR AMATEUR-SATELLITE 5.154	AMATEUR AMATEUR-SATELLITE	Amateur		EN 301 783	
		Amateur-satellite			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

18168 kHz - 18780 kHz

FIXED Mobile except aeronautical mobile	FIXED Mobile except aeronautical mobile ECA36	DSC		EN 302 885, EN 303 402	Centre frequencies at 18898.5, 18899. 18899.5 kHz (DSC) digital selective calling)
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

18780 kHz - 18900 kHz

MARITIME MOBILE	MARITIME MOBILE ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime communications		EN 303 402	Appendix 17 channelling plan
		Maritime military systems			
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

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18900 kHz - 19020 kHz

BROADCASTING (5.134) 5.146	BROADCASTING (5.134) 5.146	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

19020 kHz - 19680 kHz

FIXED	FIXED ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

19680 kHz - 19800 kHz

MARITIME MOBILE (5.132)	MARITIME MOBILE (5.132) ECA36	DSC		EN 302 885, EN 303 402	19703.5, 19704, 19704.5 kHz (DSC calling)
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime communications		EN 303 402	Appendix 17 channelling plan. Appendix 25 allotment plan. 19680.5 kHz (Maritime Safety Information)
		Maritime military systems			
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

19800 kHz - 19990 kHz

FIXED	FIXED ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		ULP-AID	ERC/REC 70-03	EN 300 330	Within frequency range 12500-20000 kHz

19990 kHz - 19995 kHz

STANDARD FREQUENCY AND TIME SIGNAL Space research 5.111	STANDARD FREQUENCY AND TIME SIGNAL Space research 5.111	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		SAR (communications)			19993 kHz (+/-3 kHz) concerning manned space vehicles
		ULP-AID		EN 300 330	Within frequency range 12500-20000 kHz

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19995 kHz - 20010 kHz

STANDARD FREQUENCY AND TIME SIGNAL (20 000 KHZ) 5.111	STANDARD FREQUENCY AND TIME SIGNAL (20 000 KHZ) 5.111	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		ULP-AID		EN 300 330	Within frequency range 12500-20000 kHz

20010 kHz - 21000 kHz

FIXED Mobile	FIXED Mobile ECA36	Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

21000 kHz - 21450 kHz

AMATEUR AMATEUR-SATELLITE	AMATEUR AMATEUR-SATELLITE	Amateur		EN 301 783	
		Amateur-satellite			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

21450 kHz - 21850 kHz

BROADCASTING	BROADCASTING	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

21850 kHz - 21870 kHz

FIXED (5.155A) 5.155	FIXED ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

21870 kHz - 21924 kHz

FIXED (5.155B)	FIXED (5.155B) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

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21924 kHz - 22000 kHz

AERONAUTICAL MOBILE (R)	AERONAUTICAL MOBILE (R) ECA36	Aeronautical communications			Appendix 27 Allotment Plan. Including HF Data Links.
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

22000 kHz - 22855 kHz

MARITIME MOBILE (5.132 5.137A) 5.156	MARITIME MOBILE (5.132 5.137A) ECA36	DSC		EN 302 885, EN 303 402	22374.5, 22375, 22375.5, 22444, 22444.5, 22445 kHz (DSC calling)
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime communications		EN 303 402	Appendix 17 channelling plan. Appendix 25 allotment plan. 22376 kHz safety information.
		Maritime military systems			

22855 kHz - 23000 kHz

FIXED 5.156	FIXED ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

23000 kHz - 23200 kHz

FIXED Mobile except aeronautical mobile (R) 5.156	FIXED Mobile except aeronautical mobile (R) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

23200 kHz - 23350 kHz

AERONAUTICAL MOBILE (OR) FIXED (5.156A)	AERONAUTICAL MOBILE (OR) FIXED (5.156A) ECA36	Aeronautical communications			
		Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

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23350 kHz - 24000 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.157)	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.157) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

24000 kHz - 24450 kHz

FIXED LAND MOBILE	FIXED LAND MOBILE ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

24450 kHz - 24600 kHz

FIXED LAND MOBILE Radiolocation (5.132A) 5.158	FIXED LAND MOBILE Radiolocation (5.132A) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

24600 kHz - 24890 kHz

FIXED LAND MOBILE	FIXED LAND MOBILE ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			

24890 kHz - 24990 kHz

AMATEUR AMATEUR-SATELLITE	AMATEUR AMATEUR-SATELLITE	Amateur		EN 301 783	
		Amateur-satellite			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

24990 kHz - 25005 kHz

STANDARD FREQUENCY AND TIME SIGNAL (25 000 KHZ)	STANDARD FREQUENCY AND TIME SIGNAL (25 000 KHZ)	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
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25005 kHz - 25010 kHz

STANDARD FREQUENCY AND TIME SIGNAL Space research	STANDARD FREQUENCY AND TIME SIGNAL Space research ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Space research			Scientific and medical space research

25010 kHz - 25070 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

25070 kHz - 25210 kHz

MARITIME MOBILE	MARITIME MOBILE	DSC		EN 302 885, EN 303 402	25208.5, 25209, 25209.5 kHz (DSC calling)
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime communications		EN 303 402	Appendix 17 channelling plan
		Maritime military systems			

25210 kHz - 25550 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

25550 kHz - 25670 kHz

RADIO ASTRONOMY 5.149	RADIO ASTRONOMY 5.149	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Radio astronomy			Continuum observations

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25670 kHz - 26100 kHz

BROADCASTING	BROADCASTING	Broadcasting		EN 302 017, EN 302 245	RR-Article 12 planning procedure. Digital systems to be introduced.
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

26100 kHz - 26175 kHz

MARITIME MOBILE (5.132)	MARITIME MOBILE (5.132) ECA36	DSC		EN 302 885, EN 303 402	26121, 26121.5, 26122 kHz (DSC calling)
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Maritime communications		EN 303 402	Appendix 17 channelling plan. Appendix 25 allotment plan. 26100.5 kHz Maritime Safety Information.
		Maritime military systems			

26175 kHz - 26200 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

26200 kHz - 26350 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE Radiolocation (5.132A) 5.133A	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE Radiolocation (5.132A) ECA36	Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

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26350 kHz - 27500 kHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE 5.150	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE 5.150 ECA36	CB radio	ECC/DEC/(11)03, ERC/REC 70-03	EN 300 433	(CEPT PR 27). Within the band 26.960-27.410 MHz
		Eurobalise	ERC/REC 70-03	EN 302 608	Centre frequency 27.095 MHz
		ISM			Within the band 26.957-27.283 MHz
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			
		Model control	ERC/REC 70-03	EN 300 220	26.995, 27.045, 27.095, 27.145, 27.195 MHz
		Non-specific SRDs	ERC/REC 70-03	EN 300 220, EN 300 330	Within the band 26.957-27.283 MHz

27500 kHz - 28 MHz

FIXED METEOROLOGICAL AIDS MOBILE	FIXED METEOROLOGICAL AIDS MOBILE ECA36	Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			

28 MHz - 29.7 MHz

AMATEUR AMATEUR-SATELLITE	AMATEUR AMATEUR-SATELLITE	Amateur		EN 301 783	
		Amateur-satellite			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz

29.7 MHz - 30.005 MHz

FIXED MOBILE	MOBILE ECA36	Aeronautical military systems			
		Inductive applications	ERC/REC 70-03	EN 300 330	Within the band 148.5 kHz - 30 MHz
		Land military systems			
		Maritime military systems			
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis.
		ULP-MMI	ERC/REC 70-03	EN 302 510	Within the band 30.0-30.005 MHz

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30.005 MHz - 30.01 MHz

FIXED MOBILE SPACE OPERATION (SATELLITE IDENTIFICATION) SPACE RESEARCH	MOBILE ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis
		ULP-MMI	ERC/REC 70-03	EN 302 510	

30.01 MHz - 37.5 MHz

FIXED MOBILE	MOBILE ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		Model control	ERC/DEC/(01)11, ERC/REC 70-03	EN 300 220	Within the band 34.995-35.225 MHz only for flying models
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis.
		ULP-MMI	ERC/REC 70-03	EN 302 510	

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37.5 MHz - 38.25 MHz

FIXED MOBILE Radio astronomy 5.149	MOBILE Radio astronomy 5.149 ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Radio astronomy			Continuum observations
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis.

38.25 MHz - 39 MHz

FIXED MOBILE	MOBILE ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis

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39 MHz - 39.5 MHz

FIXED MOBILE Radiolocation (5.132A) 5.159	MOBILE Radiolocation (5.132A) ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		Meteor scatter communications	ERC/REC/(00)04		Within the band 39.0-39.2 MHz
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis.

39.5 MHz - 39.986 MHz

FIXED MOBILE	MOBILE ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		Meteor scatter communications			Within the band 39.0-39.2 MHz
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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39.986 MHz - 40 MHz

FIXED MOBILE Space research	MOBILE Space research ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis

40 MHz - 40.02 MHz

FIXED MOBILE Earth exploration-satellite (5.159A) Space research	MOBILE Earth exploration-satellite (5.159A) Space research ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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40.02 MHz - 40.98 MHz

FIXED MOBILE Earth exploration-satellite (5.159A) 5.150	MOBILE Earth exploration-satellite (5.159A) 5.150 ECA36	Aeronautical military systems			
		ISM			Within the band 40.66-40.7 MHz
		Land military systems			
		Maritime military systems			
		Model control	ERC/DEC/(01)12, ERC/REC 70-03	EN 300 220	Centre frequencies 40.665, 40.675, 40.685, 40.695 MHz
		Non-specific SRDs	ERC/REC 70-03	EN 300 220	
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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40.98 MHz - 41.015 MHz

FIXED MOBILE Earth exploration-satellite (5.159A) Space research 5.160 5.161	MOBILE Space research ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis

41.015 MHz - 42 MHz

FIXED MOBILE Earth exploration-satellite (5.159A) 5.160 5.161 5.161A	MOBILE ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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42 MHz - 42.5 MHz

FIXED MOBILE Earth exploration-satellite (5.159A) Radiolocation (5.132A) 5.160 5.161B	FIXED MOBILE Radiolocation (5.132A) 5.161B ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis

42.5 MHz - 44 MHz

FIXED MOBILE Earth exploration-satellite (5.159A) 5.160 5.161 5.161A	MOBILE ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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44 MHz - 47 MHz

FIXED MOBILE Earth exploration-satellite (5.159A) 5.162 5.162A	MOBILE 5.162A ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis
		Wind profilers			In the range 46-68 MHz, geographical sharing with other services

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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47 MHz - 50 MHz

BROADCASTING Earth exploration-satellite (5.159A) 5.162A 5.163 5.164 5.165	LAND MOBILE 5.162A 5.164 ECA36	Earth exploration-satellite			In the range 48.5-50 MHz. Space Research/EESS
		Land military systems			
		On-site paging		EN 300 224	On site paging in the band 47.0-47.25 MHz
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Wind profilers			In the range 46-68 MHz, geographical sharing with other services

50 MHz - 52 MHz

BROADCASTING Amateur (5.166C 5.166E 5.166B) 5.162A 5.164 5.165 5.166A 5.169A 5.169B	LAND MOBILE Amateur 5.162A 5.164 5.166A 5.169B ECA36	Amateur		EN 301 783	
		Land military systems			
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Wind profilers			In the range 46-68 MHz, geographical sharing with other services

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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52 MHz - 68 MHz

BROADCASTING 5.162A 5.163 5.163 5.164 5.169A 5.169B	LAND MOBILE 5.162A 5.163 5.164 ECA36	Land military systems			
		PMR	T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	Mobile station transmit band in 68-74.8 MHz paired with base station transmit band in 77.8-84.6 MHz
		Wind profilers			In the range 46-68 MHz, geographical sharing with other services

68 MHz - 70.45 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE 5.175	MOBILE Amateur ECA9 ECA36	Amateur		EN 301 783	Within the band 69.9-70.5 MHz
		Land military systems			
		Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	Mobile station transmit band in 68-74.8 MHz paired with base station transmit band in 77.8-84.6 MHz

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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70.45 MHz - 74.8 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE 5.149 5.175 5.177 5.178 5.179	MOBILE EXCEPT AERONAUTICAL MOBILE Amateur Radio astronomy 5.149 ECA9 ECA36	Amateur		EN 301 783	Within the band 69.9-70.5 MHz
		Land military systems			
		Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	Mobile station transmit band in 68-74.8 MHz paired with base station transmit band in 77.8-84.6 MHz
		Radio astronomy			Continuum observations (inter alia solar wind monitoring in 73-74.6 MHz)

74.8 MHz - 75.2 MHz

AERONAUTICAL RADIONAVIGATION 5.180	AERONAUTICAL RADIONAVIGATION 5.180	ILS			Marker beacons
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75.2 MHz - 87.5 MHz

FIXED Mobile except aeronautical mobile 5.175 5.179 5.187	MOBILE ECA36	Land military systems			
		Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	Mobile station transmit band in 75.2-77.7 MHz paired with base station transmit band in 85.0-87.5 MHz Base station transmit band in 77.8-84.6 MHz paired with mobile station transmit band in 68-74.8 MHz

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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87.5 MHz - 100 MHz

BROADCASTING 5.190	BROADCASTING	FM sound analogue		EN 302 018, EN 303 345	Geneva Agreement GE84
		Wireless audio/multimedia	ERC/REC 70-03	EN 301 357	Within the band 87.5-108.0 MHz

100 MHz - 108 MHz

BROADCASTING 5.192 5.194	BROADCASTING	FM sound analogue		EN 302 018, EN 303 345	Geneva Agreement GE84
		Wireless audio/multimedia	ERC/REC 70-03	EN 301 357	Within the band 87.5-108.0 MHz

108 MHz - 117.975 MHz

AERONAUTICAL RADIONAVIGATION 5.197 5.197A	AERONAUTICAL MOBILE (R) AERONAUTICAL RADIONAVIGATION 5.197A	Aeronautical communications		EN 301 842	Safety and regularity of flights, below 112 MHz limited to ground based data link transmitters
		GBAS		EN 303 084	GBAS/VDB within 112-117.975 MHz
		ILS			Localiser within the band 108-112 MHz
		VOR			Within the band 108-117.975 MHz

117.975 MHz - 137 MHz

AERONAUTICAL MOBILE (R) AERONAUTICAL MOBILE-SATELLITE (5.198A 5.198B) 5.111 5.200 5.201 5.202	AERONAUTICAL MOBILE (R) 5.111 5.200 5.201 5.202 ECA5	-		EN 300 676, EN 301 841, EN 302 961	Maritime Personal Homing Beacon for search and rescue purposes. 123.1 MHz.
		Aeronautical communications		EN 300 676, EN 301 841, EN 301 842	Safety and regularity of flights. EN 301 841-3 is for ground-based equipment. 121.5 MHz. Aeronautical mobile distress communication.
		EPIRBs		EN 300 152	Band only available for distress and safety.

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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137 MHz - 137.025 MHz

METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE-SATELLITE (5.208A 5.208B 5.209) SPACE OPERATION (5.203C) SPACE RESEARCH (SPACE-TO-EARTH) Fixed Mobile except aeronautical mobile (R) 5.204 5.205 5.206 5.207 5.208	METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE MOBILE-SATELLITE (5.208A 5.208B 5.209) SPACE OPERATION (SPACE-TO-EARTH) SPACE RESEARCH (5.203C) 5.206 5.208 ECA6 ECA36	Aeronautical military systems			
		Land military systems			
		Land mobile			Mobile restricted to Aeronautical Mobile (OR), including air sport
		MSS Earth stations	ERC/DEC/(99)06	EN 301 721	Non-geostationary
		Maritime military systems			
		Satellite systems (military)			
		Weather satellites			

137.025 MHz - 137.175 MHz

METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) SPACE OPERATION (5.203C) SPACE RESEARCH (SPACE-TO-EARTH) Fixed Mobile except aeronautical mobile (R) Mobile-Satellite (5.208A 5.208B 5.209) 5.204 5.205 5.206 5.207 5.208	METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE SPACE OPERATION (SPACE-TO-EARTH) SPACE RESEARCH (5.203C) Mobile-Satellite (5.208A 5.208B 5.209) 5.206 5.208 ECA6 ECA36	Aeronautical military systems			
		Land military systems			
		Land mobile			Mobile restricted to Aeronautical Mobile (OR), including air sport
		MSS Earth stations	ERC/DEC/(99)06	EN 301 721	Non-geostationary
		Maritime military systems			
		Satellite systems (military)			
		Weather satellites			

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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137.175 MHz - 137.825 MHz

METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE-SATELLITE (5.208A 5.208B 5.209) SPACE OPERATION (5.209A 5.203C) SPACE RESEARCH (SPACE-TO-EARTH) Fixed Mobile except aeronautical mobile (R) 5.204 5.205 5.206 5.207 5.208	METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE MOBILE-SATELLITE (5.208A 5.208B 5.209) SPACE OPERATION (5.203C 5.209A) SPACE RESEARCH (SPACE-TO-EARTH) 5.206 5.208 ECA6 ECA36	Aeronautical military systems			
		Land military systems			
		Land mobile			Mobile restricted to Aeronautical Mobile (OR), including air sport
		MSS Earth stations	ERC/DEC/(99)06	EN 301 721	Non-geostationary
		Maritime military systems			
		Satellite systems (military)			
		Weather satellites			

137.825 MHz - 138 MHz

METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) SPACE OPERATION (5.203C) SPACE RESEARCH (SPACE-TO-EARTH) Fixed Mobile except aeronautical mobile (R) Mobile-Satellite (5.208A 5.208B 5.209) 5.204 5.205 5.206 5.207 5.208	METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE SPACE OPERATION (5.203C) SPACE RESEARCH (SPACE-TO-EARTH) Mobile-Satellite (5.208A 5.208B 5.209) 5.206 5.208 ECA6 ECA36	Aeronautical military systems			
		Land military systems			
		Land mobile			Mobile restricted to Aeronautical Mobile (OR), including air sport
		MSS Earth stations	ERC/DEC/(99)06	EN 301 721	Non-geostationary
		Maritime military systems			
		Satellite systems (military)			
		Weather satellites			

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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138 MHz - 143.6 MHz

AERONAUTICAL MOBILE (OR) 5.210 5.211 5.212 5.214	AERONAUTICAL MOBILE (OR) LAND MOBILE Space research (space-to-Earth) 5.211 ECA5 ECA36	Aeronautical military systems			
		Land military systems			
		Land mobile			
		Maritime military systems			
		Non-specific SRDs	ERC/REC 70-03	EN 300 220	Within the band 138.20-138.45 MHz

143.6 MHz - 143.65 MHz

AERONAUTICAL MOBILE (OR) SPACE RESEARCH (SPACE-TO-EARTH) 5.211 5.212 5.214	AERONAUTICAL MOBILE (OR) LAND MOBILE SPACE RESEARCH (SPACE-TO-EARTH) 5.211 ECA5 ECA36	Aeronautical military systems			
		Land military systems			
		Land mobile			
		Maritime military systems			

143.65 MHz - 144 MHz

AERONAUTICAL MOBILE (OR) 5.210 5.211 5.212 5.214	AERONAUTICAL MOBILE (OR) LAND MOBILE 5.211 ECA5 ECA36	Aeronautical military systems			
		Land military systems			
		Land mobile			
		Maritime military systems			

144 MHz - 146 MHz

AMATEUR AMATEUR-SATELLITE 5.216	AMATEUR AMATEUR-SATELLITE	Amateur		EN 301 783	
		Amateur-satellite			

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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146 MHz - 148 MHz

FIXED Mobile except aeronautical mobile (R)	MOBILE ECA7 ECA36	Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	

148 MHz - 149.9 MHz

FIXED MOBILE-SATELLITE (5.209) Mobile except aeronautical mobile (R) 5.218 5.218A 5.219 5.221	MOBILE MOBILE-SATELLITE (5.209) 5.218 5.218A 5.219 5.221 ECA6 ECA7 ECA36	MSS Earth stations	ERC/DEC/(99)06	EN 301 721	Non-geostationary
		Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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149.9 MHz - 150.05 MHz

MOBILE-SATELLITE (5.209 5.220)	MOBILE MOBILE-SATELLITE (5.209 5.220) ECA6 ECA36	MSS Earth stations	ERC/DEC/(99)06	EN 301 721	Non-geostationary
		Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	Single frequency applications

150.05 MHz - 153 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE RADIO ASTRONOMY 5.149	MOBILE EXCEPT AERONAUTICAL MOBILE RADIO ASTRONOMY 5.149 ECA7 ECA36	Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	
		Radio astronomy			Continuum observations (inter-alia solar research)

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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153 MHz - 154 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) Meteorological aids	MOBILE EXCEPT AERONAUTICAL MOBILE (R) ECA7 ECA36	Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	Base station transmit paired with 148.4-149.4 MHz

154 MHz - 156.4875 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.225A 5.226	MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.226 ECA7 ECA8 ECA36	Maritime communications	ECC/DEC/(19)03	EN 300 162, EN 300 698, EN 301 025, EN 301 178, EN 301 929	RR Appendix 18
		Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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156.4875 MHz - 156.5125 MHz

MARITIME MOBILE (DISTRESS AND CALLING VIA DSC) 5.226 5.227	MARITIME MOBILE (DISTRESS AND CALLING VIA DSC) 5.226 5.227 ECA7 ECA8 ECA36	Maritime communications	ECC/DEC/(19)03	EN 300 162, EN 300 698, EN 301 025, EN 301 178, EN 301 929	RR Appendix 18
		Maritime military systems			

156.5125 MHz - 156.5375 MHz

MARITIME MOBILE (DISTRESS AND CALLING VIA DSC) 5.111 5.226	MARITIME MOBILE (DISTRESS AND CALLING VIA DSC) 5.111 5.226 ECA36	AMRD Group A	ECC/DEC/(22)02		
		DSC	ECC/DEC/(19)03	EN 301 025, EN 301 929, EN 302 885, EN 303 132	RR Appendix 18. Distress, safety and calling 156.525 MHz.
		Maritime military systems			

156.5375 MHz - 156.5625 MHz

MARITIME MOBILE (DISTRESS AND CALLING VIA DSC) 5.226 5.227	MARITIME MOBILE (DISTRESS AND CALLING VIA DSC) MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.226 5.227 ECA7 ECA8 ECA36	Maritime communications	ECC/DEC/(19)03	EN 300 162, EN 300 698, EN 301 025, EN 301 178, EN 301 929	RR Appendix 18
		Maritime military systems			

156.5625 MHz - 156.7625 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.226	MOBILE EXCEPT AERONAUTICAL MOBILE (R) 5.226 ECA7 ECA8 ECA36	Maritime communications	ECC/DEC/(19)03	EN 300 162, EN 300 698, EN 301 025, EN 301 178, EN 301 929	RR Appendix 18
		Maritime military systems			

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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156.7625 MHz - 156.7875 MHz

MARITIME MOBILE Mobile-Satellite (Earth-to-space) 5.111 5.226 5.228	MARITIME MOBILE (DISTRESS AND CALLING) 5.111 5.226 5.228 ECA36	Maritime communications	ECC/DEC/(19)03	EN 301 929	RR Appendix 18. Satellite AIS Earth-to-space
		Maritime military systems			

156.7875 MHz - 156.8125 MHz

MARITIME MOBILE (DISTRESS AND CALLING) 5.111 5.226	MARITIME MOBILE (DISTRESS AND CALLING) 5.111 5.226 ECA36	Maritime communications	ECC/DEC/(19)03	EN 300 162	RR Appendix 18. Distress, safety and calling 156.8 MHz for the maritime mobile VHF radiotelephone service.
		Maritime military systems			

156.8125 MHz - 156.8375 MHz

MARITIME MOBILE Mobile-Satellite (Earth-to-space) 5.111 5.226 5.228	MARITIME MOBILE 5.111 5.226 5.228 ECA36	Maritime communications	ECC/DEC/(19)03	EN 301 929	RR Appendix 18. Satellite AIS Earth-to-space.
		Maritime military systems			

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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156.8375 MHz - 157.1875 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.226)	MOBILE EXCEPT AERONAUTICAL MOBILE (5.226 ECA7 ECA8) ECA36	Maritime communications	ECC/DEC/(19)03	EN 300 162, EN 300 698, EN 301 025, EN 301 178, EN 301 929	RR Appendix 18
		Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	

157.1875 MHz - 157.3375 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.226) Maritime mobile-satellite (5.208A 5.208B 5.228AC 5.228AB)	Maritime mobile-satellite (5.208A 5.208B 5.228AB 5.228AC) Mobile except aeronautical mobile (5.226) ECA7 ECA8 ECA36	Maritime communications	ECC/DEC/(19)03	EN 300 162, EN 300 698, EN 301 025, EN 301 178, EN 301 929	RR Appendix 18
		Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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157.3375 MHz - 161.7875 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.226)	MOBILE EXCEPT AERONAUTICAL MOBILE (5.226) ECA7 ECA8 ECA36	AMRD Group B	ECC/DEC/(22)02		Within frequency range 160.8875-160.9125 MHz
		Maritime communications	ECC/DEC/(19)03	EN 300 162, EN 300 698, EN 301 025, EN 301 178, EN 301 929	RR Appendix 18
		Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	

161.7875 MHz - 161.9375 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.226) Maritime mobile-satellite (5.208A 5.228AC 5.228AB 5.208B)	MOBILE EXCEPT AERONAUTICAL MOBILE (5.226) Maritime mobile-satellite (5.208A 5.208B 5.228AB 5.228AC) ECA7 ECA8 ECA36	Maritime communications	ECC/DEC/(19)03	EN 300 162, EN 300 698, EN 301 025, EN 301 178, EN 301 929	RR Appendix 18
		Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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161.9375 MHz - 161.9625 MHz

FIXED MARITIME MOBILE-SATELLITE (5.228AA) MOBILE EXCEPT AERONAUTICAL MOBILE 5.226	MOBILE EXCEPT AERONAUTICAL MOBILE Maritime mobile-satellite (5.228AA) 5.226 ECA7 ECA8 ECA36	Maritime communications	ECC/DEC/(19)03	EN 300 162, EN 300 698, EN 301 025, EN 301 178, EN 301 929	RR Appendix 18
		Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	

161.9625 MHz - 161.9875 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.226) Mobile-Satellite (5.228F) 5.226 5.228A 5.228B	MOBILE EXCEPT AERONAUTICAL MOBILE Mobile-Satellite (5.228F) 5.226 ECA7 ECA8	AIS		EN 303 098	Centre frequency 161.975 MHz
		AMRD Group A	ECC/DEC/(22)02		
		Maritime communications	ECC/DEC/(19)03	EN 300 162, EN 300 698, EN 301 025, EN 301 178, EN 301 929	RR Appendix 18
		Maritime military systems			

161.9875 MHz - 162.0125 MHz

FIXED MARITIME MOBILE-SATELLITE (5.228AA) MOBILE EXCEPT AERONAUTICAL MOBILE 5.226	MOBILE EXCEPT AERONAUTICAL MOBILE Maritime mobile-satellite (5.228AA) 5.226 ECA7 ECA8 ECA36	Maritime communications	ECC/DEC/(19)03	EN 300 162, EN 300 698, EN 301 025, EN 301 178, EN 301 929	RR Appendix 18
		Maritime military systems			

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162.0125 MHz - 162.0375 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE Mobile-Satellite (5.228F) 5.226 5.228A 5.228B	MOBILE EXCEPT AERONAUTICAL MOBILE 5.226 ECA7 ECA8 ECA36	AIS		EN 303 098	Centre frequency 162.025 MHz
		AMRD Group A	ECC/DEC/(22)02		
		Maritime communications	ECC/DEC/(19)03	EN 300 162, EN 300 698, EN 301 025, EN 301 178, EN 301 929	RR Appendix 18
		Maritime military systems			

162.0375 MHz - 174 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE 5.226	Mobile except aeronautical mobile 5.226 ECA7 ECA36	ALD	ECC/DEC/(05)02, ERC/REC 70-03	EN 300 422	The bands 169.400-169.475 MHz and 169.4875-169.5875 MHz.
		Maritime military systems			
		Meter reading	ECC/DEC/(05)02, ERC/REC 70-03	EN 300 220	Within the band 169.400-169.475 MHz
		Non-specific SRDs	ECC/DEC/(05)02, ERC/REC 70-03	EN 300 220	
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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174 MHz - 223 MHz

BROADCASTING 5.235	BROADCASTING LAND MOBILE 5.235	Audio PMSE	ERC/REC 25-10	EN 300 454	Radio microphones and In-ear monitors on a tuning range basis within 174-216 MHz
		Broadcasting (terrestrial)		EN 302 077, EN 302 296	Geneva Agreement 2006. TV Broadcasting T-DAB.
		Radio microphones and ALD	ERC/REC 25-10, ERC/REC 70-03	EN 300 422	On a tuning range basis within 174-216 MHz

223 MHz - 225 MHz

BROADCASTING Fixed Mobile 5.243 5.246 5.247	BROADCASTING	Broadcasting (terrestrial)		EN 302 077, EN 302 296	Geneva Agreement 2006. TV Broadcasting, T-DAB
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225 MHz - 230 MHz

BROADCASTING Fixed Mobile 5.246 5.247	BROADCASTING Land mobile ECA10 ECA36	Broadcasting (terrestrial)		EN 302 077, EN 302 296	Geneva Agreement 2006. This band is within the military tuning range 225-400 MHz. Sharing with defence on national basis. TV Broadcasting, T-DAB.
		Defence systems			

230 MHz - 235 MHz

FIXED MOBILE 5.247 5.251 5.252	MOBILE ECA10 ECA36	Defence systems			
		T-DAB			T-DAB sharing with defence on a national basis. Wiesbaden 1995 Special Arrangement, as revised in Constanta, 2007. Within the frequency range 230-240 MHz.

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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235 MHz - 240 MHz

FIXED MOBILE 5.252 5.254	MOBILE 5.254 ECA10 ECA36	Defence systems			
		T-DAB			T-DAB sharing with defence on a national basis. Wiesbaden 1995 Special Arrangement, as revised in Constanta, 2007. Within the frequency range 230-240 MHz

240 MHz - 242.95 MHz

FIXED MOBILE 5.111 5.254 5.256	MOBILE 5.254 ECA10 ECA36	Defence systems		EN 302 617	
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242.95 MHz - 243.05 MHz

FIXED MOBILE 5.111 5.254 5.256	AERONAUTICAL MOBILE 5.111 5.254 5.256	EPIRBs		EN 300 152	Band only available for distress and safety purposes 243.0 MHz
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243.05 MHz - 267 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE 5.111 5.252 5.254 5.256 5.256A	MOBILE 5.254 ECA10 ECA36	Defence systems		EN 302 617	
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267 MHz - 272 MHz

FIXED MOBILE Space Operation (space-to-Earth) 5.254 5.257	MOBILE 5.254 5.257 ECA10 ECA36	Defence systems		EN 302 617	
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272 MHz - 273 MHz

FIXED MOBILE SPACE OPERATION (SPACE-TO-EARTH) 5.254	MOBILE 5.254 ECA10 ECA36	Defence systems		EN 302 617	
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273 MHz - 312 MHz

FIXED MOBILE 5.254	MOBILE 5.254 ECA10 ECA36	Defence systems		EN 302 617	
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312 MHz - 315 MHz

FIXED MOBILE Mobile-Satellite (5.254 5.255)	MOBILE 5.254 5.255 ECA10 ECA36	Defence systems		EN 302 617	
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315 MHz - 322 MHz

FIXED MOBILE 5.254	MOBILE 5.254 ECA10 ECA36	Defence systems		EN 302 617	
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322 MHz - 328.6 MHz

FIXED MOBILE RADIO ASTRONOMY 5.149	MOBILE RADIO ASTRONOMY 5.149 ECA10 ECA36	Defence systems			
		Radio astronomy			Continuum and spectral line observations (e.g. deuterium), VLBI

328.6 MHz - 335.4 MHz

AERONAUTICAL RADIONAVIGATION 5.258 5.259	AERONAUTICAL RADIONAVIGATION 5.258	ILS			Glide path
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RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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335.4 MHz - 380 MHz

FIXED MOBILE 5.254	MOBILE 5.254 ECA7 ECA10 ECA36	Defence systems		EN 302 617	
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380 MHz - 385 MHz

FIXED MOBILE 5.254	MOBILE 5.254 ECA10 ECA36	Defence systems			
		PPDR	ECC/DEC/(06)05, ECC/DEC/(08)05, ERC/DEC/(01)19, T/R 25-08	EN 300 113, EN 301 449, EN 301 502, EN 301 511, EN 301 526, EN 302 426, EN 302 561	Within the bands 384.8-385.0 and 394.8-395.0 MHz for AGA, 384.750-384.800 MHz and 394.750-394.800 MHz may be used as preferred extension bands for AGA. Within the bands 380-380.15 and 390-390.15 MHz for DMO. Mobile station transmit paired with 390-395 MHz. PPDR sharing with defence applications. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05.

385 MHz - 387 MHz

FIXED MOBILE 5.254	MOBILE 5.254 ECA10 ECA36	Defence systems			
		PPDR	ECC/DEC/(08)05, T/R 25-08	EN 300 113, EN 301 449, EN 301 502, EN 301 511, EN 301 526, EN 302 426, EN 302 561	Mobile station transmit paired with 395-397 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05.

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387 MHz - 390 MHz

FIXED MOBILE Mobile-Satellite (5.208A 5.254 5.255 5.208B)	MOBILE ECA10 ECA36	Defence systems			
		PPDR	ECC/DEC/(08)05, T/R 25-08	EN 300 113, EN 301 449, EN 301 502, EN 301 511, EN 301 526, EN 302 426, EN 302 561	Single frequency applications in 389.9-390 MHz. Mobile station transmit paired with 397.0-399.9 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05.

390 MHz - 395 MHz

FIXED MOBILE 5.254	MOBILE 5.254 ECA10 ECA36	Defence systems			
		PPDR	ECC/DEC/(06)05, ECC/DEC/(08)05, ERC/DEC/(01)19, T/R 25-08	EN 300 113, EN 301 449, EN 301 502, EN 301 511, EN 301 526, EN 302 426, EN 302 561	Within the bands 384.8-385.0 and 394.8-395.0 MHz for AGA, 384.750-384.800 MHz and 394.750-394.800 MHz may be used as preferred extension bands. Within the bands 380-380.15 and 390-390.15 MHz for DMO. Base station transmit paired with 380-385 MHz. PPDR sharing with defence applications. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05.

395 MHz - 399.9 MHz

FIXED MOBILE 5.254	MOBILE 5.254 ECA10 ECA36	Defence systems			
		PPDR	ECC/DEC/(08)05, T/R 25-08	EN 300 113, EN 301 449, EN 301 502, EN 301 511, EN 301 526, EN 302 426, EN 302 561	Base station transmit paired with 385.0-389.9 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05.

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399.9 MHz - 400.05 MHz

MOBILE-SATELLITE (5.220 5.209 5.260A 5.260B)	MOBILE-SATELLITE (5.209 5.220)	MSS Earth stations	ERC/DEC/(99)05, ERC/DEC/(99)06	EN 301 721	
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400.15 MHz - 401 MHz

METEOROLOGICAL AIDS METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE-SATELLITE (5.208A 5.208B 5.209) SPACE RESEARCH (5.263) Space Operation (space-to-Earth) 5.262 5.264	METEOROLOGICAL AIDS METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE-SATELLITE (5.208A 5.208B 5.209) SPACE OPERATION (SPACE-TO-EARTH) SPACE RESEARCH (5.263) 5.262 5.264	MSS Earth stations	ERC/DEC/(99)05, ERC/DEC/(99)06	EN 301 721	Non-geostationary
		Sondes		EN 302 054	
		Weather satellites			

401 MHz - 402 MHz

EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE) METEOROLOGICAL AIDS METEOROLOGICAL-SATELLITE (EARTH-TO-SPACE) SPACE OPERATION (SPACE-TO-EARTH) Fixed Mobile except aeronautical mobile 5.264A 5.264B	EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE) METEOROLOGICAL AIDS METEOROLOGICAL-SATELLITE (EARTH-TO-SPACE) 5.264A 5.264B	Sondes		EN 302 054	
		ULP-AMI	ERC/DEC/(01)17, ERC/REC 70-03	EN 302 537	
		Weather satellites			Data collection platform telemetry

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402 MHz - 403 MHz

EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE) METEOROLOGICAL AIDS METEOROLOGICAL-SATELLITE (EARTH-TO-SPACE) Fixed Mobile except aeronautical mobile 5.264A 5.264B	EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE) METEOROLOGICAL AIDS METEOROLOGICAL-SATELLITE (EARTH-TO-SPACE) 5.264A 5.264B	Sondes		EN 302 054	
		ULP-AMI	ERC/DEC/(01)17, ERC/REC 70-03	EN 301 839	
		Weather satellites			Data collection platform telemetry

403 MHz - 406 MHz

METEOROLOGICAL AIDS Fixed Mobile except aeronautical mobile 5.265	METEOROLOGICAL AIDS 5.265	Sondes		EN 302 054	
		ULP-AMI	ERC/DEC/(01)17, ERC/REC 70-03	EN 301 839, EN 302 537	

406 MHz - 406.1 MHz

MOBILE-SATELLITE (EARTH-TO-SPACE) 5.265 5.266 5.267	MOBILE-SATELLITE (EARTH-TO-SPACE) 5.265 5.266 5.267	EPIRBs		EN 300 066, EN 302 152	Band only available for distress and safety purposes
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406.1 MHz - 410 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE RADIO ASTRONOMY 5.149 5.265	LAND MOBILE RADIO ASTRONOMY 5.149 5.265 ECA36	Land military systems			
		Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	Single frequency applications. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05
		Radio astronomy			Continuum observations, VLBI

410 MHz - 420 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE SPACE RESEARCH (5.268)	MOBILE EXCEPT AERONAUTICAL MOBILE ECA36	Land military systems			
		Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 301 908, EN 302 561, EN 303 039	Mobile station transmit paired with 420-430 MHz.
		PPDR	ECC/DEC/(08)05, ECC/DEC/(16)02, T/R 25-08	EN 300 113, EN 301 449, EN 301 502, EN 301 511, EN 301 526, EN 301 908, EN 302 426, EN 302 561	Mobile station transmit paired with 420-430 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05.

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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420 MHz - 430 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE Radiolocation 5.269 5.270 5.271	MOBILE EXCEPT AERONAUTICAL MOBILE Radiolocation ECA7 ECA36	Land military systems			
		Maritime military systems			
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 301 908, EN 302 561, EN 303 039	Base station transmit paired with 410-420 MHz.
		PPDR	ECC/DEC/(08)05, ECC/DEC/(16)02, T/R 25-08	EN 300 113, EN 301 449, EN 301 502, EN 301 511, EN 301 526, EN 301 908, EN 302 426, EN 302 561	Base station transmit paired with 410-420 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05.
		Radiolocation (military)			

430 MHz - 432 MHz

AMATEUR RADIOLOCATION 5.271 5.274 5.275 5.276 5.277	AMATEUR RADIOLOCATION ECA12 ECA36	Amateur		EN 301 783	Within the band 430-440 MHz
		Radiolocation (military)			
		ULP-WMCE	ERC/REC 70-03	EN 303 520	Within the band 430-440 MHz

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432 MHz - 433.05 MHz

AMATEUR RADIOLOCATION Earth exploration-satellite (5.279A) 5.138 5.271 5.276 5.277 5.280	AMATEUR RADIOLOCATION Earth exploration-satellite (5.279A) ECA12 ECA36	Active sensors (satellite)			The use of this band by sensors in the EESS (active) shall be in accordance with Recommendation ITU-R SA 1260-1
		Amateur		EN 301 783	Within the band 430-440 MHz
		Radiolocation (military)			
		ULP-WMCE	ERC/REC 70-03	EN 303 520	Within the band 430-440 MHz

433.05 MHz - 434.79 MHz

AMATEUR RADIOLOCATION Earth exploration-satellite (5.279A) 5.138 5.271 5.276 5.277 5.280 5.281	AMATEUR RADIOLOCATION Earth exploration-satellite (5.279A) Land mobile 5.138 5.280 ECA12 ECA36	Active sensors (satellite)			The use of this band by sensors in the EESS (active) shall be in accordance with Recommendation ITU-R SA 1260-1
		Amateur		EN 301 783	Within the band 430-440 MHz
		ISM			
		Non-specific SRDs	ERC/REC 70-03	EN 300 220	
		Radiolocation (military)			
		ULP-WMCE	ERC/REC 70-03	EN 303 520	Within the band 430-440 MHz

434.79 MHz - 438 MHz

AMATEUR RADIOLOCATION Earth exploration-satellite (5.279A) 5.138 5.271 5.276 5.277 5.280 5.282	AMATEUR AMATEUR-SATELLITE RADIOLOCATION Earth exploration-satellite (5.279A) ECA12 ECA36	Active sensors (satellite)			The use of this band by sensors in the EESS (active) shall be in accordance with Recommendation ITU-R SA 1260-1
		Amateur		EN 301 783	Within the band 430-440 MHz
		Amateur-satellite			Amateur Satellite Service restricted to 435-438 MHz
		Radiolocation (military)			
		ULP-WMCE	ERC/REC 70-03	EN 303 520	Within the band 430-440 MHz

438 MHz - 440 MHz

AMATEUR RADIOLOCATION 5.271 5.274 5.275 5.276 5.277 5.283	AMATEUR RADIOLOCATION ECA12 ECA36	Amateur		EN 301 783	Within the band 430-440 MHz
		Radiolocation (military)			
		ULP-WMCE	ERC/REC 70-03	EN 303 520	Within the band 430-440 MHz

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440 MHz - 450 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE Radiolocation 5.269 5.270 5.271 5.284 5.285 5.286	MOBILE EXCEPT AERONAUTICAL MOBILE Radiolocation ECA7 ECA36	Land military systems			
		Maritime military systems			
		On-site paging		EN 300 224	Call-out & answer-back
		PMR 446	ECC/DEC/(15)05, ERC/REC 70-03	EN 303 405	PMR446 in 446.0-446.2 MHz
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 302 561, EN 303 039	Single frequency operation. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05.
		Radiolocation (military)			
		Wind profilers			Geographical sharing with other services

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450 MHz - 455 MHz

FIXED MOBILE (5.286AA) 5.209 5.271 5.286 5.286A 5.286B 5.286C 5.286D 5.286E	MOBILE ECA7 ECA34	On-site paging		EN 300 224	Call-out & answer-back
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 301 908, EN 302 561, EN 303 039	Mobile station transmit paired with 460-465 MHz.
		PPDR	ECC/DEC/(08)05, ECC/DEC/(16)02, T/R 25-08	EN 300 113, EN 301 449, EN 301 502, EN 301 511, EN 301 526, EN 301 908, EN 302 426, EN 302 561	Mobile station transmit paired with 460-465 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05.

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455 MHz - 456 MHz

FIXED MOBILE (5.286AA) 5.209 5.271 5.286A 5.286B 5.286C 5.286E	MOBILE ECA7 ECA34	Land mobile			Existing public cellular networks
		On-site paging		EN 300 224	Call-out & answer-back
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 301 908, EN 302 561, EN 303 039	Mobile station transmit paired with 465-466 MHz.
		PPDR	ECC/DEC/(08)05, ECC/DEC/(16)02, T/R 25-08	EN 300 113, EN 301 449, EN 301 502, EN 301 511, EN 301 526, EN 301 908, EN 302 426, EN 302 561	Mobile station transmit paired with 465-466 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05.

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456 MHz - 459 MHz

FIXED MOBILE (5.286AA) 5.271 5.287 5.288	MOBILE 5.287 ECA7 ECA34	Land mobile			Existing public cellular networks
		On-board communications		EN 300 720	Within 457.5125-457.5875 MHz and 467.5125-467.5875 MHz
		On-site paging		EN 300 224	Call-out & answer-back
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 301 908, EN 302 561, EN 303 039	Mobile station transmit paired with 466-469 MHz.
		PPDR	ECC/DEC/(08)05, ECC/DEC/(16)02, T/R 25-08	EN 300 113, EN 301 449, EN 301 502, EN 301 511, EN 301 526, EN 301 908, EN 302 426, EN 302 561	Mobile station transmit paired with 466-469 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05.

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459 MHz - 460 MHz

FIXED MOBILE (5.286AA) 5.209 5.271 5.286A 5.286B 5.286C 5.286E	MOBILE ECA7	Land mobile			Existing public cellular networks
		On-site paging		EN 300 224	Call-out & answer-back
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 301 908, EN 302 561, EN 303 039	Mobile station transmit paired with 469-470 MHz
		PPDR	ECC/DEC/(08)05, ECC/DEC/(16)02, T/R 25-08	EN 300 113, EN 301 449, EN 301 502, EN 301 511, EN 301 526, EN 301 908, EN 302 426, EN 302 561	Mobile station transmit paired with 469-470 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05.

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460 MHz - 470 MHz

FIXED MOBILE (5.286AA) Meteorological-satellite (space-to-Earth) 5.287 5.288 5.289 5.290	MOBILE 5.287 5.289 ECA7 ECA34	Land mobile			Existing public cellular networks
		On-board communications		EN 300 720	Within 457.5125-457.5875 MHz and 467.5125-467.5875 MHz
		On-site paging		EN 300 224	Call-out & answer-back
		PMR/PAMR	ECC/DEC/(19)02, T/R 25-08	EN 300 086, EN 300 113, EN 300 219, EN 300 296, EN 300 341, EN 300 390, EN 300 471, EN 301 166, EN 301 908, EN 302 561, EN 303 039	Base station transmit paired with 450-460 MHz.
		PPDR	ECC/DEC/(08)05, ECC/DEC/(16)02, T/R 25-08	EN 300 113, EN 301 449, EN 301 502, EN 301 511, EN 301 526, EN 301 908, EN 302 426, EN 302 561	Base station transmit paired with 450-460 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05.
		Space research			Allocation to EESS is via RR 5.289. Data collection platform telecommand. Geographical sharing with other services

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470 MHz - 694 MHz

BROADCASTING 5.149 5.291A 5.294 5.295A 5.296 5.300 5.304 5.306 5.307A 5.307B 5.312	BROADCASTING 5.149 5.291A 5.296 5.306 ECA13	Audio PMSE	ERC/REC 25-10	EN 300 422, EN 300 454	Audio links and Talkback on a tuning range basis
		Broadcasting (terrestrial)		EN 302 296, EN 303 340	Geneva Agreement 2006. TV Broadcasting
		Radio astronomy			Continuum observations, VLBI
		Radio microphones and ALD	ERC/REC 25-10, ERC/REC 70-03	EN 300 422	Within the band 470-789 MHz on a tuning range basis
		Wind profilers			Limited to the band 470-494 MHz. Geographical sharing with other services

694 MHz - 790 MHz

BROADCASTING MOBILE EXCEPT AERONAUTICAL MOBILE (5.312A 5.317A 5.312B) 5.300 5.312	BROADCASTING MOBILE EXCEPT AERONAUTICAL MOBILE (5.312A 5.317A) 5.312 ECA38	Audio PMSE	ERC/REC 25-10	EN 300 422, EN 300 454	Radio microphones and In-ear monitors on a tuning range basis within the band 733-757.5 MHz
		Broadcasting (terrestrial)		EN 302 296, EN 303 340	Geneva Agreement 2006 TV Broadcasting
		MFCN	ECC/DEC/(15)01, ECC/DEC/(22)01, ECC/DEC/(22)07, ECC/REC/(15)01	EN 301 908	Within the band 703-788 MHz
		PPDR	ECC/DEC/(16)02, ECC/REC/(16)03	EN 301 908	BB-PPDR options in 698-703/753-758 MHz, 703-733/758-788 MHz and 733-736/788-791 MHz
		Radio microphones and ALD	ERC/REC 25-10, ERC/REC 70-03	EN 300 422	Within the band 470-703 MHz and 733-757.5 MHz on a tuning range basis

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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790 MHz - 862 MHz

BROADCASTING FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.317A 5.316B 5.312B) 5.312 5.319	MOBILE EXCEPT AERONAUTICAL MOBILE 5.312 5.316B 5.317A ECA13 ECA38	-			Geneva Agreement 2006
		MFCN	ECC/DEC/(09)03, ECC/DEC/(22)01, ECC/DEC/(22)07, ECC/REC/(11)04	EN 301 908	Within the band 791-862 MHz. Aerial UE are permitted in the band 832-862 MHz in ECC Decision (22)07
		PPDR	ECC/DEC/(16)02, ECC/REC/(16)03		BB-PPDR options in 698-703/753-758 MHz, 703-733/758-788 MHz and 733-736/788-791 MHz
		Radio microphones and ALD	ERC/REC 25-10, ERC/REC 70-03	EN 300 422	Within the band 821.5-832 MHz

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
862 MHz - 890 MHz BROADCASTING (5.322) FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.312B 5.317A) 5.319 5.323	MOBILE (5.317A) 5.323 ECA13 ECA29 ECA36	-			This band is identified for IMT in the RRs, but within CEPT this band is not planned for the harmonised introduction of IMT
		Alarms	ERC/REC 70-03	EN 300 220, EN 303 406	Within the bands 868.6-868.7, 869.25-869.4 and 869.65-869.7 MHz
		Audio PMSE	ERC/REC 25-10	EN 300 422	Radio microphones and In-ear monitors within the band 863-865 MHz
		GSM	ECC/REC/(05)08, ECC/REC/(08)02, ERC/DEC/(97)02	EN 301 502, EN 301 511, EN 303 609	Within the band 880-890 MHz paired with 925-935 MHz
		GSM-R	ECC/REC/(05)08	EN 301 502, EN 301 511	Within the band 876-880 MHz paired with 921-925 MHz. Railway systems
		Land military systems			The bands 870-876 MHz and 915-921 MHz are used for land military systems, specifically for unmanned systems.
		MCV	ECC/DEC/(08)08		Within the band 880-915 MHz
		MFCN	ECC/DEC/(06)13, ECC/DEC/(22)01, ECC/DEC/(22)07, ECC/REC/(08)02	EN 301 908	Within the band 880-915 MHz. Aerial UE are permitted in ECC Decision (22)07.
		Maritime military systems			
		Non-specific SRDs	ERC/REC 70-03	EN 300 220	Within the band 862-876 MHz
		RFID	ERC/REC 70-03	EN 302 208	Within the band 865-868 MHz
		RMR	ECC/DEC/(20)02, ECC/REC/(25)02	EN 301 502, EN 301 511	Within the band 874.4-880.0 MHz
		Radio microphones and ALD	ERC/REC 25-10, ERC/REC 70-03	EN 300 422, EN 301 357	Within the band 863-865 MHz
		Social alarms	ERC/REC 70-03	EN 300 220	Within the band 869.2-869.25 MHz
		Telemetry/Telecommand (military)			Within the band 890-915 MHz
		Tracking, tracing and data acquisition	ERC/REC 70-03	EN 303 204, EN 303 659	Within the bands 865-868 MHz and 870-874.4 MHz
		Wideband data transmission systems	ERC/REC 70-03	EN 304 220	Within the band 863-868 MHz

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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890 MHz - 942 MHz

BROADCASTING (5.322) FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.317A 5.312B) Radiolocation 5.323	Mobile (5.317A) Radiolocation 5.323 ECA13 ECA14 ECA29 ECA30 ECA32 ECA36	-			The band 915-925 MHz is identified for IMT in the RRs, but within CEPT this band is not planned for the harmonised introduction of IMT
		GSM	ECC/REC/(05)08, ECC/REC/(08)02, ERC/DEC/(94)01, ERC/DEC/(97)02	EN 301 502, EN 301 511, EN 303 609	Within the band 890-915 MHz paired with 935-960 MHz
		GSM-R	ECC/REC/(05)08	EN 301 502, EN 301 511	Within the bands 876-880 MHz paired with 921-925 MHz
		Land military systems			The bands 870-876 MHz and 915-921 MHz are used for land military systems, specifically for unmanned systems.
		MCV	ECC/DEC/(08)08		Within the band 880-915 MHz and 925-960 MHz
		MFCN	ECC/DEC/(06)13, ECC/DEC/(22)01, ECC/DEC/(22)07, ECC/REC/(08)02	EN 301 908	Within the band 880-915 MHz and 925-960 MHz. Aerial UE are permitted in the band 880-915 MHz in ECC Decision (22)07.
		Maritime military systems			
		Non-specific SRDs	ERC/REC 70-03	EN 300 220	Within the band 915-919.4 MHz
		RFID	ERC/REC 70-03	EN 302 208	Within the band 915-919.4 MHz
		RMR	ECC/DEC/(20)02, ECC/REC/(25)02	EN 301 502, EN 301 511	Within the band 919.4-925 MHz
		Telemetry/Telecommand (military)			
		Tracking, tracing and data acquisition	ERC/REC 70-03	EN 303 659	Within the band 915-919.4 MHz
		Wideband data transmission systems	ERC/REC 70-03	EN 304 220	Within the band 915.8-919.4 MHz

942 MHz - 960 MHz

BROADCASTING (5.322) FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.312B 5.317A) 5.323	MOBILE (5.317A) 5.323 ECA13 ECA29 ECA32	GSM	ECC/REC/(05)08, ECC/REC/(08)02, ERC/DEC/(94)01, ERC/DEC/(97)02	EN 301 502, EN 301 511, EN 303 609	Base station transmit paired with 897-915 MHz
		MCV	ECC/DEC/(08)08		
		MFCN	ECC/DEC/(06)13, ECC/REC/(08)02	EN 301 908	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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960 MHz - 1164 MHz

AERONAUTICAL MOBILE (5.327A) AERONAUTICAL RADIONAVIGATION (5.328) 5.328AA	AERONAUTICAL MOBILE (5.327A) AERONAUTICAL MOBILE-SATELLITE (R) AERONAUTICAL RADIONAVIGATION (5.328) 5.328AA ECA36	Aeronautical			Including DME and SSR
		Aeronautical military systems			Military use includes JTIDS/MIDS and TACAN within 108.7-1092.3 MHz

1164 MHz - 1215 MHz

AERONAUTICAL RADIONAVIGATION (5.328) RADIONAVIGATION-SATELLITE (5.328B) 5.328A	AERONAUTICAL RADIONAVIGATION (5.328) RADIONAVIGATION-SATELLITE (5.328B) 5.328A ECA36	Aeronautical military systems			Military use includes JTIDS/MIDS
		Aeronautical navigation			
		GALILEO		EN 303 413	Within the band 1164-1214 MHz
		GLONASS		EN 303 413	Within the band 1190.3-1213.8 MHz
		GNSS Repeater	ECC/REC/(10)02	EN 302 645	Within the band 1164-1300 MHz
		Satellite systems (military)			

1215 MHz - 1240 MHz

EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION RADIONAVIGATION-SATELLITE (5.328B 5.329 5.329A) SPACE RESEARCH (ACTIVE) 5.330 5.331 5.332	EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION RADIONAVIGATION-SATELLITE (5.328B 5.329 5.329A) SPACE RESEARCH (ACTIVE) 5.331 5.332 ECA36	Active sensors (satellite)			
		GLONASS		EN 303 413	Within the band 1237.8-1253.8 MHz
		GNSS Repeater	ECC/REC/(10)02	EN 302 645	Within the band 1164-1300 MHz
		GPS		EN 303 413	Within the band 1215.6-1239.6 MHz
		Radiolocation (civil)			Radar and Navigation systems
		Radiolocation (military)			
		Satellite systems (military)			

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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1240 MHz - 1300 MHz

EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION RADIONAVIGATION-SATELLITE (5.329 5.328B 5.329A) SPACE RESEARCH (ACTIVE) Amateur 5.282 5.330 5.331 5.332 5.332A 5.335 5.335A	EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION RADIONAVIGATION-SATELLITE (5.228B 5.329 5.329A) SPACE RESEARCH (ACTIVE) Amateur Amateur-satellite 5.282 5.331 5.332 5.335A ECA36	Active sensors (satellite)			
		Amateur	ECC/DEC/(25)01	EN 301 783	Within the band 1258-1300 MHz
		Amateur-satellite	ECC/DEC/(25)01		Within the band 1258-1300 MHz
		GALILEO	ECC/DEC/(25)01	EN 303 413	Within the band 1258-1300 MHz
		GLONASS		EN 303 413	Within the band 1237.8-1253.8 MHz
		GNSS Repeater	ECC/REC/(10)02	EN 302 645	Within the band 1164-1300 MHz
		Radiolocation (civil)			Radar and Navigation systems
		Radiolocation (military)			
		Satellite systems (military)			
		Wind profilers			Within the band 1270-1295 MHz

1300 MHz - 1350 MHz

AERONAUTICAL RADIONAVIGATION (5.337) RADIOLOCATION RADIONAVIGATION-SATELLITE (EARTH-TO-SPACE) 5.149 5.337A	AERONAUTICAL RADIONAVIGATION (5.337) RADIOLOCATION RADIONAVIGATION-SATELLITE (EARTH-TO-SPACE) 5.149 5.337A ECA36	Radio astronomy			Continuum and spectral line observations (e.g. neutral hydrogen line). VLBI.
		Radiolocation (civil)			Radar and Navigation systems
		Radiolocation (military)			
		Satellite navigation systems			
		Satellite systems (military)			

1350 MHz - 1400 MHz

FIXED MOBILE RADIOLOCATION 5.149 5.338 5.338A 5.339	FIXED MOBILE RADIOLOCATION 5.149 5.338A 5.339 ECA36	Aeronautical military systems			
		Fixed	T/R 13-01	EN 302 217	Low capacity fixed links
		Land military systems			
		Maritime military systems			
		Radio astronomy			Continuum and spectral line observations (e.g. neutral hydrogen line). VLBI
		Radio microphones and ALD	ERC/REC 25-10, ERC/REC 70-03	EN 300 422	Radio microphones and In-ear monitors
		Radiolocation (military)			

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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1400 MHz - 1427 MHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.341	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.341	Passive sensors (satellite)	ECC/DEC/(11)01		Measurement of soil moisture, salinity, ocean surface temperature, vegetation index
		Radio astronomy			Continuum and spectral line observations (e.g. neutral hydrogen line). VLBI.

1427 MHz - 1429 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.341A) SPACE OPERATION (EARTH-TO-SPACE) 5.338A 5.341	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE SPACE OPERATION (EARTH-TO-SPACE) 5.338A 5.341 ECA36	Fixed	T/R 13-01	EN 302 217	Low capacity fixed links
		Land military systems			
		MFCN	ECC/DEC/(17)06, ECC/DEC/(22)01, ECC/REC/(15)01	EN 301 908	Supplemental Downlink
		Maritime military systems			
		Telemetry/Telecommand (military)			

1429 MHz - 1452 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.341A) 5.338A 5.341 5.342	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE 5.338A 5.341 ECA36	Fixed	T/R 13-01	EN 302 217	Low capacity fixed links
		Land military systems			
		MFCN	ECC/DEC/(17)06, ECC/DEC/(22)01, ECC/REC/(15)01	EN 301 908	Supplemental Downlink
		Maritime military systems			
		Telemetry/Telecommand (military)			

1452 MHz - 1492 MHz

BROADCASTING BROADCASTING-SATELLITE (5.208B) FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.346) 5.341 5.342 5.345	BROADCASTING MOBILE EXCEPT AERONAUTICAL MOBILE Fixed 5.341 5.342 5.345	MFCN	ECC/DEC/(13)03, ECC/DEC/(22)01, ECC/REC/(15)01	EN 301 908	Supplemental Downlink
		T-DAB		EN 302 077, EN 303 345	Within the band 1452.0-1479.5 MHz. Maastricht 2002 Special Arrangement, as revised in Constanta, 2007.
		Telemetry/Telecommand (military)			

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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1492 MHz - 1518 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.341A) 5.341 5.342	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE 5.341 ECA36	Fixed	T/R 13-01	EN 302 217	Low capacity fixed links
		Land military systems			
		MFCN	ECC/DEC/(17)06, ECC/DEC/(22)01, ECC/REC/(15)01	EN 301 908	Supplemental Downlink
		Maritime military systems			
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	On a tuning range basis
		Telemetry/Telecommand (military)			

1518 MHz - 1525 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE MOBILE-SATELLITE (5.348 5.348A 5.348B 5.351A) 5.341 5.342	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE MOBILE-SATELLITE (5.348 5.348A 5.348B 5.351A) 5.341 ECA36	Audio PMSE	ERC/REC 25-10	EN 300 422	Radio microphones and In-ear monitors on a tuning range basis
		Fixed		EN 302 217	Unidirectional fixed links
		IMT-2000 satellite component			
		Land military systems			
		MSS Earth stations	ECC/DEC/(04)09, ECC/DEC/(12)01	EN 301 444, EN 301 473, EN 301 681	
		Maritime military systems			
		Radio microphones and ALD	ERC/REC 70-03	EN 300 422	On a tuning range basis
		Telemetry/Telecommand (military)			

1525 MHz - 1530 MHz

FIXED MOBILE-SATELLITE (5.208B 5.351A) SPACE OPERATION (SPACE-TO-EARTH) Earth exploration-satellite Mobile except aeronautical mobile (5.349) 5.341 5.342 5.350 5.351 5.352A 5.354	FIXED MOBILE-SATELLITE (5.208B 5.351A) SPACE OPERATION (SPACE-TO-EARTH) 5.341 5.351 5.354	Fixed		EN 302 217	Unidirectional fixed links
		IMT-2000 satellite component			
		MSS Earth stations	ECC/DEC/(12)01	EN 301 426, EN 301 444, EN 301 473, EN 301 681	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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1530 MHz - 1535 MHz

MOBILE-SATELLITE (5.208B 5.353A 5.351A) SPACE OPERATION (SPACE-TO-EARTH) Earth exploration-satellite Fixed Mobile except aeronautical mobile 5.341 5.342 5.351 5.354	MOBILE-SATELLITE (5.208B 5.351A 5.353A) SPACE OPERATION (SPACE-TO-EARTH) Earth exploration-satellite Fixed Mobile except aeronautical mobile 5.341 5.351 5.354	IMT-2000 satellite component			
		MSS Earth stations	ECC/DEC/(12)01	EN 301 426, EN 301 444, EN 301 473, EN 301 681	Priority for GMDSS Distress, urgency and safety and for AMS(R)S categories 1 to 6 communications

1535 MHz - 1559 MHz

MOBILE-SATELLITE (5.208B 5.351A) 5.341 5.351 5.353A 5.354 5.355 5.356 5.357 5.357A 5.359	MOBILE-SATELLITE (5.208B 5.351A) 5.341 5.351 5.353A 5.354 5.356 5.357 5.357A 5.359	IMT-2000 satellite component			
		MSS Earth stations	ECC/DEC/(12)01	EN 301 426, EN 301 444, EN 301 473, EN 301 681	Priority for GMDSS Distress, urgency and safety and for AMS(R)S categories 1 to 6 communications within the band 1544-1545 MHz

1559 MHz - 1610 MHz

AERONAUTICAL RADIONAVIGATION RADIONAVIGATION-SATELLITE (5.208B) RADIONAVIGATION-SATELLITE (5.328B 5.329A) 5.341	AERONAUTICAL RADIONAVIGATION RADIONAVIGATION-SATELLITE (5.208B) RADIONAVIGATION-SATELLITE (5.328B 5.329A) 5.341	GALILEO		EN 303 413	Within the band 1559.42-1591.42 MHz
		GLONASS		EN 303 413	Within the band 1592.9-1610.5 MHz
		GNSS Pseudolites	ECC/REC/(11)08		
		GNSS Repeater	ECC/REC/(10)02	EN 302 645	
		GPS		EN 303 413	Within the band 1563.42-1587.42 MHz
		IMT-2000 satellite component			
		MSS Earth stations	ECC/DEC/(09)02, ECC/DEC/(12)01	EN 301 441, EN 301 473	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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1610 MHz - 1610.6 MHz

AERONAUTICAL RADIONAVIGATION MOBILE-SATELLITE (5.351A) 5.341 5.355 5.359 5.364 5.366 5.367 5.368 5.369 5.371 5.372	AERONAUTICAL RADIONAVIGATION MOBILE-SATELLITE (5.351A) 5.341 5.359 5.364 5.366 5.367 5.368 5.371 5.372	GLONASS			Within the band 1592.9-1610.5 MHz
		IMT-2000 satellite component			
		MSS Earth stations	ECC/DEC/(09)02, ECC/DEC/(12)01	EN 301 441, EN 301 473	

1610.6 MHz - 1613.8 MHz

AERONAUTICAL RADIONAVIGATION MOBILE-SATELLITE (5.351A) RADIO ASTRONOMY 5.149 5.341 5.355 5.359 5.364 5.366 5.367 5.368 5.369 5.371 5.372	AERONAUTICAL RADIONAVIGATION MOBILE-SATELLITE (5.351A) RADIO ASTRONOMY 5.149 5.341 5.359 5.364 5.366 5.367 5.368 5.371 5.372	IMT-2000 satellite component			
		MSS Earth stations	ECC/DEC/(09)02, ECC/DEC/(12)01	EN 301 441, EN 301 473	
		Radio astronomy			Spectral line observations (e.g. hydroxyl line). VLBI

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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1613.8 MHz - 1621.35 MHz

AERONAUTICAL RADIONAVIGATION MOBILE-SATELLITE (5.351A) Mobile-Satellite (5.208B) 5.341 5.355 5.359 5.364 5.365 5.366 5.367 5.368 5.369 5.371 5.372 5.372A	AERONAUTICAL RADIONAVIGATION MOBILE-SATELLITE (5.351A) Mobile-Satellite (5.208B) 5.341 5.359 5.364 5.365 5.366 5.367 5.368 5.371 5.372	IMT-2000 satellite component			
		MSS Earth stations	ECC/DEC/(09)02, ECC/DEC/(09)04, ECC/DEC/(12)01	EN 301 426, EN 301 441, EN 301 473	

1621.35 MHz - 1626.5 MHz

AERONAUTICAL RADIONAVIGATION MARITIME MOBILE-SATELLITE (5.373A 5.373) MOBILE-SATELLITE (5.351A) Mobile-Satellite (5.208B) Mobile-satellite except maritime mobile satellite (space-to-Earth) 5.341 5.355 5.359 5.364 5.365 5.366 5.367 5.368 5.369 5.371 5.372	AERONAUTICAL RADIONAVIGATION MARITIME MOBILE-SATELLITE (5.373 5.373A) MOBILE-SATELLITE (5.351A) Mobile-Satellite (5.208B) Mobile-satellite except maritime mobile satellite (space-to-Earth) 5.341 5.359 5.364 5.365 5.366 5.367 5.368 5.371 5.372	IMT-2000 satellite component			
		MSS Earth stations	ECC/DEC/(09)02, ECC/DEC/(09)04, ECC/DEC/(12)01	EN 301 426, EN 301 441, EN 301 473	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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1626.5 MHz - 1660 MHz

MOBILE-SATELLITE (5.351A) 5.341 5.351 5.353A 5.354 5.355 5.357A 5.359 5.374 5.375 5.376	MOBILE-SATELLITE (5.351A) 5.100 5.341 5.351 5.353A 5.354 5.359 5.374 5.375 5.376	ALS	ERC/REC 70-03	EN 300 422	ALS is Assistive Listening Systems. Within 1656.5-1660.5 MHz
		IMT-2000 satellite component			
		MSS Earth stations	ECC/DEC/(12)01	EN 301 426, EN 301 473, EN 301 681	Priority for GMDSS Distress, urgency and safety and for AMS(R)S categories 1 to 6 communications within the band 1645.5-1646.5 MHz

1660 MHz - 1660.5 MHz

MOBILE-SATELLITE (5.351A) RADIO ASTRONOMY 5.149 5.341 5.351 5.354 5.376A	MOBILE-SATELLITE (5.351A) RADIO ASTRONOMY 5.149 5.341 5.351 5.354 5.376A	ALS	ERC/REC 70-03	EN 300 422	ALS is Assistive Listening Systems. Within 1656.5-1660.5 MHz
		IMT-2000 satellite component			
		MSS Earth stations	ECC/DEC/(12)01	EN 301 426, EN 301 444, EN 301 473, EN 301 681	
		Radio astronomy			Continuum and spectral line observations (e.g. hydroxyl line), VLBI

1660.5 MHz - 1668 MHz

RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) Fixed Mobile except aeronautical mobile 5.149 5.341 5.379 5.379A	RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) Fixed Mobile except aeronautical mobile 5.149 5.341 5.379A	Radio astronomy			Continuum and spectral line observations (e.g. hydroxyl line), VLBI
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RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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1668 MHz - 1668.4 MHz

MOBILE-SATELLITE (5.351A 5.379B 5.379C) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) Fixed Mobile except aeronautical mobile 5.149 5.341 5.379 5.379A	MOBILE-SATELLITE (5.351A 5.379B 5.379C) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) Fixed Mobile except aeronautical mobile 5.149 5.341 5.379A	IMT-2000 satellite component			
		Radio astronomy			Continuum and spectral line observations (e.g. hydroxyl line), VLBI

1668.4 MHz - 1670 MHz

FIXED METEOROLOGICAL AIDS MOBILE EXCEPT AERONAUTICAL MOBILE MOBILE-SATELLITE (5.351A 5.379B 5.379C) RADIO ASTRONOMY 5.149 5.341 5.379D 5.379E	FIXED METEOROLOGICAL AIDS MOBILE EXCEPT AERONAUTICAL MOBILE MOBILE-SATELLITE (5.351A 5.379B 5.379C) RADIO ASTRONOMY 5.149 5.341 5.379D 5.379E	IMT-2000 satellite component			
		Meteorology			
		Radio astronomy			Continuum and spectral line observations (e.g. hydroxyl line), VLBI

1670 MHz - 1675 MHz

FIXED METEOROLOGICAL AIDS METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE MOBILE-SATELLITE (5.351A 5.379B) 5.341 5.379D 5.379E 5.380A	METEOROLOGICAL AIDS METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE MOBILE-SATELLITE (5.351A 5.379B) Fixed 5.341 5.379D 5.379E 5.380A	IMT-2000 satellite component			
		MSS Earth stations	ECC/DEC/(04)09, ECC/DEC/(12)01	EN 301 444, EN 301 473, EN 301 681	
		Meteorology			
		Weather satellites			

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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1675 MHz - 1690 MHz

FIXED METEOROLOGICAL AIDS METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE EXCEPT AERONAUTICAL MOBILE 5.341	FIXED METEOROLOGICAL AIDS METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE EXCEPT AERONAUTICAL MOBILE 5.341 ECA36	Land military systems			
		Maritime military systems			
		Meteorological aids (military)			
		Sondes		EN 302 454	Meteorological radiosondes
		Weather satellites			Data collection platform

1690 MHz - 1700 MHz

METEOROLOGICAL AIDS METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) Fixed Mobile except aeronautical mobile 5.289 5.341 5.382	METEOROLOGICAL AIDS METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) Fixed Mobile except aeronautical mobile 5.289 5.341 ECA36	Land military systems			
		Maritime military systems			
		Meteorological aids (military)			
		Weather satellites			Data collection platform. Allocation to EESS is via RR 5.289

1700 MHz - 1710 MHz

FIXED METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE EXCEPT AERONAUTICAL MOBILE 5.289 5.341	FIXED FIXED METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE (5.384A) Mobile except aeronautical mobile 5.289 5.341 ECA36	Land military systems			
		Maritime military systems			
		Meteorological aids (military)			
		Weather satellites			Data collection platform. Allocation to EESS is via RR 5.289

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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1710 MHz - 1930 MHz

FIXED MOBILE (5.384A 5.388A) 5.149 5.341 5.385 5.386 5.387 5.388	FIXED MOBILE (5.384A 5.388 5.388A ECA38) 5.149 5.341 5.385 ECA29 ECA36	DECT	ERC/DEC/(94)03, ERC/DEC/(98)22, ERC/REC 70-03	EN 301 406	Within the band 1880-1900 MHz
		GSM	ECC/REC/(05)08, ECC/REC/(08)02, ERC/DEC/(95)03	EN 301 502, EN 301 511, EN 303 609	
		Land military systems			
		Land mobile			Mobile applications
		MCA	ECC/DEC/(06)07	EN 302 480	Within the band 1920-1980 MHz
		MCV	ECC/DEC/(08)08		Within the band 1920-1980 MHz
		MFCN	ECC/DEC/(06)01, ECC/DEC/(06)13, ECC/DEC/(22)01, ECC/DEC/(22)07, ECC/REC/(08)02, ERC/REC 01-01	EN 301 908	1920-1980 MHz, Aerial UE are permitted – See ECC Decision (22)07
		RMR	ECC/DEC/(20)02, ECC/REC/(23)01	EN 301 502, EN 301 511	Within the band 1900-1910 MHz
		Radio microphones and ALD	ERC/REC 25-10, ERC/REC 70-03	EN 300 422	Within the band 1785-1805 MHz
		UAS	ECC/REC/(24)02		Within the bands 1880-1900 MHz and 1910-1920 MHz

1930 MHz - 1970 MHz

FIXED MOBILE (5.388A) 5.388	MOBILE (5.388 5.388A ECA38) Fixed ECA29	-			This band can also be used by fixed service on a national basis
		MCA	ECC/DEC/(06)07		
		MCV	ECC/DEC/(08)08		
		MFCN	ECC/DEC/(06)01, ECC/DEC/(22)01, ECC/DEC/(22)07, ERC/REC 01-01	EN 301 908	1920-1980 MHz, Aerial UE are permitted – See ECC Decision (22)07

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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1970 MHz - 1980 MHz

FIXED MOBILE (5.388A) 5.388	MOBILE (5.388A ECA38) Fixed 5.388 ECA29	-			This band can also be used by fixed service on a national basis
		MCA	ECC/DEC/(06)07		
		MCV	ECC/DEC/(08)08		
		MFCN	ECC/DEC/(06)01, ECC/DEC/(22)01, ECC/DEC/(22)07, ERC/REC 01-01	EN 301 908	1920-1980 MHz, Aerial UE are permitted – See ECC Decision (22)07

1980 MHz - 2010 MHz

FIXED MOBILE MOBILE-SATELLITE (5.351A) 5.388 5.389A 5.389B 5.389F	MOBILE MOBILE-SATELLITE (5.351A) 5.388 5.389A	-			This band can also be used by fixed service on a national basis
		MSS Earth stations	ECC/DEC/(06)09, ECC/DEC/(06)10, ECC/DEC/(12)01	EN 301 442, EN 301 473, EN 302 574	The mobile satellite systems using this band may incorporate a complementary Ground Component (CGC)

2010 MHz - 2025 MHz

FIXED MOBILE (5.388A 5.388) 5.388	MOBILE Fixed	-			This band can also be used by fixed service on a national basis
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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2025 MHz - 2110 MHz

EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE) (SPACE-TO-SPACE) FIXED MOBILE (5.391) SPACE OPERATION (EARTH-TO-SPACE) (SPACE-TO-SPACE) SPACE RESEARCH (EARTH-TO-SPACE) (SPACE-TO-SPACE) 5.392	EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE) (SPACE-TO-SPACE) FIXED MOBILE (5.391) SPACE OPERATION (EARTH-TO-SPACE) (SPACE-TO-SPACE) SPACE RESEARCH (EARTH-TO-SPACE) (SPACE-TO-SPACE) 5.392 ECA16A ECA36	Aeronautical military systems			
		Fixed	T/R 13-01	EN 302 217	
		Land military systems			
		MSS Earth stations	ECC/REC/(24)03	EN 301 473	
		Maritime military systems			
		Space research			Satellite payload and platform telecommand
		Telemetry/Telecommand (military)			
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links

2110 MHz - 2120 MHz

FIXED MOBILE (5.388A) SPACE RESEARCH (DEEP SPACE) (EARTH-TO-SPACE) 5.388	MOBILE (5.388A) SPACE RESEARCH (DEEP SPACE) (EARTH-TO-SPACE) Fixed 5.388 ECA29	-			Satellite payload and platform telecommand for space research (deep space). This band can also be used by fixed service on a national basis
		MCA	ECC/DEC/(06)07		
		MCV	ECC/DEC/(08)08		
		MFCN	ECC/DEC/(06)01, ECC/DEC/(22)01, ERC/REC 01-01	EN 301 908	

2120 MHz - 2160 MHz

FIXED MOBILE (5.388A) 5.388	MOBILE (5.388A) Fixed 5.388 ECA29	-			This band can also be used by fixed service on a national basis
		MCA	ECC/DEC/(06)07		
		MCV	ECC/DEC/(08)08		
		MFCN	ECC/DEC/(06)01, ECC/DEC/(22)01, ERC/REC 01-01	EN 301 908	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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2160 MHz - 2170 MHz

FIXED MOBILE (5.388A) 5.388	MOBILE (5.388A) Fixed 5.388 ECA29	-			This band can also be used by fixed service on a national basis
		MCA	ECC/DEC/(06)07		
		MCV	ECC/DEC/(08)08		
		MFCN	ECC/DEC/(06)01, ECC/DEC/(22)01, ERC/REC 01-01	EN 301 908	

2170 MHz - 2200 MHz

FIXED MOBILE MOBILE-SATELLITE (5.351A) 5.388 5.389A 5.389F	MOBILE MOBILE-SATELLITE (5.351A) 5.388 5.389A	-			This band can also be used by fixed service on a national basis
		MSS Earth stations	ECC/DEC/(06)09, ECC/DEC/(06)10, ECC/DEC/(12)01, ECC/REC/(10)01	EN 301 442, EN 301 473, EN 302 574	The mobile satellite systems using this band may incorporate a Complementary Ground Component (CGC)

2200 MHz - 2290 MHz

EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH) (SPACE-TO-SPACE) FIXED MOBILE (5.391) SPACE OPERATION (SPACE-TO-EARTH) (SPACE-TO-SPACE) SPACE RESEARCH (SPACE-TO-EARTH) (SPACE-TO-SPACE) 5.392	EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH) (SPACE-TO-SPACE) FIXED MOBILE (5.391) SPACE OPERATION (SPACE-TO-EARTH) (SPACE-TO-SPACE) SPACE RESEARCH (SPACE-TO-EARTH) (SPACE-TO-SPACE) 5.392 ECA16A ECA36	Aeronautical military systems			
		Fixed	T/R 13-01	EN 302 217	
		Land military systems			
		MSS Earth stations	ECC/REC/(24)03		
		Maritime military systems			
		Radio astronomy			Continuum observations, VLBI (used by SRS)
		Space research	ECC/REC/(10)01		EESS Satellite payload and platform telemetry
		Telemetry/Telecommand (military)			
		UWB applications	ECC/DEC/(07)01, ERC/REC 70-03		
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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2290 MHz - 2300 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE SPACE RESEARCH (DEEP SPACE) (SPACE-TO-EARTH)	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE SPACE RESEARCH (DEEP SPACE) (SPACE-TO-EARTH)	Land mobile			Mobile applications
		Space research			Satellite payload and platform telemetry for space research (deep space). Continuum observations, VLBI (used by SRS)
		UWB applications	ECC/DEC/(07)01, ERC/REC 70-03		
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links

2300 MHz - 2400 MHz

FIXED MOBILE (5.384A) Amateur Radiolocation 5.395	FIXED MOBILE (5.384A) Amateur Radiolocation ECA36	Aeronautical military systems			
		Aeronautical telemetry	ERC/REC 62-02		Parts of the band are used for aeronautical telemetry on a national basis
		Amateur		EN 301 783	Within the band 2300-2450 MHz
		Land military systems			
		MFCN	ECC/DEC/(14)02, ECC/DEC/(22)01, ECC/REC/(14)04, ECC/REC/(15)04	EN 301 908	Shared use of spectrum envisaged
		Maritime military systems			
		Telemetry/Telecommand (military)			
		Video PMSE	ECC/REC/(15)04, ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links

2400 MHz - 2450 MHz

FIXED MOBILE Amateur Radiolocation 5.150 5.282	FIXED MOBILE Amateur Amateur-satellite Radiolocation 5.150 5.282	Amateur		EN 301 783	Within the band 2300-2450 MHz
		Amateur-satellite			
		ISM			
		Non-specific SRDs	ERC/REC 70-03	EN 300 440	Within the band 2400.0-2483.5 MHz
		RFID	ERC/REC 70-03	EN 300 440	Within the band 2446-2454 MHz
		Radiodetermination applications	ERC/REC 70-03	EN 300 440	Within the band 2400.0-2483.5 MHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links
		Wideband data transmission systems	ERC/REC 70-03	EN 300 328	Within the band 2400-2483.5 MHz

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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2450 MHz - 2483.5 MHz

FIXED MOBILE Radiolocation 5.150	FIXED MOBILE 5.150	ISM			
		Non-specific SRDs	ERC/REC 70-03	EN 300 440	Within the band 2400.0-2483.5 MHz
		RFID	ERC/REC 70-03	EN 300 440	Within the band 2446-2454 MHz
		Radiodetermination applications	ERC/REC 70-03	EN 300 440	Within the band 2400.0-2483.5 MHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links
		Wideband data transmission systems	ERC/REC 70-03	EN 300 328	Within the band 2400-2483.5 MHz

2483.5 MHz - 2500 MHz

FIXED MOBILE MOBILE-SATELLITE (5.351A) RADIODETERMINATION-SATELLITE (5.398) Radiolocation (5.398A) 5.150 5.368 5.372A 5.399 5.401 5.402	FIXED MOBILE MOBILE-SATELLITE (5.351A) 5.150 5.399 5.402	IMT-2000 satellite component			
		ISM			
		LP-AMI	ERC/REC 70-03	EN 301 559	Low Power Active Medical Implants and associated peripherals
		Land mobile			Mobile applications
		MBANS	ERC/REC 70-03	EN 303 203	
		MSS Earth stations	ECC/DEC/(09)02, ECC/DEC/(12)01	EN 301 441	
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links

2500 MHz - 2520 MHz

FIXED (5.410) MOBILE EXCEPT AERONAUTICAL MOBILE (5.384A 5.409A) 5.412	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.384A)	MCV	ECC/DEC/(08)08		
		MFCN	ECC/DEC/(05)05, ECC/DEC/(22)01, ECC/DEC/(22)07, ECC/REC/(11)05	EN 301 908	2500-2570 MHz, Aerial UE are permitted – See ECC Decision (22)07

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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2520 MHz - 2655 MHz

BROADCASTING-SATELLITE (5.413 5.416) FIXED (5.410) MOBILE EXCEPT AERONAUTICAL MOBILE (5.384A 5.409A) 5.339 5.412 5.418B 5.418C	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.384A ECA38) 5.339 5.418B 5.418C ECA16	MCV	ECC/DEC/(08)08		
		MFCN	ECC/DEC/(05)05, ECC/DEC/(22)01, ECC/DEC/(22)07, ECC/REC/(11)05	EN 301 908	2500-2570 MHz, Aerial UE are permitted – See ECC Decision (22)07

2655 MHz - 2670 MHz

BROADCASTING-SATELLITE (5.208B 5.413 5.416) FIXED (5.410) MOBILE EXCEPT AERONAUTICAL MOBILE (5.384A 5.409A) Earth exploration-satellite (passive) Radio astronomy Space research (passive) 5.149 5.412	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.384A) Earth exploration-satellite (passive) Radio astronomy Space research (passive) 5.149 5.208B ECA16	MCV	ECC/DEC/(08)08		
		MFCN	ECC/DEC/(05)05, ECC/DEC/(22)01, ECC/REC/(11)05	EN 301 908	
		Radio astronomy			Continuum observations, VLBI

2670 MHz - 2690 MHz

FIXED (5.410) MOBILE EXCEPT AERONAUTICAL MOBILE (5.384A 5.409A) Earth exploration-satellite (passive) Radio astronomy Space research (passive) 5.149 5.412	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.384A) Radio astronomy 5.149	MCV	ECC/DEC/(08)08		
		MFCN	ECC/DEC/(05)05, ECC/DEC/(22)01, ECC/REC/(11)05	EN 301 908	
		Radio astronomy			Continuum observations, VLBI

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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2690 MHz - 2700 MHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.422	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	Passive sensors (satellite)			
		Radio astronomy			Continuum observations, VLBI

2700 MHz - 2900 MHz

AERONAUTICAL RADIONAVIGATION (5.337) Radiolocation 5.423	AERONAUTICAL RADIONAVIGATION (5.337) Radiolocation 5.423 ECA36	Aeronautical navigation	ECC/REC/(02)09		Radar and navigation systems
		Radiolocation (civil)			
		Radiolocation (military)			
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links
		Weather radar		EN 303 347	

2900 MHz - 3100 MHz

RADIOLOCATION (5.424A) RADIONAVIGATION (5.426) 5.425 5.427	RADIOLOCATION (5.424A) RADIONAVIGATION (5.426) 5.425 5.427 ECA36	Radiolocation (civil)		EN 302 248, EN 302 752	Radar and navigation systems
		Radiolocation (military)			

3100 MHz - 3300 MHz

RADIOLOCATION Earth exploration-satellite (active) Space research (active) 5.149 5.428	RADIOLOCATION Earth exploration-satellite (active) Space research (active) 5.149 ECA36	Active sensors (satellite)			
		Radio astronomy			Spectral line observations (e.g. methine line)
		Radiolocation (civil)			Radars
		Radiolocation (military)			
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/REC/(11)09, ECC/REC/(11)10, ERC/REC 70-03	EN 302 065	Generic UWB. Location Tracking Type 2 (LT2). Location Application for Emergency Services (LAES)

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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3300 MHz - 3400 MHz

RADIOLOCATION 5.149 5.429 5.429A 5.429B 5.430	RADIOLOCATION 5.149 ECA36	Radio astronomy			Spectral line observations (e.g. methine line)
		Radiolocation (civil)			Upper limit for airborne radars 3410 MHz
		Radiolocation (military)			Upper limit for airborne radars is 3410 MHz
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/REC/(11)09, ECC/REC/(11)10, ERC/REC 70-03	EN 302 065	Generic UWB. Location Tracking Type 2 (LT2). Location Application for Emergency Services (LAES)

3400 MHz - 3600 MHz

FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE EXCEPT AERONAUTICAL MOBILE (5.430A) Radiolocation 5.431	FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE EXCEPT AERONAUTICAL MOBILE (5.430A ECA38) Amateur Radiolocation ECA36	Amateur		EN 301 783	Within the band 3400-3410 MHz
		MFCN	ECC/DEC/(11)06, ECC/DEC/(22)01, ECC/REC/(15)01, ECC/REC/(20)03, ECC/REC/(21)02	EN 301 908	
		Radiolocation (civil)			Upper limit for airborne radars is 3410 MHz
		Radiolocation (military)			Upper limit for airborne radars is 3410 MHz
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/REC/(11)09, ECC/REC/(11)10, ERC/REC 70-03	EN 302 065	Generic UWB. Location Tracking Type 2 (LT2). Location Application for Emergency Services (LAES)

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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3600 MHz - 3800 MHz

FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE EXCEPT AERONAUTICAL MOBILE (5.433B 5.434A 5.434B) 5.435A	FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE (5.434A 5.434B ECA38) ECA37	ESV	ECC/DEC/(05)09	EN 301 447	Within the band 3700-3800 MHz
		Fixed	ERC/REC 12-08	EN 302 217	Medium/high capacity fixed
		MFCN	ECC/DEC/(11)06, ECC/DEC/(22)01, ECC/REC/(15)01, ECC/REC/(20)03, ECC/REC/(21)02	EN 301 908	
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/REC/(11)09, ECC/REC/(11)10, ERC/REC 70-03	EN 302 065	Generic UWB. Location Tracking Type 2 (LT2). Location Application for Emergency Services (LAES)
		VSAT		EN 301 443	Within the band 3700-3800 MHz

3800 MHz - 4200 MHz

FIXED FIXED-SATELLITE (SPACE-TO-EARTH) Mobile	FIXED FIXED-SATELLITE (SPACE-TO-EARTH)	ESV	ECC/DEC/(05)09	EN 301 447	Within the band 3700-4200 MHz
		Fixed	ERC/REC 12-08	EN 302 217	Medium/high capacity fixed
		MFCN	ECC/DEC/(24)01		
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/REC/(11)09, ECC/REC/(11)10, ERC/REC 70-03	EN 302 065	Generic UWB. Location Tracking Type 2 (LT2). Location Application for Emergency Services (LAES)
		VSAT		EN 301 443	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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4200 MHz - 4400 MHz

AERONAUTICAL MOBILE (5.436) AERONAUTICAL RADIONAVIGATION (5.438) 5.437 5.439 5.440	AERONAUTICAL MOBILE (5.436) AERONAUTICAL RADIONAVIGATION (5.438) 5.437 5.440 ECA36	Aeronautical military systems			
		Altimeters			
		Passive sensors (satellite)			For sea surface temperature measurements
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/REC/(11)09, ECC/REC/(11)10, ERC/REC 70-03	EN 302 065	Generic UWB. Location Tracking Type 2 (LT2). Location Application for Emergency Services (LAES)
		WAIC			Wireless Avionics Intra-Communications

4400 MHz - 4500 MHz

FIXED MOBILE	FIXED MOBILE ECA20 ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		Telemetry/Telecommand (military)			
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/REC/(11)09, ECC/REC/(11)10, ERC/REC 70-03	EN 302 065	Generic UWB. Location Tracking Type 2 (LT2). Location Application for Emergency Services (LAES)

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4500 MHz - 4800 MHz

FIXED FIXED-SATELLITE (5.441) MOBILE	FIXED FIXED-SATELLITE (5.441) MOBILE ECA20 ECA36	Aeronautical military systems			
		FSS Earth stations			FSS not to be implemented in NATO Europe. Fixed-Satellite frequency plan in 4500-4800 MHz
		Land military systems			
		Maritime military systems			
		TLPR	ERC/REC 70-03	EN 302 372	
		Telemetry/Telecommand (military)			
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/REC/(11)09, ECC/REC/(11)10, ERC/REC 70-03	EN 302 065	Generic UWB. Location Tracking Type 2 (LT2). Location Application for Emergency Services (LAES)

4800 MHz - 4990 MHz

FIXED MOBILE (5.442 5.440A 5.441A 5.441B) Radio astronomy 5.149 5.339 5.443	FIXED MOBILE (5.440A 5.441A 5.441B 5.442) Radio astronomy 5.149 5.339 ECA20 ECA36	Aeronautical military systems			
		BBDR	ECC/REC/(08)04	EN 302 625	Within the band 4940-4990 MHz. Optimal band for BBDR within the PPDR uses
		Land military systems			
		Maritime military systems			
		Passive sensors (satellite)			Space Research and EESS (passive) above 4950 MHz in some countries
		Radio astronomy			Continuum and spectral line observations, (e.g. formaldehyde line), VLBI
		TLPR	ERC/REC 70-03	EN 302 372	
		Telemetry/Telecommand (military)			

4990 MHz - 5000 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE RADIO ASTRONOMY Space research (passive) 5.149	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE RADIO ASTRONOMY 5.149 ECA20 ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		Radio astronomy			Continuum observations, VLBI
		TLPR	ERC/REC 70-03	EN 302 372	
		Telemetry/Telecommand (military)			

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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5000 MHz - 5010 MHz

AERONAUTICAL MOBILE-SATELLITE (5.443AA) AERONAUTICAL RADIONAVIGATION RADIONAVIGATION-SATELLITE (EARTH-TO-SPACE)	AERONAUTICAL MOBILE-SATELLITE (5.443AA) AERONAUTICAL RADIONAVIGATION RADIONAVIGATION-SATELLITE (EARTH-TO-SPACE) Radio astronomy Space research (passive)	GALILEO			For future use by Galileo
		Radio astronomy			Continuum observation, VLBI
		Satellite navigation systems			Aeronautical Radionavigation and FSS envisaged in some countries
		TLPR	ERC/REC 70-03	EN 302 372	

5010 MHz - 5030 MHz

AERONAUTICAL MOBILE-SATELLITE (5.443AA) AERONAUTICAL RADIONAVIGATION RADIONAVIGATION-SATELLITE (5.328B 5.443B)	AERONAUTICAL MOBILE-SATELLITE (5.443AA) AERONAUTICAL RADIONAVIGATION RADIONAVIGATION-SATELLITE (5.328B 5.443B) Radio astronomy Space research (passive)	GALILEO			
		Radio astronomy			Continuum observation, VLBI
		Satellite navigation systems			Aeronautical Radionavigation and FSS envisaged in some countries
		TLPR	ERC/REC 70-03	EN 302 372	

5030 MHz - 5091 MHz

AERONAUTICAL MOBILE (5.443C) AERONAUTICAL MOBILE-SATELLITE (5.443D) AERONAUTICAL RADIONAVIGATION 5.444	AERONAUTICAL MOBILE (5.443C) AERONAUTICAL MOBILE-SATELLITE (5.443D) AERONAUTICAL RADIONAVIGATION 5.444	MLS			Aeronautical Radionavigation envisaged in some countries. FSS in use in some countries
		TLPR	ERC/REC 70-03	EN 302 372	

5091 MHz - 5150 MHz

AERONAUTICAL MOBILE (5.444B) AERONAUTICAL MOBILE-SATELLITE (5.443AA) AERONAUTICAL RADIONAVIGATION FIXED-SATELLITE (5.444A) 5.444	AERONAUTICAL MOBILE (5.444B) AERONAUTICAL MOBILE-SATELLITE (5.443AA) AERONAUTICAL RADIONAVIGATION FIXED-SATELLITE (5.444A) 5.444	-			FSS in use in some countries
		TLPR	ERC/REC 70-03	EN 302 372	

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5150 MHz - 5250 MHz

AERONAUTICAL RADIONAVIGATION (5.446D) FIXED-SATELLITE (5.447A) MOBILE EXCEPT AERONAUTICAL MOBILE (5.446B 5.446A) 5.446 5.446C 5.447 5.447B 5.447C	AERONAUTICAL RADIONAVIGATION FIXED-SATELLITE (5.447A) MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.446B) 5.446 5.446C 5.447 5.447B 5.447C	Aeronautical telemetry			
		BBDR	ECC/REC/(08)04	EN 302 625	Temporary use by PPDR users
		Feeder links			Feeder links for MSS. Aeronautical Radionavigation and FSS envisaged in some countries
		RLAN	ECC/DEC/(04)08, ERC/REC 70-03	EN 301 893	WAS/RLANs within the bands 5150-5350 MHz and 5470-5725 MHz
		TLPR	ERC/REC 70-03	EN 302 372	

5250 MHz - 5255 MHz

EARTH EXPLORATION-SATELLITE (ACTIVE) MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.447F) RADIOLOCATION SPACE RESEARCH (5.447D) 5.447E 5.448 5.448A	EARTH EXPLORATION-SATELLITE (ACTIVE) MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.447F) RADIOLOCATION SPACE RESEARCH (5.447D) 5.448A ECA22 ECA36	-			Position fixing
		Active sensors (satellite)			
		Maritime radar			Shipborne and VTS radar
		RLAN	ECC/DEC/(04)08, ERC/REC 70-03	EN 301 893	WAS/RLANs within the bands 5150-5350 MHz and 5470-5725 MHz
		Radiolocation (military)			
		TLPR	ERC/REC 70-03	EN 302 372	
		Weather radar		EN 303 347	Ground based and airborne

5255 MHz - 5350 MHz

EARTH EXPLORATION-SATELLITE (ACTIVE) MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.447F) RADIOLOCATION SPACE RESEARCH (ACTIVE) 5.447E 5.448 5.448A	EARTH EXPLORATION-SATELLITE (ACTIVE) MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.447F) RADIOLOCATION SPACE RESEARCH (ACTIVE) 5.448A ECA22 ECA36	-			Position fixing
		Active sensors (satellite)			
		Maritime radar			Shipborne and VTS radar
		RLAN	ECC/DEC/(04)08, ERC/REC 70-03	EN 301 893	WAS/RLANs within the bands 5150-5350 MHz and 5470-5725 MHz
		Radiolocation (military)			
		TLPR	ERC/REC 70-03	EN 302 372	
		Weather radar		EN 303 347	Ground based and airborne

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5350 MHz - 5460 MHz

AERONAUTICAL RADIONAVIGATION (5.449) EARTH EXPLORATION-SATELLITE (5.448B) RADIOLOCATION (5.448D) SPACE RESEARCH (5.448C)	AERONAUTICAL RADIONAVIGATION (5.449) EARTH EXPLORATION-SATELLITE (5.448B) RADIOLOCATION (5.448D) SPACE RESEARCH (5.448C) ECA22 ECA36	-			Position fixing
		Active sensors (satellite)			
		Maritime radar			Shipborne and VTS radar
		Radiolocation (military)			
		TLPR	ERC/REC 70-03	EN 302 372	
		Weather radar		EN 303 347	Ground based and airborne

5460 MHz - 5470 MHz

EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION (5.448D) RADIONAVIGATION (5.449) SPACE RESEARCH (ACTIVE) 5.448B	EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION (5.448D) RADIONAVIGATION (5.449) SPACE RESEARCH (ACTIVE) 5.448B ECA22 ECA36	-			Position fixing
		Active sensors (satellite)			
		Maritime radar			Shipborne and VTS radar
		Radiolocation (military)			
		TLPR	ERC/REC 70-03	EN 302 372	
		Weather radar		EN 303 347	Ground based and airborne

5470 MHz - 5570 MHz

EARTH EXPLORATION-SATELLITE (ACTIVE) MARITIME RADIONAVIGATION MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.450A) RADIOLOCATION (5.450B) SPACE RESEARCH (ACTIVE) 5.448B 5.450 5.451	EARTH EXPLORATION-SATELLITE (ACTIVE) MARITIME RADIONAVIGATION MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.450A) RADIOLOCATION (5.450B) SPACE RESEARCH (ACTIVE) 5.448B ECA22 ECA36	-			Position fixing
		Active sensors (satellite)			
		Maritime radar			Shipborne and VTS radar
		RLAN	ECC/DEC/(04)08, ERC/REC 70-03	EN 301 893	WAS/RLANs within the bands 5150-5350 MHz and 5470-5725 MHz
		Radiolocation (military)			
		TLPR	ERC/REC 70-03	EN 302 372	
		Weather radar		EN 303 347	Ground based and airborne

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5570 MHz - 5650 MHz

MARITIME RADIONAVIGATION MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.450A) RADIOLOCATION (5.450B) 5.450 5.451 5.452	MARITIME RADIONAVIGATION MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.450A) RADIOLOCATION (5.450B) 5.452 ECA22 ECA36	-			Position fixing
		Maritime radar			Shipborne and VTS radar
		RLAN	ECC/DEC/(04)08, ERC/REC 70-03	EN 301 893	WAS/RLANs within the bands 5150-5350 MHz and 5470-5725 MHz
		Radiolocation (military)			
		TLPR	ERC/REC 70-03	EN 302 372	
		Weather radar		EN 303 347	Ground based

5650 MHz - 5725 MHz

MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.450A) RADIOLOCATION Amateur Space research (deep space) 5.282 5.451 5.453 5.454 5.455	MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.450A) RADIOLOCATION Amateur Amateur-satellite (Earth-to-space) 5.282 ECA22 ECA23 ECA36	-			Position fixing
		Amateur		EN 301 783	Within the band 5650-5850 MHz
		Amateur-satellite			Within the band 5650-5670 MHz
		Maritime radar			Shipborne and VTS radar
		RLAN	ECC/DEC/(04)08, ERC/REC 70-03	EN 301 893	WAS/RLANs within the bands 5150-5350 MHz and 5470-5725 MHz
		Radiolocation (military)			
		TLPR	ERC/REC 70-03	EN 302 372	
		Weather radar		EN 303 347	Ground based and airborne

5725 MHz - 5830 MHz

FIXED-SATELLITE (EARTH-TO-SPACE) RADIOLOCATION Amateur 5.150 5.451 5.453 5.455	FIXED-SATELLITE (EARTH-TO-SPACE) RADIOLOCATION Amateur Fixed Mobile 5.150 ECA17 ECA22 ECA36	Amateur		EN 301 783	Within the band 5650-5850 MHz
		BFWA	ECC/REC/(06)04	EN 302 502	Within the band 5725-5875 MHz
		ISM			Within the band 5725-5875 MHz
		Non-specific SRDs	ERC/REC 70-03	EN 300 440	Within the band 5725-5875 MHz
		Radiolocation (military)			
		TLPR	ERC/REC 70-03	EN 302 372	
		TTT	ERC/REC 70-03	EN 300 674	Within the band 5795-5805 MHz. TTT in the band 5805-5815 MHz on a national basis
		WIA	ERC/REC 70-03	EN 303 258	Within the band 5725-5875 MHz
		Weather radar		EN 303 347	Ground based and airborne

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5830 MHz - 5850 MHz

FIXED-SATELLITE (EARTH-TO-SPACE) RADIOLOCATION Amateur Amateur-satellite (space-to-Earth) 5.150 5.451 5.453 5.455	FIXED-SATELLITE (EARTH-TO-SPACE) RADIOLOCATION Amateur Amateur-satellite (space-to-Earth) Fixed Mobile 5.150 ECA22 ECA23 ECA36	-			Within the band 5725-5875 MHz
		Amateur		EN 301 783	Within the band 5650-5850 MHz
		Amateur-satellite			Within the band 5830-5850 MHz
		BFWA			Within the band 5725-5875 MHz
		ISM			Within the band 5725-5875 MHz
		Non-specific SRDs	ERC/REC 70-03	EN 300 440	Within the band 5725-5875 MHz
		Radiolocation (military)			
		TLPR	ERC/REC 70-03	EN 302 372	
		WIA	ERC/REC 70-03	EN 303 258	Within the band 5725-5875 MHz
		Weather radar		EN 303 347	Ground based and airborne

5850 MHz - 5925 MHz

FIXED FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE 5.150	FIXED FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE 5.150	BFWA	ECC/REC/(06)04	EN 302 502	Within the band 5725-5875 MHz
		ISM			Within the band 5725-5875 MHz
		ITS	ECC/DEC/(08)01, ECC/REC/(08)01, ERC/REC 70-03	EN 302 571	Safety related applications within the band 5875-5935 MHz. Non-Safety applications within the band 5855-5875 MHz.
		MBR	ECC/REC/(17)03	EN 303 276	Within 5852-5872 MHz and 5880-5900 MHz
		Non-specific SRDs	ERC/REC 70-03	EN 300 440	Within the band 5725-5875 MHz
		TLPR	ERC/REC 70-03	EN 302 372	
		VSAT		EN 301 443	
		WIA	ERC/REC 70-03	EN 303 258	Within the band 5725-5875 MHz

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5925 MHz - 6700 MHz

FIXED (5.457) FIXED-SATELLITE (5.457A 5.457B) MOBILE (5.457C 5.457E) 5.149 5.440 5.458	FIXED FIXED-SATELLITE (5.457A) MOBILE (5.457E) Earth exploration-satellite (passive) 5.149 5.440 5.458	-			
		ESV	ECC/DEC/(05)09	EN 301 447	Within the band 5925-6425 MHz
		FSS Earth stations			
		Fixed	ECC/REC/(14)06, ERC/REC 14-01, ERC/REC 14-02	EN 302 217	Point-to-point
		ITS	ECC/DEC/(08)01, ERC/REC 70-03		Urban rail systems only 5925–5935 MHz. Safety related applications within the band 5875-5935 MHz.
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Passive sensors (satellite)			For sea surface temperature, sea surface wind speed and soil moisture measurements
		RLAN	ECC/DEC/(20)01	EN 303 687	Within the band 5945-6425 MHz
		Radio astronomy			Spectral line observations (e.g. methanol line), VLBI.
		TLPR	ERC/REC 70-03	EN 302 372	
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB as well as UWB on-board aircraft regulation within the band 6.0- 8.5 GHz
		VSAT		EN 301 443	

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6700 MHz - 7075 MHz

FIXED FIXED-SATELLITE (5.441) MOBILE (5.457E) 5.458 5.458A 5.458B	FIXED FIXED-SATELLITE (5.441) MOBILE (5.457E) Earth exploration-satellite (passive) 5.458 5.458A 5.458B	Feeder links			
		Fixed	ECC/REC/(14)06, ERC/REC 14-02	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Passive sensors (satellite)			For sea surface temperature, sea surface wind speed and soil moisture measurements
		TLPR	ERC/REC 70-03	EN 302 372	
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz
		VSAT		EN 301 443	
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link within 7-8.5 GHz.

7075 MHz - 7145 MHz

FIXED MOBILE (5.457E) 5.458	FIXED MOBILE (5.457E) Earth exploration-satellite (passive) 5.458	Fixed	ECC/REC/(02)06, ECC/REC/(14)06, ERC/REC 14-02	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Passive sensors (satellite)			For sea surface temperature, sea surface wind speed and soil moisture measurements
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.

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7145 MHz - 7190 MHz

FIXED MOBILE SPACE RESEARCH (DEEP SPACE) (EARTH-TO-SPACE) 5.458	FIXED MOBILE SPACE RESEARCH (DEEP SPACE) (EARTH-TO-SPACE) Space Operation (Earth-to-space) 5.458	Fixed	ECC/REC/(02)06	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB. On-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.

7190 MHz - 7235 MHz

EARTH EXPLORATION-SATELLITE (5.460A 5.460B) FIXED MOBILE SPACE RESEARCH (5.460) 5.458	EARTH EXPLORATION-SATELLITE (5.460A 5.460B) FIXED MOBILE SPACE RESEARCH (5.460) 5.458	Fixed	ECC/REC/(02)06	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Passive sensors (satellite)			For sea surface temperature, sea surface wind speed and soil moisture measurements
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB. On-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.

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7235 MHz - 7250 MHz

EARTH EXPLORATION-SATELLITE (5.460A) FIXED MOBILE 5.458	EARTH EXPLORATION-SATELLITE (5.460A) FIXED Space research (Earth-to-space) 5.458	Fixed	ECC/REC/(02)06	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Passive sensors (satellite)			For sea surface temperature, sea surface wind speed and soil moisture measurements
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.

7250 MHz - 7300 MHz

FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE 5.461	FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE 5.461 ECA36	Fixed	ECC/REC/(02)06	EN 302 217	Point-to-point. FIXED and MOBILE services not to be implemented in most NATO countries
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Land military systems			
		MSS Earth stations			Mobile satellite applications within the band 7250-7375 MHz
		Satellite systems (military)			
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.

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7300 MHz - 7375 MHz

FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE EXCEPT AERONAUTICAL MOBILE 5.461	FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE EXCEPT AERONAUTICAL MOBILE 5.461 ECA36	Fixed	ECC/REC/(02)06	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Land military systems			
		MSS Earth stations			Mobile satellite applications within the band 7250-7375 MHz
		Maritime military systems			
		Satellite systems (military)			
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB. On-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.

7375 MHz - 7450 MHz

FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MARITIME MOBILE-SATELLITE (5.461AA 5.461AB) MOBILE EXCEPT AERONAUTICAL MOBILE 5.461AC	FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MARITIME MOBILE-SATELLITE (5.461AA 5.461AB) MOBILE EXCEPT AERONAUTICAL MOBILE 5.461AC ECA36	Fixed	ECC/REC/(02)06	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Land military systems			
		MSS Earth stations			
		Maritime military systems			
		Satellite systems (military)			
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB. On-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.

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7450 MHz - 7550 MHz

FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MARITIME MOBILE-SATELLITE (5.461AA 5.461AB) METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE EXCEPT AERONAUTICAL MOBILE 5.461A 5.461AC	FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MARITIME MOBILE-SATELLITE (5.461AA 5.461AB) METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) MOBILE EXCEPT AERONAUTICAL MOBILE 5.461A 5.461AC ECA36	Fixed	ECC/REC/(02)06	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Land military systems			
		Maritime military systems			
		Satellite systems (military)			
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.
		Weather satellites			Limited to geostationary systems

7550 MHz - 7750 MHz

FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MARITIME MOBILE-SATELLITE (5.461AA 5.461AB) MOBILE EXCEPT AERONAUTICAL MOBILE 5.461AC	FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MARITIME MOBILE-SATELLITE (5.461AA 5.461AB) MOBILE EXCEPT AERONAUTICAL MOBILE 5.461AC ECA36	Fixed	ECC/REC/(02)06	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Land military systems			
		Maritime military systems			
		Satellite systems (military)			
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB. On-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.

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7750 MHz - 7900 MHz

FIXED METEOROLOGICAL-SATELLITE (5.461B) MOBILE EXCEPT AERONAUTICAL MOBILE	FIXED METEOROLOGICAL-SATELLITE (5.461B) MOBILE EXCEPT AERONAUTICAL MOBILE	Fixed	ECC/REC/(02)06	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.
		Weather satellites			Limited to non-geostationary systems

7900 MHz - 8025 MHz

FIXED FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE 5.461	FIXED FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE 5.461 ECA36	Fixed	ECC/REC/(02)06	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Land military systems			
		MSS Earth stations			Mobile satellite applications
		Satellite systems (military)			
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB. On-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.

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8025 MHz - 8175 MHz

EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH) FIXED FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE (5.463) 5.462A	EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH) FIXED FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE (5.463) 5.462A ECA36	Earth exploration-satellite			Satellite payload telemetry
		Fixed	ECC/REC/(02)06	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Land military systems			
		Land mobile			Mobile applications within the band 8025-8200 MHz
		Satellite systems (military)			
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.

8175 MHz - 8215 MHz

EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH) FIXED FIXED-SATELLITE (EARTH-TO-SPACE) METEOROLOGICAL-SATELLITE (EARTH-TO-SPACE) MOBILE (5.463) 5.462A	EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH) FIXED FIXED-SATELLITE (EARTH-TO-SPACE) METEOROLOGICAL-SATELLITE (EARTH-TO-SPACE) MOBILE (5.463) 5.462A ECA36	Earth exploration-satellite			Satellite payload telemetry
		Fixed	ECC/REC/(02)06	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Land military systems			
		Land mobile			Mobile applications within the band 8025-8200 MHz
		Satellite systems (military)			
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB. On-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.

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8215 MHz - 8400 MHz

EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH) FIXED FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE (5.463) 5.462A	EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH) FIXED FIXED-SATELLITE (EARTH-TO-SPACE) 5.462A 5.463	Earth exploration-satellite			Satellite payload telemetry
		Fixed	ECC/REC/(02)06	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Land military systems			
		Radio astronomy			Continuum observations, VLBI (used by SRS)
		Satellite systems (military)			
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.

8400 MHz - 8500 MHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE SPACE RESEARCH (5.465 5.466)	FIXED SPACE RESEARCH (5.465) Radiolocation	Fixed	ECC/REC/(02)06	EN 302 217	Point-to-point
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Space research			Satellite payload telemetry. The band 8400-8450 MHz is limited to deep space applications. Continuum observations, VLBI (used by SRS)
		UWB applications	ECC/DEC/(06)04, ECC/DEC/(07)01, ECC/DEC/(12)03, ERC/REC 70-03	EN 302 065	Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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8500 MHz - 8550 MHz

RADIOLOCATION 5.468 5.469	RADIOLOCATION 5.469 ECA24 ECA36	Aeronautical military systems			
		Aeronautical navigation		EN 303 064	Civil and military e.g. airfield approach
		Radiolocation (civil)		EN 303 135	Shipborne, land and airborne surveillance
		Radiolocation (military)			Shipborne, land and airborne surveillance
		TLPR	ERC/REC 70-03	EN 302 372	
		UWB applications	ECC/DEC/(06)04, ERC/REC 70-03	EN 302 065	Generic UWB

8550 MHz - 8650 MHz

EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION SPACE RESEARCH (ACTIVE) 5.468 5.469 5.469A	EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION SPACE RESEARCH (ACTIVE) 5.469 5.469A ECA24 ECA36	Active sensors (satellite)			
		Aeronautical military systems			
		Aeronautical navigation		EN 303 064	Civil and military e.g. airfield approach
		Radiolocation (civil)		EN 303 135	Shipborne, land and airborne surveillance
		Radiolocation (military)			Shipborne, land and airborne surveillance
		TLPR	ERC/REC 70-03	EN 302 372	
		UWB applications	ECC/DEC/(06)04, ERC/REC 70-03	EN 302 065	Generic UWB

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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8650 MHz - 8750 MHz

RADIOLOCATION 5.468 5.469	RADIOLOCATION 5.469 ECA24 ECA36	Aeronautical military systems			
		Aeronautical navigation		EN 303 064	Civil and military e.g. airfield approach
		Radiolocation (civil)		EN 303 135	Shipborne, land and airborne surveillance
		Radiolocation (military)			Shipborne, land and airborne surveillance
		TLPR	ERC/REC 70-03	EN 302 372	
		UWB applications	ECC/DEC/(06)04, ERC/REC 70-03	EN 302 065	Generic UWB

8750 MHz - 8850 MHz

AERONAUTICAL RADIONAVIGATION (5.470) RADIOLOCATION 5.471	AERONAUTICAL RADIONAVIGATION (5.470) RADIOLOCATION Space research 5.471 ECA24 ECA36	Aeronautical military systems			
		Aeronautical navigation		EN 303 064	Civil and military e.g. airfield approach
		Radiolocation (civil)		EN 303 135	Shipborne, land and airborne surveillance
		Radiolocation (military)			Shipborne, land and airborne surveillance
		TLPR	ERC/REC 70-03	EN 302 372	
		UWB applications	ECC/DEC/(06)04, ERC/REC 70-03	EN 302 065	Generic UWB

8850 MHz - 9000 MHz

MARITIME RADIONAVIGATION (5.472) RADIOLOCATION 5.473	MARITIME RADIONAVIGATION (5.472) RADIOLOCATION Space research 5.473 ECA24 ECA36	Aeronautical military systems			
		Aeronautical navigation		EN 303 064	Civil and military e.g. airfield approach
		Radiolocation (civil)		EN 303 135	Shipborne, land and airborne surveillance
		Radiolocation (military)			Shipborne, land and airborne surveillance
		TLPR	ERC/REC 70-03	EN 302 372	
		UWB applications	ECC/DEC/(06)04, ERC/REC 70-03	EN 302 065	Generic UWB

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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9000 MHz - 9200 MHz

AERONAUTICAL RADIONAVIGATION (5.337) Radiolocation 5.471 5.473A	AERONAUTICAL RADIONAVIGATION (5.337) RADIOLOCATION Space research 5.471 5.473A ECA24 ECA36	Aeronautical military systems			
		Aeronautical navigation		EN 303 064	Civil and military e.g. airfield approach
		Radiolocation (civil)		EN 303 135, EN 303 213	Shipborne, land and airborne surveillance. EN 303 213-1 X-band sensors
		Radiolocation (military)			Shipborne, land and airborne surveillance
		TLPR	ERC/REC 70-03	EN 302 372	

9200 MHz - 9300 MHz

EARTH EXPLORATION-SATELLITE (5.474A 5.474B 5.474C) MARITIME RADIONAVIGATION (5.472) RADIOLOCATION 5.473 5.474 5.474D	EARTH EXPLORATION-SATELLITE (5.474A 5.474B 5.474C) MARITIME RADIONAVIGATION (5.472) RADIOLOCATION Space research 5.473 5.474 5.474D ECA24 ECA36	Aeronautical military systems			
		Aeronautical navigation		EN 303 064	Civil and military e.g. airfield approach
		Radiodetermination applications	ERC/REC 70-03	EN 300 440	Within the band 9200-9975 MHz;
		Radiolocation (civil)		EN 303 135	Shipborne, land and airborne surveillance
		Radiolocation (military)			Shipborne, land and airborne surveillance
		Synthetic aperture radar			
		TLPR	ERC/REC 70-03	EN 302 372	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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9300 MHz - 9500 MHz

EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION RADIONAVIGATION (5.475) SPACE RESEARCH (ACTIVE) 5.427 5.474 5.475 5.475A 5.475B 5.476A	EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION RADIONAVIGATION (5.475) SPACE RESEARCH (ACTIVE) 5.427 5.474 5.475 5.475A 5.475B 5.476A ECA24 ECA36	Aeronautical military systems			
		Aeronautical navigation		EN 303 064	Civil and military e.g. airfield approach
		Radiodetermination applications	ERC/REC 70-03	EN 300 440	Within the band 9200-9975 MHz;
		Radiolocation (civil)		EN 302 194, EN 302 248, EN 302 752, EN 303 135, EN 303 213	Shipborne, land and airborne surveillance EN 303 213-6-1 X-band sensors
		Radiolocation (military)			Shipborne, land and airborne surveillance
		Satellite systems (military)			
		TLPR	ERC/REC 70-03	EN 302 372	
		Weather radar		EN 303 347	Shipborne, land and airborne surveillance

9500 MHz - 9800 MHz

EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION RADIONAVIGATION SPACE RESEARCH (ACTIVE) 5.476A	EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION SPACE RESEARCH (ACTIVE) 5.476A ECA24 ECA36	Active sensors (satellite)			
		Aeronautical military systems			
		Aeronautical navigation		EN 303 064	Civil and military e.g. airfield approach
		Radiodetermination applications	ERC/REC 70-03	EN 300 440	Within the band 9200-9975 MHz
		Radiolocation (civil)		EN 303 135	Shipborne, land and airborne surveillance
		Radiolocation (military)			Shipborne, land and airborne surveillance
		Satellite systems (military)			
		TLPR	ERC/REC 70-03	EN 302 372	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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9800 MHz - 9900 MHz

RADIOLOCATION Earth exploration-satellite (active) Fixed Space research (active) 5.477 5.478 5.478A 5.478B	RADIOLOCATION Earth exploration-satellite (active) Space research (active) 5.478 5.478A 5.478B ECA24 ECA36	Aeronautical military systems			
		Aeronautical navigation		EN 303 064	Civil and military e.g. airfield approach
		Radiodetermination applications	ERC/REC 70-03	EN 300 440	Within the band 9200-9975 MHz;
		Radiolocation (civil)		EN 303 135	Shipborne, land and airborne surveillance
		Radiolocation (military)			Shipborne, land and airborne surveillance
		Satellite systems (military)			
		TLPR	ERC/REC 70-03	EN 302 372	

9900 MHz - 10000 MHz

EARTH EXPLORATION-SATELLITE (5.474A 5.474B 5.474C) RADIOLOCATION Fixed 5.477 5.478 5.479	EARTH EXPLORATION-SATELLITE (5.474A 5.474B 5.474C) RADIOLOCATION Fixed 5.477 5.478 5.479	Aeronautical military systems			
		Aeronautical navigation		EN 303 064	Civil and military e.g. Airfield approach
		Radiodetermination applications	ERC/REC 70-03	EN 300 440	Within the band 9200-9975 MHz
		Radiolocation (civil)		EN 303 135	Shipborne, land and airborne surveillance
		Radiolocation (military)			Shipborne, land and airborne surveillance
		Satellite systems (military)			
		Synthetic aperture radar			
		TLPR	ERC/REC 70-03	EN 302 372	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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10000 MHz - 10400 MHz

EARTH EXPLORATION-SATELLITE (5.474A 5.474B 5.474C) FIXED MOBILE RADIOLOCATION Amateur 5.474D 5.479	EARTH EXPLORATION-SATELLITE (5.474A 5.474B 5.474C) FIXED MOBILE RADIOLOCATION Amateur 5.474D 5.479 ECA17A ECA36	Aeronautical military systems			
		Amateur		EN 301 783	Within the band 10-10.5 GHz
		FWA		EN 302 326	Including Point-to-Multipoint
		Fixed	ERC/REC 12-05	EN 302 217	
		Land military systems			
		Maritime military systems			
		Radiolocation (civil)			
		Radiolocation (military)			
		Synthetic aperture radar			
		TLPR	ERC/REC 70-03	EN 302 372	
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Temporary point-to-point video link

10400 MHz - 10450 MHz

FIXED MOBILE RADIOLOCATION Amateur	FIXED RADIOLOCATION Amateur Mobile ECA17 ECA17A ECA36	Aeronautical military systems			
		Amateur		EN 301 783	Within the band 10-10.5 GHz
		Land military systems			
		Maritime military systems			
		Radiolocation (civil)			Low power radars in certain subbands
		Radiolocation (military)			
		TLPR	ERC/REC 70-03	EN 302 372	
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Temporary point-to-point video link

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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10450 MHz - 10.5 GHz

RADIOLOCATION Amateur Amateur-satellite 5.481	FIXED MOBILE RADIOLOCATION Amateur Amateur-satellite 5.481 ECA17 ECA17A ECA23 ECA36	Aeronautical military systems			
		Amateur		EN 301 783	Within the band 10-10.5 GHz
		Amateur-satellite			
		Land military systems			
		Maritime military systems			
		Radiolocation (civil)			
		Radiolocation (military)			
		TLPR	ERC/REC 70-03	EN 302 372	
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Temporary point-to-point video link

10.5 GHz - 10.55 GHz

FIXED MOBILE Radiolocation	FIXED MOBILE Radiolocation ECA17A	Fixed	ERC/REC 12-05	EN 302 217, EN 302 326	Including Point-to-Multipoint
		Radiodetermination applications	ERC/REC 70-03	EN 300 440	Within the band 10.5-10.6 GHz;
		TLPR	ERC/REC 70-03	EN 302 372	
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Temporary point-to-point video link

10.55 GHz - 10.6 GHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE Radiolocation	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE Radiolocation ECA17A	Fixed	ERC/REC 12-05	EN 302 217, EN 302 326	Including Point-to-Multipoint
		Radiodetermination applications	ERC/REC 70-03	EN 300 440	Within the band 10.5-10.6 GHz,
		TLPR	ERC/REC 70-03	EN 302 372	
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Temporary point-to-point video link

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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10.6 GHz - 10.68 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) Mobile except aeronautical mobile Radiolocation 5.149 5.482 5.482A	EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) Mobile except aeronautical mobile Radiolocation 5.149 5.482 5.482A ECA17	Fixed	ECC/DEC/(10)01, ERC/REC 12-05	EN 302 217, EN 302 326	Including Point-to-Multipoint
		Passive sensors (satellite)	ECC/DEC/(10)01		Surface emissivity and precipitation measurements
		Radio astronomy			Continuum observations, VLBI
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Portable video links; Temporary point-to-point video link

10.68 GHz - 10.7 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.483	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	Passive sensors (satellite)			Surface emissivity and precipitation measurement
		Radio astronomy			Continuum observations, VLBI

10.7 GHz - 10.95 GHz

FIXED FIXED-SATELLITE (5.484) FIXED-SATELLITE (5.441) MOBILE EXCEPT AERONAUTICAL MOBILE	FIXED FIXED-SATELLITE (5.484) FIXED-SATELLITE (5.441) MOBILE EXCEPT AERONAUTICAL MOBILE Mobile-Satellite (space-to-Earth)	AES	ECC/DEC/(05)11, ECC/DEC/(19)04	EN 302 186	
		ESIM	ECC/DEC/(18)04, ECC/DEC/(18)05	EN 302 448, EN 302 977, EN 303 980, EN 303 981	
		ESV	ECC/DEC/(05)10	EN 302 340	
		FSS Earth stations	ERC/DEC/(00)08	EN 301 427, EN 301 430, EN 302 448	Within the band 10.7-10.95/11.2-11.45 GHz in accordance with App 30B of RR - VSAT
		Fixed	ERC/DEC/(00)08, ERC/REC 12-06	EN 302 217	Limited to high capacity fixed links
		HEST	ECC/DEC/(06)03	EN 301 428, EN 301 459	
		NGSO FSS	ECC/DEC/(17)04	EN 303 980, EN 303 981	
		VSAT	ECC/DEC/(03)04	EN 301 428	SNG

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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10.95 GHz - 11.2 GHz

FIXED FIXED-SATELLITE (5.484) FIXED-SATELLITE (5.484A 5.484B) MOBILE EXCEPT AERONAUTICAL MOBILE	FIXED FIXED-SATELLITE (5.484) FIXED-SATELLITE (5.484A 5.484B) MOBILE EXCEPT AERONAUTICAL MOBILE	AES	ECC/DEC/(05)11, ECC/DEC/(19)04	EN 302 186	
		ESIM	ECC/DEC/(18)04, ECC/DEC/(18)05	EN 302 448, EN 302 977, EN 303 980, EN 303 981	
		ESV	ECC/DEC/(05)10	EN 302 340	
		Fixed	ERC/DEC/(00)08, ERC/REC 12-06	EN 302 217	Limited to high capacity fixed links
		HEST	ECC/DEC/(06)03	EN 301 428, EN 301 459	
		NGSO FSS	ECC/DEC/(17)04	EN 303 980, EN 303 981	
		VSAT	ECC/DEC/(03)04	EN 301 428	SNG

11.2 GHz - 11.45 GHz

FIXED FIXED-SATELLITE (5.484) FIXED-SATELLITE (5.441) MOBILE EXCEPT AERONAUTICAL MOBILE	FIXED FIXED-SATELLITE (5.484) FIXED-SATELLITE (5.441) MOBILE EXCEPT AERONAUTICAL MOBILE	AES	ECC/DEC/(05)11, ECC/DEC/(19)04	EN 302 186	
		ESIM	ECC/DEC/(18)04, ECC/DEC/(18)05	EN 302 448, EN 302 977, EN 303 980, EN 303 981	
		ESV	ECC/DEC/(05)10	EN 302 340	
		Fixed	ERC/DEC/(00)08, ERC/REC 12-06	EN 302 217	Limited to high capacity fixed links
		HEST	ECC/DEC/(06)03	EN 301 428, EN 301 459	
		NGSO FSS	ECC/DEC/(17)04	EN 303 980, EN 303 981	
		VSAT	ECC/DEC/(03)04	EN 301 428	SNG

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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11.45 GHz - 11.7 GHz

FIXED FIXED-SATELLITE (5.484) FIXED-SATELLITE (5.484A 5.484B) MOBILE EXCEPT AERONAUTICAL MOBILE	FIXED FIXED-SATELLITE (5.484) FIXED-SATELLITE (5.484A 5.484B) MOBILE EXCEPT AERONAUTICAL MOBILE	AES	ECC/DEC/(05)11, ECC/DEC/(19)04	EN 302 186	
		ESIM	ECC/DEC/(18)04, ECC/DEC/(18)05	EN 302 448, EN 302 977, EN 303 980, EN 303 981	
		ESV	ECC/DEC/(05)10	EN 302 340	
		Fixed	ERC/DEC/(00)08, ERC/REC 12-06	EN 302 217	Limited to high capacity fixed links
		HEST	ECC/DEC/(06)03	EN 301 428, EN 301 459	
		NGSO FSS	ECC/DEC/(17)04	EN 303 980, EN 303 981	
		VSAT	ECC/DEC/(03)04	EN 301 428	SNG

11.7 GHz - 12.5 GHz

BROADCASTING BROADCASTING-SATELLITE (5.492) FIXED Mobile except aeronautical mobile 5.487 5.487A	BROADCASTING-SATELLITE (5.492) MOBILE EXCEPT AERONAUTICAL MOBILE 5.487 5.487A ECA28	AES	ECC/DEC/(19)04	EN 302 186	
		Broadcasting (satellite)	ERC/DEC/(00)08	EN 302 340, EN 302 448	In accordance with App 30 of RR. SIT within the band 12.4 - 12.5 GHz
		ESIM	ECC/DEC/(18)04, ECC/DEC/(18)05	EN 302 448, EN 302 977, EN 303 980, EN 303 981	
		HEST	ECC/DEC/(06)03	EN 301 428, EN 301 459	
		NGSO FSS	ECC/DEC/(17)04	EN 303 980, EN 303 981	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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12.5 GHz - 12.75 GHz

FIXED-SATELLITE (EARTH-TO-SPACE) FIXED-SATELLITE (5.484A 5.484B) 5.494 5.495 5.496	FIXED-SATELLITE (EARTH-TO-SPACE) FIXED-SATELLITE (5.484A 5.484B) 5.496	AES	ECC/DEC/(05)11, ECC/DEC/(19)04	EN 302 186	
		ESIM	ECC/DEC/(18)04, ECC/DEC/(18)05	EN 302 448, EN 302 977, EN 303 980, EN 303 981	
		ESV	ECC/DEC/(05)10	EN 302 340	
		HEST	ECC/DEC/(06)03	EN 301 428, EN 301 459	
		NGSO FSS	ECC/DEC/(17)04	EN 303 980, EN 303 981	

12.75 GHz - 13.25 GHz

FIXED FIXED-SATELLITE (5.441 5.496A) MOBILE Space research (deep space) (space-to-Earth)	FIXED FIXED-SATELLITE (5.441 5.496A)	AES	ECC/DEC/(19)04	EN 302 186	
		FSS Earth stations		EN 301 430	
		Fixed	ERC/REC 12-02	EN 302 217	

13.25 GHz - 13.4 GHz

AERONAUTICAL RADIONAVIGATION (5.497) EARTH EXPLORATION-SATELLITE (ACTIVE) SPACE RESEARCH (ACTIVE) 5.498A 5.499	AERONAUTICAL RADIONAVIGATION (5.497) EARTH EXPLORATION-SATELLITE (ACTIVE) SPACE RESEARCH (ACTIVE) 5.498A ECA26	Active sensors (satellite)			Altimeters, scatterometers, precipitation radars
		Airborne doppler navigation aids			
		Maritime radar			Ship berthing radars

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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13.4 GHz - 13.65 GHz

EARTH EXPLORATION-SATELLITE (ACTIVE) FIXED-SATELLITE (5.499A 5.499B) RADIOLOCATION SPACE RESEARCH (5.499C 5.499D) Standard frequency and time signal-satellite (Earth-to-space) 5.499E 5.500 5.501 5.501B	EARTH EXPLORATION-SATELLITE (ACTIVE) FIXED-SATELLITE (5.499A 5.499B) RADIOLOCATION SPACE RESEARCH (5.499C 5.499D) 5.501B ECA26 ECA36	-			Data relay satellites
		Active sensors (satellite)			Altimeters, scatterometers, precipitation radars
		Airborne doppler navigation aids			
		FSS Earth stations			
		Maritime radar			Ship berthing radars
		Radiodetermination applications	ERC/REC 70-03	EN 300 440	Within the band 13.4-14.0 GHz
		Radiolocation (military)			

13.65 GHz - 13.75 GHz

EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION SPACE RESEARCH (5.501A) Standard frequency and time signal-satellite (Earth-to-space) 5.499 5.500 5.501 5.501B	EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION SPACE RESEARCH (5.501A) 5.501B ECA26 ECA36	-			Data relay satellites
		Active sensors (satellite)			Altimeters, scatterometers, precipitation radars
		Airborne doppler navigation aids			
		Maritime radar			Ship berthing radars
		Radiodetermination applications	ERC/REC 70-03	EN 300 440	Within the band 13.4-14.0 GHz
		Radiolocation (military)			

13.75 GHz - 14 GHz

FIXED-SATELLITE (5.484A) RADIOLOCATION Earth exploration-satellite Space research Standard frequency and time signal-satellite (Earth-to-space) 5.499 5.500 5.501 5.502 5.503	FIXED-SATELLITE (5.484A) RADIOLOCATION Space research 5.502 5.503 ECA26 ECA36	-			Data relay satellites
		FSS Earth stations		EN 301 430	minimum antenna size imposed according to 5.502
		Maritime radar			Navigation radars, ship berthing radars
		Passive sensors (satellite)			Future VLBI measurements
		Radiodetermination applications	ERC/REC 70-03	EN 300 440	Within the band 13.4-14.0 GHz
		Radiolocation (military)			

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14 GHz - 14.25 GHz

FIXED-SATELLITE (5.457A 5.457B 5.484A 5.506 5.506B 5.484B) RADIONAVIGATION (5.504) Mobile-Satellite (5.504B 5.504C 5.506A) Space research 5.504A 5.505	FIXED-SATELLITE (5.457A 5.457B 5.484A 5.484B 5.506 5.506B) Mobile-Satellite (5.504B 5.504C 5.506A) Space research 5.504	AES	ECC/DEC/(05)11	EN 302 186	
		ESIM	ECC/DEC/(18)04, ECC/DEC/(18)05	EN 302 448, EN 302 977, EN 303 980, EN 303 981	
		ESV	ECC/DEC/(05)10	EN 302 340	
		HEST	ECC/DEC/(06)03	EN 301 428, EN 301 459	
		MSS Earth stations		EN 301 427, EN 302 977	
		NGSO FSS	ECC/DEC/(17)04	EN 303 980, EN 303 981	
		VSAT	ECC/DEC/(03)04, ERC/REC 13-03	EN 301 428, EN 301 430	Low density carriers, including VSATs and digital SNG, are encouraged to use this band

14.25 GHz - 14.3 GHz

FIXED-SATELLITE (5.457A 5.457B 5.484A 5.484B 5.506 5.506B) RADIONAVIGATION (5.504) Mobile-Satellite (5.504B 5.506A 5.508A) Space research 5.504A 5.505 5.508	FIXED-SATELLITE (5.457A 5.457B 5.484A 5.484B 5.506 5.506B) Mobile-Satellite (5.504B 5.506A 5.508A) Space research 5.504	AES	ECC/DEC/(05)11	EN 302 186	
		ESIM	ECC/DEC/(18)04, ECC/DEC/(18)05	EN 302 448, EN 302 977, EN 303 980, EN 303 981	
		ESV	ECC/DEC/(05)10	EN 302 340	
		MSS Earth stations		EN 301 427, EN 302 977	
		NGSO FSS	ECC/DEC/(17)04	EN 303 980, EN 303 981	
		VSAT	ECC/DEC/(03)04, ERC/REC 13-03	EN 301 428, EN 301 430	SNG

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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14.3 GHz - 14.4 GHz

FIXED FIXED-SATELLITE (5.506 5.457A 5.506B 5.484A 5.457B 5.484B) MOBILE EXCEPT AERONAUTICAL MOBILE Mobile-Satellite (5.504B 5.506A 5.509A) Radionavigation-satellite 5.504A	FIXED-SATELLITE (5.457A 5.457B 5.484A 5.484B 5.506 5.506B) Mobile-Satellite (5.504B 5.506A 5.509A)	AES	ECC/DEC/(05)11	EN 302 186	
		ESIM	ECC/DEC/(18)04, ECC/DEC/(18)05	EN 302 448, EN 302 977, EN 303 980, EN 303 981	
		ESV	ECC/DEC/(05)10	EN 302 340	
		FSS Earth stations		EN 302 340	Fixed links to be coordinated with Fixed Satellite Services on a national basis
		MSS Earth stations		EN 301 427, EN 302 977	
		NGSO FSS	ECC/DEC/(17)04	EN 303 980, EN 303 981	
		VSAT	ECC/DEC/(03)04, ERC/REC 13-03	EN 301 428, EN 301 430	SNG

14.4 GHz - 14.47 GHz

FIXED FIXED-SATELLITE (5.457A 5.457B 5.484A 5.506 5.506B 5.484B) MOBILE EXCEPT AERONAUTICAL MOBILE Mobile-Satellite (5.504B 5.506A 5.509A) Space research (space-to-Earth) 5.504A	FIXED-SATELLITE (5.457A 5.457B 5.484A 5.484B 5.506 5.506B) Mobile-Satellite (5.504B 5.506A 5.509A) 5.504A	AES	ECC/DEC/(05)11	EN 302 186	
		ESIM	ECC/DEC/(18)04, ECC/DEC/(18)05	EN 302 448, EN 302 977, EN 303 980, EN 303 981	
		ESV	ECC/DEC/(05)10	EN 302 340	
		FSS Earth stations		EN 302 340	Fixed links to be coordinated with Fixed Satellite Services on a national basis
		MSS Earth stations		EN 301 427, EN 302 977	
		NGSO FSS	ECC/DEC/(17)04	EN 303 980, EN 303 981	
		VSAT	ECC/DEC/(03)04, ERC/REC 13-03	EN 301 428, EN 301 430	SNG

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14.47 GHz - 14.5 GHz

FIXED FIXED-SATELLITE (5.457A 5.457B 5.484A 5.506 5.506B) MOBILE EXCEPT AERONAUTICAL MOBILE Mobile-Satellite (5.504B 5.506A 5.509A) Radio astronomy 5.149 5.504A	FIXED-SATELLITE (5.457A 5.484A 5.506) Mobile-Satellite (5.504B 5.506A 5.509A) Radio astronomy 5.149 5.504A	AES	ECC/DEC/(05)11	EN 302 186	
		ESIM	ECC/DEC/(18)04, ECC/DEC/(18)05	EN 302 448, EN 302 977, EN 303 980, EN 303 981	
		ESV	ECC/DEC/(05)10	EN 302 340	
		FSS Earth stations		EN 302 340	Fixed links to be coordinated with Fixed Satellite Service on a national basis
		MSS Earth stations		EN 301 427, EN 302 977	
		NGSO FSS	ECC/DEC/(17)04	EN 303 980, EN 303 981	
		Radio astronomy			Spectral line observations, VLBI
		VSAT	ERC/REC 13-03	EN 301 428, EN 301 430	SNG

14.5 GHz - 14.75 GHz

FIXED FIXED-SATELLITE (5.510 5.509B 5.509C 5.509D 5.509E 5.509F) MOBILE Space research (5.509G)	FIXED MOBILE Radio astronomy ECA20 ECA36	Aeronautical military systems			
		Fixed	ERC/REC 12-07	EN 302 217	
		Land military systems			
		Maritime military systems			
		Radio astronomy			VLBI (when compatible with primary use)

14.75 GHz - 14.8 GHz

FIXED FIXED-SATELLITE (5.510) MOBILE Space research (5.509G)	FIXED MOBILE Radio astronomy ECA20 ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		Radio astronomy			VLBI (when compatible with primary use)

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14.8 GHz - 15.35 GHz

FIXED MOBILE SPACE RESEARCH (5.510A) 5.339	FIXED MOBILE Radio astronomy 5.339 ECA20 ECA36	Aeronautical military systems			
		Fixed	ERC/REC 12-07	EN 302 217	
		Land military systems			
		Maritime military systems			
		Radio astronomy			VLBI (when compatible with primary use)

15.35 GHz - 15.4 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.511	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	Passive sensors (satellite)			
		Radio astronomy			Continuum observations, VLBI

15.4 GHz - 15.41 GHz

AERONAUTICAL RADIONAVIGATION RADIOLOCATION (5.511E 5.511F)	RADIOLOCATION (5.511E 5.511F) Aeronautical radionavigation	Airborne doppler navigation aids			Doppler radar low power sensing
		Radiolocation (civil)			Ground movement radars

15.41 GHz - 15.43 GHz

AERONAUTICAL RADIONAVIGATION RADIOLOCATION (5.511F 5.511E) Aeronautical mobile (5.511G)	RADIOLOCATION (5.511E 5.511F) Aeronautical radionavigation	Airborne doppler navigation aids			Doppler radar low power sensing
		Radiolocation (civil)			Ground movement radars

15.43 GHz - 15.63 GHz

AERONAUTICAL RADIONAVIGATION FIXED-SATELLITE (5.511A) RADIOLOCATION (5.511E 5.511F) Aeronautical mobile (5.511G) 5.511C	AERONAUTICAL RADIONAVIGATION FIXED-SATELLITE (EARTH-TO-SPACE) RADIOLOCATION (5.511E 5.511F) 5.511C	Airborne doppler navigation aids			Doppler radar low power sensing
		FSS Earth stations			MSS feeder links
		Radiolocation (civil)			Ground movement radars

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15.63 GHz - 15.7 GHz

AERONAUTICAL RADIONAVIGATION RADIOLOCATION (5.511E 5.511F) Aeronautical mobile (5.511G)	AERONAUTICAL RADIONAVIGATION RADIOLOCATION (5.511E 5.511F)	Airborne doppler navigation aids			Doppler radar low power sensing
		Radiolocation (civil)			Ground movement radars

15.7 GHz - 16.6 GHz

RADIOLOCATION 5.512 5.513	RADIOLOCATION ECA36	Radiolocation (military)			
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16.6 GHz - 17.1 GHz

RADIOLOCATION Space research (deep space) (Earth-to-space) 5.512 5.513	RADIOLOCATION Space research (deep space) (Earth-to-space) ECA36	Radiolocation (military)			
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17.1 GHz - 17.2 GHz

RADIOLOCATION 5.512 5.513	RADIOLOCATION Mobile ECA36	GBSAR	ERC/REC 70-03	EN 303 661	
		Radiolocation (military)			

17.2 GHz - 17.3 GHz

EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION SPACE RESEARCH (ACTIVE) 5.512 5.513 5.513A	EARTH EXPLORATION-SATELLITE (ACTIVE) MOBILE RADIOLOCATION SPACE RESEARCH (ACTIVE) 5.513A ECA36	GBSAR	ERC/REC 70-03	EN 303 661	
		Radiolocation (military)			

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17.3 GHz - 17.7 GHz

FIXED-SATELLITE (5.516) FIXED-SATELLITE (5.516A 5.516B) Radiolocation 5.514	FIXED-SATELLITE (5.516) FIXED-SATELLITE (5.516A 5.516B) Radiolocation ECA36	FSS Earth stations	ECC/DEC/(05)08		High Density FSS
		Feeder links			Feeder links for the BSS service. Appendix 30A of RR
		GSO ESOMPs	ECC/DEC/(13)01	EN 303 978	
		NGSO ESOMPs	ECC/DEC/(15)04	EN 303 979	
		Radiolocation (military)			

17.7 GHz - 18.1 GHz

FIXED FIXED-SATELLITE (5.516) FIXED-SATELLITE (5.484A 5.517A 5.517B) MOBILE	FIXED FIXED-SATELLITE (5.516 5.517A 5.517B) FIXED-SATELLITE (5.484A)	FSS Earth stations	ERC/DEC/(00)07		
		Feeder links			Feeder links for the BSS service. Appendix 30A of RR
		Fixed	ERC/DEC/(00)07, ERC/REC 12-03	EN 302 217	
		GSO ESOMPs	ECC/DEC/(13)01	EN 303 978	
		NGSO ESOMPs	ECC/DEC/(15)04	EN 303 979	

18.1 GHz - 18.4 GHz

FIXED FIXED-SATELLITE (5.520) FIXED-SATELLITE (5.484A 5.517A 5.517B 5.516B) INTER-SATELLITE (5.521A) MOBILE 5.519 5.521	FIXED FIXED-SATELLITE (5.520) FIXED-SATELLITE (5.484A 5.517A 5.517B) INTER-SATELLITE (5.521A) METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH) 5.519	FSS Earth stations	ERC/DEC/(00)07		
		Feeder links			Feeder links for the BSS service
		Fixed	ERC/DEC/(00)07, ERC/REC 12-03	EN 302 217	
		GSO ESOMPs	ECC/DEC/(13)01	EN 303 978	
		NGSO ESOMPs	ECC/DEC/(15)04	EN 303 979	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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18.4 GHz - 18.6 GHz

FIXED FIXED-SATELLITE (5.484A 5.517A 5.517B 5.516B) INTER-SATELLITE (5.521A) MOBILE	FIXED FIXED-SATELLITE (5.484A 5.517A 5.517B) INTER-SATELLITE (5.521A)	FSS Earth stations	ERC/DEC/(00)07		
		Fixed	ERC/DEC/(00)07, ERC/REC 12-03	EN 302 217	
		GSO ESOMPs	ECC/DEC/(13)01	EN 303 978	
		NGSO ESOMPs	ECC/DEC/(15)04	EN 303 979	

18.6 GHz - 18.8 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED FIXED-SATELLITE (5.522B 5.517A) MOBILE EXCEPT AERONAUTICAL MOBILE Space research (passive) 5.522A 5.522C	EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED FIXED-SATELLITE (5.522B) 5.522A	FSS Earth stations	ERC/DEC/(00)07		
		Fixed	ERC/DEC/(00)07, ERC/REC 12-03	EN 302 217	
		GSO ESOMPs	ECC/DEC/(13)01	EN 303 978	
		NGSO ESOMPs	ECC/DEC/(15)04	EN 303 979	
		Passive sensors (satellite)			Surface emissivity, snow, sea, ice and precipitation

18.8 GHz - 19.3 GHz

FIXED FIXED-SATELLITE (5.523A 5.517A 5.517B 5.516B) INTER-SATELLITE (5.521A) MOBILE	FIXED FIXED-SATELLITE (5.517A 5.517B 5.523A) INTER-SATELLITE (5.521A)	FSS Earth stations	ERC/DEC/(00)07		
		Fixed	ERC/DEC/(00)07, ERC/REC 12-03	EN 302 217	
		GSO ESOMPs	ECC/DEC/(13)01	EN 303 978	
		NGSO ESOMPs	ECC/DEC/(15)04	EN 303 979	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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19.3 GHz - 19.7 GHz

FIXED FIXED-SATELLITE (5.523B 5.523C 5.523D 5.523E 5.517A) INTER-SATELLITE (5.521A 5.523DA) MOBILE	FIXED FIXED-SATELLITE (5.517A 5.523B 5.523C 5.523D 5.523E)	FSS Earth stations	ERC/DEC/(00)07		
		Fixed	ERC/DEC/(00)07, ERC/REC 12-03	EN 302 217	
		GSO ESOMPs	ECC/DEC/(13)01	EN 303 978	
		NGSO ESOMPs	ECC/DEC/(15)04	EN 303 979	

19.7 GHz - 20.1 GHz

FIXED-SATELLITE (5.484A 5.516B 5.527A 5.484B 5.517B) INTER-SATELLITE (5.521A) Mobile-Satellite (space-to-Earth) 5.524	FIXED-SATELLITE (5.484A 5.484B 5.516B 5.517B 5.527A) INTER-SATELLITE (5.521A) Mobile-Satellite (space-to-Earth)	FSS Earth stations	ECC/DEC/(05)08		High Density FSS
		GSO ESOMPs	ECC/DEC/(13)01	EN 303 978	
		HEST	ECC/DEC/(06)03	EN 301 428, EN 301 459	
		MSS Earth stations			For uncoordinated Earth stations SUT
		NGSO ESOMPs	ECC/DEC/(15)04	EN 303 979	

20.1 GHz - 20.2 GHz

FIXED-SATELLITE (5.484A 5.516B 5.527A 5.484B 5.517B) INTER-SATELLITE (5.521A) MOBILE-SATELLITE (SPACE-TO- EARTH) 5.524 5.525 5.526 5.527 5.528	FIXED-SATELLITE (5.484A 5.484B 5.516B 5.517B 5.527A) INTER-SATELLITE (5.521A) MOBILE-SATELLITE (SPACE-TO- EARTH) 5.525 5.526 5.527 5.528	FSS Earth stations	ECC/DEC/(05)08		High Density FSS
		GSO ESOMPs	ECC/DEC/(13)01	EN 303 978	
		HEST	ECC/DEC/(06)03	EN 301 428, EN 301 459	
		MSS Earth stations			For uncoordinated Earth stations SUT
		NGSO ESOMPs	ECC/DEC/(15)04	EN 303 979	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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20.2 GHz - 21.2 GHz

FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE-SATELLITE (SPACE-TO-EARTH) Standard frequency and time signal-satellite (space-to-Earth) 5.524 5.529A	FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE-SATELLITE (SPACE-TO-EARTH) ECA36	MSS Earth stations			For uncoordinated Earth stations
		Satellite systems (military)			

21.2 GHz - 21.4 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED MOBILE SPACE RESEARCH (PASSIVE)	EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED MOBILE SPACE RESEARCH (PASSIVE)	Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Temporary point-to-point video link
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21.4 GHz - 22 GHz

BROADCASTING-SATELLITE (5.208B) FIXED MOBILE 5.530A 5.530B	BROADCASTING-SATELLITE (5.208B) 5.530A 5.530B	Broadcasting (satellite)			
		SRR	ECC/DEC/(04)10, ERC/REC 70-03		Within the frequency band 21.65-22 GHz
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Temporary point-to-point video link

22 GHz - 22.2 GHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.531D 5.531A 5.531B 5.531F 5.531C) 5.149	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.149 ECA17A	Fixed	T/R 13-02	EN 302 217, EN 302 326	
		Radio astronomy			Continuum and spectral line observations (e.g. water line), VLBI
		SRR	ECC/DEC/(04)10, ERC/REC 70-03		
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Temporary point-to-point video link

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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22.2 GHz - 22.21 GHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE 5.149	FIXED	Fixed	T/R 13-02	EN 302 217, EN 302 326	
		Radio astronomy			Continuum and spectral line observations (e.g. water line), VLBI
		SRR	ECC/DEC/(04)10, ERC/REC 70-03		
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Temporary point-to-point video link

22.21 GHz - 22.5 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED MOBILE EXCEPT AERONAUTICAL MOBILE RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.149 5.532	FIXED MOBILE EXCEPT AERONAUTICAL MOBILE RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) Earth exploration-satellite (passive) Mobile (ECA39) 5.149 5.532 ECA17A	Fixed	T/R 13-02	EN 302 217, EN 302 326	
		Radio astronomy			Continuum and spectral line observations (e.g. water line), VLBI
		SRR	ECC/DEC/(04)10, ERC/REC 70-03		
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Temporary point-to-point video link

22.5 GHz - 22.55 GHz

FIXED MOBILE	FIXED MOBILE (ECA39) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) ECA17A	Fixed	T/R 13-02	EN 302 217, EN 302 326	
		Radio astronomy			Continuum and spectral line observations (e.g. water line), VLBI
		SRR	ECC/DEC/(04)10, ERC/REC 70-03		
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Temporary point-to-point video link

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22.55 GHz - 23.15 GHz

FIXED INTER-SATELLITE (5.338A) MOBILE SPACE RESEARCH (5.532A) 5.149	FIXED INTER-SATELLITE (5.338A) MOBILE (ECA39) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) ECA17A	Fixed	T/R 13-02	EN 302 217, EN 302 326	
		Radio astronomy			Continuum and spectral line observations (e.g. water line), VLBI
		SRR	ECC/DEC/(04)10, ERC/REC 70-03		
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Temporary point-to-point video link

23.15 GHz - 23.55 GHz

FIXED INTER-SATELLITE (5.338A) MOBILE	FIXED INTER-SATELLITE (5.338A) MOBILE (ECA39)	Fixed	T/R 13-02	EN 302 217, EN 302 326	
		SRR	ECC/DEC/(04)10, ERC/REC 70-03		
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Temporary point-to-point video link

23.55 GHz - 23.6 GHz

FIXED MOBILE	FIXED INTER-SATELLITE MOBILE (ECA39)	Fixed	T/R 13-02	EN 302 217, EN 302 326	
		SRR	ECC/DEC/(04)10, ERC/REC 70-03		
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Temporary point-to-point video link

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23.6 GHz - 24 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	Passive sensors (satellite)			Measurement of water vapour, liquid water, clouds for atmospheric sounding
		Radio astronomy			Continuum and spectral line observations (e.g. ammonia line). VLBI
		SRR	ECC/DEC/(04)10, ERC/REC 70-03		
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Temporary point-to-point video link

24 GHz - 24.05 GHz

AMATEUR AMATEUR-SATELLITE 5.150	AMATEUR AMATEUR-SATELLITE 5.150	Amateur		EN 301 783	Within the band 24-24.25 GHz
		Amateur-satellite			
		ISM			Within the band 24-24.25 GHz
		Non-specific SRDs	ERC/REC 70-03	EN 300 440	Within the band 24-24.25 GHz
		SRR	ECC/DEC/(04)10, ERC/REC 70-03		
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Temporary point-to-point video link

24.05 GHz - 24.25 GHz

RADIOLOCATION Amateur Earth exploration-satellite (active) 5.150	RADIOLOCATION Amateur Earth exploration-satellite (active) Fixed Mobile 5.150 ECA36	Active sensors (satellite)			Rain radars from satellites
		Amateur		EN 301 783	Within the band 24-24.25 GHz
		ISM			Within the band 24-24.25 GHz
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Non-specific SRDs	ERC/REC 70-03	EN 300 440	Within the band 24-24.25 GHz
		Radiodetermination applications		EN 300 440	
		Radiolocation (military)			
		SRR	ECC/DEC/(04)10, ERC/REC 70-03		
		TLPR	ERC/REC 70-03	EN 302 372	
		TTT	ERC/REC 70-03	EN 302 858	Automotive radars
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Temporary point-to-point video link

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24.25 GHz - 24.45 GHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.338A 5.532AB)	FIXED MOBILE (5.338A 5.532AB) ECA17A	Fixed	T/R 13-02	EN 302 217, EN 302 326	Unidirectional fixed links
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		MFCN	ECC/DEC/(18)06, ECC/DEC/(22)01, ECC/REC/(23)02		
		SRR	ECC/DEC/(04)10, ERC/REC 70-03	EN 302 288	New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 24.25-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type-approval application has been submitted and has been granted before 1 January 2018
		TLPR	ERC/REC 70-03	EN 302 372	
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Temporary point-to-point video link

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24.45 GHz - 24.5 GHz

FIXED INTER-SATELLITE MOBILE EXCEPT AERONAUTICAL MOBILE (5.338A 5.532AB)	FIXED MOBILE (5.338A 5.532AB) ECA17A	Fixed	T/R 13-02	EN 302 217, EN 302 326	Unidirectional fixed links
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		MFCN	ECC/DEC/(18)06, ECC/DEC/(22)01, ECC/REC/(23)02		
		SRR	ECC/DEC/(04)10, ERC/REC 70-03	EN 302 288	New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type-approval application has been submitted and has been granted before 1 January 2018
		TLPR	ERC/REC 70-03	EN 302 372	
		Video PMSE	ERC/REC 25-10	EN 302 064	Cordless Cameras; Temporary point-to-point video link

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24.5 GHz - 24.65 GHz

FIXED INTER-SATELLITE MOBILE EXCEPT AERONAUTICAL MOBILE (5.338A 5.532AB)	FIXED MOBILE (5.338A 5.532AB)	FWA	ECC/REC/(11)01	EN 302 326	CRS paired with 25.5-26.5 GHz for FDD systems
		Fixed	T/R 13-02	EN 302 217, EN 302 326	
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		MFCN	ECC/DEC/(18)06, ECC/DEC/(22)01, ECC/REC/(23)02		
		SRR	ECC/DEC/(04)10, ERC/REC 70-03	EN 302 288	New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type-approval application has been submitted and has been granted before 1 January 2018
		TLPR	ERC/REC 70-03	EN 302 372	

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24.65 GHz - 24.75 GHz

FIXED FIXED-SATELLITE (5.532B) INTER-SATELLITE MOBILE EXCEPT AERONAUTICAL MOBILE (5.338A 5.532AB)	FIXED FIXED-SATELLITE (5.532B) MOBILE (5.338A 5.532AB)	FWA	ECC/REC/(11)01	EN 302 326	CRS paired with 25.5-26.5 GHz for FDD systems
		Fixed	T/R 13-02	EN 302 217, EN 302 326	
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		MFCN	ECC/DEC/(18)06, ECC/DEC/(22)01, ECC/REC/(20)01, ECC/REC/(23)02		
		SRR	ECC/DEC/(04)10, ERC/REC 70-03	EN 302 288	New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type-approval application has been submitted and has been granted before 1 January 2018
		TLPR	ERC/REC 70-03	EN 302 372	

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24.75 GHz - 25.25 GHz

FIXED FIXED-SATELLITE (5.532B) INTER-SATELLITE MOBILE EXCEPT AERONAUTICAL MOBILE (5.338A 5.532AB)	FIXED FIXED-SATELLITE (5.532B) INTER-SATELLITE MOBILE (5.338A 5.532AB)	FWA	ECC/REC/(11)01	EN 302 326	CRS paired with 25.5-26.5 GHz for FDD systems
		Fixed	T/R 13-02	EN 302 217, EN 302 326	
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		MFCN	ECC/DEC/(18)06, ECC/DEC/(22)01, ECC/REC/(20)01, ECC/REC/(23)02		
		SRR	ECC/DEC/(04)10, ERC/REC 70-03	EN 302 288	New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type-approval application has been submitted and has been granted before 1 January 2018
		TLPR	ERC/REC 70-03	EN 302 372	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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25.25 GHz - 25.5 GHz

FIXED (5.534A) INTER-SATELLITE (5.536) MOBILE (5.338A 5.532AB) Standard frequency and time signal-satellite (Earth-to-space)	FIXED INTER-SATELLITE (5.536) MOBILE (5.338A 5.532AB) ECA36	Aeronautical military systems			
		FWA	ECC/REC/(11)01	EN 302 326	CRS paired with 25.5-26.5 GHz for FDD systems
		Fixed	T/R 13-02	EN 302 217, EN 302 326	
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Land military systems			
		MFCN	ECC/DEC/(18)06, ECC/DEC/(22)01, ECC/REC/(23)02		
		Maritime military systems			
		SRR	ECC/DEC/(04)10, ERC/REC 70-03	EN 302 288	New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type-approval application has been submitted and has been granted before 1 January 2018
		TLPR	ERC/REC 70-03	EN 302 372	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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25.5 GHz - 26.5 GHz

EARTH EXPLORATION-SATELLITE (5.536B) FIXED (5.534A) INTER-SATELLITE (5.536) MOBILE (5.338A 5.532AB) SPACE RESEARCH (5.536C) Standard frequency and time signal-satellite (Earth-to-space) 5.536A	FIXED INTER-SATELLITE (5.536) MOBILE (5.338A 5.532AB) SPACE RESEARCH (5.536C) Earth exploration-satellite (5.536B) 5.536A ECA36	Aeronautical military systems			
		FWA	ECC/REC/(11)01	EN 302 326	TS should be paired with 24.5-25.5 GHz for FDD systems
		Fixed	T/R 13-02	EN 302 217, EN 302 326	
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Land military systems			
		MFCN	ECC/DEC/(18)06, ECC/DEC/(22)01, ECC/REC/(19)01, ECC/REC/(23)02		
		Maritime military systems			
		SRR	ECC/DEC/(04)10, ERC/REC 70-03	EN 302 288	New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 24.25-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type-approval application has been submitted and has been granted before 1 January 2018
		Space research	ECC/REC/(19)01		Satellite payload telemetry
		TLPR	ERC/REC 70-03	EN 302 372	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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26.5 GHz - 27 GHz

EARTH EXPLORATION-SATELLITE (5.536B) FIXED (5.534A) INTER-SATELLITE (5.536) MOBILE (5.338A 5.532AB) SPACE RESEARCH (5.536C) Standard frequency and time signal-satellite (Earth-to-space) 5.536A	FIXED INTER-SATELLITE (5.536) MOBILE (5.338A 5.532AB) SPACE RESEARCH (5.536C) Earth exploration-satellite (5.536B) 5.536A ECA36	Land military systems			
		MFCN	ECC/DEC/(18)06, ECC/DEC/(22)01, ECC/REC/(19)01, ECC/REC/(23)02		
		SRR	ECC/DEC/(04)10, ERC/REC 70-03	EN 302 288	New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 24.25-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type approval application has been submitted and has been granted before 1 January 2018
		Space research	ECC/REC/(19)01		Satellite payload telemetry
		TLPR	ERC/REC 70-03	EN 302 372	

27 GHz - 27.5 GHz

FIXED INTER-SATELLITE (5.536) MOBILE (5.338A 5.532AB)	FIXED INTER-SATELLITE (5.536) MOBILE (5.338A 5.532AB) Earth exploration-satellite (space-to-Earth) ECA36	Land military systems			
		MFCN	ECC/DEC/(18)06, ECC/DEC/(22)01, ECC/REC/(19)01, ECC/REC/(23)02		

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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27.5 GHz - 28.5 GHz

FIXED (5.537A) FIXED-SATELLITE (5.484A 5.516B 5.539 5.517A 5.517B) INTER-SATELLITE (5.521A) MOBILE 5.538 5.540	FIXED FIXED-SATELLITE (5.484A 5.516B 5.517A 5.517B 5.539) INTER-SATELLITE (5.521A) 5.538 5.540	FSS Earth stations	ECC/DEC/(05)01		The Earth-to-Space direction for uncoordinated Earth stations within the band 27.5-27.8285 GHz. The Space-to-Earth direction is limited to beacons for uplink power control 27.5-27.501 GHz
		FWA	ECC/DEC/(05)01, ECC/REC/(11)01	EN 302 326	CRS paired with 28.5-29.5 GHz for FDD systems.
		Feeder links			Feeder links to be used for Broadcasting satellites (HDTV) 27.5-29.5 GHz
		Fixed	ECC/DEC/(05)01, T/R 13-02	EN 302 217, EN 302 326	For frequency arrangement between FS and FSS see ECC/DEC/(05)01. CRS paired with 28.5-29.5 GHz for FDD systems.
		GSO ESOMPs	ECC/DEC/(13)01	EN 303 978	
		NGSO ESOMPs	ECC/DEC/(15)04	EN 303 979	
		NGSO FSS	ECC/DEC/(05)01	EN 303 699	

28.5 GHz - 29.1 GHz

FIXED FIXED-SATELLITE (5.484A 5.516B 5.523A 5.539 5.517A 5.517B) INTER-SATELLITE (5.521A) MOBILE Earth exploration-satellite (5.541) 5.540	FIXED FIXED-SATELLITE (5.484A 5.516B 5.517A 5.517B 5.523A 5.539) INTER-SATELLITE (5.521A) Earth exploration-satellite (5.541) 5.540	FSS Earth stations	ECC/DEC/(05)01		Uncoordinated Earth stations within the band 28.4445-28.8365 GHz
		FWA	ECC/DEC/(05)01, ECC/REC/(11)01	EN 302 326	TS paired with 27.5-28.5 GHz for FDD systems.
		Feeder links			Feeder links to be used for Broadcasting satellites (HDTV) 27.5-29.5 GHz
		Fixed	ECC/DEC/(05)01, T/R 13-02	EN 302 217, EN 302 326	For frequency arrangement between FS and FSS see ECC/DEC/(05)01. TS paired with 27.5-28.5 GHz for FDD systems.
		GSO ESOMPs	ECC/DEC/(13)01	EN 303 978	
		NGSO ESOMPs	ECC/DEC/(15)04	EN 303 979	
		NGSO FSS	ECC/DEC/(05)01	EN 303 699	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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29.1 GHz - 29.5 GHz

FIXED FIXED-SATELLITE (5.516B 5.523C 5.523E 5.535A 5.539 5.541A 5.517A) INTER-SATELLITE (5.521A) MOBILE Earth exploration-satellite (5.541) 5.540	FIXED FIXED-SATELLITE (5.516B 5.517A 5.523C 5.523E 5.535A 5.539 5.541A) INTER-SATELLITE (5.521A) Earth exploration-satellite (5.541) 5.540	FSS Earth stations	ECC/DEC/(05)01		Uncoordinated Earth stations within the band 29.4525-29.5 GHz
		FWA	ECC/DEC/(05)01, ECC/REC/(11)01	EN 302 326	TS paired with 27.5-28.5 GHz for FDD systems.
		Feeder links			Feeder links to be used for Broadcasting satellites (HDTV) 27.5-29.5 GHz
		Fixed	ECC/DEC/(05)01, T/R 13-02	EN 302 217, EN 302 326	Within the band 29.0605-29.4525 GHz. TS paired with 27.5-28.5 GHz for FDD systems.
		GSO ESOMPs	ECC/DEC/(13)01	EN 303 978	

29.5 GHz - 29.9 GHz

FIXED-SATELLITE (5.484A 5.516B 5.539 5.484B 5.527A 5.517B) INTER-SATELLITE (5.521A) Earth exploration-satellite (5.541) Mobile-Satellite (Earth-to-space) 5.540 5.542	FIXED-SATELLITE (5.484A 5.484B 5.516B 5.517B 5.527A 5.539) INTER-SATELLITE (5.521A) Earth exploration-satellite (5.541) Mobile-Satellite (Earth-to-space) 5.540	FSS Earth stations	ECC/DEC/(05)08		High Density FSS
		GSO ESOMPs	ECC/DEC/(13)01	EN 303 978	
		HEST	ECC/DEC/(06)03	EN 301 459	
		MSS Earth stations			
		NGSO ESOMPs	ECC/DEC/(15)04	EN 303 979	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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29.9 GHz - 30 GHz

FIXED-SATELLITE (5.484A 5.516B 5.539 5.484B 5.527A 5.517B) INTER-SATELLITE (5.521A) MOBILE-SATELLITE (EARTH-TO-SPACE) Earth exploration-satellite (5.541 5.543) 5.525 5.526 5.527 5.538 5.540 5.542	EARTH EXPLORATION-SATELLITE (5.541 5.543) FIXED-SATELLITE (5.484A 5.484B 5.516B 5.517B 5.527A 5.539) INTER-SATELLITE (5.521A) MOBILE-SATELLITE (EARTH-TO-SPACE) 5.525 5.526 5.527 5.538 5.540	FSS Earth stations	ECC/DEC/(05)08		High Density FSS
		GSO ESOMPs	ECC/DEC/(13)01	EN 303 978	
		HEST	ECC/DEC/(06)03	EN 301 459	
		MSS Earth stations			
		NGSO ESOMPs	ECC/DEC/(15)04	EN 303 979	

30 GHz - 31 GHz

FIXED-SATELLITE (5.338A) MOBILE-SATELLITE (EARTH-TO-SPACE) Standard frequency and time signal-satellite (space-to-Earth) 5.529A 5.542	FIXED-SATELLITE (5.338A) MOBILE-SATELLITE (EARTH-TO-SPACE) ECA36	FSS Earth stations			For uncoordinated Earth stations
		MSS Earth stations			
		Satellite systems (military)			

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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31 GHz - 31.3 GHz

FIXED (5.338A 5.543B 5.543A) MOBILE Space research (5.544 5.545) Standard frequency and time signal-satellite (space-to-Earth) 5.149	FIXED (5.338A 5.543B) MOBILE 5.149	Fixed	ECC/REC/(02)02	EN 302 217, EN 302 326	
		Radio astronomy			Continuum observations

31.3 GHz - 31.5 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	Passive sensors (satellite)	ECC/DEC/(10)02		Measurement of sea ice, water vapour, oil spills, liquid water, clouds, surface temperature, emissivity and atmospheric attenuation. Reference window for the 50-60 GHz range
		Radio astronomy			Continuum observations

31.5 GHz - 31.8 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) Fixed Mobile except aeronautical mobile 5.149 5.546	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) Fixed Mobile except aeronautical mobile 5.149 5.546	Fixed			
		Passive sensors (satellite)			Measurement of sea ice, water vapour, oil spills, liquid water, clouds, surface temperature. Emissivity and atmospheric attenuation. Reference window for the 50-60 GHz range
		Radio astronomy			Continuum observations

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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31.8 GHz - 32 GHz

FIXED (5.547A) RADIONAVIGATION SPACE RESEARCH (DEEP SPACE) (SPACE-TO-EARTH) 5.547 5.547B 5.548	FIXED (5.547A) RADIONAVIGATION SPACE RESEARCH (DEEP SPACE) (SPACE-TO-EARTH) 5.547 5.548	FWA	ECC/REC/(11)01	EN 302 326	Point-to-Point and Point-to-Multipoint
		Fixed	ERC/REC/(01)02	EN 302 217	High Density FS

32 GHz - 32.3 GHz

FIXED (5.547A) RADIONAVIGATION SPACE RESEARCH (DEEP SPACE) (SPACE-TO-EARTH) 5.547 5.547C 5.548	FIXED (5.547A) RADIONAVIGATION SPACE RESEARCH (DEEP SPACE) (SPACE-TO-EARTH) 5.547 5.548	FWA	ECC/REC/(11)01	EN 302 326	Point-to-Point and Point-to-Multipoint
		Fixed	ERC/REC/(01)02	EN 302 217	High Density FS

32.3 GHz - 33 GHz

FIXED (5.547A) INTER-SATELLITE RADIONAVIGATION 5.547 5.547D 5.548	FIXED (5.547A) INTER-SATELLITE RADIONAVIGATION 5.547 5.548	FWA	ECC/REC/(11)01	EN 302 326	Point-to-Point and Point-to-Multipoint
		Fixed	ERC/REC/(01)02	EN 302 217	High Density FS

33 GHz - 33.4 GHz

FIXED (5.547A) RADIONAVIGATION 5.547 5.547E	FIXED (5.547A) INTER-SATELLITE RADIONAVIGATION 5.547	FWA	ECC/REC/(11)01	EN 302 326	Point-to-Point and Point-to-Multipoint
		Fixed	ERC/REC/(01)02	EN 302 217	High Density FS
		Radiolocation (military)			

33.4 GHz - 34.2 GHz

RADIOLOCATION 5.549	RADIOLOCATION ECA36	Radiolocation (military)			
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RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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34.2 GHz - 34.7 GHz

RADIOLOCATION SPACE RESEARCH (DEEP SPACE) (EARTH-TO-SPACE) 5.549	RADIOLOCATION SPACE RESEARCH (DEEP SPACE) (EARTH-TO-SPACE) ECA36	Radiolocation (military)			
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34.7 GHz - 35.2 GHz

RADIOLOCATION Space research 5.549	RADIOLOCATION Space research ECA36	Radiolocation (military)			
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35.2 GHz - 35.5 GHz

METEOROLOGICAL AIDS RADIOLOCATION 5.549	METEOROLOGICAL AIDS RADIOLOCATION ECA36	Active sensors (satellite)			Rain radar from satellites
		Radiolocation (military)			

35.5 GHz - 36 GHz

EARTH EXPLORATION-SATELLITE (ACTIVE) METEOROLOGICAL AIDS RADIOLOCATION SPACE RESEARCH (ACTIVE) 5.549 5.549A	EARTH EXPLORATION-SATELLITE (ACTIVE) METEOROLOGICAL AIDS RADIOLOCATION SPACE RESEARCH (ACTIVE) 5.549A ECA36	Active sensors (satellite)			
		Radiolocation (military)			

36 GHz - 37 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED MOBILE SPACE RESEARCH (PASSIVE) 5.149 5.550A	EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED MOBILE SPACE RESEARCH (PASSIVE) Radio astronomy 5.149 5.550A	Passive sensors (satellite)			EESS surface emissivity, snow, sea ice and precipitation
		Radio astronomy			Spectral line observations (Hydrogen cyanide and Hydroxyl lines) 36.43-36.50 GHz

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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37 GHz - 37.5 GHz

FIXED MOBILE EXCEPT AERONAUTICAL MOBILE (5.550B) SPACE OPERATION (SPACE-TO-EARTH) 5.547	FIXED SPACE RESEARCH (SPACE-TO-EARTH) 5.547	Fixed	T/R 12-01		Major use by civil Fixed Service systems. High Density fixed links
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37.5 GHz - 38 GHz

FIXED FIXED-SATELLITE (5.550C 5.550CA) MOBILE EXCEPT AERONAUTICAL MOBILE (5.550B) SPACE RESEARCH (SPACE-TO-EARTH) Earth exploration-satellite (space-to-Earth) 5.547	FIXED FIXED-SATELLITE (5.550C 5.550CA) SPACE RESEARCH (SPACE-TO-EARTH) Earth exploration-satellite (space-to-Earth) 5.547	FSS Earth stations	ERC/DEC/(00)02		Uncoordinated Earth stations shall not claim protection from the Fixed Service
		Fixed	T/R 12-01		Major use by civil Fixed Service systems. High Density fixed links

38 GHz - 39.5 GHz

FIXED (5.550D) FIXED-SATELLITE (5.550C) MOBILE (5.550B) Earth exploration-satellite (space-to-Earth) 5.547	FIXED (5.550D) FIXED-SATELLITE (5.550C) Earth exploration-satellite (space-to-Earth) 5.547	FSS Earth stations	ERC/DEC/(00)02		Uncoordinated Earth stations shall not claim protection from the Fixed Service
		Fixed	T/R 12-01		Major use by civil Fixed Service systems. High Density fixed links

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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39.5 GHz - 40 GHz

FIXED FIXED-SATELLITE (5.516B 5.550C) MOBILE (5.550B) MOBILE-SATELLITE (SPACE-TO-EARTH) Earth exploration-satellite (space-to-Earth) 5.547 5.550E	FIXED FIXED-SATELLITE (5.516B 5.550C) MOBILE MOBILE-SATELLITE (SPACE-TO-EARTH) Earth exploration-satellite (space-to-Earth) 5.547 5.550E	FSS Earth stations	ECC/REC/(22)02, ERC/DEC/(00)02		
		MSS Earth stations	ERC/DEC/(00)02		

40 GHz - 40.5 GHz

EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE) FIXED FIXED-SATELLITE (5.516B 5.550C) MOBILE (5.550B) MOBILE-SATELLITE (SPACE-TO-EARTH) SPACE RESEARCH (EARTH-TO-SPACE) Earth exploration-satellite (space-to-Earth) 5.550E	FIXED FIXED-SATELLITE (5.516B) MOBILE MOBILE-SATELLITE (SPACE-TO-EARTH) SPACE RESEARCH (EARTH-TO-SPACE) Earth exploration-satellite (space-to-Earth) 5.550E	FSS Earth stations	ECC/REC/(22)02, ERC/DEC/(00)02		
		MSS Earth stations	ERC/DEC/(00)02		

40.5 GHz - 41 GHz

BROADCASTING BROADCASTING-SATELLITE FIXED FIXED-SATELLITE (5.550C) LAND MOBILE (5.550B) Aeronautical mobile Maritime mobile 5.547	BROADCASTING BROADCASTING-SATELLITE FIXED LAND MOBILE (5.550B) Aeronautical mobile Maritime mobile 5.547	FSS Earth stations	ECC/DEC/(23)01, ECC/REC/(22)01		
		MFCN	ECC/DEC/(22)06, ECC/REC/(22)01, ECC/REC/(22)02		
		Point-to-Point	ECC/REC/(01)04	EN 302 217	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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41 GHz - 42.5 GHz

BROADCASTING BROADCASTING-SATELLITE FIXED FIXED-SATELLITE (5.550C) LAND MOBILE (5.550B) Aeronautical mobile Maritime mobile 5.547 5.551H 5.551I	BROADCASTING BROADCASTING-SATELLITE FIXED FIXED-SATELLITE (5.550C) LAND MOBILE (5.550B) Aeronautical mobile Maritime mobile 5.547 5.551H 5.551I	FSS Earth stations	ECC/DEC/(23)01, ECC/REC/(22)01		
		MFCN	ECC/DEC/(22)06, ECC/REC/(22)01, ECC/REC/(22)02		
		Point-to-Point	ECC/REC/(01)04	EN 302 217	

42.5 GHz - 43.5 GHz

FIXED FIXED-SATELLITE (5.552) MOBILE EXCEPT AERONAUTICAL MOBILE (5.550B) RADIO ASTRONOMY 5.149 5.547	FIXED FIXED-SATELLITE (5.552) MOBILE EXCEPT AERONAUTICAL MOBILE RADIO ASTRONOMY 5.149 5.547	FSS Earth stations	ECC/DEC/(23)01, ECC/REC/(22)01		
		MFCN	ECC/DEC/(22)06, ECC/REC/(22)01, ECC/REC/(22)02		
		Point-to-Point	ECC/REC/(01)04	EN 302 217	
		Radio astronomy			Continuum and spectral line observations (e.g. silicon monoxide line), VLBI

43.5 GHz - 45.5 GHz

MOBILE (5.553 5.553A) MOBILE-SATELLITE RADIONAVIGATION RADIONAVIGATION-SATELLITE 5.554	MOBILE (5.553) MOBILE-SATELLITE Fixed-satellite 5.554 ECA36	Aeronautical military systems			
		Land military systems			
		Maritime military systems			
		Satellite systems (military)			

45.5 GHz - 47 GHz

MOBILE (5.553 5.553A) MOBILE-SATELLITE RADIONAVIGATION RADIONAVIGATION-SATELLITE 5.554	MOBILE (5.553) MOBILE-SATELLITE RADIONAVIGATION RADIONAVIGATION-SATELLITE 5.554	-			
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RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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47 GHz - 47.2 GHz

AMATEUR AMATEUR-SATELLITE	AMATEUR AMATEUR-SATELLITE	Amateur			
		Amateur-satellite			

47.2 GHz - 47.5 GHz

FIXED FIXED-SATELLITE (5.552 5.550C) MOBILE (5.553B) 5.552A	FIXED FIXED-SATELLITE (5.550C 5.552) MOBILE (5.553B) 5.552A	FSS Earth stations	ECC/DEC/(21)01		Coordinated gateway Earth stations
		Feeder links			For 40 GHz Broadcasting satellites

47.5 GHz - 47.9 GHz

FIXED FIXED-SATELLITE (5.552 5.550C) FIXED-SATELLITE (5.516B 5.554A) MOBILE (5.553B)	FIXED FIXED-SATELLITE (5.550C 5.552) FIXED-SATELLITE (5.516B 5.554A) MOBILE (5.553B)	FSS Earth stations	ECC/DEC/(05)08, ECC/DEC/(21)01		High Density FSS
		Feeder links			For 40 GHz Broadcasting satellites

47.9 GHz - 48.2 GHz

FIXED FIXED-SATELLITE (5.552 5.550C) MOBILE (5.553B) 5.552A	FIXED FIXED-SATELLITE (5.550C 5.552) MOBILE (5.553B) 5.552A	FSS Earth stations	ECC/DEC/(21)01		Coordinated gateway Earth stations
		Feeder links			For 40 GHz Broadcasting satellites

48.2 GHz - 48.54 GHz

FIXED FIXED-SATELLITE (5.552 5.550C) FIXED-SATELLITE (5.516B 5.554A 5.555B) MOBILE	FIXED FIXED-SATELLITE (5.550C 5.552) FIXED-SATELLITE (5.516B 5.554A 5.555B) MOBILE	FSS Earth stations	ECC/DEC/(05)08, ECC/DEC/(21)01		High Density FSS
		Feeder links			For 40 GHz Broadcasting satellites
		Fixed	ERC/REC 12-11	EN 302 217	Within the band 48.5-50.2 GHz and 50.9-52.6 GHz

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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48.54 GHz - 49.44 GHz

FIXED FIXED-SATELLITE (5.552 5.550C) MOBILE 5.149 5.340 5.555	FIXED FIXED-SATELLITE (5.550C 5.552) MOBILE RADIO ASTRONOMY 5.149 5.340 5.555 ECA17A	FSS Earth stations	ECC/DEC/(21)01		
		Feeder links			48.5-49.2 GHz for 40 GHz Broadcasting satellites
		Fixed	ERC/REC 12-11	EN 302 217	Within the band 48.5-50.2 GHz and 50.9-52.6 GHz
		Radio astronomy			Spectral line observations (e.g. carbon monosulphide line)

49.44 GHz - 50.2 GHz

FIXED FIXED-SATELLITE (5.552 5.338A 5.550C) FIXED-SATELLITE (5.516B 5.554A 5.555B) MOBILE	FIXED FIXED-SATELLITE (5.338A 5.550C 5.552) FIXED-SATELLITE (5.516B 5.554A 5.555B) MOBILE ECA17A	FSS Earth stations	ECC/DEC/(05)08, ECC/DEC/(21)01		High Density FSS
		Fixed	ERC/REC 12-11	EN 302 217	Within the band 48.5-50.2 GHz and 50.9-52.6 GHz

50.2 GHz - 50.4 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) SPACE RESEARCH (PASSIVE) 5.340	EARTH EXPLORATION-SATELLITE (PASSIVE) SPACE RESEARCH (PASSIVE) 5.340	Passive sensors (satellite)			Atmospheric temperature sounding. Terrestrial passive radiometers. Reference window for the 52.6-59.3 GHz band
		Radio astronomy			Continuum and spectral line observations

50.4 GHz - 51.4 GHz

FIXED FIXED-SATELLITE (5.338A 5.550C) MOBILE Mobile-Satellite (Earth-to-space)	FIXED FIXED-SATELLITE (5.338A 5.550C) Mobile-Satellite (Earth-to-space)	FSS Earth stations	ECC/DEC/(21)01		Coordinated gateway Earth stations
		Fixed	ERC/REC 12-11	EN 302 217	Within the band 48.5-50.2 GHz and 50.9-52.6 GHz

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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51.4 GHz - 52.4 GHz

FIXED (5.338A) FIXED-SATELLITE (5.555C) MOBILE 5.547 5.556	FIXED (5.338A) FIXED-SATELLITE (5.555C) MOBILE RADIO ASTRONOMY 5.547 5.556	FSS Earth stations	ECC/DEC/(21)01		Coordinated gateway Earth stations
		Fixed	ERC/REC 12-11	EN 302 217	Within the band 48.5-50.2 GHz and 50.9-52.6 GHz
		Radio astronomy			Continuum and spectral line observations

52.4 GHz - 52.6 GHz

FIXED (5.338A) MOBILE 5.547 5.556	FIXED (5.338A) MOBILE RADIO ASTRONOMY (5.547 5.556)	Fixed	ERC/REC 12-11	EN 302 217	Within the band 48.5-50.2 GHz and 50.9-52.6 GHz
		Radio astronomy			Continuum and spectral line observations

52.6 GHz - 54.25 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) SPACE RESEARCH (PASSIVE) 5.340 5.556	EARTH EXPLORATION-SATELLITE (PASSIVE) SPACE RESEARCH (PASSIVE) 5.340 5.556	Passive sensors (satellite)			Atmospheric temperature sounding. Terrestrial passive radiometers
		Radio astronomy			Continuum and spectral line observations

54.25 GHz - 55.78 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) INTER-SATELLITE (5.556A) SPACE RESEARCH (PASSIVE) 5.556B	EARTH EXPLORATION-SATELLITE (PASSIVE) SPACE RESEARCH (PASSIVE)	Passive sensors (satellite)			Atmospheric temperature sounding. Terrestrial passive radiometers
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RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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55.78 GHz - 56.9 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED (5.557A) INTER-SATELLITE (5.556A) MOBILE (5.558) SPACE RESEARCH (PASSIVE) 5.547 5.557	EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED (5.557A) INTER-SATELLITE (5.556A) SPACE RESEARCH (PASSIVE) 5.547 5.558	Fixed	ERC/REC 12-12	EN 302 217	High density fixed links
		Passive sensors (satellite)			Atmospheric temperature sounding

56.9 GHz - 57 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED INTER-SATELLITE (5.558A) MOBILE (5.558) SPACE RESEARCH (PASSIVE) 5.547 5.557	EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED INTER-SATELLITE (5.558A) MOBILE (5.558) SPACE RESEARCH (PASSIVE) 5.547	Fixed	ERC/REC 12-12	EN 302 217	High density fixed links
		Passive sensors (satellite)			Atmospheric temperature sounding

57 GHz - 58.2 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED INTER-SATELLITE (5.556A) MOBILE (5.558) SPACE RESEARCH (PASSIVE) 5.547 5.557	EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED INTER-SATELLITE (5.556A) MOBILE (5.558) SPACE RESEARCH (PASSIVE) 5.547	Fixed		EN 302 217	Un-coordinated deployment. High density fixed links
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Non-specific SRDs	ERC/REC 70-03	EN 305 550	Within the band 57-64 GHz
		Passive sensors (satellite)			Atmospheric temperature sounding
		TLPR	ERC/REC 70-03	EN 302 372	
		Wideband data transmission systems	ERC/REC 70-03	EN 302 567, EN 303 722, EN 303 753	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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58.2 GHz - 59 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED MOBILE SPACE RESEARCH (PASSIVE) 5.547 5.556	EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.547 5.556 ECA6 ECA19	Fixed		EN 302 217	Un-coordinated deployment. High density fixed links
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Non-specific SRDs	ERC/REC 70-03	EN 305 550	Within the band 57-64 GHz
		Passive sensors (satellite)			Atmospheric temperature sounding. Terrestrial passive radiometers
		Radio astronomy			Continuum and spectral line observations
		TLPR	ERC/REC 70-03	EN 302 372	
		Wideband data transmission systems	ERC/REC 70-03	EN 302 567, EN 303 722, EN 303 753	

59 GHz - 59.3 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED INTER-SATELLITE (5.556A) MOBILE (5.558) RADIOLOCATION (5.559) SPACE RESEARCH (PASSIVE)	EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED INTER-SATELLITE (5.556A) MOBILE (5.558) RADIOLOCATION (5.559) SPACE RESEARCH (PASSIVE)	Fixed		EN 302 217	High density fixed links
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Non-specific SRDs	ERC/REC 70-03	EN 305 550	Within the band 57-64 GHz
		Passive sensors (satellite)			Atmospheric temperature sounding. Terrestrial passive radiometers
		TLPR	ERC/REC 70-03	EN 302 372	
		Wideband data transmission systems	ERC/REC 70-03	EN 302 567, EN 303 722, EN 303 753	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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59.3 GHz - 64 GHz

FIXED INTER-SATELLITE MOBILE (5.558) RADIOLOCATION (5.559) 5.138	FIXED INTER-SATELLITE MOBILE (5.558) RADIOLOCATION (5.559) 5.138	Fixed		EN 302 217	High density fixed links
		ISM			Within the band 61.0-61.5 GHz
		ITS	ECC/DEC/(09)01	EN 302 686	Within the band 63.72- 65.88 GHz
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Non-specific SRDs	ERC/REC 70-03	EN 305 550	Within the band 57-64 GHz
		TLPR	ERC/REC 70-03	EN 302 372	
		Wideband data transmission systems	ERC/REC 70-03	EN 302 567, EN 303 722, EN 303 753	

64 GHz - 65 GHz

FIXED INTER-SATELLITE MOBILE EXCEPT AERONAUTICAL MOBILE 5.547 5.556	FIXED INTER-SATELLITE MOBILE EXCEPT AERONAUTICAL MOBILE 5.547 5.556	Fixed		EN 302 217	High density fixed links
		ITS	ECC/DEC/(09)01, ERC/REC 70-03	EN 302 686	Within the band 63.72 - 65.88 GHz
		Radio astronomy			Continuum and spectral line observations
		Wideband data transmission systems	ERC/REC 70-03	EN 302 567, EN 303 722, EN 303 753	

65 GHz - 66 GHz

EARTH EXPLORATION-SATELLITE FIXED INTER-SATELLITE MOBILE EXCEPT AERONAUTICAL MOBILE SPACE RESEARCH 5.547	EARTH EXPLORATION-SATELLITE FIXED INTER-SATELLITE MOBILE EXCEPT AERONAUTICAL MOBILE SPACE RESEARCH 5.547	Fixed		EN 302 217	High density fixed links
		ITS	ECC/DEC/(09)01, ERC/REC 70-03	EN 302 686	Within the band 63.72 - 65.88 GHz
		Land mobile			Broadband mobile systems for connection to IBCN paired with 62-63 GHz
		Wideband data transmission systems	ERC/REC 70-03	EN 302 567, EN 303 722, EN 303 753	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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66 GHz - 71 GHz

INTER-SATELLITE MOBILE (5.553 5.558) MOBILE-SATELLITE RADIONAVIGATION RADIONAVIGATION-SATELLITE 5.554 5.559AA	INTER-SATELLITE MOBILE (5.553 5.558) MOBILE-SATELLITE RADIONAVIGATION RADIONAVIGATION-SATELLITE 5.554	-			Future civil systems
		Wideband data transmission systems	ERC/REC 70-03	EN 302 567, EN 303 722, EN 303 753	

71 GHz - 74 GHz

FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE MOBILE-SATELLITE (SPACE-TO-EARTH)	FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE MOBILE-SATELLITE (SPACE-TO-EARTH)	Fixed	ECC/REC/(05)07	EN 302 217	
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74 GHz - 75.5 GHz

BROADCASTING BROADCASTING-SATELLITE FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE Space research (space-to-Earth) 5.561	BROADCASTING BROADCASTING-SATELLITE FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE Space research (space-to-Earth) 5.561	Fixed	ECC/REC/(05)07	EN 302 217	
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Space research			VLBI measurements within the band 74-84 GHz
		TLPR	ERC/REC 70-03	EN 302 372	

75.5 GHz - 76 GHz

BROADCASTING BROADCASTING-SATELLITE FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE Space research (space-to-Earth) 5.561	BROADCASTING BROADCASTING-SATELLITE FIXED FIXED-SATELLITE (SPACE-TO-EARTH) Amateur Amateur-satellite 5.561 ECA35	Amateur			Within the band 75.5-81.5 GHz
		Amateur-satellite			Within the band 75.5-81.5 GHz
		Fixed	ECC/REC/(05)07	EN 302 217	
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Space research			VLBI
		TLPR	ERC/REC 70-03	EN 302 372	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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76 GHz - 77.5 GHz

RADIO ASTRONOMY RADIOLOCATION Amateur Amateur-satellite Space research (space-to-Earth) 5.149	RADIO ASTRONOMY RADIOLOCATION Amateur Amateur-satellite Space research (space-to-Earth) 5.149	Amateur			Within the band 75.5-81.5 GHz
		Amateur-satellite			Within the band 75.5-81.5 GHz
		GBSAR	ECC/DEC/(21)02, ERC/REC 70-03	EN 303 661	Within frequency band 76-77 GHz
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Radio astronomy			Continuum and spectral line observations
		Radiolocation (civil)			
		Railway applications	ERC/REC 70-03	EN 301 091	Obstruction/vehicle detection at level crossings
		SRR	ECC/DEC/(04)03, ERC/REC 70-03	EN 302 264	
		TLPR	ERC/REC 70-03	EN 302 372	
		TTT	ECC/DEC/(16)01, ERC/REC 70-03	EN 301 091, EN 303 360	Within the band 76-77 GHz. Rotorcraft Radar

77.5 GHz - 78 GHz

AMATEUR AMATEUR-SATELLITE RADIOLOCATION (5.559B) Radio astronomy Space research (space-to-Earth) 5.149	AMATEUR AMATEUR-SATELLITE RADIOLOCATION (5.559B) Space research (space-to-Earth) 5.149	Amateur			Within the band 75.5-81.5 GHz
		Amateur-satellite			Within the band 75.5-81.5 GHz
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Radio astronomy			Continuum and spectral line observations
		SRR	ECC/DEC/(04)03, ERC/REC 70-03	EN 302 264	
		TLPR	ERC/REC 70-03	EN 302 372	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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78 GHz - 79 GHz

RADIOLOCATION Amateur Amateur-satellite Radio astronomy Space research (space-to-Earth) 5.149 5.560	RADIOLOCATION Amateur Amateur-satellite Radio astronomy Space research (space-to-Earth) 5.149 5.560	Amateur			Within the band 75.5-81.5 GHz
		Amateur-satellite			Within the band 75.5-81.5 GHz
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Radio astronomy			Continuum and spectral line observations
		Radiolocation (civil)			
		SRR	ECC/DEC/(04)03, ERC/REC 70-03	EN 302 264	
		TLPR	ERC/REC 70-03	EN 302 372	

79 GHz - 81 GHz

RADIO ASTRONOMY RADIOLOCATION Amateur Amateur-satellite Space research (space-to-Earth) 5.149	RADIO ASTRONOMY RADIOLOCATION Amateur Amateur-satellite 5.149	Amateur			Within the band 75.5-81.5 GHz
		Amateur-satellite			Within the band 75.5-81.5 GHz
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Radio astronomy			Continuum and spectral line observations
		Radiolocation (civil)			
		SRR	ECC/DEC/(04)03, ERC/REC 70-03	EN 302 264	
		TLPR	ERC/REC 70-03	EN 302 372	

81 GHz - 84 GHz

FIXED (5.338A) FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE MOBILE-SATELLITE (EARTH-TO-SPACE) RADIO ASTRONOMY Space research (space-to-Earth) 5.149 5.561A	FIXED (5.338A) FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE MOBILE-SATELLITE (EARTH-TO-SPACE) RADIO ASTRONOMY Space research (space-to-Earth) 5.149 5.561A	Amateur			Within the band 75.5-81.5 GHz
		Amateur-satellite			Within the band 75.5-81.5 GHz
		Fixed	ECC/REC/(05)07	EN 302 217	
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	
		Radio astronomy			Continuum and spectral line observations
		TLPR	ERC/REC 70-03	EN 302 372	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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84 GHz - 86 GHz

FIXED (5.338A) FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE RADIO ASTRONOMY 5.149	FIXED (5.338A) FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE RADIO ASTRONOMY 5.149	Fixed	ECC/REC/(05)07	EN 302 217	
		LPR	ECC/DEC/(11)02, ERC/REC 70-03	EN 302 729	Within the band 84-85 GHz
		Radio astronomy			Continuum and spectral line observations
		TLPR	ERC/REC 70-03	EN 302 372	Within the band 84-85 GHz

86 GHz - 92 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	Passive sensors (satellite)			Measurement of clouds, oil spills, ice, snow, rain, reference window for the temperature sounding near 118 GHz
		Radio astronomy			Continuum and spectral line observations. VLBI

92 GHz - 94 GHz

FIXED (5.338A) MOBILE RADIO ASTRONOMY RADIOLOCATION 5.149	FIXED (5.338A) MOBILE RADIO ASTRONOMY RADIOLOCATION 5.149	Fixed	ECC/REC/(14)01, ECC/REC/(18)02		
		Radio astronomy			Continuum and spectral line observations

94 GHz - 94.1 GHz

EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION SPACE RESEARCH (ACTIVE) Radio astronomy 5.562 5.562A	EARTH EXPLORATION-SATELLITE (ACTIVE) RADIOLOCATION SPACE RESEARCH (ACTIVE) Radio astronomy 5.562 5.562A	Active sensors (satellite)			Cloud radars
		Radio astronomy			Continuum and spectral line observations
		Space research			

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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94.1 GHz - 95 GHz

FIXED MOBILE RADIO ASTRONOMY RADIOLOCATION 5.149	FIXED MOBILE RADIO ASTRONOMY RADIOLOCATION 5.149	Fixed	ECC/REC/(14)01, ECC/REC/(18)02		
		Radio astronomy			Continuum and spectral line observations

95 GHz - 100 GHz

FIXED MOBILE RADIO ASTRONOMY RADIOLOCATION RADIONAVIGATION RADIONAVIGATION-SATELLITE 5.149 5.554	FIXED MOBILE RADIO ASTRONOMY RADIOLOCATION RADIONAVIGATION RADIONAVIGATION-SATELLITE 5.149 5.554	Fixed	ECC/REC/(18)02		
		Radio astronomy			Continuum and spectral line observations

100 GHz - 102 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.341	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.341	Passive sensors (satellite)			Limb sounding of atmospheric constituents
		Radio astronomy			Continuum and spectral line observations

102 GHz - 105 GHz

FIXED MOBILE RADIO ASTRONOMY 5.149 5.341	FIXED MOBILE RADIO ASTRONOMY 5.149 5.341	Fixed	ECC/REC/(18)02		
		Radio astronomy			Continuum and spectral line observations

105 GHz - 109.5 GHz

FIXED MOBILE RADIO ASTRONOMY SPACE RESEARCH (5.562B) 5.149 5.341	FIXED MOBILE RADIO ASTRONOMY SPACE RESEARCH (5.562B) 5.149 5.341	Fixed	ECC/REC/(18)02		
		Radio astronomy			Continuum and spectral line observations

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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109.5 GHz - 111.8 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.341	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.341	Radio astronomy			Continuum and spectral line observations
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111.8 GHz - 114.25 GHz

FIXED MOBILE RADIO ASTRONOMY SPACE RESEARCH (5.562B) 5.149 5.341	FIXED MOBILE RADIO ASTRONOMY SPACE RESEARCH (5.562B) 5.149 5.341	Fixed	ECC/REC/(18)02		
		Radio astronomy			Continuum and spectral line observations

114.25 GHz - 116 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.341	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.341	Radio astronomy			Continuum and spectral line observations
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116 GHz - 119.98 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) INTER-SATELLITE (5.562C) SPACE RESEARCH (PASSIVE) 5.341	EARTH EXPLORATION-SATELLITE (PASSIVE) INTER-SATELLITE (5.562C) 5.341	Passive sensors (satellite)			Passive sensing as part of the oxygen absorption band with peak at 118.75 GHz
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

119.98 GHz - 120.02 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) INTER-SATELLITE (5.562C) SPACE RESEARCH (PASSIVE) 5.341	EARTH EXPLORATION-SATELLITE (PASSIVE) INTER-SATELLITE (5.562C) 5.341	Passive sensors (satellite)			Passive sensing as part of the oxygen absorption band with peak at 118.75 GHz
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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120.02 GHz - 122.25 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) INTER-SATELLITE (5.562C) SPACE RESEARCH (PASSIVE) 5.138	EARTH EXPLORATION-SATELLITE (PASSIVE) INTER-SATELLITE (5.562C) SPACE RESEARCH (PASSIVE) 5.138	Non-specific SRDs	ERC/REC 70-03	EN 305 550	Within the band 122-123 GHz
		Passive sensors (satellite)			Passive sensing as part of the oxygen absorption band with peak at 118.75 GHz
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

122.25 GHz - 123 GHz

FIXED INTER-SATELLITE MOBILE (5.558) Amateur 5.138	FIXED INTER-SATELLITE MOBILE (5.558) Amateur Amateur-satellite 5.138	Amateur			
		Amateur-satellite			
		Non-specific SRDs	ERC/REC 70-03	EN 305 550	Within the band 122-123 GHz
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

123 GHz - 130 GHz

FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE-SATELLITE (SPACE-TO-EARTH) RADIONAVIGATION RADIONAVIGATION-SATELLITE Radio astronomy 5.149 5.554	FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE-SATELLITE (SPACE-TO-EARTH) RADIONAVIGATION RADIONAVIGATION-SATELLITE Radio astronomy 5.149 5.554	Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

130 GHz - 134 GHz

EARTH EXPLORATION-SATELLITE (5.562E) FIXED INTER-SATELLITE MOBILE (5.558) RADIO ASTRONOMY 5.149 5.562A	EARTH EXPLORATION-SATELLITE (5.562E) FIXED INTER-SATELLITE MOBILE (5.558) RADIO ASTRONOMY 5.149 5.562A	Fixed	ECC/REC/(18)01		
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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134 GHz - 136 GHz

AMATEUR AMATEUR-SATELLITE Radio astronomy	AMATEUR AMATEUR-SATELLITE Radio astronomy	Amateur			Within the band 134-141 GHz
		Amateur-satellite			Within the band 134-141 GHz
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

136 GHz - 141 GHz

RADIO ASTRONOMY RADIOLOCATION Amateur Amateur-satellite 5.149	RADIO ASTRONOMY RADIOLOCATION Amateur Amateur-satellite 5.149	Amateur			Within the band 134-141 GHz
		Amateur-satellite			Within the band 134-141 GHz
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

141 GHz - 148.5 GHz

FIXED MOBILE RADIO ASTRONOMY RADIOLOCATION 5.149	FIXED MOBILE RADIO ASTRONOMY RADIOLOCATION 5.149	Fixed	ECC/REC/(18)01		
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

148.5 GHz - 151.5 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	Passive sensors (satellite)			Harmonised reference window for passive sensor observations
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

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151.5 GHz - 155.5 GHz

FIXED MOBILE RADIO ASTRONOMY RADIOLOCATION 5.149	FIXED MOBILE RADIO ASTRONOMY RADIOLOCATION 5.149	Fixed	ECC/REC/(18)01		
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

155.5 GHz - 158.5 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED MOBILE RADIO ASTRONOMY SPACE RESEARCH (5.562B) 5.149	EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED MOBILE RADIO ASTRONOMY SPACE RESEARCH (5.562B) 5.149	Fixed	ECC/REC/(18)01		
		Passive sensors (satellite)			Protection until 1.1.2018
		Radio astronomy			Spectral line and wide band continuum observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

158.5 GHz - 164 GHz

FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE MOBILE-SATELLITE (SPACE-TO-EARTH)	FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE MOBILE-SATELLITE (SPACE-TO-EARTH)	Fixed	ECC/REC/(18)01		
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

164 GHz - 167 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	Passive sensors (satellite)			Passive sensing of the water vapour absorption line whose peak is at 183.31 GHz. Atmospheric limb sounding of the 164.38 GHz CO line
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

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167 GHz - 174.5 GHz

FIXED FIXED-SATELLITE (SPACE-TO-EARTH) INTER-SATELLITE MOBILE (5.558) 5.149	FIXED FIXED-SATELLITE (SPACE-TO-EARTH) INTER-SATELLITE MOBILE (5.558) 5.149	Fixed	ECC/REC/(18)01		
		Radio astronomy			Within the band 168-174.5 GHz. Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

174.5 GHz - 174.8 GHz

FIXED INTER-SATELLITE MOBILE (5.558)	FIXED INTER-SATELLITE MOBILE (5.558)	Fixed	ECC/REC/(18)01		
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

174.8 GHz - 182 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) INTER-SATELLITE (5.562H) SPACE RESEARCH (PASSIVE)	EARTH EXPLORATION-SATELLITE (PASSIVE) INTER-SATELLITE (5.562H) SPACE RESEARCH (PASSIVE)	Passive sensors (satellite)			Passive sensing of the water vapour absorption line whose peak is at 183.31 GHz
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

182 GHz - 185 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	Passive sensors (satellite)			Passive sensing of the water vapour absorption line whose peak is at 183.31 GHz
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

185 GHz - 190 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) INTER-SATELLITE (5.562H) SPACE RESEARCH (PASSIVE)	EARTH EXPLORATION-SATELLITE (PASSIVE) INTER-SATELLITE (5.562H) SPACE RESEARCH (PASSIVE)	Passive sensors (satellite)			Passive sensing of the water vapour absorption line whose peak is at 183.31 GHz
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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190 GHz - 191.8 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) SPACE RESEARCH (PASSIVE) 5.340	EARTH EXPLORATION-SATELLITE (PASSIVE) SPACE RESEARCH (PASSIVE) 5.340	Passive sensors (satellite)			Passive sensing of the water vapour absorption line whose peak is at 183.31 GHz
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

191.8 GHz - 200 GHz

FIXED INTER-SATELLITE MOBILE (5.558) MOBILE-SATELLITE RADIONAVIGATION RADIONAVIGATION-SATELLITE 5.149 5.341 5.554	FIXED INTER-SATELLITE MOBILE (5.558) MOBILE-SATELLITE RADIONAVIGATION RADIONAVIGATION-SATELLITE 5.149 5.341 5.554	Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

200 GHz - 202 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.341 5.563A	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.341 5.563A	Earth exploration-satellite			(EESS) Atmospheric limb sounding and atmospheric remote sensing of nitrous oxide at 201 GHz
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

202 GHz - 209 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.341 5.563A	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.341 5.563A	Earth exploration-satellite			(EESS) Atmospheric limb sounding and atmospheric remote sensing of water vapour at 203.4 GHz and ozone at 208.5 GHz
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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209 GHz - 217 GHz

FIXED FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE RADIO ASTRONOMY 5.149 5.341	FIXED FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE RADIO ASTRONOMY 5.149 5.341	Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

217 GHz - 226 GHz

FIXED FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE RADIO ASTRONOMY SPACE RESEARCH (5.562B) 5.149 5.341	FIXED FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE RADIO ASTRONOMY SPACE RESEARCH (5.562B) 5.149 5.341	Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

226 GHz - 231.5 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340	Passive sensors (satellite)			Atmospheric limb sounding. Reference window for higher frequency water vapour measurements
		Radio astronomy			Continuum and spectral line observations (e.g. CO line), VLBI
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

231.5 GHz - 232 GHz

FIXED MOBILE Radiolocation	FIXED MOBILE Radiolocation	Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	
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RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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232 GHz - 235 GHz

FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE Radiolocation	FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE Radiolocation	Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	
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235 GHz - 238 GHz

EARTH EXPLORATION-SATELLITE (5.563AA) FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE SPACE RESEARCH (PASSIVE) 5.563A 5.563B	EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED-SATELLITE (SPACE-TO-EARTH) SPACE RESEARCH (PASSIVE) 5.563A 5.563B	Passive sensors (satellite)			Passive sensing limited to microwave sounding
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

238 GHz - 239.2 GHz

FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE RADIOLOCATION RADIONAVIGATION RADIONAVIGATION-SATELLITE	FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE RADIOLOCATION RADIONAVIGATION RADIONAVIGATION-SATELLITE	Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	
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239.2 GHz - 240 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) FIXED-SATELLITE (SPACE-TO-EARTH) RADIOLOCATION RADIONAVIGATION RADIONAVIGATION-SATELLITE	FIXED FIXED-SATELLITE (SPACE-TO-EARTH) MOBILE RADIOLOCATION RADIONAVIGATION RADIONAVIGATION-SATELLITE	Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	
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RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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240 GHz - 241 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIOLOCATION	FIXED MOBILE RADIOLOCATION	Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	
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241 GHz - 242.2 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY RADIOLOCATION Amateur Amateur-satellite 5.149	RADIO ASTRONOMY RADIOLOCATION Amateur Amateur-satellite 5.138 5.149	Amateur			
		Amateur-satellite			Within the band 241-250 GHz
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

242.2 GHz - 244.2 GHz

RADIO ASTRONOMY RADIOLOCATION Amateur Amateur-satellite 5.138 5.149	RADIO ASTRONOMY RADIOLOCATION Amateur Amateur-satellite 5.138 5.149	Amateur			
		Amateur-satellite			
		Non-specific SRDs	ERC/REC 70-03	EN 305 550	Within the band 244-246 GHz
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

244.2 GHz - 247.2 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY RADIOLOCATION Amateur Amateur-satellite (5.138 5.149) 5.138 5.149	RADIO ASTRONOMY RADIOLOCATION Amateur Amateur-satellite 5.138 5.149	Amateur			
		Amateur-satellite			
		Non-specific SRDs	ERC/REC 70-03	EN 305 550	Within the band 244-246 GHz
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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247.2 GHz - 248 GHz

RADIO ASTRONOMY RADIOLOCATION Amateur Amateur-satellite 5.149	RADIO ASTRONOMY RADIOLOCATION Amateur Amateur-satellite 5.138 5.149	Amateur			
		Amateur-satellite			
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

248 GHz - 250 GHz

AMATEUR AMATEUR-SATELLITE Radio astronomy 5.149	AMATEUR AMATEUR-SATELLITE Radio astronomy 5.149	Amateur			
		Amateur-satellite			Within the band 241-250 GHz
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

250 GHz - 252 GHz

EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.563A	EARTH EXPLORATION-SATELLITE (PASSIVE) RADIO ASTRONOMY SPACE RESEARCH (PASSIVE) 5.340 5.563A	Earth exploration-satellite			(EESS) Limb sounding of nitrous oxide near 251 GHz
		Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	

252 GHz - 265 GHz

FIXED MOBILE MOBILE-SATELLITE (EARTH-TO-SPACE) RADIO ASTRONOMY RADIONAVIGATION RADIONAVIGATION-SATELLITE 5.149 5.554	FIXED MOBILE MOBILE-SATELLITE (EARTH-TO-SPACE) RADIO ASTRONOMY RADIONAVIGATION RADIONAVIGATION-SATELLITE 5.149 5.554	Radio astronomy			Continuum and spectral line observations
		Radiodetermination applications	ECC/DEC/(22)03, ERC/REC 70-03	EN 305 550	within frequency range 116-260 GHz

RR Region 1	European Common Allocations	Application	CEPT Deliverables	Standard	Note
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265 GHz - 275 GHz

FIXED FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE RADIO ASTRONOMY 5.149 5.563A	FIXED FIXED-SATELLITE (EARTH-TO-SPACE) MOBILE RADIO ASTRONOMY 5.149 5.563A	Radio astronomy			Continuum and spectral line observations
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275 GHz - 3000 GHz

Not allocated 5.564A 5.565	Not allocated 5.564A 5.565	Fixed			Within the frequency bands 275-296 GHz, 306-313 GHz, 318-333 GHz and 356-450 GHz
		Land mobile			Within the frequency bands 275-296 GHz, 306-313 GHz, 318-333 GHz and 356-450 GHz

ECA Footnotes

Footnote Number	Footnote Content
ECA5	In parts of this band aeronautical stations and aircraft stations utilise the preferred 8.33 kHz channel spacing for non secure communications requirements.
ECA6	The mobile-satellite service is limited to low earth orbiting satellites.
ECA7	This band can also be used by low capacity fixed links in rural areas on a national basis. These links need to be coordinated with mobile service and require full protection.
ECA8	Any use of low capacity fixed links shall be avoided in areas where such use might cause harmful interference to the maritime mobile VHF radiocommunication service.
ECA9	CEPT administrations may authorise all or parts of the band 69.9-70.5 MHz to the amateur service on a secondary basis.
ECA10	The range 225-399.9 MHz is essential to NATO and is in military use for land mobile, mobile-satellite, Air/Ground/Air and specific maritime and terrestrial communications, including ITU Region 2. This NATO UHF band 225-400 MHz is the only harmonised and commonly available resource managed by NATO on a daily basis in and for NATO nations. It is recognised that 380-385 MHz and 390-395 MHz are currently shared with narrowband Public Protection and Disaster Relief (PPDR) applications.
ECA12	The applicable RR 5 footnotes in column 1 remain in force. Administrations are however urged to aim for the fullest possible harmonisation with the ITU Table of Allocations and ECA.
ECA13	CEPT administrations are urged to take all practical steps to clear the band 645-960 MHz of the assignments to the aeronautical radionavigation service.
ECA14	Radiolocation limited to military requirements for naval ship borne radars.
ECA16	Use of the band by the mobile service is limited to tactical radio relay and Video links applications.
ECA16A	Use of the band by the mobile service is limited to tactical radio relay and SAP/SAB applications.
ECA17	In the sub-bands 5755-5765 MHz, 10.36-10.37 GHz, 10.45-10.46 GHz the amateur service operates on a secondary basis. In making assignments to other services, CEPT administrations are requested wherever possible to maintain these sub-bands in such a way as to facilitate the reception of amateur emissions with minimal power flux densities.
ECA17A	Use of the band by the mobile service is limited to Video links.
ECA19	This band is allocated to the radio astronomy service. CEPT administrations are urged to take all practicable steps to protect the radio astronomy service from harmful interference. Emissions from space or airborne stations in this and adjacent bands can cause serious harmful interference.
ECA20	This fixed service band is designated for common use by civil and non civil users. Any user priorities in respect of preferred channels or sub-bands are to be determined after discussions between interested parties.
ECA22	The band 5250-5850 MHz is utilised for a variety of radiodetermination applications falling within the radionavigation and radiolocation services. This band will be subject to further detailed consideration.
ECA23	In the sub-bands 5660-5670 MHz (earth to space), 5830-5850 MHz (space to earth) and 10.45-10.50 GHz the amateur-satellite additionally operates on a secondary and non interference basis to other services. In making assignments to other services, CEPT administrations are requested wherever possible to maintain these allocations in such a way as to facilitate the reception of amateur emissions with minimal power flux densities.
ECA24	The band 8500-10000 MHz is utilised for a variety of radiodetermination applications falling within the radionavigation and radiolocation services. This band will be subject to further detailed consideration in conjunction with the band 5250-5850 MHz (see ECA22).
ECA26	The band 13.25-14.0 GHz is utilised for a variety of radiodetermination applications falling within the radionavigation and radiolocation services. This band will be subject to further detailed consideration.
ECA28	CEPT administrations shall not deploy new fixed service systems in the band 11.7-12.5 GHz (ERC/DEC(00)08).
ECA29	The frequency bands 890-915 / 935-960 MHz, 880-890 / 925-935 MHz, 1710-1785 / 1805-1880 MHz, 1920-1980 MHz and 2110-2170 MHz are reserved for public cellular mobile use only. Other services such as the fixed service should only be allowed in the above bands where coexistence with public mobile systems is possible i.e. in sparsely populated or rural areas where the frequency band is not needed for mobile cellular systems.

Footnote Number	Footnote Content
ECA30	National administrations should consider co-ordination zones around the EISCAT sites when using the band 925-935 MHz for mobile services including international planning for military services. Short Range Devices should not use this band.
ECA32	The bands 880-915 MHz and 925-960 MHz are currently used for GSM (2nd generation terrestrial mobile system) in most CEPT member countries and by IMT, depending on the market demands and national licensing schemes.
ECA34	Parts of the bands 450-457.5/460-467.5 MHz may also be used for existing and evolving public cellular networks on a national basis.
ECA35	In Europe the band 75.5-76 GHz is also allocated to the Amateur and Amateur Satellite services.
ECA36	A frequency band, which has been harmonised by NATO and NATO member nations for military use as defined in the NATO Joint Civil/Military Frequency Agreement (NJFA) 2014. Note: NATO Joint Civil/Military Frequency Agreement (NJFA) - Extract for Public Disclosure – 14 February 2017
ECA37	In Europe the allocation to the mobile service is limited to the band 3400-3800 MHz.
ECA38	Administrations may choose at national level to allow MFCN for the command and control and payload links of UAS within the current MFCN bands. Administrations are requested to ensure protection of other existing systems and services in these frequency bands
ECA39	Administrations shall avoid deployment of high-density mobile systems incl. high-density fixed wireless access in the 22.0-23.6 GHz frequency band (ECC/DEC/(18)06)

Radio Regulations Footnotes

Footnote Number	Footnote Content
5.54A	Use of the 8.3-11.3 kHz frequency band by stations in the meteorological aids service is limited to passive use only. In the band 9-11.3 kHz, meteorological aids stations shall not claim protection from stations of the radionavigation service submitted for notification to the Bureau prior to 1 January 2013. For sharing between stations of the meteorological aids service and stations in the radionavigation service submitted for notification after this date, the most recent version of Recommendation ITU-R RS.1881 should be applied. (WRC-12)
5.54B	Additional allocation: in Algeria, Saudi Arabia, Bahrain, Egypt, the United Arab Emirates, the Russian Federation, Iran (Islamic Republic of), Iraq, Kuwait, Lebanon, Morocco, Qatar, the Syrian Arab Republic, Sudan and Tunisia, the frequency band 8.3-9 kHz is also allocated to the radionavigation, fixed and mobile services on a primary basis. (WRC-15)
5.55	Additional allocation: in Armenia, the Russian Federation, Georgia, Kyrgyzstan, Tajikistan and Turkmenistan, the frequency band 14-17 kHz is also allocated to the radionavigation service on a primary basis. (WRC-15)
5.56	The stations of services to which the frequency bands 14-19.95 kHz and 20.05-70 kHz and in Region 1 also the frequency bands 72-84 kHz and 86-90 kHz are allocated may transmit standard frequency and time signals. Such stations shall be afforded protection from harmful interference. In Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Kyrgyzstan, Tajikistan and Turkmenistan, the frequencies 25 kHz and 50 kHz will be used for this purpose under the same conditions. (WRC-23)
5.57	The use of the bands 14-19.95 kHz, 20.05-70 kHz and 70-90 kHz (72-84 kHz and 86-90 kHz in Region 1) by the maritime mobile service is limited to coast radiotelegraph stations (A1A and F1B only). Exceptionally, the use of class J2B or J7B emissions is authorized subject to the necessary bandwidth not exceeding that normally used for class A1A or F1B emissions in the band concerned.
5.58	Additional allocation: in Armenia, Azerbaijan, the Russian Federation, Georgia, Kyrgyzstan, Tajikistan and Turkmenistan, the frequency band 67-70 kHz is also allocated to the radionavigation service on a primary basis. (WRC-23)
5.60	In the bands 70-90 kHz (70-86 kHz in Region 1) and 110-130 kHz (112-130 kHz in Region 1), pulsed radionavigation systems may be used on condition that they do not cause harmful interference to other services to which these bands are allocated.
5.62	Administrations which operate stations in the radionavigation service in the band 90-110 kHz are urged to coordinate technical and operating characteristics in such a way as to avoid harmful interference to the services provided by these stations.
5.64	Only classes A1A or F1B, A2C, A3C, F1C or F3C emissions are authorized for stations of the fixed service in the bands allocated to this service between 90 kHz and 160 kHz (148.5 kHz in Region 1) and for stations of the maritime mobile service in the bands allocated to this service between 110 kHz and 160 kHz (148.5 kHz in Region 1). Exceptionally, class J2B or J7B emissions are also authorized in the bands between 110 kHz and 160 kHz (148.5 kHz in Region 1) for stations of the maritime mobile service.
5.66	Different category of service: in Germany, the allocation of the band 115-117.6 kHz to the fixed and maritime mobile services is on a primary basis (see No. 5.33) and to the radionavigation service on a secondary basis (see No. 5.32).
5.67	Additional allocation: in Kyrgyzstan and Turkmenistan, the frequency band 130-148.5 kHz is also allocated to the radionavigation service on a secondary basis. Within and between these countries this service shall have an equal right to operate. (WRC-19)
5.67A	Stations in the amateur service using frequencies in the band 135.7-137.8 kHz shall not exceed a maximum radiated power of 1 W (e.i.r.p.) and shall not cause harmful interference to stations of the radionavigation service operating in countries listed in No. 5.67. WRC-07)
5.67B	The use of the frequency band 135.7-137.8 kHz in Algeria, Egypt, Iraq, Lebanon, Syrian Arab Republic, Sudan, South Sudan and Tunisia is limited to the fixed and maritime mobile services. The amateur service shall not be used in the above-mentioned countries in the frequency band 135.7-137.8 kHz, and this should be taken into account by the countries authorizing such use. (WRC-19)
5.68	Alternative allocation: in Congo (Rep of the), the Dem. Rep. of the Congo and South Africa, the frequency band 160-200 kHz is allocated to the fixed service on a primary basis. (WRC-15)
5.69	Additional allocation: in Somalia, the band 200-255 kHz is also allocated to the aeronautical radionavigation service on a primary basis.
5.70	Alternative allocation: in Angola, Botswana, Burundi, the Central African Rep., Congo (Rep. of the), Eswatini, Ethiopia, Kenya, Lesotho, Madagascar, Malawi, Mozambique, Namibia, Nigeria, Oman, the Dem. Rep. of the Congo, South Africa, Tanzania, Chad, Zambia and Zimbabwe, the frequency band 200-283.5 kHz is allocated to the aeronautical radionavigation service on a primary basis. (WRC-19)

Footnote Number	Footnote Content
5.73	The band 285-325 kHz (283.5-325 kHz in Region 1) in the maritime radionavigation service may be used to transmit supplementary navigational information using narrow-band techniques, on condition that no harmful interference is caused to radiobeacon stations operating in the radionavigation service. (WRC-97)
5.74	Additional allocation: in Region 1, the frequency band 285.3-285.7 kHz is also allocated to the maritime radionavigation service (other than radiobeacons) on a primary basis.
5.75	Different category of service: in Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Moldova, Kyrgyzstan, Tajikistan, Turkmenistan, Ukraine and the Black Sea areas of Romania, the allocation of the band 315-325 kHz to the maritime radionavigation service is on a primary basis under the condition that in the Baltic Sea area, the assignment of frequencies in this band to new stations in the maritime or aeronautical radionavigation services shall be subject to prior consultation between the administrations concerned. (WRC-07)
5.76	The frequency 410 kHz is designated for radio direction-finding in the maritime radionavigation service. The other radionavigation services to which the band 405-415 kHz is allocated shall not cause harmful interference to radio direction-finding in the band 406.5-413.5 kHz
5.77	Different category of service: in Australia, China, the French overseas communities of Region 3, Korea (Rep. of), India, Iran (Islamic Republic of), Japan, Pakistan, Papua New Guinea, the Dem. People's Rep. of Korea and Sri Lanka, the allocation of the frequency band 415-495 kHz to the aeronautical radionavigation service is on a primary basis. In Armenia, Azerbaijan, Belarus, the Russian Federation, Kazakhstan, Latvia, Uzbekistan and Kyrgyzstan, the allocation of the frequency band 435-495 kHz to the aeronautical radionavigation service is on a primary basis. Administrations in all the aforementioned countries shall take all practical steps necessary to ensure that aeronautical radionavigation stations in the frequency band 435-495 kHz do not cause interference to reception by coast stations of transmissions from ship stations on frequencies designated for ship stations on a worldwide basis. (WRC-19)
5.79	In the maritime mobile service, the frequency bands 415-495 kHz and 505-526.5 kHz are limited to radiotelegraphy and may also be used for the NAVDAT system in accordance with the most recent version of Recommendation ITU-R M.2010, subject to agreement between interested and affected administrations. NAVDAT transmitting stations are limited to coast stations. (WRC-19)
5.79A	When establishing coast stations in the NAVTEX service on the frequencies 490 kHz, 518 kHz and 4209.5 kHz, administrations are strongly recommended to coordinate the operating characteristics in accordance with the procedures of the International Maritime Organization (IMO) (see Resolution 339 (Rev.WRC-07)). (WRC-07)
5.80	In Region 2, the use of the band 435-495 kHz by the aeronautical radionavigation service is limited to non-directional beacons not employing voice transmission
5.80A	The maximum equivalent isotropically radiated power (e.i.r.p.) of stations in the amateur service using frequencies in the band 472-479 kHz shall not exceed 1 W. Administrations may increase this limit of e.i.r.p. to 5 W in portions of their territory which are at a distance of over 800 km from the borders of Algeria, Saudi Arabia, Azerbaijan, Bahrain, Belarus, China, Comoros, Djibouti, Egypt, United Arab Emirates, the Russian Federation, Iran (Islamic Republic of), Iraq, Jordan, Kazakhstan, Kuwait, Lebanon, Libya, Morocco, Mauritania, Oman, Uzbekistan, Qatar, Syrian Arab Republic, Kyrgyzstan, Somalia, Sudan, Tunisia, Ukraine and Yemen. In this frequency band, stations in the amateur service shall not cause harmful interference to, or claim protection from, stations of the aeronautical radionavigation service. (WRC-12)
5.80B	The use of the frequency band 472-479 kHz in Algeria, Saudi Arabia, Azerbaijan, Bahrain, Belarus, China, Comoros, Djibouti, Egypt, United Arab Emirates, the Russian Federation, Iraq, Jordan, Kazakhstan, Kuwait, Lebanon, Libya, Mauritania, Oman, Uzbekistan, Qatar, Syrian Arab Republic, Kyrgyzstan, Somalia, Sudan, Tunisia and Yemen is limited to the maritime mobile and aeronautical radionavigation services. The amateur service shall not be used in the above-mentioned countries in this frequency band, and this should be taken into account by the countries authorizing such use. (WRC-12)
5.82	In the maritime mobile service, the frequency 490 kHz is to be used exclusively for the transmission by coast stations of navigational and meteorological warnings and urgent information to ships, by means of narrow-band direct-printing telegraphy. The conditions for use of the frequency 490 kHz are prescribed in Articles 31 and 52. In using the band 415-495 kHz for the aeronautical radionavigation service, administrations are requested to ensure that no harmful interference is caused to the frequency 490 kHz. (WRC-12)
5.82C	The frequency band 495-505 kHz is used for the international NAVDAT system as described in the most recent version of Recommendation ITU-R M.2010. NAVDAT transmitting stations are limited to coast stations. (WRC-19)
5.82D	When establishing coast stations in the NAVDAT system on the frequencies 500 kHz and 4 226 kHz, the conditions for the use of the frequencies 500 kHz and 4 226 kHz are prescribed in Articles 31 and 52. Administrations are strongly recommended to coordinate the NAVDAT systems operating characteristics in accordance with the procedures of the International Maritime Organization (IMO) (see Resolution 364 (WRC-23)). (WRC-23)
5.84	The conditions for the use of the frequency 518 kHz by the maritime mobile service are prescribed in Articles 31 and 52. (WRC-07)
5.87	Additional allocation: in Angola, Botswana, Eswatini, Lesotho, Malawi, Mozambique, Namibia and Niger, the frequency band 526.5-535 kHz is also allocated to the mobile service on a secondary basis. (WRC-19)

Footnote Number	Footnote Content
5.87A	Additional allocation: in Uzbekistan, the band 526.5-1606.5 kHz is also allocated to the radionavigation service on a primary basis. Such use is subject to agreement obtained under No. 9.21 with administrations concerned and limited to ground-based radiobeacons in operation on 27 October 1997 until the end of their lifetime. (WRC-97)
5.90	In the band 1605-1705 kHz, in cases where a broadcasting station of Region 2 is concerned, the service area of the maritime mobile stations in Region 1 shall be limited to that provided by ground-wave propagation
5.92	Some countries of Region 1 use radiodetermination systems in the bands 1606.5-1625 kHz, 1635-1800 kHz, 1850-2160 kHz, 2194-2300 kHz, 2502-2850 kHz and 3500-3800 kHz, subject to agreement obtained under No. 9.21. The radiated mean power of these stations shall not exceed 50 W.
5.93	Additional allocation: in Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Hungary, Kazakhstan, Latvia, Lithuania, Mongolia, Nigeria, Uzbekistan, Poland, Kyrgyzstan, Slovakia, Tajikistan, Chad, Turkmenistan and Ukraine, the frequency bands 1625-1635 kHz, 1800-1810 kHz and 2160-2170 kHz are also allocated to the fixed and land mobile services on a primary basis, subject to agreement obtained under No. 9.21. (WRC-15)
5.96	In Germany, Armenia, Austria, Azerbaijan, Belarus, Croatia, Denmark, Estonia, the Russian Federation, Finland, Georgia, Hungary, Iceland, Ireland, Israel, Kazakhstan, Latvia, Liechtenstein, Lithuania, Malta, Moldova, Norway, Uzbekistan, Poland, Kyrgyzstan, Slovakia, the Czech Rep., the United Kingdom, Sweden, Switzerland, Tajikistan, Turkmenistan and Ukraine, administrations may allocate up to 200 kHz to their amateur service in the frequency bands 1715-1800 kHz and 1850-2000 kHz. However, when allocating the frequency bands within this range to their amateur service, administrations shall, after prior consultation with administrations of neighbouring countries, take such steps as may be necessary to prevent harmful interference from their amateur service to the fixed and mobile services of other countries. The mean power of any amateur station shall not exceed 10 W. (WRC-15)
5.98	Alternative allocation: in Armenia, Azerbaijan, Belarus, Belgium, Cameroon, Congo (Rep. of the), Denmark, Eritrea, Spain, Ethiopia, the Russian Federation, Georgia, Greece, Italy, Kazakhstan, Lebanon, Lithuania, the Syrian Arab Republic, Türkiye, Kyrgyzstan, Somalia, Tajikistan, Tunisia and Turkmenistan, the frequency band 1 810- 1 830 kHz is allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-23)
5.99	Additional allocation: in Saudi Arabia, Austria, Egypt, Iraq, Libya, Uzbekistan, Romania, Slovakia, Slovenia, Chad, and Togo, the frequency band 1 810-1 830 kHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-23)
5.100	In Region 1, the authorization to use the band 1810-1830 kHz by the amateur service in countries situated totally or partially north of 40° N shall be given only after consultation with the countries mentioned in Nos. 5.98 and 5.99 to define the necessary steps to be taken to prevent harmful interference between amateur stations and stations of other services operating in accordance with Nos. 5.98 and 5.99.
5.103	In Region 1, in making assignments to stations in the fixed and mobile services in the bands 1850-2045 kHz, 2194-2498 kHz, 2502-2625 kHz and 2650-2850 kHz, administrations should bear in mind the special requirements of the maritime mobile service.
5.104	In Region 1, the use of the band 2025-2045 kHz by the meteorological aids service is limited to oceanographic buoy stations.
5.107	Additional allocation: in Saudi Arabia, Eritrea, Eswatini, Ethiopia, Iraq, Libya and Somalia, the frequency band 2 160-2 170 kHz is also allocated to the fixed and mobile, except aeronautical mobile (R), services on a primary basis. The mean power of stations in these services shall not exceed 50 W. (WRC-19)
5.108	The carrier frequency 2182 kHz is an international distress and calling frequency for radiotelephony. The conditions for the use of the band 2173.5-2190.5 kHz are prescribed in Articles 31 and 52. (WRC-07)
5.109	The frequencies 2187.5 kHz, 4207.5 kHz, 6312 kHz, 8414.5 kHz, 12577 kHz and 16804.5 kHz are international distress frequencies for digital selective calling. The conditions for the use of these frequencies are prescribed in Article 31.
5.110	The frequencies 2 174.5 kHz, 4 177.5 kHz, 6 268 kHz, 8 376.5 kHz, 12 520 kHz and 16 695 kHz are used for the automatic connection system (ACS), as described in the most recent version of Recommendation ITU-R M.541. (WRC-23)
5.111	The carrier frequencies 2182 kHz, 3023 kHz, 5680 kHz, 8364 kHz and the frequencies 121.5 MHz, 156.525 MHz, 156.8 MHz and 243 MHz may also be used, in accordance with the procedures in force for terrestrial radiocommunication services, for search and rescue operations concerning manned space vehicles. The conditions for the use of the frequencies are prescribed in Article 31. The same applies to the frequencies 10003 kHz, 14993 kHz and 19993 kHz, but in each of these cases emissions must be confined in a band of ± 3 kHz about the frequency. (WRC-07)
5.112	Alternative allocation: in Sri Lanka, the frequency band 2 194-2 300 kHz is allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-19)

Footnote Number	Footnote Content
5.113	For the conditions for the use of the bands 2300-2495 kHz (2498 kHz in Region 1), 3200-3400 kHz, 4750-4995 kHz and 5005-5060 kHz by the broadcasting service, see Nos. 5.16 to 5.20, 5.21 and 23.3 to 23.10.
5.114	Alternative allocation: in Iraq, the frequency band 2 502-2 625 kHz is allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-19)
5.115	The carrier (reference) frequencies 3023 kHz and 5680 kHz may also be used, in accordance with Article 31 by stations of the maritime mobile service engaged in coordinated search and rescue operations. (WRC-07)
5.116	Administrations are urged to authorize the use of the band 3155-3195 kHz to provide a common worldwide channel for low power wireless hearing aids. Additional channels for these devices may be assigned by administrations in the bands between 3155 kHz and 3400 kHz to suit local needs. It should be noted that frequencies in the range 3000 kHz to 4000 kHz are suitable for hearing aid devices which are designed to operate over short distances within the induction field.
5.117	Alternative allocation: in Liberia, Sri Lanka and Togo, the frequency band 3 155-3 200 kHz is allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-23)
5.118	Additional allocation: in the United States, Mexico and Peru, the frequency band 3 230-3 400 kHz is also allocated to the radiolocation service on a secondary basis. (WRC-19)
5.123	Additional allocation: in Botswana, Eswatini, Lesotho, Malawi, Mozambique, Namibia, South Africa, Zambia and Zimbabwe, the frequency band 3 900-3 950 kHz is also allocated to the broadcasting service on a primary basis, subject to agreement obtained under No. 9.21. (WRC-19)
5.126	In Region 3, the stations of those services to which the band 3 995-4 005 kHz is allocated may transmit standard frequency and time signals
5.127	The use of the band 4000-4063 kHz by the maritime mobile service is limited to ship stations using radiotelephony (see No. 52.220 and Appendix 17).
5.128	Frequencies in the bands 4 063-4 123 kHz and 4 130-4 438 kHz may be used exceptionally by stations in the fixed service, communicating only within the boundary of the country in which they are located, with a mean power not exceeding 50 W, on condition that harmful interference is not caused to the maritime mobile service. In addition, in Afghanistan, Argentina, Armenia, Belarus, Botswana, Burkina Faso, the Central African Rep., China, the Russian Federation, Georgia, India, Kazakhstan, Mali, Niger, Pakistan, Kyrgyzstan, Tajikistan, Chad, Turkmenistan and Ukraine, in the frequency bands 4 063-4 123 kHz, 4 130-4 133 kHz and 4 408-4 438 kHz, stations in the fixed service, with a mean power not exceeding 1 kW, can be operated on condition that they are situated at least 600 km from the coast and that harmful interference is not caused to the maritime mobile service. (WRC-19)
5.130	The conditions for the use of the carrier frequencies 4125 kHz and 6215 kHz are prescribed in Articles 31 and 52. (WRC-07)
5.131	The frequency 4209.5 kHz is used exclusively for the transmission by coast stations of meteorological and navigational warnings and urgent information to ships by means of narrow-band direct-printing techniques. (WRC-97)
5.132	The frequencies 4 210 kHz, 6 314 kHz, 8 416.5 kHz, 12 579 kHz, 16 806.5 kHz, 19 680.5 kHz, 22 376 kHz and 26 100.5 kHz are the international frequencies for the transmission of maritime safety information (MSI) (see Appendices 15 and 17). (WRC-23)
5.132A	Stations in the radiolocation service shall not cause harmful interference to, or claim protection from, stations operating in the fixed or mobile services. Applications of the radiolocation service are limited to oceanographic radars operating in accordance with Resolution 612 (Rev.WRC-12) (WRC-12)
5.132B	Alternative allocation: in Armenia, Belarus, Moldova and Kyrgyzstan, the frequency band 4 438- 4 488 kHz is allocated to the fixed and mobile, except aeronautical mobile (R), services on a primary basis. (WRC-19)
5.133	Different category of service: in Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Kazakhstan, Latvia, Lithuania, Niger, Uzbekistan, Kyrgyzstan, Tajikistan, Turkmenistan and Ukraine, the allocation of the band 5130-5250 kHz to the mobile, except aeronautical mobile, service is on a primary basis (see No. 5.33). (WRC-12)
5.133A	Alternative allocation: in Armenia, Belarus, Moldova and Kyrgyzstan, the frequency bands 5 250-5 275 kHz and 26 200-26 350 kHz are allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-19)
5.133B	Stations in the amateur service using the frequency band 5 351.5-5 366.5 kHz shall not exceed a maximum radiated power of 15 W (e.i.r.p.). However, in Region 2 in Mexico, stations in the amateur service using the frequency band 5 351.5-5 366.5 kHz shall not exceed a maximum radiated power of 20 W (e.i.r.p.). In the following Region 2 countries: Antigua and Barbuda, Argentina, Bahamas, Barbados, Belize, Bolivia, Brazil, Chile, Colombia, Costa Rica, Cuba, Dominican Republic, Dominica, El Salvador, Ecuador, Grenada, Guatemala, Guyana, Haiti, Honduras, Jamaica, Nicaragua, Panama, Paraguay, Peru, Saint Lucia, Saint Kitts and Nevis, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago, Uruguay, Venezuela, as well as the overseas countries and territories within the Kingdom of the Netherlands in Region 2, stations in the amateur service using the frequency band 5 351.5-5 366.5 kHz shall not exceed a maximum radiated power of 25 W (e.i.r.p.). (WRC-19)

Footnote Number	Footnote Content
5.134	The use of the frequency bands 5 900-5 950 kHz, 7 300-7 350 kHz, 9 400-9 500 kHz, 11 600-11 650 kHz, 12 050-12 100 kHz, 13 570-13 600 kHz, 13 800-13 870 kHz, 15 600-15 800 kHz, 17 480-17 550 kHz and 18 900-19 020 kHz by the broadcasting service is subject to the application of the procedure of Article 12. Administrations are encouraged to use these frequency bands to facilitate the introduction of digitally modulated emissions in accordance with the provisions of Resolution 517 (Rev.WRC-19). (WRC-19)
5.136	Additional allocation: Frequencies in the band 5900-5950 kHz may be used by stations in the following services, communicating only within the boundary of the country in which they are located: fixed service (in all three Regions), land mobile service (in Region 1), mobile except aeronautical mobile (R) service (in Regions 2 and 3), on condition that harmful interference is not caused to the broadcasting service. When using frequencies for these services, administrations are urged to use the minimum power required and to take account of the seasonal use of frequencies by the broadcasting service published in accordance with the Radio Regulations. (WRC-07)
5.137	On condition that harmful interference is not caused to the maritime mobile service, the bands 6200-6213.5 kHz and 6220.5-6525 kHz may be used exceptionally by stations in the fixed service, communicating only within the boundary of the country in which they are located, with a mean power not exceeding 50 W. At the time of notification of these frequencies, the attention of the Bureau will be drawn to the above conditions.
5.137A	The frequencies 6 337.5 kHz, 8 443 kHz, 12 663.5 kHz, 16 909.5 kHz and 22 450.5 kHz are the regional frequencies for the transmission of maritime safety information (MSI) by means of the NAVDAT system (see Appendices 15 and 17). (WRC-23)
5.138	The following bands: 6765-6795 kHz (centre frequency 6780 kHz), 433.05-434.79 MHz (centre frequency 433.92 MHz) in Region 1 except in the countries mentioned in No. 5.280, 61-61.5 GHz (centre frequency 61.25 GHz), 122-123 GHz (centre frequency 122.5 GHz), and 244-246 GHz (centre frequency 245 GHz) are designated for industrial, scientific and medical (ISM) applications. The use of these frequency bands for ISM applications shall be subject to special authorisation by the administration concerned, in agreement with other administrations whose radiocommunication services might be affected. In applying this provision, administrations shall have due regard to the latest relevant ITU-R Recommendations.
5.140	Additional allocation: in Angola, Iraq, Somalia and Togo, the frequency band 7000-7050 kHz is also allocated to the fixed service on a primary basis. (WRC-15)
5.141	Alternative allocation: in Egypt, Eritrea, Ethiopia, Guinea, Libya, Madagascar and Niger, the band 7000-7050 kHz is allocated to the fixed service on a primary basis. (WRC-12)
5.141A	Additional allocation: in Uzbekistan and Kyrgyzstan, the bands 7000-7100 kHz and 7100-7200 kHz are also allocated to the fixed and land mobile services on a secondary basis. (WRC-03)
5.141B	Additional allocation: in Algeria, Saudi Arabia, Australia, Bahrain, Botswana, Brunei Darussalam, China, Comoros, Korea (Rep. of), Diego Garcia, Djibouti, Egypt, United Arab Emirates, Eritrea, Guinea, Indonesia, Iran (Islamic Republic of), Japan, Jordan, Kuwait, Libya, Mali, Morocco, Mauritania, Niger, New Zealand, Oman, Papua New Guinea, Qatar, the Syrian Arab Republic, the Dem. People's Rep. of Korea, Singapore, Sudan, South Sudan, Tunisia, Viet Nam and Yemen, the frequency band 7 100-7 200 kHz is also allocated to the fixed and the mobile, except aeronautical mobile (R), services on a primary basis. (WRC-19)
5.143	Additional allocation: frequencies in the band 7300-7350 kHz may be used by stations in the fixed service and in the land mobile service, communicating only within the boundary of the country in which they are located, on condition that harmful interference is not caused to the broadcasting service. When using frequencies for these services, administrations are urged to use the minimum power required and to take account of the seasonal use of frequencies by the broadcasting service published in accordance with the Radio Regulations. (WRC-07)
5.143A	In Region 3, frequencies in the band 7 350-7 450 kHz may be used by stations in the fixed service on a primary basis and land mobile service on a secondary basis, communicating only within the boundary of the country in which they are located, on condition that harmful interference is not caused to the broadcasting service. When using frequencies for these services, administrations are urged to use the minimum power required and to take account of the seasonal use of frequencies by the broadcasting service published in accordance with the Radio Regulations. (WRC-12)
5.143B	In Region 1, frequencies in the band 7350-7450 kHz may be used by stations in the fixed and land mobile services communicating only within the boundary of the country in which they are located on condition that harmful interference is not caused to the broadcasting service. The total radiated power of each station shall not exceed 24 dBW. (WRC-12)
5.143C	Additional allocation: in Algeria, Saudi Arabia, Bahrain, Comoros, Djibouti, Egypt, United Arab Emirates, Iran (Islamic Republic of), Jordan, Kuwait, Libya, Morocco, Mauritania, Niger, Oman, Qatar, the Syrian Arab Republic, Sudan, South Sudan, Tunisia and Yemen, the bands 7350-7400 kHz and 7400-7450 kHz are also allocated to the fixed service on a primary basis. (WRC-12)
5.143D	In Region 2, frequencies in the band 7 350-7 400 kHz may be used by stations in the fixed service and in the land mobile service, communicating only within the boundary of the country in which they are located, on condition that harmful interference is not caused to the broadcasting service. When using frequencies for these services, administrations are urged to use the minimum power required and to take account of the seasonal use of frequencies by the broadcasting service published in accordance with the Radio Regulations. (WRC-12)

Footnote Number	Footnote Content
5.144	In Region 3, the stations of those services to which the band 7 995-8 005 kHz is allocated may transmit standard frequency and time signals.
5.145	The conditions for the use of the carrier frequencies 8291 kHz, 12290 kHz and 16420 kHz are prescribed in Articles 31 and 52. (WRC-07)
5.145A	Stations in the radiolocation service shall not cause harmful interference to, or claim protection from, stations operating in the fixed service. Applications of the radiolocation service are limited to oceanographic radars operating in accordance with Resolution 612 (Rev.WRC-12) (WRC-12)
5.145B	Alternative allocation: in Armenia, Belarus, Moldova and Kyrgyzstan, the frequency bands 9 305- 9 355 kHz and 16 100-16 200 kHz are allocated to the fixed service on a primary basis. (WRC-19)
5.146	Additional allocation: Frequencies in the bands 9400-9500 kHz, 11600-11650 kHz, 12050-12100 kHz, 15600-15800 kHz, 17480-17550 kHz and 18900-19020 kHz may be used by stations in the fixed service, communicating only within the boundary of the country in which they are located, on condition that harmful interference is not caused to the broadcasting service. When using frequencies in the fixed service, administrations are urged to use the minimum power required and to take account of the seasonal use of frequencies by the broadcasting service published in accordance with the Radio Regulations. (WRC-07)
5.147	On condition that harmful interference is not caused to the broadcasting service, frequencies in the bands 9775-9900 kHz, 11650-11700 kHz and 11975-12050 kHz may be used by stations in the fixed service communicating only within the boundary of the country in which they are located, each station using a total radiated power not exceeding 24 dBW.
5.149	In making assignments to stations of other services to which the bands: 13360-13410 kHz, 25550-25670 kHz, 37.5-38.25 MHz, 73-74.6 MHz in Regions 1 and 3, 150.05-153 MHz in Region 1, 322-328.6 MHz, 406.1-410 MHz, 608-614 MHz in Regions 1 and 3, 1330-1400 MHz, 1610.6-1613.8 MHz, 1660-1670 MHz, 1718.8-1722.2 MHz, 2655-2690 MHz, 3260-3267 MHz, 3332-3339 MHz, 3345.8-3352.5 MHz, 4825-4835 MHz, 4950-4990 MHz, 4990-5000 MHz, 6650-6675.2 MHz, 10.6-10.68 GHz, 14.47-14.5 GHz, 22.01-22.21 GHz, 22.21-22.5 GHz, 22.81-22.86 GHz, 23.07-23.12 GHz, 31.2-31.3 GHz, 31.5-31.8 GHz in Regions 1 and 3, 36.43-36.5 GHz, 42.5-43.5 GHz, 48.94-49.04 GHz, 76-86 GHz, 92-94 GHz, 94.1-100 GHz, 102-109.5 GHz, 111.8-114.25 GHz, 128.33-128.59 GHz, 129.23-129.49 GHz, 130-134 GHz, 136-148.5 GHz, 151.5-158.5 GHz, 168.59-168.93 GHz, 171.11-171.45 GHz, 172.31-172.65 GHz, 173.52-173.85 GHz, 195.75-196.15 GHz, 209-226 GHz, 241-250 GHz, 252-275 GHz are allocated, administrations are urged to take all practicable steps to protect the radio astronomy service from harmful interference. Emissions from spaceborne or airborne stations can be particularly serious sources of interference to the radio astronomy service (see Nos. 4.5 and 4.6 and Article 29). (WRC-07)
5.149A	Alternative allocation: in Armenia, Belarus, Moldova and Kyrgyzstan, the frequency band 13 450-13 550 kHz is allocated to the fixed service on a primary basis and to the mobile, except aeronautical mobile (R), service on a secondary basis. (WRC-19)
5.150	The following bands: 13553-13567 kHz (centre frequency 13560 kHz), 26957-27283 kHz (centre frequency 27120 kHz), 40.66-40.70 MHz (centre frequency 40.68 MHz), 902-928 MHz in Region 2 (centre frequency 915 MHz), 2400-2500 MHz (centre frequency 2450 MHz), 5725-5875 MHz (centre frequency 5800 MHz), and 24-24.25 GHz (centre frequency 24.125 GHz) are also designated for industrial, scientific and medical (ISM) applications. Radiocommunication services operating within these bands must accept harmful interference which may be caused by these applications. ISM equipment operating in these bands is subject to the provisions of No. 15.13.
5.151	Additional allocation: Frequencies in the bands 13570-13600 kHz and 13800-13870 kHz may be used by stations in the fixed service and in the mobile except aeronautical mobile (R) service, communicating only within the boundary of the country in which they are located, on the condition that harmful interference is not caused to the broadcasting service. When using frequencies in these services, administrations are urged to use the minimum power required and to take account of the seasonal use of frequencies by the broadcasting service published in accordance with the Radio Regulations.(WRC-07)
5.152	Additional allocation: in Armenia, Azerbaijan, China, Ivory Coast, Georgia, Iran (Islamic Republic of), Kazakhstan, Uzbekistan, Kyrgyzstan, the Russian Federation, Tajikistan, Turkmenistan and Ukraine, the band 14250-14350 kHz is also allocated to the fixed service on a primary basis. Stations of the fixed service shall not use a radiated power exceeding 24 dBW. (WRC-03)
5.154	Additional allocation: in Armenia, Azerbaijan, Georgia, Kazakhstan, Kyrgyzstan, the Russian Federation, Tajikistan, Turkmenistan and Ukraine, the band 18068-18168 kHz is also allocated to the fixed service on a primary basis for use within their boundaries, with a peak envelope power not exceeding 1 kW. (WRC-03)
5.155	Additional allocation: in Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Moldova, Uzbekistan, Kyrgyzstan, Slovakia, Tajikistan, Turkmenistan and Ukraine, the frequency band 21 850-21 870 kHz is also allocated to the aeronautical mobile (R) service on a primary basis. (WRC-23)
5.155A	In Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Moldova, Uzbekistan, Kyrgyzstan, Slovakia, Tajikistan, Turkmenistan and Ukraine, the use of the frequency band 21 850-21 870 kHz by the fixed service is limited to provision of services related to aircraft flight safety. (WRC-23)

Footnote Number	Footnote Content
5.155B	The band 21870-21924 kHz is used by the fixed service for provision of services related to aircraft flight safety.
5.156	Additional allocation: in Nigeria, the band 22720-23200 kHz is also allocated to the meteorological aids service (radiosondes) on a primary basis.
5.156A	The use of the band 23200-23350 kHz by the fixed service is limited to provision of services related to aircraft flight safety.
5.157	The use of the band 23350-24000 kHz by the maritime mobile service is limited to inter-ship radiotelegraphy.
5.158	Alternative allocation: in Armenia, Belarus, Moldova and Kyrgyzstan, the frequency band 24 450-24 600 kHz is allocated to the fixed and land mobile services on a primary basis. (WRC-19)
5.159	Alternative allocation: in Armenia, Belarus, Moldova and Kyrgyzstan, the frequency band 39-39.5 MHz is allocated to the fixed and mobile services on a primary basis. (WRC-19)
5.159A	The use of the frequency band 40-50 MHz by the Earth exploration-satellite service (active) shall be in accordance with the geographical area restrictions and the operational and technical conditions defined in Resolution 677 (WRC-23). The provisions of this footnote in no way diminish the obligation of the Earth exploration-satellite service (active) to operate as a secondary service in accordance with Nos. 5.29 and 5.30. (WRC-23)
5.160	Additional allocation: in Botswana, Burundi, the Dem. Rep. of the Congo and Rwanda, the band 41-44 MHz is also allocated to the aeronautical radionavigation service on a primary basis. (WRC-12)
5.161	Additional allocation: in Iran (Islamic Republic of) and Japan, the band 41-44 MHz is also allocated to the radiolocation service on a secondary basis.
5.161A	Additional allocation: in Korea (Rep. of), the United States and Mexico, the frequency bands 41.015-41.665 MHz and 43.35-44 MHz are also allocated to the radiolocation service on a primary basis. Stations in the radiolocation service shall not cause harmful interference to, or claim protection from, stations operating in the fixed or mobile services. Applications of the radiolocation service are limited to oceanographic radars operating in accordance with Resolution 612 (Rev.WRC-12). (WRC-19)
5.161B	Alternative allocation: in Albania, Germany, Armenia, Austria, Belarus, Belgium, Bosnia and Herzegovina, Cyprus, Vatican, Croatia, Denmark, Spain, Estonia, Finland, France, Greece, Hungary, Ireland, Iceland, Italy, Latvia, Liechtenstein, Lithuania, Luxembourg, North Macedonia, Malta, Moldova, Monaco, Montenegro, Norway, Uzbekistan, Netherlands, Portugal, Kyrgyzstan, Slovakia, Czech Rep., Romania, United Kingdom, San Marino, Slovenia, Sweden, Switzerland, Turkey and Ukraine, the frequency band 42-42.5 MHz is allocated to the fixed and mobile services on a primary basis. (WRC-19)
5.162	Additional allocation: in Australia, the band 44-47 MHz is also allocated to the broadcasting service on a primary basis. (WRC-12)
5.162A	Additional allocation: in Germany, Australia, Austria, Belgium, Bosnia and Herzegovina, China, Vatican, Korea (Rep. of), Denmark, Spain, Estonia, the Russian Federation, Finland, France, Indonesia, Ireland, Iceland, Italy, Japan, Latvia, Liechtenstein, Lithuania, Luxembourg, North Macedonia, Monaco, Montenegro, Norway, the Netherlands, Poland, Portugal, the Dem. People's Rep. of Korea, the Czech Rep., the United Kingdom, Serbia, Slovenia, Sweden and Switzerland, the frequency band 46-68 MHz is also allocated to the radiolocation service on a secondary basis. This use is limited to the operation of wind profiler radars in accordance with Resolution 217 (Rev.WRC-23). (WRC-23)
5.163	Additional allocation: in Armenia, Belarus, the Russian Federation, Georgia, Kazakhstan, Latvia, Moldova, Uzbekistan, Kyrgyzstan, Tajikistan, Turkmenistan and Ukraine, the frequency bands 47-48.5 MHz and 56.5-58 MHz are also allocated to the fixed and land mobile services on a secondary basis. (WRC-19)
5.164	Additional allocation: in Albania, Algeria, Germany, Austria, Belgium, Bosnia and Herzegovina, Botswana, Bulgaria, Ivory Coast, Croatia, Denmark, Spain, Estonia, Eswatini, Finland, France, Gabon, Greece, Hungary, Ireland, Israel, Italy, Jordan, Lebanon, Libya, Liechtenstein, Lithuania, Luxembourg, Madagascar, Mali, Malta, Morocco, Mauritania, Monaco, Montenegro, Nigeria, Norway, the Netherlands, Poland, Syrian Arab Republic, Slovakia, Czech Rep., Romania, the United Kingdom, Serbia, Slovenia, Sweden, Switzerland, Chad, Togo, Tunisia and Turkey, the frequency band 47-68 MHz, in South Africa the frequency band 47-50 MHz, and in Latvia the frequency bands 48.5-56.5 MHz and 58-68 MHz, are also allocated to the land mobile service on a primary basis. However, stations of the land mobile service in the countries mentioned in connection with each frequency band referred to in this footnote shall not cause harmful interference to, or claim protection from, existing or planned broadcasting stations of countries other than those mentioned in connection with the frequency band. (WRC-19)
5.165	Additional allocation: in Angola, Cameroon, Congo (Rep. of the), Egypt, Madagascar, Mozambique, Niger, Somalia, Sudan, South Sudan, Tanzania and Chad, the frequency band 47- 68 MHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-19)
5.166A	Different category of service: in Austria, Cyprus, the Vatican, Croatia, Denmark, Spain, Finland, Hungary, Latvia, the Netherlands, the Czech Republic, the United Kingdom, Slovakia and Slovenia, the frequency band 50.0-50.5 MHz is allocated to the amateur service on a primary basis. Stations in the amateur service in these countries shall not cause harmful interference to, or claim protection from, stations of the broadcasting, fixed and mobile services operating in accordance with the Radio Regulations in the frequency band 50.0-50.5 MHz in the countries not listed in this provision. For a station of these services, the protection criteria in No. 5.169B shall also apply. In Region 1, with the exception of those countries listed in No. 5.169, wind profiler radars operating in the radiolocation service under No. 5.162A are authorized to operate on the basis of equality with stations in the amateur service in the frequency band 50.0-50.5 MHz. (WRC-19)

Footnote Number	Footnote Content
5.166B	In Region 1, stations in the amateur service operating on a secondary basis shall not cause harmful interference to, or claim protection from, stations of the broadcasting service. The field strength generated by an amateur station in Region 1 in the frequency band 50-52 MHz shall not exceed a calculated value of +6 dB(μ V/m) at a height of 10 m above ground for more than 10% of time along the border of a country with operational analogue broadcasting stations in Region 1 and of neighbouring countries with broadcasting stations in Region 3 listed in Nos. 5.167 and 5.168. (WRC-19)
5.166C	In Region 1, stations in the amateur service in the frequency band 50-52 MHz, with the exception of those countries listed in No. 5.169, shall not cause harmful interference to, or claim protection from, wind profiler radars operating in the radiolocation service under No. 5.162A. (WRC-19)
5.166E	In the Russian Federation, only the frequency band 50.080-50.280 MHz is allocated to the amateur service on a secondary basis. The protection criteria for the other services in the countries not listed in this provision are specified in Nos. 5.166B and 5.169B. (WRC-19)
5.169A	Alternative allocation: in the following countries in Region 1: Angola, Saudi Arabia, Bahrain, Burkina Faso, Burundi, the United Arab Emirates, Gambia, Jordan, Kenya, Kuwait, Mauritius, Mozambique, Oman, Uganda, Qatar, South Sudan and Tanzania, the frequency band 50- 54 MHz is allocated to the amateur service on a primary basis. In Guinea-Bissau, the frequency band 50.0-50.5 MHz is allocated to the amateur service on a primary basis. In Djibouti, the frequency band 50-52 MHz is allocated to the amateur service on a primary basis. With the exception of those countries listed in No. 5.169, stations in the amateur service operating in Region 1 under this footnote, in all or part of the frequency band 50-54 MHz, shall not cause harmful interference to, or claim protection from, stations of other services operating in accordance with the Radio Regulations in Algeria, Egypt, Iran (Islamic Republic of), Iraq, Israel, Libya, Palestine*, the Syrian Arab Republic, the Dem. People's Republic of Korea, Sudan and Tunisia. The field strength generated by an amateur station in the frequency band 50-54 MHz shall not exceed a value of +6 dB(μ V/m) at a height of 10 m above ground for more than 10% of time along the borders of listed countries requiring protection. (WRC-19)
5.169B	Except countries listed under No. 5.169, stations in the amateur service used in Region 1, in all or part of the 50-54 MHz frequency band, shall not cause harmful interference to, or claim protection from, stations of other services used in accordance with the Radio Regulations in Algeria, Armenia, Azerbaijan, Belarus, Egypt, Russian Federation, Iran (Islamic Republic of), Iraq, Kazakhstan, Kyrgyzstan, Libya, Uzbekistan, Palestine*, the Syrian Arab Republic, Sudan, Tunisia and Ukraine. The field strength generated by an amateur station in the frequency band 50-54 MHz shall not exceed a value of +6 dB(μ V/m) at a height of 10 m above ground for more than 10% of time along the borders of the countries listed in this provision. (WRC-19)
5.175	Alternative allocation: in Armenia, Belarus, the Russian Federation, Kazakhstan, Moldova, Uzbekistan, Kyrgyzstan, Tajikistan, Turkmenistan and Ukraine, the frequency bands 68-73 MHz and 76-87.5 MHz are allocated to the broadcasting service on a primary basis. In Latvia and Lithuania, the frequency bands 68-73 MHz and 76-87.5 MHz are allocated to the broadcasting and mobile, except aeronautical mobile, services on a primary basis. In Mongolia, the frequency band 76-87.5 MHz is allocated to the broadcasting service on a primary basis; the stations of the broadcasting service shall not cause harmful interference to, or claim protection from, existing or planned fixed and mobile stations in the neighbouring countries. The services to which these frequency bands are allocated in other countries and the broadcasting service in the countries listed above are subject to agreements with the neighbouring countries concerned. (WRC-23)
5.177	Additional allocation: in Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Kazakhstan, Uzbekistan, Kyrgyzstan, Tajikistan, Turkmenistan and Ukraine, the band 73-74 MHz is also allocated to the broadcasting service on a primary basis, subject to agreement obtained under No. 9.21. (WRC-07)
5.178	Additional allocation: in Colombia, Cuba, El Salvador, Guatemala, Guyana, Honduras and Nicaragua, the band 73-74.6 MHz is also allocated to the fixed and mobile services on a secondary basis. (WRC-12)
5.179	Additional allocation: in Armenia, Azerbaijan, Belarus, China, the Russian Federation, Georgia, Kazakhstan, Lithuania, Mongolia, Kyrgyzstan, Tajikistan, Turkmenistan and Ukraine, the bands 74.6-74.8 MHz and 75.2-75.4 MHz are also allocated to the aeronautical radionavigation service, on a primary basis, for ground-based transmitters only. (WRC-12)
5.180	The frequency 75 MHz is assigned to marker beacons. Administrations shall refrain from assigning frequencies close to the limits of the guardband to stations of other services which, because of their power or geographical position, might cause harmful interference or otherwise place a constraint on marker beacons. Every effort should be made to improve further the characteristics of airborne receivers and to limit the power of transmitting stations close to the limits 74.8 MHz and 75.2 MHz.
5.187	Alternative allocation: in Albania, the band 81-87.5 MHz is allocated to the broadcasting service on a primary basis and used in accordance with the decisions contained in the Final Acts of the Special Regional Conference (Geneva, 1960).
5.190	Additional allocation: in Monaco, the band 87.5-88 MHz is also allocated to the land mobile service on a primary basis, subject to agreement obtained under No. 9.21. (WRC-97)
5.192	Additional allocation: in China and Korea (Rep. of), the band 100-108 MHz is also allocated to the fixed and mobile services on a primary basis. (WRC-97)

Footnote Number	Footnote Content
5.194	Additional allocation: in Kyrgyzstan, Somalia and Turkmenistan, the frequency band 104-108 MHz is also allocated to the mobile, except aeronautical mobile (R), service on a secondary basis. (WRC-19)
5.197	Additional allocation: in the Syrian Arab Republic, the band 108-111.975 MHz is also allocated to the mobile service on a secondary basis, subject to agreement obtained under No. 9.21. In order to ensure that harmful interference is not caused to stations of the aeronautical radionavigation service, stations of the mobile service shall not be introduced in the band until it is no longer required for the aeronautical radionavigation service by any administration which may be identified in the application of the procedures invoked under No. 9.21 (WRC-12)
5.197A	Additional allocation: the frequency band 108-117.975 MHz is also allocated on a primary basis to the aeronautical mobile (R) service, limited to systems operating in accordance with recognized international aeronautical standards. Such use shall be in accordance with Resolution 413 (Rev.WRC-23). The use of the frequency band 108-112 MHz by the aeronautical mobile (R) service shall be limited to systems composed of ground-based transmitters and associated receivers that provide navigational information in support of air navigation functions in accordance with recognized international aeronautical standards. (WRC-23)
5.198A	The use of the frequency band 117.975-137 MHz by the aeronautical mobile-satellite (R) service is subject to coordination under No. 9.11A. No. 9.16 does not apply. Such use shall be limited to non-geostationary-satellite systems operated in accordance with international aeronautical standards. Resolution 406 (WRC-23) applies. (WRC-23)
5.198B	The use of the frequency band 117.975-137 MHz by the aeronautical mobile (R) service shall have priority over use by the aeronautical mobile-satellite (R) service. (WRC-23)
5.200	In the frequency band 117.975-137 MHz, the frequency 121.5 MHz is the aeronautical emergency frequency and, where required, the frequency 123.1 MHz is the aeronautical frequency auxiliary to 121.5 MHz. Mobile stations of the maritime mobile service may communicate on these frequencies under the conditions laid down in Article 31 for distress and safety purposes with stations of the aeronautical mobile service and the aeronautical mobile satellite service. (WRC-23)
5.201	Additional allocation: in Saudi Arabia, Armenia, Azerbaijan, Bahrain, Egypt, Estonia, the Russian Federation, Georgia, Hungary, Iran (Islamic Republic of), Iraq (Republic of), Japan, Kazakhstan, Mali, Mongolia, Mozambique, Uzbekistan, Papua New Guinea, Poland, Qatar, Kyrgyzstan, Romania, Senegal, Somalia, Tajikistan and Turkmenistan, the frequency band 132-136 MHz is also allocated to the aeronautical mobile (OR) service on a primary basis. In assigning frequencies to stations of the aeronautical mobile (OR) service, the administration shall take account of the frequencies assigned to stations in the aeronautical mobile (R) service. (WRC-23)
5.202	Additional allocation: in Saudi Arabia, Armenia, Azerbaijan, Bahrain, the United Arab Emirates, the Russian Federation, Georgia, Iran (Islamic Republic of), Jordan, Mali, Oman, Uzbekistan, Poland, the Syrian Arab Republic, Kyrgyzstan, Romania, Senegal, Tajikistan and Turkmenistan, the frequency band 136-137 MHz is also allocated to the aeronautical mobile (OR) service on a primary basis. In assigning frequencies to stations of the aeronautical mobile (OR) service, the administration shall take account of the frequencies assigned to stations in the aeronautical mobile (R) service. (WRC-23)
5.203C	The use of the space operation service (space-to-Earth) with non-geostationary satellite short-duration mission systems in the frequency band 137-138 MHz is subject to Resolution 660 (WRC-19). Resolution 32 (WRC-19) applies. These systems shall not cause harmful interference to, or claim protection from, the existing services to which the frequency band is allocated on a primary basis. (WRC-19)
5.204	Different category of service: in Afghanistan, Saudi Arabia, Bahrain, Bangladesh, Brunei Darussalam, China, Cuba, the United Arab Emirates, India, Indonesia, Iran (Islamic Republic of), Iraq, Kuwait, Montenegro, Oman, Pakistan, the Philippines, Qatar, Singapore, Thailand and Yemen, the frequency band 137-138 MHz is allocated to the fixed and mobile, except aeronautical mobile (R), services on a primary basis (see No. 5.33). (WRC-19)
5.205	Different category of service: in Israel and Jordan, the allocation of the band 137-138 MHz to the fixed and mobile, except aeronautical mobile, services is on a primary basis (see No. 5.33).
5.206	Different category of service: in Armenia, Azerbaijan, Belarus, Bulgaria, Egypt, the Russian Federation, Finland, France, Georgia, Greece, Kazakhstan, Lebanon, Moldova, Mongolia, Uzbekistan, Poland, Kyrgyzstan, the Syrian Arab Republic, Slovakia, the Czech Republic, Romania, Tajikistan, Turkmenistan and Ukraine, the allocation of the band 137-138 MHz to the aeronautical mobile (OR) service is on a primary basis (see No. 5.33). (WRC-2000)
5.207	Additional allocation: in Australia, the band 137-144 MHz is also allocated to the broadcasting service on a primary basis until that service can be accommodated within regional broadcasting allocations.
5.208	The use of the band 137-138 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. (WRC-97)

Footnote Number	Footnote Content
5.208A	In making assignments to space stations in the mobile-satellite service in the frequency bands 137-138 MHz, 387-390 MHz and 400.15-401 MHz and in the maritime mobile-satellite service (space-to-Earth) in the frequency bands 157.1875-157.3375 MHz and 161.7875-161.9375 MHz, administrations shall take all practicable steps to protect the radio astronomy service in the frequency bands 150.05-153 MHz, 322-328.6 MHz, 406.1-410 MHz and 608-614 MHz from harmful interference from unwanted emissions as shown in the most recent version of Recommendation ITU-R RA.769. (WRC-19)
5.208B	In the frequency bands: 137-138 MHz, 157.1875-157.3375 MHz, 161.7875-161.9375 MHz, 387-390 MHz, 400.15-401 MHz, 1 452-1 492 MHz, 1 525-1 610 MHz, 1 613.8-1 626.5 MHz, 2 655-2 690 MHz, 21.4-22 GHz, Resolution 739 (Rev.WRC-19) applies. (WRC-19) *This provision was previously numbered as No. 5.347A. It was renumbered to preserve the sequential order.
5.209	The use of the bands 137-138 MHz, 148-150.05 MHz, 399.9-400.05 MHz, 400.15-401 MHz, 454-456 MHz and 459-460 MHz by the mobile-satellite service is limited to non-geostationary-satellite systems. (WRC-97)
5.209A	The use of the frequency band 137.175-137.825 MHz by non-geostationary satellite systems in the space operation service identified as short-duration mission in accordance with Appendix 4 is not subject to No. 9.11A. (WRC-19)
5.210	Additional allocation: in Italy and the United Kingdom, the frequency bands 138-143.6 MHz and 143.65-144 MHz are also allocated to the space research service (space-to-Earth) on a secondary basis. (WRC-23)
5.211	Additional allocation: in Germany, Saudi Arabia, Austria, Bahrain, Belgium, Denmark, the United Arab Emirates, Spain, Finland, Greece, Guinea, Ireland, Israel, Kenya, Kuwait, Lebanon, Liechtenstein, Luxembourg, North Macedonia, Mali, Malta, Montenegro, Norway, the Netherlands, Qatar, Slovakia, the United Kingdom, Serbia, Slovenia, Somalia, Sweden, Switzerland, Tanzania, Tunisia and Turkey, the frequency band 138-144 MHz is also allocated to the maritime mobile and land mobile services on a primary basis. (WRC-19)
5.212	Alternative allocation: in Angola, Botswana, Cameroon, the Central African Rep., Congo (Rep. of the), Eswatini, Gabon, Gambia, Ghana, Guinea, Iraq, Jordan, Lesotho, Liberia, Libya, Malawi, Mozambique, Namibia, Niger, Oman, Uganda, Syrian Arab Republic, the Dem. Rep. of the Congo, Rwanda, Sierra Leone, South Africa, Chad, Togo, Zambia and Zimbabwe, the frequency band 138-144 MHz is allocated to the fixed and mobile services on a primary basis. (WRC-19)
5.214	Additional allocation: in Eritrea, Ethiopia, Kenya, North Macedonia, Montenegro, Serbia, Somalia, Sudan, South Sudan and Tanzania, the frequency band 138-144 MHz is also allocated to the fixed service on a primary basis. (WRC-19)
5.216	Additional allocation: in China, the band 144-146 MHz is also allocated to the aeronautical mobile (OR) service on a secondary basis.
5.218	Additional allocation: the band 148-149.9 MHz is also allocated to the space operation service (Earth-to-space) on a primary basis, subject to agreement obtained under No. 9.21. The bandwidth of any individual transmission shall not exceed ± 25 kHz.
5.218A	The frequency band 148-149.9 MHz in the space operation service (Earth-to-space) may be used by non-geostationary satellite systems with short-duration missions. Non-geostationary satellite systems in the space operation service used for a short-duration mission in accordance with Resolution COM5/5 (WRC-19) of the Radio Regulations are not subject to agreement under No. 9.21. At the stage of coordination, the provisions of Nos. 9.17 and 9.18 also apply. In the frequency band 148-149.9 MHz, non-geostationary satellite systems with short-duration missions shall not cause unacceptable interference to, or claim protection from, existing primary services within this frequency band, or impose additional constraints on the space operation and mobile-satellite services. In addition, earth stations in non-geostationary satellite systems in the space operation service with short-duration missions in the frequency band 148-149.9 MHz shall ensure that the power flux-density does not exceed -149 dB(W/(m ² $\square$ 4 kHz)) for more than 1% of time at the border of the territory of the following countries: Armenia, Azerbaijan, Belarus, China, Korea (Rep. of), Cuba, Russian Federation, India, Iran (Islamic Republic of), Japan, Kazakhstan, Malaysia, Uzbekistan, Kyrgyzstan, Thailand and Viet Nam. In case this power flux-density limit is exceeded, agreement under No. 9.21 is required to be obtained from countries mentioned in this footnote. (WRC-19)
5.219	The use of the frequency band 148-149.9 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. The mobile-satellite service shall not constrain the development and use of the fixed, mobile and space operation services in the frequency band 148-149.9 MHz. The use of the frequency band 148-149.9 MHz by non-geostationary-satellite systems in the space operation service identified as short-duration mission is not subject to No. 9.11A. (WRC-19)
5.220	The use of the bands 149.9-150.05 MHz and 399.9-400.05 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. (WRC-15)
5.221	Stations of the mobile-satellite service in the frequency band 148-149.9 MHz shall not cause harmful interference to, or claim protection from, stations of the fixed or mobile services operating in accordance with the Table of Frequency Allocations in the following countries: Albania, Algeria, Germany, Saudi Arabia, Australia, Austria, Bahrain, Bangladesh, Barbados, Belarus, Belgium, Benin, Bosnia and Herzegovina, Botswana, Brunei Darussalam, Bulgaria, Cameroon, China, Cyprus, Congo (Rep. of the), Korea (Rep. of), Côte d'Ivoire, Croatia, Cuba, Denmark, Djibouti, Egypt, the United Arab Emirates, Eritrea, Spain, Estonia, Eswatini, Ethiopia, the Russian Federation, Finland, France, Gabon, Georgia, Ghana, Greece, Guinea, Guinea Bissau, Hungary, India, Iran (Islamic Republic of), Ireland, Iceland, Israel, Italy, Jamaica, Japan, Jordan, Kazakhstan, Kenya, Kuwait, Lesotho, Latvia, Lebanon, Libya, Liechtenstein, Lithuania, Luxembourg, North Macedonia, Malaysia, Mali, Malta, Mauritania, Moldova, Mongolia, Montenegro, Mozambique, Namibia, Norway, New Zealand, Oman, Uganda, Uzbekistan, Pakistan, Panama, Papua New Guinea, Paraguay, the Netherlands, the Philippines, Poland, Portugal, Qatar, the Syrian Arab Republic, Türkiye, Kyrgyzstan, Dem. People's Rep. of Korea, Slovakia, Romania, the United Kingdom, Senegal, Serbia, Sierra Leone, Singapore, Slovenia, Somalia, Sudan, Sri Lanka, South Africa, Sweden, Switzerland, Tanzania, Chad, Togo, Tonga, Trinidad and Tobago, Tunisia, Ukraine, Viet Nam, Yemen, Zambia and Zimbabwe. (WRC-23)

Footnote Number	Footnote Content
5.225A	Additional allocation: in Algeria, Armenia, Azerbaijan, Belarus, China, France, Iran (Islamic Republic of), Kazakhstan, Uzbekistan, Kyrgyzstan, Tajikistan, Turkmenistan, Ukraine and Viet Nam, the frequency band 154-156 MHz is also allocated to the radiolocation service on a primary basis. The usage of the frequency band 154-156 MHz by the radiolocation service shall be limited to space-object detection systems operating from terrestrial locations. The operation of stations in the radiolocation service in the frequency band 154-156 MHz shall be subject to agreement obtained under No. 9.21. For the identification of potentially affected administrations in Region 1, the instantaneous field-strength value of 12 dB(μV/m) for 10% of the time produced at 10 m above ground level in the 25 kHz reference frequency band at the border of the territory of any other administration shall be used. For the identification of potentially affected administrations in Region 3, the interference-to-noise ratio (I/N) value of -6 dB (N = -161 dBW/4 kHz), or -10 dB for applications with greater protection requirements, such as public protection and disaster relief (PPDR (N = -161 dBW/4 kHz)), for 1% of the time produced at 60 m above ground level at the border of the territory of any other administration shall be used. In the frequency bands 156.7625-156.8375 MHz, 156.5125-156.5375 MHz, 161.9625-161.9875 MHz, 162.0125-162.0375 MHz, out-of-band e.i.r.p. of space surveillance radars shall not exceed -16 dBW. Frequency assignments to the radiolocation service under this allocation in Ukraine shall not be used without the agreement of Moldova. (WRC-12)
5.226	The frequency 156.8 MHz is the international distress, safety and calling frequency for the maritime mobile VHF radiotelephone service. The conditions for the use of this frequency and the band 156.7625-156.8375 MHz are contained in Article 31 and Appendix 18. The frequency 156.525 MHz is the international distress, safety and calling frequency for the maritime mobile VHF radiotelephone service using digital selective calling (DSC). The conditions for the use of this frequency and the band 156.4875-156.5625 MHz are contained in Articles 31 and 52, and in Appendix 18. In the bands 156-156.4875 MHz, 156.5625-156.7625 MHz, 156.8375-157.45 MHz, 160.6-160.975 MHz and 161.475-162.05 MHz, each administration shall give priority to the maritime mobile service on only such frequencies as are assigned to stations of the maritime mobile service by the administration (see Articles 31 and 52, and Appendix 18). Any use of frequencies in these bands by stations of other services to which they are allocated should be avoided in areas where such use might cause harmful interference to the maritime mobile VHF radiocommunication service. However, the frequencies 156.8 MHz and 156.525 MHz and the frequency bands in which priority is given to the maritime mobile service may be used for radiocommunications on inland waterways subject to agreement between interested and affected administrations and taking into account current frequency usage and existing agreements. (WRC-07)
5.227	Additional allocation: the bands 156.4875-156.5125 MHz and 156.5375-156.5625 MHz are also allocated to the fixed and land mobile services on a primary basis. The use of these bands by the fixed and land mobile services shall not cause harmful interference to nor claim protection from the maritime mobile VHF radiocommunication service. (WRC-07)
5.228	The use of the frequency bands 156.7625-156.7875 MHz and 156.8125-156.8375 MHz by the mobile-satellite service (Earth-to-space) is limited to the reception of automatic identification system (AIS) emissions of long-range AIS broadcast messages (Message 27, see the most recent version of Recommendation ITU R M.1371). With the exception of AIS emissions, emissions in these frequency bands by systems operating in the maritime mobile service for communications shall not exceed 1 W. (WRC-12)
5.228A	The frequency bands 161.9625-161.9875 MHz and 162.0125-162.0375 MHz may be used by aircraft stations for the purpose of search and rescue operations and other safety-related communications. (WRC-12)
5.228AA	The use of the frequency bands 161.9375-161.9625 MHz and 161.9875-162.0125 MHz by the maritime mobile-satellite (Earth-to-space) service is limited to the systems which operate in accordance with Appendix 18. (WRC-15)
5.228AB	The use of the frequency bands 157.1875-157.3375 MHz and 161.7875-161.9375 MHz by the maritime mobile-satellite service (Earth-to-space) is limited to non-GSO satellite systems operating in accordance with Appendix 18. (WRC-19)
5.228AC	The use of the frequency bands 157.1875-157.3375 MHz and 161.7875-161.9375 MHz by the maritime mobile-satellite service (space-to-Earth) is limited to non-GSO satellite systems operating in accordance with Appendix 18. Such use is subject to agreement obtained under No. 9.21 with respect to the terrestrial services in Azerbaijan, Belarus, China, Korea (Rep. of), Cuba, the Russian Federation, the Syrian Arab Republic, the Dem. People's Rep. of Korea, South Africa and Viet Nam. (WRC-19)
5.228B	The use of the frequency bands 161.9625-161.9875 MHz and 162.0125-162.0375 MHz by the fixed and land mobile services shall not cause harmful interference to, or claim protection from, the maritime mobile service. (WRC-12)
5.228F	The use of the frequency bands 161.9625-161.9875 MHz and 162.0125-162.0375 MHz by the mobile-satellite service (Earth-to-space) is limited to the reception of automatic identification system emissions from stations operating in the maritime mobile service. (WRC-12)
5.235	Additional allocation: in Germany, Austria, Belgium, Denmark, Spain, Finland, France, Israel, Italy, Liechtenstein, Malta, Monaco, Norway, the Netherlands, the United Kingdom, Sweden and Switzerland, the band 174 - 223 MHz is also allocated to the land mobile service on a primary basis. However, the stations of the land mobile service shall not cause harmful interference to, or claim protection from, broadcasting stations, existing or planned, in countries other than those listed in this footnote.

Footnote Number	Footnote Content
5.243	Additional allocation: in Somalia, the band 216-225 MHz is also allocated to the aeronautical radionavigation service on a primary basis, subject to not causing harmful interference to existing or planned broadcasting services in other countries.
5.246	Alternative allocation: in Spain, France, Israel and Monaco, the band 223-230 MHz is allocated to the broadcasting and land mobile services on a primary basis (see No. 5.33) on the basis that, in the preparation of frequency plans, the broadcasting service shall have prior choice of frequencies; and allocated to the fixed and mobile, except land mobile, services on a secondary basis. However, the stations of the land mobile service shall not cause harmful interference to, or claim protection from, existing or planned broadcasting stations in Morocco and Algeria.
5.247	Additional allocation: in Saudi Arabia, Bahrain, the United Arab Emirates, Jordan, Oman, Qatar and Syrian Arab Republic, the band 223-235 MHz is also allocated to the aeronautical radionavigation service on a primary basis.
5.251	Additional allocation: in Nigeria, the band 230-235 MHz is also allocated to the aeronautical radionavigation service on a primary basis, subject to agreement obtained under No. 9.21.
5.252	Alternative allocation: in Botswana, Eswatini, Lesotho, Malawi, Mozambique, Namibia, South Africa, Zambia and Zimbabwe, the frequency bands 230-238 MHz and 246-254 MHz are allocated to the broadcasting service on a primary basis, subject to agreement obtained under No. 9.21. (WRC-19)
5.254	The bands 235-322 MHz and 335.4-399.9 MHz may be used by the mobile-satellite service, subject to agreement obtained under No. 9.21, on condition that stations in this service do not cause harmful interference to those of other services operating or planned to be operated in accordance with the Table of Frequency Allocations except for the additional allocation made in footnote No. 5.256A. (WRC-03)
5.255	The bands 312-315 MHz (Earth-to-space) and 387-390 MHz (space-to-Earth) in the mobile-satellite service may also be used by non-geostationary-satellite systems. Such use is subject to coordination under No. 9.11A.
5.256	The frequency 243 MHz is the frequency in this band for use by survival craft stations and equipment used for survival purposes. (WRC-07)
5.256A	Additional allocation: in China, the Russian Federation and Kazakhstan, the frequency band 258-261 MHz is also allocated to the space research service (Earth-to-space) and space operation service (Earth-to-space) on a primary basis. Stations in the space research service (Earth-to-space) and space operation service (Earth-to-space) shall not cause harmful interference to, or claim protection from, or constrain the use and development of the mobile service systems and mobile-satellite service systems operating in the frequency band. Stations in space research service (Earth-to-space) and space operation service (Earth-to-space) shall not constrain the future development of fixed service systems of other countries. (WRC-15)
5.257	The band 267-272 MHz may be used by administrations for space telemetry in their countries on a primary basis, subject to agreement obtained under No. 9.21.
5.258	The use of the band 328.6-335.4 MHz by the aeronautical radionavigation service is limited to Instrument Landing Systems (glide path).
5.259	Additional allocation: in Egypt and the Syrian Arab Republic, the band 328.6-335.4 MHz is also allocated to the mobile service on a secondary basis, subject to agreement obtained under No. 9.21. In order to ensure that harmful interference is not caused to stations of the aeronautical radionavigation service, stations of the mobile service shall not be introduced in the band until it is no longer required for the aeronautical radionavigation service by any administration which may be identified in the application of the procedure invoked under No. 9.21. (WRC-12)
5.260A	In the frequency band 399.9-400.05 MHz, the maximum e.i.r.p. of any emission of earth stations in the mobile-satellite service shall not exceed 5 dBW in any 4 kHz band and the maximum e.i.r.p. of each earth station in the mobile-satellite service shall not exceed 5 dBW in the whole 399.9-400.05 MHz frequency band. Until 22 November 2022, this limit shall not apply to satellite systems for which complete notification information has been received by the Radiocommunication Bureau by 22 November 2019 and that have been brought into use by that date. After 22 November 2022, these limits shall apply to all systems within the mobile-satellite service operating in this frequency band. In the frequency band 399.99-400.02 MHz, the e.i.r.p. limits as specified above shall apply after 22 November 2022 to all systems within the mobile-satellite service. Administrations are requested that their mobile-satellite service satellite links in the 399.99-400.02 MHz frequency band comply with the e.i.r.p. limits as specified above, after 22 November 2019. (WRC-19)
5.260B	In the frequency band 400.02-400.05 MHz, the provisions of No. 5.260A are not applicable for telecommand uplinks within the mobile-satellite service. (WRC-19)
5.262	Additional allocation: in Saudi Arabia, Armenia, Azerbaijan, Bahrain, Belarus, Botswana, Colombia, Cuba, Egypt, the United Arab Emirates, Ecuador, the Russian Federation, Georgia, Hungary, Iran (Islamic Republic of), Iraq, Israel, Jordan, Kazakhstan, Kuwait, Liberia, Malaysia, Moldova, Oman, Uzbekistan, Pakistan, the Philippines, Qatar, the Syrian Arab Republic, Kyrgyzstan, Singapore, Somalia, Tajikistan, Chad, Turkmenistan and Ukraine, the band 400.05-401 MHz is also allocated to the fixed and mobile services on a primary basis. (WRC-12)
5.263	The band 400.15-401 MHz is also allocated to the space research service in the space-to-space direction for communications with manned space vehicles. In this application, the space research service will not be regarded as a safety service.

Footnote Number	Footnote Content
5.264	The use of the band 400.15-401 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. The power flux-density limit indicated in Annex 1 of Appendix 5 shall apply until such time as a competent world radiocommunication conference revises it.
5.264A	In the frequency band 401-403 MHz, the maximum e.i.r.p. of any emission of each earth station in the meteorological-satellite service and the Earth exploration-satellite service shall not exceed 22 dBW in any 4 kHz band for geostationary systems and non-geostationary systems with an orbit of apogee equal or greater than 35 786 km. The maximum e.i.r.p. of any emission of each earth station in the meteorological satellite service and the Earth exploration-satellite service shall not exceed 7 dBW in any 4 kHz band for non-geostationary systems with an orbit of apogee lower than 35 786 km. The maximum e.i.r.p. of each earth station in the meteorological-satellite service and the Earth exploration-satellite service shall not exceed 22 dBW for geostationary systems and nongeostationary systems with an orbit of apogee equal or greater than 35 786 km in the whole 401-403 MHz frequency band. The maximum e.i.r.p. of each earth station in the meteorological-satellite service and the Earth exploration-satellite service shall not exceed 7 dBW for non-geostationary systems with an orbit of apogee lower than 35 786 km in the whole 401-403 MHz frequency band. Until 22 November 2029, these limits shall not apply to satellite systems for which complete notification information has been received by the Radiocommunication Bureau by 22 November 2019 and that have been brought into use by that date. After 22 November 2029, these limits shall apply to all systems within the meteorological-satellite service and the Earth exploration-satellite service operating in this frequency band. (WRC-19)
5.264B	Non-geostationary-satellite systems in the meteorological-satellite service and the Earth exploration-satellite service for which complete notification information has been received by the Radiocommunication Bureau no later than 28 April 2007 are exempt from provisions of No. 5.264A and may continue to operate in the frequency band 401.898-402.522 MHz on a primary basis without exceeding a maximum e.i.r.p. level of 12 dBW. (WRC-23)
5.265	In the frequency band 403-410 MHz, Resolution 205 (Rev.WRC-19) applies. (WRC-19)
5.266	The use of the band 406-406.1 MHz by the mobile-satellite service is limited to low power satellite emergency position-indicating radiobeacons (see also Article 31). (WRC-07)
5.267	Any emission capable of causing harmful interference to the authorised uses of the band 406-406.1 MHz is prohibited.
5.268	Use of the frequency band 410-420 MHz by the space research service is limited to space-to-space communications links with an orbiting, manned space vehicle. The power flux-density at the surface of the Earth produced by emissions from transmitting stations of the space research service (space-to-space) in the frequency band 410-420 MHz shall not exceed $-153 \text{ dB(W/m}^2\text{)}$ for $0^\circ \leq \delta \leq 5^\circ$, $-153 + 0.077 (\delta - 5) \text{ dB(W/m}^2\text{)}$ for $5^\circ \leq \delta \leq 70^\circ$ and $-148 \text{ dB(W/m}^2\text{)}$ for $70^\circ \leq \delta \leq 90^\circ$, where δ is the angle of arrival of the radio-frequency wave and the reference bandwidth is 4 kHz. In this frequency band, stations of the space research service (space-to-space) service shall not claim protection from, nor constrain the use and development of, stations of the fixed and mobile services. No. 4.10 does not apply. (WRC-15)
5.269	Different category of service: in Australia, Brazil, the United States, India, Japan and the United Kingdom, the allocation of the frequency bands 420-430 MHz and 440-450 MHz to the radiolocation service is on a primary basis (see No. 5.33). (WRC-23)
5.270	Additional allocation: in Australia, the United States, Jamaica and the Philippines, the bands 420-430 MHz and 440-450 MHz are also allocated to the amateur service on a secondary basis.
5.271	Additional allocation: in Belarus, China, India, Kyrgyzstan and Turkmenistan, the band 420-460 MHz is also allocated to the aeronautical radionavigation service (radio altimeters) on a secondary basis. (WRC-07)
5.274	Alternative allocation: in Denmark, Norway, Sweden and Chad, the bands 430-432 MHz and 438-440 MHz are allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-12)
5.275	Additional allocation: in Croatia, Estonia, Finland, Libya, North Macedonia, Montenegro and Serbia, the frequency bands 430-432 MHz and 438-440 MHz are also allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-19)
5.276	Additional allocation: in Afghanistan, Algeria, Saudi Arabia, Bahrain, Bangladesh, Brunei Darussalam, Burkina Faso, Djibouti, Egypt, the United Arab Emirates, Ecuador, Eritrea, Ethiopia, Greece, Guinea, India, Indonesia, Iran (Islamic Republic of), Iraq, Israel, Italy, Jordan, Kenya, Kuwait, Libya, Malaysia, Niger, Nigeria, Oman, Pakistan, the Philippines, Qatar, the Syrian Arab Republic, the Dem. People's Rep. of Korea, Singapore, Somalia, Sudan, Switzerland, Thailand, Togo, Turkey and Yemen, the frequency band 430-440 MHz is also allocated to the fixed service on a primary basis and the frequency bands 430-435 MHz and 438-440 MHz are also allocated, except in Equador, to the mobile, except aeronautical mobile, service on a primary basis. (WRC-15)
5.277	Additional allocation: in Angola, Armenia, Azerbaijan, Belarus, Cameroon, Congo (Rep. of the), Djibouti, the Russian Federation, Georgia, Hungary, Israel, Kazakhstan, Mali, Uzbekistan, Poland, the Dem. Rep. of the Congo, Kyrgyzstan, Slovakia, Romania, Rwanda, Tajikistan, Chad, Turkmenistan and Ukraine, the frequency band 430-440 MHz is also allocated to the fixed service on a primary basis. (WRC-19)

Footnote Number	Footnote Content
5.279A	The use of the frequency band 432-438 MHz by sensors in the Earth exploration-satellite service (active) shall be in accordance with Recommendation ITU-R RS.1260-2. Additionally, the Earth exploration-satellite service (active) in the frequency band 432-438 MHz shall not cause harmful interference to the aeronautical radionavigation service in China. The provisions of this footnote in no way diminish the obligation of the Earth exploration-satellite service (active) to operate as a secondary service in accordance with Nos. 5.29 and 5.30. (WRC-19)
5.280	In Germany, Austria, Bosnia and Herzegovina, Croatia, Liechtenstein, North Macedonia, Montenegro, Portugal, Serbia, Slovenia and Switzerland, the frequency band 433.05-434.79 MHz (centre frequency 433.92 MHz) is designated for industrial, scientific and medical (ISM) applications. Radiocommunication services of these countries operating within this frequency band must accept harmful interference which may be caused by these applications. ISM equipment operating in this frequency band is subject to the provisions of No. 15.13. (WRC-19)
5.281	Additional allocation: in the French Overseas Departments in Region 2 and India, the band 433.75-434.25 MHz is also allocated to the space operation service (Earth-to-space) on a primary basis. In France and in Brazil, the band is allocated to the same service on a secondary basis.
5.282	In the bands 435-438 MHz, 1260-1270 MHz, 2400-2450 MHz, 3400-3410 MHz (in Regions 2 and 3 only) and 5650-5670 MHz, the amateur-satellite service may operate subject to not causing harmful interference to other services operating in accordance with the Table (see No. 5.43). Administrations authorising such use shall ensure that any harmful interference caused by emissions from a station in the amateur-satellite service is immediately eliminated in accordance with the provisions of No. 25.11. The use of the bands 1260-1270 MHz and 5650-5670 MHz by the amateur-satellite service is limited to the Earth-to-space direction.
5.283	Additional allocation: in Austria, the band 438-440 MHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis.
5.284	Additional allocation: in Canada, the band 440-450 MHz is also allocated to the amateur service on a secondary basis.
5.285	Different category of service: in Canada, the allocation of the band 440-450 MHz to the radiolocation service is on a primary basis (see No. 5.33).
5.286	The band 449.75-450.25 MHz may be used for the space operation service (Earth-to-space) and the space research service (Earth-to-space), subject to agreement obtained under No. 9.21.
5.286A	The use of the bands 454-456 MHz and 459-460 MHz by the mobile-satellite service is subject to coordination under 9.11A. (WRC-97)
5.286AA	The frequency band 450-470 MHz is identified for use by administrations wishing to implement International Mobile Telecommunications (IMT) - see Resolution 224 (Rev.WRC-19). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. (WRC-19)
5.286B	The use of the band 454-455 MHz in the countries listed in No. 5.286D, 455-456 MHz and 459-460 MHz in Region 2, and 454-456 MHz and 459-460 MHz in the countries listed in No. 5.286E, by stations in the mobile-satellite service, shall not cause harmful interference to, or claim protection from, stations of the fixed or mobile services operating in accordance with the Table of Frequency Allocations. (WRC-97)
5.286C	The use of the band 454-455 MHz in the countries listed in No. 5.286D, 455-456 MHz and 459-460 MHz in Region 2, and 454-456 MHz and 459-460 MHz in the countries listed in No. 5.286E, by stations in the mobile-satellite service, shall not constrain the development and use of the fixed and mobile services operating in accordance with the Table of Frequency Allocations. (WRC-97)
5.286D	Additional allocation: in Canada, the United States and Panama, the band 454-455 MHz is also allocated to the mobile-satellite service (Earth-to-space) on a primary basis. (WRC-07)
5.286E	Additional allocation: in Cabo Verde, Nepal and Nigeria, the bands 454-456 MHz and 459-460 MHz are also allocated to the mobile-satellite (Earth-to-space) service on a primary basis. (WRC-23)
5.287	Use of the frequency bands 457.5125-457.5875 MHz and 467.5125-467.5875 MHz by the maritime mobile service is limited to on-board communication stations. The characteristics of the equipment and the channelling arrangement shall be in accordance with Recommendation ITU-R M.1174-4. The use of these frequency bands in territorial waters is subject to the national regulations of the administration concerned. (WRC-19)
5.288	In the territorial waters of the United States and the Philippines, the preferred frequencies for use by on-board communication stations shall be 457.525 MHz, 457.550 MHz, 457.575 MHz and 457.600 MHz paired, respectively, with 467.750 MHz, 467.775 MHz, 467.800 MHz and 467.825 MHz. The characteristics of the equipment used shall conform to those specified in Recommendation ITU-R M.1174-4. (WRC-19)
5.289	Earth exploration-satellite service applications, other than the meteorological-satellite service, may also be used in the bands 460-470 MHz and 1690-1710 MHz for space-to-Earth transmissions subject to not causing harmful interference to stations operating in accordance with the Table.
5.290	Different category of service: in Afghanistan, Azerbaijan, Belarus, China, the Russian Federation, Kyrgyzstan, Tajikistan, and Turkmenistan, the allocation of the band 460-470 MHz to the meteorological-satellite service (space-to Earth) is on a primary basis (see No. 5.33), subject to agreement obtained under No. 9.21. (WRC-12)

Footnote Number	Footnote Content
5.291A	Additional allocation: in Germany, Austria, Denmark, Estonia, Liechtenstein, Serbia and Switzerland, the frequency band 470-494 MHz is also allocated to the radiolocation service on a secondary basis. This use is limited to the operation of wind profiler radars in accordance with Resolution 217 (Rev.WRC-23). (WRC-23)
5.294	Additional allocation: in Saudi Arabia, Cameroon, Ivory Coast, Egypt, Ethiopia, Israel, Libya, Palestine*, the Syrian Arab Republic, Chad and Yemen, the frequency band 470-582 MHz is also allocated to the fixed service on a secondary basis. (WRC-23) *Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995.
5.295A	Additional allocation: in Albania, Germany, Andorra, Austria, Belgium, Bosnia and Herzegovina, Bulgaria, Cyprus, Vatican, Croatia, Denmark, Estonia, Finland, France, Georgia, Greece, Hungary, Ireland, Iceland, Latvia, Liechtenstein, Lithuania, Luxembourg, North Macedonia, Malta, Moldova, Monaco, Montenegro, Norway, Uzbekistan, Kingdom of the Netherlands, Poland, Portugal, Türkiye, Slovakia, the Czech Republic, Romania, the United Kingdom, San Marino, Serbia, Slovenia, Sweden, Switzerland and Ukraine, the frequency band 470-694 MHz is allocated to the mobile, except aeronautical mobile, service on a secondary basis, subject to agreement obtained under No. 9.21. For the protection of the broadcasting service, stations in the mobile service shall not create a field strength for more than 1% of the time at the highest of the clutter height or 10 m above ground level at the border of the territory of any other administration that exceeds the field strength value as calculated using § 4.1.3.2 of Annex 2 to the GE06 Agreement with regard to allowance for multiple interference, Table A.1.10 and the methodology given in the GE06 Agreement. These limits may be exceeded on the territory of any country whose administration has so agreed. This allocation shall in no way adversely affect the broadcast development or undermine new entries of the broadcasting service to the GE06 Plan. (WRC-23)
5.296	Additional allocation: in Albania, Algeria, Germany, Angola, Saudi Arabia, Austria, Bahrain, Belgium, Benin, Bosnia and Herzegovina, Botswana, Bulgaria, Burkina Faso, Burundi, Cameroon, Vatican, Congo (Rep. of the), Ivory Coast, Croatia, Denmark, Djibouti, Egypt, United Arab Emirates, Spain, Estonia, Eswatini, Finland, France, Gabon, Gambia, Georgia, Ghana, Hungary, Iraq, Ireland, Iceland, Israel, Italy, Jordan, Kenya, Kuwait, Lesotho, Latvia, Lebanon, Libya, Liechtenstein, Lithuania, Luxembourg, North Macedonia, Malawi, Mali, Malta, Morocco, Mauritius, Mauritania, Moldova, Monaco, Mozambique, Namibia, Niger, Nigeria, Norway, Oman, Uganda, Palestine*, the Netherlands, Poland, Portugal, Qatar, the Syrian Arab Republic, Türkiye, Slovakia, the Czech Republic, Romania, the United Kingdom, Rwanda, San Marino, Senegal, Serbia, Sudan, South Africa, Sweden, Switzerland, Tanzania, Chad, Togo, Tunisia, Ukraine, Zambia and Zimbabwe, the frequency band 470-694 MHz is also allocated on a secondary basis to the land mobile service, intended for applications ancillary to broadcasting and programme-making. Stations of the land mobile service in the countries listed in this footnote shall not cause harmful interference to existing or planned stations operating in accordance with the Table in countries other than those listed in this footnote. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference and taking into account the Israeli Palestinian Interim Agreement of 28 September 1995
5.300	Additional allocation: in Saudi Arabia, Cameroon, Egypt, the United Arab Emirates, Iraq, Israel, Jordan, Libya, Oman, Palestine*, Qatar, the Syrian Arab Republic and Sudan, the frequency band 582-790 MHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a secondary basis. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference, and taking into account the Israeli Palestinian Interim Agreement of 28 September 1995.
5.304	Additional allocation: in the African Broadcasting Area (see Nos. 5.10 to 5.13), the band 606-614 MHz is also allocated to the radio astronomy service on a primary basis.
5.306	Additional allocation: in Region 1, except in the African Broadcasting Area (see Nos. 5.10 to 5.13), and in Region 3, the band 608-614 MHz is also allocated to the radio astronomy service on a secondary basis.
5.307A	Additional allocation: in Saudi Arabia, Bahrain, Egypt, the United Arab Emirates, Iraq, Jordan, Kuwait, Oman, Palestine*, Qatar and the Syrian Arab Republic, the frequency band 614-694 MHz is allocated to the mobile, except aeronautical mobile, service on a primary basis and identified for International Mobile Telecommunications (IMT) – see Resolution 224 (Rev.WRC-23) subject to the agreement obtained under No. 9.21. Stations in the mobile service shall not create a field strength for more than 1% of the time at the highest of the clutter height or 10 m above ground level at the border of the territory of any other administration that exceeds the field strength value as calculated using paragraph 4.1.3.2 of Annex 2 to the GE06 Agreement with regard to allowance for multiple interference, Table A.1.10 and the methodology given in the GE06 Agreement. Stations in the mobile service of the countries listed in this footnote shall not cause harmful interference to, or claim protection from the existing and future broadcasting stations of the neighbouring countries operating in accordance with the GE06 Plan. This identification does not preclude the use of these frequency bands by any application of the services to which they are allocated and does not establish priority in the Radio Regulations and shall in no way adversely affect the development of the existing and future broadcasting service in accordance with the GE06 Agreement. For countries party to the GE06 Agreement, the use of stations in the mobile service is also subject to the successful application of the procedures of that Agreement. This allocation does not establish priority in the Radio Regulations and shall allow the implementation and development of the broadcasting service in accordance with the GE06 Agreement. The countries listed in this footnote and located in the African Broadcasting Area should ensure protection of the radio astronomy service within the frequency band 606-614 MHz, as allocated in No. 5.304, consistent with the most recent version of Recommendation ITU-R RA.769. The countries listed in this footnote, which are neighbouring to the countries listed in No. 5.312, should ensure the protection of the aeronautical radionavigation service in the frequency band 645-862 MHz. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference, and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995.

Footnote Number	Footnote Content
5.307B	Additional allocation: in Gambia, Mauritania, Namibia, Nigeria, Senegal, Somalia, Tanzania and Chad, the frequency band 614-694 MHz is allocated to the mobile service on a secondary basis. For the protection of the broadcasting service, stations in the mobile service shall not create a field strength for more than 1% of the time at the highest of the clutter height or 10 m above ground level at the border of the territory of any other administration that exceeds the field strength value as calculated using paragraph 4.1.3.2 of Annex 2 to the GE06 Agreement with regard to allowance for multiple interference, Table A.1.10 and the methodology given in the GE06 Agreement. This allocation shall in no way adversely affect the broadcast development or undermine new entries of the broadcasting service to the GE06 Plan. Additional measures shall be used by administrations implementing stations in the mobile services to protect stations in the broadcasting service of neighbouring administrations such as a distance limitation from the border of a neighbouring country. (WRC-23)
5.312	Additional allocation: in Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Kazakhstan, Uzbekistan, Kyrgyzstan, Tajikistan, Turkmenistan and Ukraine, the frequency band 645-862 MHz, and in Bulgaria the frequency bands 726-753 MHz, 778-811 MHz and 822-852 MHz, are also allocated to the aeronautical radionavigation service on a primary basis. (WRC-23)
5.312A	In Region 1, the use of the frequency band 694-790 MHz by the mobile, except aeronautical mobile, service is subject to the provisions of Resolution 760 (Rev.WRC-23). See also Resolution 224 (Rev.WRC-23). (WRC-23)
5.312B	The frequency band 698-960 MHz, or portions thereof, in Region 2, and the frequency band 694-960 MHz, or portions thereof, in Region 1, are identified for use by high-altitude platform stations as International Mobile Telecommunications (IMT) base stations (HIBS). This identification does not preclude the use of these frequency bands by any application of the services to which they are allocated and does not establish priority in the Radio Regulations. Resolution 213 (WRC-23) shall apply. HIBS shall not claim protection from existing primary services. No. 5.43A does not apply, see resolves 2 of Resolution 213 (WRC-23). Such use of HIBS in the frequency bands 694-728 MHz, 830-835 MHz and 805.3-806.9 MHz is limited to reception by HIBS. (WRC-23)
5.316B	In Region 1, the allocation to the mobile, except aeronautical mobile, service in the frequency band 790-862 MHz is subject to agreement obtained under No. 9.21 with respect to the aeronautical radionavigation service in countries mentioned in No. 5.312. For countries party to the GE06 Agreement, the use of stations of the mobile service is also subject to the successful application of the procedures of that Agreement. Resolutions 224 (Rev.WRC-23) and 749 (Rev.WRC-23) shall apply, as appropriate. (WRC-23)
5.317A	The parts of the frequency band 698-960 MHz in Region 2 and the frequency bands 694-790 MHz in Region 1 and 790-960 MHz in Regions 1 and 3 which are allocated to the mobile service on a primary basis are identified for use by administrations wishing to implement International Mobile Telecommunications (IMT) – see Resolutions 224 (Rev.WRC-23), 760 (Rev.WRC-23) and 749 (Rev.WRC-23), where applicable. This identification does not preclude the use of these frequency bands by any application of the services to which they are allocated and does not establish priority in the Radio Regulations. (WRC-23)
5.319	Additional Allocation: In Belarus, the Russian Federation and Ukraine, the bands 806-840 MHz (Earth-to-space) and 856-890 MHz (space-to-Earth) are also allocated to the mobile-satellite, except aeronautical mobile satellite (R), service. The use of these bands by this service shall not cause harmful interference to, or claim protection from, services in other countries operating in accordance with the Table of Frequency Allocations and is subject to special agreements between the administrations concerned.
5.322	In Region 1, in the frequency band 862-960 MHz, stations of the broadcasting service shall be operated only in the African Broadcasting Area (see Nos. 5.10 to 5.13) excluding Algeria, Burundi, Djibouti, Egypt, Spain, Lesotho, Libya, Morocco, Malawi, Namibia, Nigeria, South Africa, Tanzania, Zimbabwe and Zambia, subject to agreement obtained under No. 9.21. (WRC-23)
5.323	Additional allocation: in Armenia, Azerbaijan, Belarus, the Russian Federation, Kazakhstan, Uzbekistan, Kyrgyzstan, Tajikistan, Turkmenistan and Ukraine, the frequency band 862-960 MHz, in Bulgaria the frequency bands 862-880 MHz and 915-925 MHz, and in Romania the frequency bands 862-880 MHz and 915-925 MHz, are also allocated to the aeronautical radionavigation service on a primary basis. Such use is subject to agreement obtained under No. 9.21 with administrations concerned and limited to ground-based radiobeacons in operation on 27 October 1997 until the end of their lifetime. (WRC-19)
5.327A	The use of the frequency band 960-1164 MHz by the aeronautical mobile (R) service is limited to systems that operate in accordance with recognized international aeronautical standards. Such use shall be in accordance with Resolution 417 (Rev. WRC-15). (WRC-15)
5.328	The use of the band 960-1215 MHz by the aeronautical radionavigation service is reserved on a worldwide basis for the operation and development of airborne electronic aids to air navigation and any directly associated ground-based facilities. (WRC-2000)
5.328A	Stations in the radionavigation-satellite service in the band 1164-1215 MHz shall operate in accordance with the provisions of Resolution 609 (Rev.WRC-07) and shall not claim protection from stations in the aeronautical radionavigation service in the band 960-1215 MHz. No. 5.43A does not apply. The provisions of No. 21.18 shall apply. (WRC-07)
5.328AA	The frequency band 1 087.7-1 092.3 MHz is also allocated to the aeronautical mobile-satellite (R) service (Earth-to-space) on a primary basis, limited to the space station reception of Automatic Dependent Surveillance-Broadcast (ADS-B) emissions from aircraft transmitters that operate in accordance with recognized international aeronautical standards. Stations operating in the aeronautical mobile-satellite (R) service shall not claim protection from stations operating in the aeronautical radionavigation service. Resolution 425 (Rev.WRC-19) shall apply. (WRC-19)

Footnote Number	Footnote Content
5.328B	The use of the bands 1164-1300 MHz, 1559-1610 MHz and 5010-5030 MHz by systems and networks in the radionavigation-satellite service for which complete coordination or notification information, as appropriate, is received by the Radiocommunication Bureau after 1 January 2005 is subject to the application of the provisions of Nos. 9.12, 9.12A and 9.13. Resolution 610 (WRC-03) shall also apply; however, in the case of radionavigation-satellite service (space-to-space) networks and systems, Resolution 610 (WRC-03) shall only apply to transmitting space stations. In accordance with No. 5.329A, for systems and networks in the radionavigation-satellite service (space-to-space) in the bands 1215-1300 MHz and 1559-1610 MHz, the provisions of Nos. 9.7, 9.12, 9.12A and 9.13 shall only apply with respect to other systems and networks in the radionavigation-satellite service (space-to-space). (WRC-07)
5.329	Use of the radionavigation-satellite service in the frequency band 1 215-1 300 MHz shall be subject to the condition that no harmful interference is caused to, and no protection is claimed from, the radionavigation service authorized under No. 5.331. Furthermore, the use of the radionavigation-satellite service in the frequency band 1 215- 1 300 MHz shall be subject to the condition that no harmful interference is caused to the radiolocation service. No. 5.43 shall not apply in respect of the radiolocation service. Resolution 608 (Rev.WRC-19) shall apply. (WRC-19)
5.329A	Use of systems in the radionavigation-satellite service (space-to-space) operating in the bands 1215-1300 MHz and 1559-1610 MHz is not intended to provide safety service applications, and shall not impose any additional constraints on radionavigation-satellite service (space-to-Earth) systems or on other services operating in accordance with the Table of Frequency Allocations. (WRC-07)
5.330	Additional allocation: in Angola, Saudi Arabia, Bahrain, Bangladesh, Cameroon, China, Djibouti, Egypt, the United Arab Emirates, Eritrea, Ethiopia, Guyana, India, Indonesia, Iran (Islamic Republic of), Iraq, Israel, Japan, Jordan, Kuwait, Nepal, Oman, Pakistan, Palestine*, the Philippines, Qatar, the Syrian Arab Republic, Somalia, Sudan, South Sudan, Chad, Togo and Yemen, the frequency band 1 215-1 300 MHz is also allocated to the fixed and mobile services on a primary basis. (WRC-23)
5.331	Additional allocation: in Algeria, Germany, Saudi Arabia, Australia, Austria, Bahrain, Belarus, Belgium, Benin, Bosnia and Herzegovina, Brazil, Burkina Faso, Burundi, Cameroon, China, Korea (Rep. of), Croatia, Denmark, Djibouti, Egypt, the United Arab Emirates, Estonia, the Russian Federation, Finland, France, Ghana, Greece, Guinea, Equatorial Guinea, Hungary, India, Indonesia, Iran (Islamic Republic of), Iraq, Ireland, Israel, Jordan, Kenya, Kuwait, Lesotho, Latvia, Lebanon, Liechtenstein, Lithuania, Luxembourg, North Macedonia, Madagascar, Mali, Mauritania, Montenegro, Nigeria, Norway, Oman, Pakistan, Palestine*, the Kingdom of the Netherlands, Poland, Portugal, Qatar, the Syrian Arab Republic, Turkiye, Dem. People's Rep. of Korea, Slovakia, the United Kingdom, Serbia, Slovenia, Somalia, Sudan, South Sudan, Sri Lanka, South Africa, Sweden, Switzerland, Thailand, Togo, Venezuela and Viet Nam, the frequency band 1 215-1 300 MHz is also allocated to the radionavigation service on a primary basis. In Canada and the United States, the frequency band 1 240-1 300 MHz is also allocated to the radionavigation service, and use of the radionavigation service shall be limited to the aeronautical radionavigation service. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995.
5.332	In the band 1215-1260 MHz, active spaceborne sensors in the earth exploration-satellite and space research services shall not cause harmful interference to, claim protection from, or otherwise impose constraints on operation or development of the radiolocation service, the radionavigation-satellite service and other services allocated on a primary basis. (WRC-2000)
5.332A	Administrations authorizing operation of the amateur and amateur-satellite services in the frequency band 1 240-1 300 MHz, or portions thereof, shall ensure that the amateur and amateur-satellite services do not cause harmful interference to radionavigation-satellite service (space-to-Earth) receivers in accordance with No. 5.29 (see the most recent version of Recommendation ITU-R M.2164). The authorizing administration, upon receipt of a report of harmful interference caused by a station of the amateur or amateur-satellite services, shall take all necessary steps to rapidly eliminate such interference. (WRC-23)
5.335	In Canada and the United States in the band 1 240-1 300 MHz, active spaceborne sensors in the Earth exploration-satellite and space research services shall not cause interference to, claim protection from, or otherwise impose constraints on operation or development of the aeronautical radionavigation service. (WRC-97)
5.335A	In the band 1260-1300 MHz, active spaceborne sensors in the Earth exploration-satellite and space research services shall not cause harmful interference to, claim protection from, or otherwise impose constraints on operation or development of the radiolocation service and other services allocated by footnotes on a primary basis. (WRC-2000)
5.337	The use of the bands 1300-1350 MHz, 2700-2900 MHz and 9000-9200 MHz by the aeronautical radionavigation service is restricted to ground-based radars and to associated airborne transponders which transmit only on frequencies in these bands and only when actuated by radars operating in the same band.
5.337A	The use of the band 1300-1350 MHz by earth stations in the radionavigation-satellite service and by stations in the radiolocation service shall not cause harmful interference to, nor constrain the operation and development of, the aeronautical-radionavigation service. (WRC-2000)
5.338	In Kyrgyzstan, Slovakia and Turkmenistan, existing installations of the radionavigation service may continue to operate in the band 1350-1400 MHz. (WRC-12)

Footnote Number	Footnote Content
5.338A	In the frequency bands 1 350-1 400 MHz, 1 427-1 452 MHz, 22.55-23.55 GHz, 24.25-27.5 GHz, 30-31.3 GHz, 49.7-50.2 GHz, 50.4-50.9 GHz, 51.4-52.6 GHz, 81-86 GHz and 92-94 GHz, Resolution 750 (Rev.WRC-19) applies. (WRC-19)
5.339	The bands 1370-1400 MHz, 2640-2655 MHz, 4950-4990 MHz and 15.20-15.35 GHz are also allocated to the space research (passive) and earth exploration-satellite (passive) services on a secondary basis.
5.340	All emissions are prohibited in the following bands: 1400-1427 MHz 2690-2700 MHz, except those provided for by No. 5.422 10.68-10.7 GHz, except those provided for by No. 5.483 15.35-15.4 GHz, except those provided for by No. 5.511 23.6-24 GHz 31.3-31.5 GHz 31.5-31.8 GHz, in Region 2 48.94-49.04 GHz, from airborne stations 50.2-50.4 GHz (1) 52.6-54.25 GHz 86-92 GHz 100-102 GHz 109.5-111.8 GHz 114.25-116 GHz 148.5-151.5 GHz 164-167 GHz 182-185 GHz 190-191.8 GHz 200-209 GHz, 226-231.5 GHz 250-252 GHz. (WRC-03) / (1) 5.340 The allocation to the Earth exploration-satellite service (passive) and the space research service (passive) in the band 50.2-50.4 GHz should not impose undue constraints on the use of the adjacent bands by the primary allocated services in those bands. (WRC-97)
5.341	In the bands 1400-1727 MHz, 101-120 GHz and 197-220 GHz, passive research is being conducted by some countries in a programme for the search for intentional emissions of extraterrestrial origin.
5.341A	In Region 1, the frequency bands 1427-1452 MHz and 1492-1518 MHz are identified for use by administrations wishing to implement International Mobile Telecommunications (IMT) in accordance with Resolution 223 (Rev.WRC-15). This identification does not preclude the use of these frequency bands by any other application of the services to which it is allocated and does not establish priority in the Radio Regulations. The use of IMT stations is subject to agreement obtained under No. 9.21 with respect to the aeronautical mobile service used for aeronautical telemetry in accordance with No. 5.342. (WRC-15)
5.342	Additional allocation: in Armenia, Azerbaijan, Belarus, the Russian Federation, Uzbekistan, Kyrgyzstan and Ukraine, the frequency band 1429-1535 MHz also allocated to the aeronautical mobile service on a primary basis exclusively for the purposes of aeronautical telemetry within the national territory. As of 1 April 2007, the use of the band 1452-1492 MHz is subject to agreement between the administrations concerned. (WRC-15)
5.345	Use of the frequency band 1 452-1 492 MHz by the broadcasting-satellite service, and by the broadcasting service, is limited to digital audio broadcasting and is subject to the provisions of Resolution 528 (Rev.WRC-19). (WRC-19)
5.346	In Algeria, Angola, Saudi Arabia, Bahrain, Benin, Botswana, Burkina Faso, Burundi, Cameroon, Central African Republic, Congo (Rep. of the), Ivory Coast, Djibouti, Egypt, United Arab Emirates, Eswatini, Gabon, Gambia, Ghana, Guinea, Iraq, Jordan, Kenya, Kuwait, Lesotho, Lebanon, Liberia, Madagascar, Malawi, Mali, Morocco, Mauritius, Mauritania, Mozambique, Namibia, Niger, Nigeria, Oman, Uganda, Palestine**, Qatar, Dem. Rep. of the Congo, Rwanda, Senegal, Seychelles, Somalia, Sudan, South Sudan, South Africa, Tanzania, Chad, Togo, Tunisia, Zambia, and Zimbabwe, the frequency band 1 452-1 492 MHz is identified for use by administrations listed above wishing to implement International Mobile Telecommunications (IMT) in accordance with Resolution 223 (Rev.WRC-23). This identification does not preclude the use of this frequency band by any other application of the services to which it is allocated and does not establish priority in the Radio Regulations. The use of this frequency band for the implementation of IMT is subject to agreement obtained under No. 9.21 with respect to the aeronautical mobile service used for aeronautical telemetry in accordance with No. 5.342. See also Resolution 761 (Rev.WRC-19). (WRC-23) ** The use by Palestine of the allocation to the mobile service in the frequency band 1 452-1 492 MHz identified for IMT is noted, pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995.
5.348	The use of the band 1518-1525 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. In the band 1518-1525 MHz stations in the mobile-satellite service shall not claim protection from the stations in the fixed service. No. 5.43A does not apply. (WRC-03)
5.348A	In the band 1518-1525 MHz, the coordination threshold in terms of the power flux-density levels at the surface of the Earth in application of No. 9.11A for space stations in the mobile-satellite (space-to-Earth) service, with respect to the land mobile service use for specialized mobile radios or used in conjunction with public switched telecommunication networks (PSTN) operating within the territory of Japan, shall be -150 dB(W/m2) in any 4 kHz band for all angles of arrival, instead of those given in Table 52 of Appendix 5. In the band 1518-1525 MHz stations in the mobile-satellite service shall not claim protection from stations in the mobile service in the territory of Japan. No. 5.43A does not apply. (WRC-03)
5.348B	In the band 1518-1525 MHz, stations in the mobile-satellite service shall not claim protection from aeronautical mobile telemetry stations in the mobile service in the territory of the United States (see Nos. 5.343 and 5.344) and in the countries listed in No. 5.342. No. 5.43A does not apply. (WRC-03)
5.349	Different category of service: in Saudi Arabia, Azerbaijan, Bahrain, Cameroon, Djibouti, Egypt, Iran (Islamic Republic of), Iraq, Israel, Kuwait, Lebanon, North Macedonia, Morocco, Qatar, Syrian Arab Republic, Kyrgyzstan, Turkmenistan and Yemen, the allocation of the frequency band 1 525-1 530 MHz to the mobile, except aeronautical mobile, service is on a primary basis (see No. 5.33). (WRC-23)

Footnote Number	Footnote Content
5.350	Additional allocation: in Kyrgyzstan and Turkmenistan, the frequency band 1 525-1 530 MHz is also allocated to the aeronautical mobile service on a primary basis. (WRC-19)
5.351	The bands 1525-1544 MHz, 1545-1559 MHz, 1626.5-1645.5 MHz and 1646.5-1660.5 MHz shall not be used for feeder links of any service. In exceptional circumstances, however, an earth station at a specified fixed point in any of the mobile-satellite services may be authorised by an administration to communicate via space stations using these bands.
5.351A	For the use of the frequency bands 1 518-1 544 MHz, 1 545-1 559 MHz, 1 610-1 645.5 MHz, 1 646.5-1 660.5 MHz, 1 668-1 675 MHz, 1 980-2 010 MHz, 2 170-2 200 MHz, 2 483.5-2 520 MHz and 2 670-2 690 MHz by the mobile-satellite service, see Resolutions 212 (Rev.WRC-23) and 225 (Rev.WRC-23). (WRC-23)
5.352A	In the frequency band 1 525-1 530 MHz, stations in the mobile-satellite service, except stations in the maritime mobile-satellite service, shall not cause harmful interference to, or claim protection from, stations of the fixed service in Algeria, Saudi Arabia, Egypt, Guinea, India, Israel, Italy, Jordan, Kuwait, Mali, Morocco, Mauritania, Nigeria, Oman, Pakistan, the Philippines, Qatar, Syrian Arab Republic, Viet Nam and Yemen notified prior to 1 April 1998. (WRC-19)
5.353A	In applying the procedures of Section II of Article 9 to the mobile-satellite service in the frequency bands 1 530-1 544 MHz and 1 626.5-1 645.5 MHz, priority shall be given to accommodating the spectrum requirements for distress, urgency and safety communications of the global maritime distress and safety system (GMDSS). Maritime mobile-satellite distress, urgency and safety communications shall have priority access and immediate availability over all other mobile satellite communications operating within a network. Mobile-satellite systems shall not cause unacceptable interference to, or claim protection from, distress, urgency and safety communications of the GMDSS. Account shall be taken of the priority of safety-related communications in the other mobile-satellite services. The provisions of Resolution 222 (Rev.WRC-23) shall apply. (WRC-23)
5.354	The use of the bands 1525-1559 MHz and 1626.5-1660.5 MHz by the mobile-satellite services is subject to coordination under No. 9.11A.
5.355	Additional allocation: in Bahrain, Bangladesh, the Dem. Rep. of the Congo, Djibouti, Egypt, Eritrea, Iraq, Israel, Kuwait, Qatar, Syrian Arab Republic, Somalia, Sudan, South Sudan, Chad, Togo and Yemen, the bands 1540-1559 MHz, 1610-1645.5 MHz and 1646.5-1660 MHz are also allocated to the fixed service on a secondary basis. (WRC-12)
5.356	The use of the band 1544-1545 MHz by the mobile-satellite service (space-to-Earth) is limited to distress and safety communications (see Article 31).
5.357	Transmissions in the band 1545-1555 MHz from terrestrial aeronautical stations directly to aircraft stations, or between aircraft stations, in the aeronautical mobile (R) service are also authorised when such transmissions are used to extend or supplement the satellite-to-aircraft links.
5.357A	In applying the procedures of Section II of Article 9 to the mobile-satellite service in the frequency bands 1 545-1 555 MHz and 1 646.5-1 656.5 MHz, priority shall be given to accommodating the spectrum requirements of the aeronautical mobile-satellite (R) service providing transmission of messages with priority 1 to 6 in Article 44. Aeronautical mobile-satellite (R) service communications with priority 1 to 6 in Article 44 shall have priority access and immediate availability, by pre-emption if necessary, over all other mobile-satellite communications operating within a network. Mobile-satellite systems shall not cause unacceptable interference to, or claim protection from, aeronautical mobile-satellite (R) service communications with priority 1 to 6 in Article 44. Account shall be taken of the priority of safety-related communications in the other mobile-satellite services. The provisions of Resolution 222 (Rev.WRC-23) shall apply. (WRC-23)
5.359	Additional allocation: in Germany, Saudi Arabia, Armenia, Azerbaijan, Belarus, Cameroon, the Russian Federation, Georgia, Guinea, Guinea-Bissau, Jordan, Kazakhstan, Kuwait, Lithuania, Mauritania, Uganda, Uzbekistan, Pakistan, Poland, the Syrian Arab Republic, Kyrgyzstan, the Dem. People's Rep. of Korea, Romania, Tajikistan, Tunisia and Turkmenistan, the frequency bands 1 550-1 559 MHz, 1 610-1 645.5 MHz and 1 646.5-1 660 MHz are also allocated to the fixed service on a primary basis. Administrations are urged to make all practicable efforts to avoid the implementation of new fixed-service stations in these frequency bands. (WRC-23)
5.364	The use of the band 1610-1626.5 MHz by the mobile-satellite service (Earth-to-space) and by the radiodetermination-satellite service (Earth-to-space) is subject to coordination under No. 9.11A. A mobile earth station operating in either of the services in this band shall not produce a peak e.i.r.p. density in excess of -15 dB(W/4 kHz) in the part of the band used by systems operating in accordance with the provisions of No. 5.366 (to which No. 4.10 applies), unless otherwise agreed by the affected administrations. In the part of the band where such systems are not operating, the mean e.i.r.p. density of a mobile earth station shall not exceed -3 dB(W/4 kHz). Stations of the mobile-satellite service shall not claim protection from stations in the aeronautical radionavigation service, stations operating in accordance with the provisions of No. 5.366 and stations in the fixed service operating in accordance with the provisions of No. 5.359. Administrations responsible for the coordination of mobile-satellite networks shall make all practicable efforts to ensure protection of stations operating in accordance with the provisions of No. 5.366.
5.365	The use of the band 1613.8-1626.5 MHz by the mobile-satellite service (space-to-Earth) is subject to coordination under No. 9.11A.

Footnote Number	Footnote Content
5.366	The band 1610-1626.5 MHz is reserved on a worldwide basis for the use and development of airborne electronic aids to air navigation and any directly associated ground-based or satellite-borne facilities. Such satellite use is subject to agreement obtained under No. 9.21.
5.367	Additional allocation: the bands 1610-1626.5 MHz is also allocated to the aeronautical mobile-satellite (R) service on a primary basis, subject to agreement obtained under No. 9.21.
5.368	The provisions of No. 4.10 do not apply with respect to the radiodetermination-satellite and mobile-satellite services in the frequency band 1 610-1 626.5 MHz. However, No. 4.10 applies in the frequency band 1 610-1 626.5 MHz with respect to the aeronautical radionavigation-satellite service when operating in accordance with No. 5.366, the aeronautical mobile-satellite (R) service when operating in accordance with No. 5.367, and in the frequency bands 1 614.4225-1 618.725 MHz or 1 616.3-1 620.38 MHz (Earth-to-space) (see resolves 5 of Resolution 365 (WRC-23)) and 1 621.35-1 626.5 MHz with respect to the maritime mobile-satellite service when used for the global maritime distress and safety system (GMDSS). In applying the procedure of Section II of Article 9, the provisions of No. 4.10 do not apply for the frequency bands 1 614.4225-1 618.725 MHz or 1 616.3-1 620.38 MHz (Earth-to-space) (see resolves 5 of Resolution 365 (WRC-23)) and 2 483.59-2 499.91 MHz (space-to-Earth) for the maritime mobile-satellite service when used for the GMDSS with satellite networks or systems for which complete coordination information has been received by the Radiocommunication Bureau before 20 November 2023. Resolution 365 (WRC-23) applies. (WRC-23)
5.369	Different category of service: in Angola, Australia, China, Eritrea, Ethiopia, India, Iran (Islamic Republic of), Israel, Lebanon, Liberia, Madagascar, Mali, Pakistan, Papua New Guinea, Syrian Arab Republic, the Dem. Rep. of the Congo, Sudan, South Sudan, Togo and Zambia, the allocation of the band 1610-1626.5 MHz to the radiodetermination-satellite service (Earth-to-space) is on a primary basis (see No. 5.33), subject to agreement obtained under No. 9.21 from countries not listed in this provision. (WRC-12)
5.371	Additional allocation: in Region 1, the bands 1610-1626.5 MHz (Earth-to-space) is also allocated to the radiodetermination-satellite service on a secondary basis, subject to agreement obtained under No. 9.21. (WRC-12)
5.372	Harmful interference shall not be caused to stations of the radio astronomy service using the frequency band 1 610.6-1 613.8 MHz by stations of the radiodetermination-satellite and mobile-satellite services (No. 29.13 applies). The equivalent power flux-density (epfd) produced in the frequency band 1 610.6-1 613.8 MHz by all space stations of a non-geostationary-satellite system in the mobile-satellite service (space-to-Earth) operating in frequency band 1 613.8-1 626.5 MHz shall be in compliance with the protection criteria provided in Recommendations ITU-R RA.769-2 and ITU-R RA.1513-2, using the methodology given in Recommendation ITU-R M.1583-1, and the radio astronomy antenna pattern described in Recommendation ITU-R RA.1631-0. (WRC-19)
5.372A	The maritime mobile-satellite service in the frequency bands 1 614.4225-1 618.725 MHz or 1 616.3-1 620.38 MHz (Earth-to-space) (see resolves 5 of Resolution 365 (WRC-23)) and 2 483.59-2 499.91 MHz (space-to-Earth) when they are used for the global maritime distress and safety system (GMDSS) is limited to the geostationary-satellite networks identified in Resolution 365 (WRC-23) and their associated earth stations located within a service area from 75°E to 135°E longitude and from 10°N to 55°N latitude. Resolution 365 (WRC-23) applies. (WRC-23)
5.373	Maritime mobile earth stations receiving in the frequency band 1 621.35-1 626.5 MHz shall not impose additional constraints on earth stations operating in the maritime mobile-satellite service or maritime earth stations of the radiodetermination-satellite service operating in accordance with the Radio Regulations in the frequency band 1 610-1 621.35 MHz or on earth stations operating in the maritime mobile-satellite service operating in accordance with the Radio Regulations in the frequency band 1 626.5-1 660.5 MHz, unless otherwise agreed between the notifying administrations. (WRC-19)
5.373A	Maritime mobile earth stations receiving in the frequency band 1 621.35-1 626.5 MHz shall not impose constraints on the assignments of earth stations of the mobile-satellite service (Earth-to-space) and the radiodetermination-satellite service (Earth-to-space) in the frequency band 1 621.35-1 626.5 MHz in networks for which complete coordination information has been received by the Radiocommunication Bureau before 28 October 2019. (WRC-19)
5.374	Mobile earth stations in the mobile-satellite service operating in the bands 1631.5-1634.5 MHz and 1656.5-1660 MHz shall not cause harmful interference to the stations in the fixed service operating in the countries listed in No. 5.359. (WRC-97)
5.375	The use of the frequency band 1 645.5-1 646.5 MHz by the mobile-satellite service (Earth-to-space) and for inter-satellite links is limited to distress, urgency and safety communications (see Article 31). (WRC-23)
5.376	Transmissions in the band 1646.5-1656.5 MHz from aircraft stations in the aeronautical mobile (R) service directly to terrestrial aeronautical stations, or between aircraft stations, are also authorised when such transmissions are used to extend or supplement the aircraft-to-satellite links.
5.376A	Mobile earth stations operating in the band 1660-1660.5 MHz shall not cause harmful interference to stations in the radio astronomy service. (WRC-97)
5.379	Additional allocation: in Bangladesh, India, Indonesia, Nigeria and Pakistan, the band 1 660.5-1 668.4 MHz is also allocated to the meteorological aids service on a secondary basis.

Footnote Number	Footnote Content
5.379A	Administrations are urged to give all practicable protection in the band 1660.5-1668.4 MHz for future research in radio astronomy, particularly by eliminating air-to-ground transmissions in the meteorological aids service in the band 1664.4-1668.4 MHz as soon as practicable.
5.379B	The use of the frequency band 1 668-1 675 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. (WRC-23)
5.379C	In order to protect the radio astronomy service in the band 1668-1670 MHz, the aggregate power flux-density values produced by mobile earth stations in a network of the mobile-satellite service operating in this band shall not exceed -181 dB(W/m ²) in 10 MHz and -194 dB(W/m ²) in any 20 kHz at any radio astronomy station recorded in the Master International Frequency Register, for more than 2% of integration periods of 2 000 s. (WRC-03)
5.379D	For sharing of the frequency band 1 668.4-1 675 MHz between the mobile-satellite service and the fixed and mobile services, Resolution 744 (Rev.WRC-23) shall apply. (WRC-23)
5.379E	In the band 1668.4-1675 MHz, stations in the mobile-satellite service shall not cause harmful interference to stations in the meteorological aids service in China, Iran (Islamic Republic of), Japan and Uzbekistan. In the band 1668.4-1675 MHz, administrations are urged not to implement new systems in the meteorological aids service and are encouraged to migrate existing meteorological aids service operations to other bands as soon as practicable. (WRC-03)
5.380A	In the band 1670-1675 MHz, stations in the mobile-satellite service shall not cause harmful interference to, nor constrain the development of, existing earth stations in the meteorological-satellite service notified before 1 January 2004. Any new assignment to these earth stations in this band shall also be protected from harmful interference from stations in the mobile-satellite service. (WRC-07)
5.382	Different category of service: in Saudi Arabia, Armenia, Azerbaijan, Bahrain, Belarus, Congo (Rep. of the), Egypt, the United Arab Emirates, Eritrea, Ethiopia, the Russian Federation, Guinea, Iraq, Israel, Jordan, Kazakhstan, Kuwait, Lebanon, North Macedonia, Mauritania, Moldova, Mongolia, Oman, Uzbekistan, Poland, Qatar, the Syrian Arab Republic, Kyrgyzstan, Somalia, Tajikistan, Turkmenistan, Ukraine and Yemen, the allocation of the frequency band 1 690-1 700 MHz to the fixed and mobile, except aeronautical mobile, services is on a primary basis (see No. 5.33), and in the Dem. People's Rep. of Korea, the allocation of the frequency band 1 690-1 700 MHz to the fixed service is on a primary basis (see No. 5.33) and to the mobile, except aeronautical mobile, service on a secondary basis. (WRC-19)
5.384A	The frequency bands 1710-1885 MHz, 2300-2400 MHz or 2500-2690 MHz, and portion thereof, are identified for use by administrations wishing to implement International Mobile Telecommunications(IMT)in accordance with Resolution 223 (Rev.WRC-15). This identification does not preclude the use of these frequency bands by any application of the services to which they are allocated and does not establish priority in the Radio Regulations. (WRC-15)
5.385	Additional allocation: the band 1718.8-1722.2 MHz is also allocated to the radio astronomy service on a secondary basis for spectral line observations. (WRC-2000)
5.386	Additional allocation: the band 1750-1850 MHz is also allocated to the space operation (Earth-to-space) and space research (Earth-to-space) services in Region 2, (except in Mexico) in Australia, Guam, India, Indonesia and Japan on a primary basis, subject to agreement obtained under No. 9.21, having particular regard to troposcatter systems. (WRC-15)
5.387	Additional allocation: in Belarus, Georgia, Kyrgyzstan, Romania, Tajikistan and Turkmenistan, the frequency band 1 770-1 790 MHz is also allocated to the meteorological-satellite service on a primary basis, subject to agreement obtained under No. 9.21. (WRC-23)
5.388	The frequency bands 1 885-2 025 MHz and 2 110-2 200 MHz are intended for use, on a worldwide basis, by administrations wishing to implement International Mobile Telecommunications (IMT). Such use does not preclude the use of these frequency bands by other services to which they are allocated. The frequency bands should be made available for IMT in accordance with Resolution 212 Rev.WRC-23) (see also Resolution 223 (Rev.WRC-23)). (WRC-23)
5.388A	The frequency bands 1 710-1 980 MHz, 2 010-2 025 MHz and 2 110-2 170 MHz in Regions 1 and 3 and the frequency bands 1 710-1 980 MHz and 2 110-2 160 MHz in Region 2 are identified for the use by high altitude platform stations as International Mobile Telecommunications (IMT) base stations (HIBS). This identification does not preclude the use of these frequency bands by any application of the services to which they are allocated and does not establish priority in the Radio Regulations. Resolution 221 (Rev.WRC-23) shall apply. HIBS shall not claim protection from existing primary services. No. 5.43A does not apply. Such use of HIBS in the frequency bands 1 710-1 785 MHz in Regions 1 and 2, and 1 710-1 815 MHz in Region 3 is limited to reception by HIBS, and in the frequency band 2 110-2 170 MHz is limited to transmission from HIBS. (WRC-23)
5.389A	The use of the frequency bands 1 980-2 010 MHz and 2 170-2 200 MHz by the mobile-satellite service is subject to coordination under No. 9.11A and to the provisions of Resolution 716 (Rev.WRC-23). (WRC-23)
5.389B	The use of the frequency band 1 980-1 990 MHz by the mobile-satellite service shall not cause harmful interference to or constrain the development of the fixed and mobile services in Argentina, Brazil, Canada, Chile, Ecuador, the United States, Honduras, Jamaica, Mexico, Paraguay, Peru, Suriname, Trinidad and Tobago, Uruguay and Venezuela. (WRC-19)

Footnote Number	Footnote Content
5.389F	In Algeria, Cape Verde, Egypt, Iran (Islamic Republic of), Mali, Syrian Arab Republic and Tunisia, the use of the frequency bands 1 980-2 010 MHz and 2 170-2 200 MHz by the mobile-satellite service shall neither cause harmful interference to the fixed and mobile services, nor hamper the development of those services prior to 1 January 2005, nor shall the former service request protection from the latter services. (WRC-19)
5.391	In making assignments to the mobile service in the frequency bands 2025-2110 MHz and 2200-2290 MHz, administrations shall not introduce high-density mobile systems, as described in Recommendation ITU-R SA.1154-0, and shall take that Recommendation into account for the introduction of any other type of mobile system. (WRC-15)
5.392	Administrations are urged to take all practicable measures to ensure that space-to-space transmissions between two or more non-geostationary satellites, in the space research, space operations and Earth exploration-satellite services in the bands 2025-2110 MHz and 2200-2290 MHz, shall not impose any constraints on Earth-to-space, space-to-Earth and other space-to-space transmissions of those services and in those bands between geostationary and non-geostationary satellites.
5.395	In France and Turkey, the use of the band 2310-2360 MHz by the aeronautical mobile service for telemetry has priority over other uses by the mobile service. (WRC-03)
5.398	In respect of the radiodetermination-satellite service in the band 2483.5-2500 MHz, the provisions of No. 4.10 do not apply.
5.398A	Different category of service: In Armenia, Azerbaijan, Belarus, the Russian Federation, Kazakhstan, Uzbekistan, Kyrgyzstan, Tajikistan and Ukraine, the band 2483.5-2500 MHz is allocated on a primary basis to the radiolocation service. The radiolocation stations in these countries shall not cause harmful interference to, or claim protection from, stations of the fixed, mobile and mobile-satellite services operating in accordance with the Radio Regulations in the frequency band 2483.5-2500 MHz. (WRC-12)
5.399	Except for cases referred to in No. 5.401, stations of the radiodetermination-satellite service operating in the frequency band 2483.5-2500 MHz for which notification information is received by the Bureau after 17 February 2012, and the service area of which includes Armenia, Azerbaijan, Belarus, the Russian Federation, Kazakhstan, Uzbekistan, Kyrgyzstan, Tajikistan and Ukraine, shall not cause harmful interference to, and shall not claim protection from stations of the radiolocation service operating in these countries in accordance with No.5.398A. (WRC-12)
5.401	In Angola, Australia, Bangladesh, China, Eritrea, Eswatini, Ethiopia, India, Lebanon, Liberia, Libya, Madagascar, Mali, Pakistan, Papua New Guinea, Syrian Arab Republic, Dem. Rep. of the Congo, Sudan, Togo and Zambia, the frequency band 2 483.5-2 500 MHz was already allocated on a primary basis to the radiodetermination-satellite service before WRC-12, subject to agreement obtained under No. 9.21 from countries not listed in this provision. Systems in the radiodetermination-satellite service for which complete coordination information has been received by the Radiocommunication Bureau before 18 February 2012 will retain their regulatory status, as of the date of receipt of the coordination request information. (WRC-19)
5.402	The use of the band 2483.5-2500 MHz by the mobile-satellite and the radiodetermination-satellite services is subject to the coordination under No. 9.11A. Administrations are urged to take all practicable steps to prevent harmful interference to the radio astronomy service from emissions in the 2483.5-2500 MHz band, especially those caused by second-harmonic radiation that would fall into the 4990-5000 MHz band allocated to the radio astronomy service worldwide.
5.409A	The frequency band 2 500-2 690 MHz in Regions 1 and 2, and the frequency band 2 500-2 655 MHz in Region 3 are identified for use by high-altitude platform stations as International Mobile Telecommunications (IMT) base stations (HIBS). This identification does not preclude the use of these frequency bands by any application of the services to which they are allocated and does not establish priority in the Radio Regulations. Resolution 218 (WRC-23) shall apply. HIBS shall not claim protection from existing primary services. No. 5.43A does not apply. Such use of HIBS in the frequency bands 2 500-2 510 MHz in Regions 1 and 2, and 2 500-2 535 MHz in Region 3 is limited to reception by HIBS. (WRC-23)
5.410	The band 2500-2690 MHz may be used for tropospheric scatter systems in Region 1, subject to agreement obtained under No. 9.21. No. 9.21 does not apply to tropospheric scatter links situated entirely outside Region 1. Administrations shall make all practicable efforts to avoid developing new tropospheric scatter systems in this band. When planning new tropospheric scatter radio-relay links in this band, all possible measures shall be taken to avoid directing the antennas of these links towards the geostationary-satellite orbit.(WRC-12)
5.412	Alternative allocation:in Kyrgyzstan and Turkmenistan, the band 2 500-2 690 MHz is allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-12)
5.413	In the design of systems in the broadcasting-satellite service in the bands between 2500 MHz and 2690 MHz, administrations are urged to take all necessary steps to protect the radio astronomy service in the band 2690-2700 MHz.
5.416	The use of the band 2520-2670 MHz by the broadcasting-satellite service is limited to national and regional systems for community reception, subject to agreement obtained under No. 9.21. The provisions of No. 9.19 shall be applied by administrations in this band in their bilateral and multilateral negotiations. (WRC-07)
5.418B	Use of the band 2630-2655 MHz by non geostationary-satellite systems in the broadcasting-satellite service (sound), pursuant to No. 5.418, for which complete Appendix 4 coordination information, or notification information, has been received after 2 June 2000, is subject to the application of the provisions of No. 9.12. (WRC-03)

Footnote Number	Footnote Content
5.418C	Use of the band 2630-2655 MHz by geostationary-satellite networks for which complete Appendix 4 coordination information, or notification information, has been received after 2 June 2000 is subject to the application of the provisions of No. 9.13 with respect to non geostationary-satellite systems in the broadcasting-satellite service (sound), pursuant to No. 5.418 and No. 22.2 does not apply. (WRC-03)
5.422	Additional allocation: in Saudi Arabia, Armenia, Azerbaijan, Bahrain, Belarus, Brunei Darussalam, Congo (Rep. of the), Ivory Coast, Cuba, Djibouti, Egypt, the United Arab Emirates, Eritrea, Ethiopia, Gabon, Georgia, Guinea, Guinea-Bissau, Iran (Islamic Republic of), Iraq, Israel, Jordan, Kuwait, Lebanon, Mauritania, Mongolia, Montenegro, Nigeria, Oman, Pakistan, the Philippines, Qatar, Syrian Arab Republic, Kyrgyzstan, the Dem. Rep. of the Congo, Romania, Somalia, Tajikistan, Tunisia, Turkmenistan, Ukraine and Yemen, the band 2690-2700 MHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. Such use is limited to equipment in operation by 1 January 1985. (WRC-12)
5.423	In the band 2700-2900 MHz, ground-based radars used for meteorological purposes are authorised to operate on a basis of equality with stations of the aeronautical radionavigation service.
5.424A	In the band 2900-3100 MHz, stations in the radiolocation service shall not cause harmful interference to, nor claim protection from, radar systems in the radionavigation service. (WRC-03)
5.425	In the band 2900-3100 MHz, the use of the shipborne interrogator-transponder system (SIT) shall be confined to the sub-band 2930-2950 MHz.
5.426	The use of the band 2900-3100 MHz by the aeronautical radionavigation service is limited to ground-based radars.
5.427	In the bands 2900-3100 MHz and 9300-9500 MHz, the response from radar transponders shall not be capable of being confused with the response from radar beacons (racons) and shall not cause interference to ship or aeronautical radars in the radionavigation service, having regard, however, to No. 4.9.
5.428	Additional allocation: in Kyrgyzstan and Turkmenistan, the frequency band 3 100-3 300 MHz is also allocated to the radionavigation service on a primary basis. (WRC-19)
5.429	Additional allocation: in Saudi Arabia, Bahrain, Bangladesh, Benin, Brunei Darussalam, Cambodia, Cameroon, China, Congo (Rep. of the), Korea (Rep. of), Ivory Coast, Djibouti, Egypt, the United Arab Emirates, India, Indonesia, Iran (Islamic Republic of), Iraq, Japan, Jordan, Kenya, Kuwait, Lao P.D.R., Lebanon, Libya, Malaysia, Mongolia, Myanmar, New Zealand, Oman, Uganda, Pakistan, Palestine*, Qatar, the Syrian Arab Republic, the Dem. Rep. of the Congo, the Dem. People's Rep. of Korea, Singapore, Somalia, Sudan, Thailand, Viet Nam and Yemen, the frequency band 3 300-3 400 MHz is also allocated to the fixed and mobile services on a primary basis. Mongolia, New Zealand and the countries bordering the Mediterranean shall not claim protection for their fixed and mobile services from the radiolocation service. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference, and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995.
5.429A	Additional allocation: in Angola, Benin, Botswana, Burkina Faso, Burundi, Djibouti, Eswatini, Ghana, Guinea, Guinea-Bissau, Lesotho, Liberia, Malawi, Mauritania, Mozambique, Namibia, Niger, Nigeria, Rwanda, Sudan, South Sudan, South Africa, Tanzania, Chad, Togo, Zambia and Zimbabwe, the frequency band 3 300-3 400 MHz is allocated to the mobile, except aeronautical mobile, service on a primary basis. Stations in the mobile service operating in the frequency band 3 300-3 400 MHz shall not cause harmful interference to, or claim protection from, stations operating in the radiolocation service. (WRC-19)
5.429B	In the following countries of Region 1: Angola, Benin, Botswana, Burkina Faso, Burundi, Cabo Verde, Cameroon, Central African Republic, Comoros, Congo (Rep. of the), Ivory Coast, Djibouti, Egypt, Eritrea, Eswatini, Ethiopia, Gambia, Ghana, Guinea, Guinea-Bissau, Equatorial Guinea, Kenya, Lesotho, Liberia, Madagascar, Malawi, Mauritius, Mauritania, Mongolia, Mozambique, Namibia, Niger, Nigeria, Uganda, the Dem. Rep. of the Congo, Rwanda, Sao Tome and Principe, Senegal, Seychelles, Sierra Leone, Somalia, Sudan, South Sudan, South Africa, Tanzania, Chad, Togo, Zambia and Zimbabwe, the frequency band 3 300-3 400 MHz is identified for the implementation of International Mobile Telecommunications (IMT). The use of this frequency band shall be in accordance with Resolution 223 (Rev.WRC-23). The use of the frequency band 3 300-3 400 MHz by IMT stations in the mobile service shall not cause harmful interference to, or claim protection from, systems in the radiolocation service, and administrations wishing to implement IMT shall obtain the agreement of neighbouring countries to protect operations within the radiolocation service. This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. (WRC-23)
5.430	Additional allocation: in Kyrgyzstan and Turkmenistan, the frequency band 3 300-3 400 MHz is also allocated to the radionavigation service on a primary basis. (WRC-19)
5.430A	The allocation of the frequency band 3400-3600 MHz to the mobile, except aeronautical mobile, service subject to agreement obtained under No. 9.21. This frequency band is identified for International Mobile Telecommunications (IMT). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. The provisions of Nos. 9.17 and 9.18 shall also apply in the coordination phase. Before an administration brings into use a (base or mobile) station of the mobile service in this frequency band it shall ensure that the power flux-density (pfd) produced at 3 m above ground does not exceed -154.5 dBW/(m ² · 4 kHz) for more than 20% of time at the border of the territory of any other administration. This limit may be exceeded on the territory of any country whose administration has so agreed. In order to ensure that the pfd limit at the border of the territory of any other administration is met, the calculations and verification shall be made, taking into account all relevant information, with the mutual agreement of both administrations (the administration responsible for the terrestrial station and the administration responsible for the earth station), and with the assistance of the Bureau if so requested. In case of disagreement, the calculation and verification of the pfd shall be made by the Bureau, taking into account the information referred to above. Stations of the mobile service in the frequency band 3400-3600 MHz shall not claim more protection from space stations than that provided in Table 21-4 of the Radio Regulations (Edition of 2004). (WRC-15)

Footnote Number	Footnote Content
5.431	Additional allocation: in Germany, the frequency band 3 400-3 475 MHz is also allocated to the amateur service on a secondary basis. (WRC-19)
5.433B	In Angola, Botswana, Guinea, Lesotho, Malawi and South Sudan, the frequency band 3 600-3 700 MHz is identified for International Mobile Telecommunications (IMT). This identification does not preclude the use of the frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. The conditions of No. 5.434A shall apply. (WRC-23)
5.434A	The use of the frequency band 3 600-3 800 MHz by the mobile, except aeronautical mobile, service on a primary basis in Region 1 is subject to agreement obtained under No. 9.21 if the power flux-density (pfd) limit below is exceeded. The provisions of Nos. 9.17 and 9.18 shall also apply in the coordination phase. Before an administration in Region 1 brings into use a station in the mobile service in the frequency band 3 600-3 800 MHz, for the protection of stations in the fixed and fixed-satellite services, it shall ensure that the pfd produced at 3 m above ground does not exceed $-154.5 \text{ dB(W/(m}^2 \text{ 4 kHz))}$ for more than 20% of the time at the border of the territory of any other administration. Stations in the mobile service operating in the frequency band 3 600-3 800 MHz shall not claim more protection from space stations than that provided in Table 21-4 of the Radio Regulations. (WRC-23)
5.434B	In Algeria, Saudi Arabia, Azerbaijan, Bahrain, Belarus, Benin, Burkina Faso, Burundi, Cameroon, Central African Rep., Comoros, Congo (Rep. of the), Ivory Coast, Djibouti, Egypt, United Arab Emirates, Eswatini, Gabon, Gambia, Ghana, Guinea, Iraq, Jordan, Kazakhstan, Kenya, Kuwait, Lebanon, Liberia, Libya, Madagascar, Mali, Morocco, Mauritius, Mauritania, Mozambique, Namibia, Niger, Nigeria, Oman, Uganda, Uzbekistan, Palestine*, Qatar, the Syrian Arab Republic, the Dem. Rep. of the Congo, Rwanda, Sao Tome and Principe, Senegal, Sierra Leone, Somalia, Sudan, South Africa, Tanzania, Chad, Togo, Tunisia, Yemen, Zambia and Zimbabwe, the frequency band 3 600-3 800 MHz is identified for International Mobile Telecommunications (IMT). This identification does not preclude the use of the frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. The conditions of No. 5.434A shall apply. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference, and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995.
5.435A	Different category of service: In Angola, Botswana, Guinea, Lesotho, Malawi and South Sudan, the frequency band 3 700-3 800 MHz is allocated to the mobile service on a secondary basis. (WRC-23)
5.436	Use of the frequency band 4 200-4 400 MHz by stations in the aeronautical mobile (R) service is reserved exclusively for wireless avionics intra-communication systems that operate in accordance with recognized international aeronautical standards. Such use shall be in accordance with Resolution 424 (Rev.WRC-23). (WRC-23)
5.437	Passive sensing in the Earth exploration-satellite and space research services may be authorized in the frequency band 4200-4400 MHz on a secondary basis. (WRC-15)
5.438	Use of the frequency band 4200-4400 MHz by the aeronautical radionavigation service is reserved exclusively for radio altimeters installed on board aircraft and for the associated transponders on the ground. (WRC-15)
5.439	Additional allocation: in Iran (Islamic Republic of), the band 4200-4400 MHz is also allocated to the fixed service on a secondary basis. (WRC-12)
5.440	The standard frequency and time signal-satellite service may be authorised to use the frequency 4202 MHz for space-to-Earth transmissions and the frequency 6427 MHz for Earth-to-space transmissions. Such transmissions shall be confined within the limits of $\pm 2 \text{ MHz}$ of these frequencies, subject to agreement obtained under No. 9.21.
5.440A	In Region 2 (except Brazil, Cuba, French overseas departments and communities, Guatemala, Paraguay, Uruguay and Venezuela), and in Australia, the band 4400-4940 MHz may be used for aeronautical mobile telemetry for flight testing by aircraft stations (see No. 1.83). Such use shall be in accordance with Resolution 416 (WRC-07) and shall not cause harmful interference to, nor claim protection from, the fixed-satellite and fixed service. Any such use does not preclude the use of this band by other mobile service applications or by other services to which this band is allocated on a co-primary basis and does not establish priority in the Radio Regulations. (WRC-07)
5.441	The use of the bands 4500-4800 MHz (space-to-Earth), 6725-7025 MHz (Earth-to-space) by the fixed-satellite service shall be in accordance with the provisions of Appendix 30B. The use of the bands 10.7-10.95 GHz (space-to-Earth), 11.2-11.45 GHz (space-to-Earth) and 12.75-13.25 GHz (Earth-to-space) by geostationary-satellite systems in the fixed-satellite service shall be in accordance with the provisions of Appendix 30B. The use of the bands 10.7-10.95 GHz (space-to Earth), 11.2-11.45 GHz (space-to-Earth) and 12.75-13.25 GHz (Earth-to-space) by a non-geostationary-satellite system in the fixed-satellite service is subject to application of the provisions of No. 9.12 for coordination with other non-geostationary-satellite systems in the fixed-satellite service. Non-geostationary-satellite system in the fixed-satellite service shall not claim protection from geostationary-satellite networks in the fixed-satellite service operating in accordance with the Radio Regulations, irrespective of the dates of receipt by the Bureau of the complete coordination or notification information, as appropriate, for the non-geostationary-satellite system in the fixed-satellite service and of the complete coordination or notification information, as appropriate, for the geostationary-satellite networks, and No. 5.43A does not apply. Non-geostationary-satellite systems in the fixed-satellite service in the above bands shall be operated in such a way that any unacceptable interference that may occur during their operation shall be rapidly eliminated. (WRC-2000)

Footnote Number	Footnote Content
5.441A	In Brazil, Paraguay and Uruguay, the frequency band 4 800-4 900 MHz, or portions thereof, is identified for the implementation of International Mobile Telecommunications (IMT). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. The use of this frequency band for the implementation of IMT is subject to agreement obtained with neighbouring countries, and IMT stations shall not claim protection from stations of other applications of the mobile service. Such use shall be in accordance with Resolution 223 (Rev.WRC-19). (WRC-19)
5.441B	In Angola, Argentina, Armenia, Azerbaijan, Benin, Botswana, Brazil, Burkina Faso, Burundi, Cabo Verde, Cambodia, Cameroon, Chile, China, Colombia, Congo (Rep. of the), Ivory Coast, Djibouti, Eswatini, Russian Federation, Gabon, Ghana, Guinea, Iran (Islamic Republic of), Iraq, Kazakhstan, Lao P.D.R., Lesotho, Liberia, Madagascar, Malawi, Mali, Mongolia, Namibia, Niger, Uganda, Uzbekistan, the Dem. Rep. of the Congo, Kyrgyzstan, the Dem. People's Rep. of Korea, South Sudan, South Africa, Chad, Togo, Viet Nam, Zambia and Zimbabwe, the frequency band 4 800-4 990 MHz, or portions thereof, is identified for use by administrations wishing to implement International Mobile Telecommunications (IMT). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. The use of IMT stations is subject to agreement obtained under No. 9.21 with concerned administrations, and IMT stations shall not claim protection from stations of other applications of the mobile service. In addition, before an administration brings into use an IMT station in the mobile service, it shall ensure that the power flux-density (pfd) produced by this station does not exceed $-155 \text{ dB(W/(m}^2 \cdot 1 \text{ MHz))}$ produced up to 19 km above sea level at 20 km from the coast, defined as the low-water mark, as officially recognized by the coastal State. Resolution 223 (Rev.WRC-23) applies. (WRC-23)
5.442	In the bands 4825-4835 MHz and 4950-4990 MHz, the allocation to the mobile service is restricted to the mobile, except aeronautical mobile, service. In Region 2 (except Brazil, Cuba, Guatemala, Mexico, Paraguay, Uruguay and Venezuela), and in Australia, the band 4825-4835 MHz is also allocated to the aeronautical mobile service, limited to aeronautical mobile telemetry for flight testing by aircraft stations. Such use shall be in accordance with Resolution 416 (WRC-07) and shall not cause harmful interference to the fixed service. (WRC-07)
5.443	Different category of service: in Argentina, Australia and Canada, the allocation of the bands 4 825-4 835 MHz and 4 950-4 990 MHz to the radio astronomy service is on a primary basis (see No. 5.33).
5.443AA	In the frequency bands 5000-5030 MHz and 5091-5150 MHz, the aeronautical mobile-satellite (R) service is subject to agreement obtained under No. 9.21. The use of these bands by the aeronautical mobile-satellite (R) service is limited to internationally standardized aeronautical systems. (WRC-12)
5.443B	In order not to cause harmful interference to the microwave landing system operating above 5030 MHz, the aggregate power flux-density produced at the Earth's surface in the frequency band 5030-5150 MHz by all the space stations within any radionavigation-satellite service system (space-to-Earth) operating in the frequency band 5 010-5 030 MHz shall not exceed $-124.5 \text{ dB(W/m}^2)$ in a 150 kHz band. In order not to cause harmful interference to the radio astronomy service in the frequency band 4990-5000 MHz, radionavigation-satellite service systems operating in the frequency band 5 010-5 030 MHz shall comply with the limits in the frequency band 4990-5000 MHz defined in Resolution 741 (Rev.WRC-15). (WRC-15)
5.443C	The use of the frequency band 5030-5091 MHz by the aeronautical mobile (R) service is limited to internationally standardized aeronautical systems. Unwanted emissions from the aeronautical mobile (R) service in the frequency band 5030-5091 MHz shall be limited to protect RNSS system downlinks in the adjacent 5010-5030 MHz band. Until such time that an appropriate value is established in a relevant ITU-R Recommendation, the e.i.r.p. density limit of -75 dBW/MHz in the frequency band 5010-5030 MHz for any AM(R)S station unwanted emission should be used. (WRC-12)
5.443D	In the frequency band 5030-5091 MHz, the aeronautical mobile-satellite (R) service is subject to coordination under No. 9.11A. The use of this frequency band by the aeronautical mobile-satellite (R) service is limited to internationally standardized aeronautical systems.
5.444	The frequency band 5030-5150 MHz is to be used for the operation of the international standard system (microwave landing system) for precision approach and landing. In the frequency band 5030-5091 MHz, the requirements of this system shall take precedence over other uses of this frequency band. For the use of the frequency band 5091-5150 MHz, No. 5.444A and Resolution 114 (Rev.WRC-15) apply. (WRC-15)
5.444A	The use of allocation to the fixed-satellite service (Earth-to-space) in the frequency band 5091-5150 MHz is limited to feeder links of non-geostationary satellite systems in the mobile-satellite service and is subject to coordination under No. 9.11A. The use of the frequency band 5091-5150 MHz by feeder links of non-geostationary satellite systems in the mobile-satellite service shall be subject to application of Resolution 114 (Rev. WRC-15). Moreover, to ensure that the aeronautical radionavigation service is protected from harmful interference, coordination is required for feeder-link earth stations of the non geostationary satellite systems in the mobile-satellite service which are separated by less than 450 km from the territory of an administration operating ground stations in the aeronautical radionavigation service. (WRC-15)
5.444B	The use of the frequency band 5 091-5 150 MHz by the aeronautical mobile service is limited to: – systems operating in the aeronautical mobile (R) service and in accordance with international aeronautical standards, limited to surface applications at airports. Such use shall be in accordance with Resolution 748 (Rev.WRC-19); – aeronautical telemetry transmissions from aircraft stations (see No. 1.83) in accordance with Resolution 418 (Rev.WRC-19). (WRC-19)

Footnote Number	Footnote Content
5.446	Additional allocation: in the countries listed in No. 5.369, the frequency band 5150-5216 MHz is also allocated to the radiodetermination-satellite service (space-to-Earth) on a primary basis, subject to agreement obtained under No. 9.21. In Region 2 (except in Mexico), the frequency band is also allocated to the radiodetermination-satellite service (space-to-Earth) on a primary basis. In Regions 1 and 3, except those countries listed in No. 5.369 and Bangladesh, the frequency band is also allocated to the radiodetermination-satellite service (space-to-Earth) on a secondary basis. The use by the radiodetermination-satellite service is limited to feeder links in conjunction with the radiodetermination-satellite service operating in the frequency bands 1610-1626.5 MHz and/or 2483.5-2500 MHz. The total power flux-density at the Earth's surface shall in no case exceed -159 dB(W/m ²) in any 4 kHz band for all angles of arrival. (WRC-15)
5.446A	The use of the frequency bands 5 150-5 350 MHz and 5 470-5 725 MHz by the stations in the mobile, except aeronautical mobile, service shall be in accordance with Resolution 229 (Rev.WRC-23). (WRC-23)
5.446B	In the band 5150-5250 MHz, stations in the mobile service shall not claim protection from earth stations in the fixed-satellite service. No. 5.43A does not apply to the mobile service with respect to fixed-satellite service earth stations. (WRC-03)
5.446C	Additional allocation: in Region 1 (except in Algeria, Saudi Arabia, Bahrain, Egypt, United Arab Emirates, Iraq, Jordan, Kuwait, Lebanon, Morocco, Oman, Qatar, Syrian Arab Republic, Sudan, South Sudan and Tunisia), the frequency band 5 150-5 250 MHz is also allocated to the aeronautical mobile service on a primary basis, limited to aeronautical telemetry transmissions from aircraft stations (see No. 1.83), in accordance with Resolution 418 (Rev.WRC-19). These stations shall not claim protection from other stations operating in accordance with Article 5. No. 5.43A does not apply. (WRC-19)
5.446D	Additional allocation: in Brazil, the band 5 150-5 250 MHz is also allocated to the aeronautical mobile service on a primary basis, limited to aeronautical telemetry transmissions from aircraft stations (see No. 1.83), in accordance with Resolution 418 (Rev.WRC-19). (WRC-19)
5.447	Additional allocation: in Ivory Coast, Egypt, Lebanon, the Syrian Arab Republic and Tunisia, the frequency band 5 150-5 250 MHz is also allocated to the mobile service, on a primary basis, subject to agreement obtained under No. 9.21. In this case, the provisions of Resolution 229 (Rev.WRC-23) do not apply. (WRC-23)
5.447A	The allocation to the fixed-satellite service (Earth-to-space) is limited to feeder links of non-geostationary-satellite systems in the mobile-satellite service and is subject to coordination under No. 9.11A.
5.447B	Additional allocation: the band 5150-5216 MHz is also allocated to the fixed-satellite service (space-to-Earth) on a primary basis. This allocation is limited to feeder links of non-geostationary-satellite systems in the mobile-satellite service and is subject to provisions of No. 9.11A. The power flux-density at the Earth's surface produced by space stations of the fixed-satellite service operating in the space-to-Earth direction in the band 5150-5216 MHz shall in no case exceed -164 dB(W/m ²) in any 4 kHz band for all angles of arrival.
5.447C	Administrations responsible for fixed-satellite service networks in the band 5150-5250 MHz operated under Nos. 5.447A and 5.447B shall coordinate on an equal basis in accordance with No. 9.11A with administrations responsible for non-geostationary-satellite networks operated under No. 5.446 and brought into use prior to 17 November 1995. Satellite networks operated under No. 5.446 brought into use after 17 November 1995 shall not claim protection from, and shall not cause harmful interference to, stations of the fixed-satellite service operated under Nos. 5.447A and 5.447B.
5.447D	The allocation of the band 5250-5255 MHz to the space research service on a primary basis is limited to active spaceborne sensors. Other uses of the band by the space research service are on a secondary basis. (WRC-97)
5.447E	Additional allocation: The frequency band 5 250-5 350 MHz is also allocated to the fixed service on a primary basis in the following countries in Region 3: Australia, Korea (Rep. of), India, Indonesia, Iran (Islamic Republic of), Japan, Malaysia, Papua New Guinea, the Philippines, Dem. People's Rep. of Korea, Sri Lanka, Thailand and Viet Nam. The use of this frequency band by the fixed service is intended for the implementation of fixed wireless access systems and shall comply with Recommendation ITU-R F.1613-0. In addition, the fixed service shall not claim protection from the radiodetermination, Earth exploration-satellite (active) and space research (active) services, but the provisions of No. 5.43A do not apply to the fixed service with respect to the Earth exploration-satellite (active) and space research (active) services. After implementation of fixed wireless access systems in the fixed service with protection for the existing radiodetermination systems, no more stringent constraints should be imposed on the fixed wireless access systems by future radiodetermination implementations. (WRC-15)
5.447F	In the frequency band 5 250-5 350 MHz, stations in the mobile service shall not claim protection from the radiolocation service, the Earth exploration-satellite service (active) and the space research service (active). The radiolocation service, the Earth exploration-satellite service (active) and the space research service (active) shall not impose more stringent conditions upon the mobile service than those stipulated in Resolution 229 (Rev.WRC-23). (WRC-23)
5.448	Additional allocation: in Kyrgyzstan, Romania and Turkmenistan, the frequency band 5 250-5 350 MHz is also allocated to the radionavigation service on a primary basis. (WRC-19)

Footnote Number	Footnote Content
5.448A	The Earth exploration-satellite (active) and space research (active) services in the frequency band 5 250-5 350 MHz shall not claim protection from the radiolocation service. No. 5.43A does not apply. (WRC-03)
5.448B	The Earth exploration-satellite service (active) operating in the band 5350-5570 MHz and space research service (active) operating in the band 5460-5570 MHz shall not cause harmful interference to the aeronautical radionavigation service in the band 5350-5460 MHz, the radionavigation service in the band 5460-5470 MHz and the maritime radionavigation service in the band 5470-5570 MHz. (WRC-03)
5.448C	The space research service (active) operating in the band 5350-5460 MHz shall not cause harmful interference to nor claim protection from other services to which this band is allocated. (WRC-03)
5.448D	In the frequency band 5350-5470 MHz, stations in the radiolocation service shall not cause harmful interference to, nor claim protection from, radar systems in the aeronautical radionavigation service operating in accordance with No. 5.449. (WRC-03)
5.449	The use of the band 5350-5470 MHz by the aeronautical radionavigation service is limited to airborne radars and associated airborne beacons.
5.450	Additional allocation: in Austria, Azerbaijan, Iran (Islamic Republic of), Kyrgyzstan, Romania, Turkmenistan and Ukraine, the band 5470-5650 MHz is also allocated to the aeronautical radionavigation service on a primary basis. (WRC-12)
5.450A	In the frequency band 5 470-5 725 MHz, stations in the mobile service shall not claim protection from radiodetermination services. The radiodetermination services shall not impose more stringent conditions upon the mobile service than those stipulated in Resolution 229 (Rev.WRC-23). (WRC-23)
5.450B	In the frequency band 5470-5650 MHz, stations in the radiolocation service, except ground-based radars used for meteorological purposes in the band 5 600-5 650 MHz, shall not cause harmful interference to, nor claim protection from, radar systems in the maritime radionavigation service. (WRC-03)
5.451	Additional allocation: in the United Kingdom, the band 5470-5850 MHz is also allocated to the land mobile service on a secondary basis. The power limits specified in Nos. 21.2, 21.3, 21.4 and 21.5 shall apply in the band 5725-5850 MHz.
5.452	Between 5600 MHz and 5650 MHz, ground-based radars used for meteorological purposes are authorised to operate on a basis of equality with stations of the maritime radionavigation service.
5.453	Additional allocation: in Saudi Arabia, Bahrain, Bangladesh, Brunei Darussalam, Cameroon, China, Congo (Rep. of the), Korea (Rep. of), Ivory Coast, Djibouti, Egypt, the United Arab Emirates, Eswatini, Gabon, Guinea, Equatorial Guinea, India, Indonesia, Iran (Islamic Republic of), Iraq, Japan, Jordan, Kenya, Kuwait, Lebanon, Libya, Madagascar, Malaysia, Niger, Nigeria, Oman, Uganda, Pakistan, the Philippines, Qatar, the Syrian Arab Republic, the Dem. People's Rep. of Korea, Singapore, Somalia, Sri Lanka, Tanzania, Chad, Thailand, Togo, Viet Nam and Yemen, the frequency band 5 650-5 850 MHz is also allocated to the fixed and mobile services on a primary basis. In this case, the provisions of Resolution 229 (Rev.WRC-23) do not apply. In addition, in Afghanistan, Angola, Benin, Bhutan, Botswana, Burkina Faso, Burundi, Dem. Rep. of the Congo, Fiji, Ghana, Kiribati, Lesotho, Malawi, Maldives, Mauritius, Micronesia, Mongolia, Mozambique, Myanmar, Namibia, Nauru, New Zealand, Papua New Guinea, Rwanda, Solomon Islands, South Sudan, South Africa, Tonga, Vanuatu, Zambia and Zimbabwe, the frequency band 5 725-5 850 MHz is allocated to the fixed service on a primary basis, and stations operating in the fixed service shall not cause harmful interference to and shall not claim protection from other primary services in the frequency band. (WRC-23)
5.454	Different category of service: in Azerbaijan, the Russian Federation, Georgia, Kyrgyzstan, Tajikistan and Turkmenistan, the allocation of the band 5670-5725 MHz to the space research service is on a primary basis (see No. 5.33). (WRC-12)
5.455	Additional allocation: in Armenia, Azerbaijan, Belarus, Cuba, the Russian Federation, Georgia, Hungary, Kazakhstan, Moldova, Uzbekistan, Kyrgyzstan, Romania, Tajikistan, Turkmenistan and Ukraine, the frequency band 5 670-5 850 MHz is also allocated to the fixed service on a primary basis. (WRC-19)
5.457	In Australia, Burkina Faso, Ivory Coast, Mali and Nigeria, the allocation to the fixed service in the bands 6440-6520 MHz (HAPS-to-ground direction) and 6560-6640 MHz (ground-to-HAPS direction) may also be used by gateway links for high-altitude platform stations (HAPS) within the territory of these countries. Such use is limited to operation in HAPS gateway links and shall not cause harmful interference to, and shall not claim protection from, existing services, and shall be in compliance with Resolution 150 (WRC-12). Existing services shall not be constrained in future development by HAPS gateway links. The use of HAPS gateway links in these bands requires explicit agreement with other administrations whose territories are located within 1000 kilometres from the border of an administration intending to use the HAPS gateway links. (WRC-12)
5.457A	In the frequency bands 5 925-6 425 MHz and 14-14.5 GHz, earth stations located on board vessels may communicate with space stations of the fixed-satellite service. Such use shall be in accordance with Resolution 902 (Rev.WRC-23). In the frequency band 5 925-6 425 MHz, earth stations located on board vessels and communicating with space stations of the fixed-satellite service may employ transmit antennas with minimum diameter of 1.2 m and operate without prior agreement of any administration if located at least 330 km away from the low-water mark as officially recognized by the coastal State. All other provisions of Resolution 902 (Rev.WRC-23) shall apply. (WRC-23)

Footnote Number	Footnote Content
5.457B	In the frequency bands 5 925-6 425 MHz and 14-14.5 GHz, earth stations located on board vessels may operate with the characteristics and under the conditions contained in Resolution 902 (Rev.WRC-23) in Algeria, Saudi Arabia, Bahrain, Comoros, Djibouti, Egypt, United Arab Emirates, Jordan, Kuwait, Libya, Morocco, Mauritania, Oman, Qatar, the Syrian Arab Republic, Sudan, Tunisia and Yemen, in the maritime mobile-satellite service on a secondary basis. Such use shall be in accordance with Resolution 902 (Rev.WRC-23). (WRC-23)
5.457C	In Region 2 (except Brazil, Cuba, French overseas departments and communities, Guatemala, Mexico, Paraguay, Uruguay and Venezuela), the frequency band 5925-6700 MHz may be used for aeronautical mobile telemetry for flight testing by aircraft stations (see No. 1.83). Such use shall be in accordance with Resolution 416 (WRC-07) and shall not cause harmful interference to, or claim protection from, the fixed-satellite and fixed services. Any such use does not preclude the use of this frequency band by other mobile service applications or by other services to which this frequency band is allocated on a co-primary basis and does not establish priority in the Radio Regulations. (WRC-15)
5.457E	The frequency bands 6 425-7 125 MHz in Region 1 and 7 025-7 125 MHz in Region 3 are identified for use by administrations wishing to implement the terrestrial component of International Mobile Telecommunications (IMT). This identification does not preclude the use of these frequency bands by any application of the services to which they are allocated and does not establish priority in the Radio Regulations. Resolution 220 (WRC-23) applies. The frequency bands are also used for the implementation of wireless access systems (WAS), including radio local area networks (RLANs). (WRC-23)
5.458	In the band 6425-7075 MHz, passive microwave sensor measurements are carried out over the oceans. In the band 7075-7250 MHz, passive microwave sensor measurements are carried out. Administrations should bear in mind the needs of the Earth exploration-satellite (passive) and space research (passive) services in their future planning of the bands 6425-7025 MHz and 7075-7250 MHz.
5.458A	In making assignments in the band 6700-7075 MHz to space stations of the fixed-satellite service, administrations are urged to take all practicable steps to protect spectral line observations of the radio astronomy service in the band 6650-6675.2 MHz from harmful interference from unwanted emissions.
5.458B	The space-to-Earth allocation to the fixed-satellite service in the band 6700-7075 MHz is limited to feeder links for non-geostationary satellite systems of the mobile-satellite service and is subject to coordination under No. 9.11A. The use of the band 6700-7075 MHz (space-to-Earth) by feeder links for non-geostationary satellite systems in the mobile-satellite service is not subject to No. 22.2.
5.460	No emissions from space research service (Earth-to-space) systems intended for deep space shall be effected in the frequency band 7190-7235 MHz. Geostationary satellites in the space research service operating in the frequency band 7190-7235 MHz shall not claim protection from existing and future stations of the fixed and mobile services and No. 5.43A does not apply. (WRC-15)
5.460A	The use of the frequency band 7190-7250 MHz (Earth-to-space) by the Earth exploration-satellite service shall be limited to tracking, telemetry and command for the operation of spacecraft. Space stations operating in the Earth exploration-satellite service (Earth-to-space) in the frequency band 7190-7250 MHz shall not claim protection from existing and future stations in the fixed and mobile services, and No. 5.43A does not apply. No. 9.17 applies. Additionally, to ensure protection of the existing and future deployment of fixed and mobile services, the location of earth stations supporting spacecraft in the Earth exploration-satellite service in non-geostationary orbits or geostationary orbit shall maintain a separation distance of at least 10 km and 50 km, respectively, from the respective border(s) of neighbouring countries, unless a shorter distance is otherwise agreed between the corresponding administrations. (WRC-15)
5.460B	Space stations on the geostationary orbit operating in the Earth exploration-satellite service (Earth-to-space) in the frequency band 7190-7235 MHz shall not claim protection from existing and future stations of the space research service, and No. 5.43A does not apply. (WRC-15)
5.461	Additional allocation: the frequency bands 7 250-7 375 MHz (space-to-Earth) and 7 900-8 025 MHz (Earth-to-space) are also allocated to the mobile-satellite service on a primary basis, subject to agreement obtained under No. 9.21, with the exception that No. 9.21 shall not apply to the geostationary-satellite networks in the mobile-satellite service for which complete coordination information is received by the Bureau as of 1 January 2025 with respect to non-geostationary-satellite systems for which complete coordination or notification information, according to the case, is received by the Bureau as of 1 January 2025. Non-geostationary-satellite systems for which complete coordination or notification information, according to the case, is received by the Bureau as of 1 January 2025 shall not cause unacceptable interference to and shall not claim protection from geostationary-satellite networks in the mobile-satellite service operating in accordance with these Regulations. No. 5.43A does not apply. (WRC-23)
5.461A	The use of the band 7450-7550 MHz by the meteorological-satellite service (space-to-Earth) is limited to geostationary-satellite systems. Non-geostationary meteorological-satellite systems in this band notified before 30 November 1997 may continue to operate on a primary basis until the end of their lifetime. (WRC-97)
5.461AA	The use of the frequency band 7375-7750 MHz by the maritime mobile-satellite service is limited to geostationary-satellite networks. (WRC-15)
5.461AB	In the frequency band 7375-7750 MHz, earth stations in the maritime mobile-satellite service shall not claim protection from, nor constrain the use and development of, stations in the fixed and mobile, except aeronautical mobile, services. No. 5.43A does not apply. (WRC-15)

Footnote Number	Footnote Content
5.461AC	In the frequency band 7 375-7 750 MHz, non-geostationary-satellite systems operating in the fixed-satellite service for which complete coordination or notification information, according to the case, is received by the Bureau as of 1 January 2025 shall not cause unacceptable interference to and shall not claim protection from geostationary-satellite networks in the maritime mobile-satellite service operating in accordance with these Regulations. No. 5.43A does not apply. (WRC-23)
5.461B	The use of the band 7750-7900 MHz by the meteorological-satellite service (space-to-Earth) is limited to non-geostationary satellite systems. (WRC-12)
5.462A	In Regions 1 and 3 (except for Japan), in the band 8025-8400 MHz, the Earth exploration-satellite service using geostationary satellites shall not produce a power flux-density in excess of the following provisional values for angles of arrival (θ), without the consent of the affected administration: - 135 dB(W/m ²) in a 1 MHz band for $0^\circ \leq \theta < 5^\circ$ - 135 + 0.5 ($\theta - 5$) dB(W/m ²) in a 1 MHz band for $5^\circ \leq \theta < 25^\circ$ - 125 dB(W/m ²) in a 1 MHz band for $25^\circ \leq \theta < 90^\circ$ (WRC-12)
5.463	Aircraft stations are not permitted to transmit in the band 8025-8400 MHz. (WRC-97)
5.465	In the space research service, the use of the band 8400-8450 MHz is limited to deep space.
5.466	Different category of service: in Singapore and Sri Lanka, the allocation of the band 8400-8500 MHz to the space research service is on a secondary basis (see No. 5.32). (WRC-12)
5.468	Additional allocation: in Saudi Arabia, Bahrain, Bangladesh, Brunei Darussalam, Burundi, Cameroon, China, Congo (Rep. of the), Djibouti, Egypt, the United Arab Emirates, Eswatini, Gabon, Guyana, Indonesia, Iran (Islamic Republic of), Iraq, Jamaica, Jordan, Kenya, Kuwait, Lebanon, Libya, Malaysia, Mali, Morocco, Mauritania, Nepal, Nigeria, Oman, Uganda, Pakistan, Qatar, Syrian Arab Republic, the Dem. People's Rep. of Korea, Senegal, Singapore, Somalia, Sudan, Chad, Togo, Tunisia and Yemen, the frequency band 8 500-8 750 MHz is also allocated to the fixed and mobile services on a primary basis. (WRC-19)
5.469	Additional allocation: in Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Hungary, Lithuania, Uzbekistan, Poland, Kyrgyzstan, the Czech Rep., Romania, Tajikistan, Turkmenistan and Ukraine, the frequency band 8 500-8 750 MHz is also allocated to the land mobile and radionavigation services on a primary basis. (WRC-23)
5.469A	In the band 8550-8650 MHz, stations in the earth exploration-satellite service (active) and space research service (active) shall not cause harmful interference to, or constrain the use and development of, stations of the radiolocation service. (WRC-97)
5.470	The use of the band 8750-8850 MHz by the aeronautical radionavigation service is limited to airborne Doppler navigation aids on a centre frequency of 8800 MHz.
5.471	Additional allocation: in Algeria, Germany, Bahrain, Belgium, China, Egypt, the United Arab Emirates, France, Greece, Indonesia, Iran (Islamic Republic of), Libya, the Netherlands, Qatar, and Sudan, the frequency bands 8825-8850 MHz and 9000-9200 MHz are also allocated to the maritime radionavigation service, on a primary basis, for use by shore-based radars only. (WRC-15)
5.472	In the bands 8850-9000 MHz and 9200-9225 MHz, the maritime radionavigation service is limited to shore-based radars.
5.473	Additional allocation: in Armenia, Austria, Azerbaijan, Belarus, Cuba, the Russian Federation, Georgia, Hungary, Uzbekistan, Poland, Kyrgyzstan, Romania, Tajikistan, Turkmenistan and Ukraine, the frequency bands 8 850-9 000 MHz and 9 200-9 300 MHz are also allocated to the radionavigation service on a primary basis. (WRC-19)
5.473A	In the band 9000-9200 MHz, stations operating in the radiolocation service shall not cause harmful interference to, nor claim protection from, systems identified in No. 5.337 operating in the aeronautical radionavigation service, or radar systems in the maritime radionavigation service operating in this band on a primary basis in the countries listed in No. 5.471. (WRC-07)
5.474	In the band 9200-9500 MHz, search and rescue transponders (SART) may be used, having due regard to the appropriate ITU-R Recommendation (see also Article 31).
5.474A	The use of the frequency bands 9 200-9 300 MHz and 9 900-10 400 MHz by the Earth exploration-satellite service (active) is limited to systems requiring necessary bandwidth greater than 600 MHz that cannot be fully accommodated within the frequency band 9 300-9 900 MHz. Such use is subject to agreement to be obtained under No. 9.21 from Algeria, Saudi Arabia, Bahrain, Egypt, Indonesia, Iran (Islamic Republic of), Lebanon and Tunisia. An administration that has not replied under No. 9.52 is considered as not having agreed to the coordination request. In this case, the notifying administration of the satellite system operating in the Earth exploration-satellite service (active) may request the assistance of the Bureau under Sub-Section IID of Article 9. (WRC-15)
5.474B	Stations operating in the Earth exploration-satellite (active) service shall comply with Recommendation ITU-R RS.2066-0. (WRC-15)
5.474C	Stations operating in the Earth exploration-satellite (active) service shall comply with Recommendation ITU-R RS.2065-0. (WRC-15)
5.474D	Stations operating in the Earth exploration-satellite service (active) shall not cause harmful interference to, or claim protection from, stations of the maritime radionavigation and radiolocation services in the frequency band 9 200-9 300 MHz, the radionavigation and radiolocation services in the frequency band 9 900-10 000 MHz and the radiolocation service in the frequency band 10.0-10.4 GHz. (WRC-15)

Footnote Number	Footnote Content
5.475	The use of the band 9300-9500 MHz by the aeronautical radionavigation service is limited to airborne weather radars and ground-based radars. In addition, ground-based radar beacons in the aeronautical radionavigation service are permitted in the band 9300-9320 MHz on condition that harmful interference is not caused to the maritime radionavigation service. (WRC-07)
5.475A	The use of the band 9300-9500 MHz by the Earth exploration-satellite service (active) and the space research service (active) is limited to systems requiring necessary bandwidth greater than 300 MHz that cannot be fully accommodated within the 9500-9800 MHz band. (WRC-07)
5.475B	In the band 9300-9500 MHz, stations operating in the radiolocation service shall not cause harmful interference to, nor claim protection from, radars operating in the radionavigation service in conformity with the Radio Regulations. Ground-based radars used for meteorological purposes have priority over other radiolocation uses. (WRC-07)
5.476A	In the band 9300-9800 MHz, stations in the Earth exploration-satellite service (active) and space research service (active) shall not cause harmful interference to, nor claim protection from, stations of the radionavigation and radiolocation services. (WRC-07)
5.477	Different category of service: in Algeria, Saudi Arabia, Bahrain, Bangladesh, Brunei Darussalam, Cameroon, Djibouti, Egypt, the United Arab Emirates, Eritrea, Ethiopia, Guyana, India, Indonesia, Iran (Islamic Republic of), Iraq, Jamaica, Japan, Jordan, Kuwait, Lebanon, Liberia, Malaysia, Nigeria, Oman, Pakistan, Qatar, Syrian Arab Republic, the Dem. People's Rep. of Korea, Singapore, Somalia, Sudan, South Sudan, Trinidad and Tobago, Uganda and Yemen, the allocation of the band 9800-10000 MHz to the fixed service is on a primary basis (see No. 5.33). (WRC-15)
5.478	Additional allocation: in Azerbaijan, Kyrgyzstan, Romania, Turkmenistan and Ukraine, the frequency band 9 800-10 000 MHz is also allocated to the radionavigation service on a primary basis. (WRC-19)
5.478A	The use of the band 9800-9900 MHz by the Earth exploration-satellite service (active) and space research service (active) is limited to systems requiring necessary bandwidth greater than 500 MHz that cannot be fully accommodated within the 9300-9800 MHz band. (WRC-07)
5.478B	In the band 9800-9900 MHz, stations in the Earth exploration-satellite service (active) and the space research service (active) shall not cause harmful interference to, nor claim protection from stations of the fixed service to which this band is allocated on a secondary basis. (WRC-12)
5.479	The band 9975-10025 MHz is also allocated to the meteorological-satellite service on a secondary basis for use by weather radars.
5.481	Additional allocation: in Algeria, Germany, Angola, Brazil, China, Colombia, Costa Rica, Ivory Coast, Cuba, Djibouti, the Dominican Republic, Egypt, El Salvador, Ecuador, Spain, Guatemala, Hungary, Jamaica, Japan, Kenya, Morocco, Mexico, Nigeria, Oman, Uzbekistan, Pakistan, Palestine*, Paraguay, Peru, the Dem. People's Rep. of Korea, Romania, Somalia, Suriname, Tunisia and Uruguay, the frequency band 10.45-10.5 GHz is also allocated to the fixed and mobile services on a primary basis. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference, and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995.
5.482	In the band 10.6-10.68 GHz, the power delivered to the antenna of stations of the fixed and mobile, except aeronautical mobile, services shall not exceed -3 dBW. This limit may be exceeded, subject to agreement obtained under No. 9.21. However, in Algeria, Saudi Arabia, Armenia, Azerbaijan, Bahrain, Bangladesh, Belarus, Egypt, United Arab Emirates, Georgia, India, Indonesia, Iran (Islamic Republic of), Iraq, Jordan, Libyan Arab Jamahiriya, Kazakhstan, Kuwait, Lebanon, Morocco, Mauritania, Moldova, Nigeria, Oman, Uzbekistan, Pakistan, Philippines, Qatar, Syrian Arab Republic, Kyrgyzstan, Singapore, Tajikistan, Tunisia, Turkmenistan and Viet Nam, this restriction on the fixed and mobile, except aeronautical mobile, service is not applicable. (WRC-07)
5.482A	For sharing of the band 10.6-10.68 GHz between the Earth exploration-satellite (passive) service and the fixed and mobile, except aeronautical mobile, services, Resolution 751 (WRC-07) applies. (WRC-07)
5.483	Additional allocation: in Saudi Arabia, Armenia, Azerbaijan, Bahrain, Belarus, China, Colombia, Korea (Rep. of), Egypt, the United Arab Emirates, Georgia, Iran (Islamic Republic of), Iraq, Israel, Jordan, Kazakhstan, Kuwait, Lebanon, Mongolia, Qatar, Kyrgyzstan, the Dem. People's Rep. of Korea, Tajikistan, Turkmenistan and Yemen, the frequency band 10.68-10.7 GHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. Such use is limited to equipment in operation by 1 January 1985. (WRC-19)
5.484	In Region 1, the use of the band 10.7-11.7 GHz by the fixed-satellite service (Earth-to-space) is limited to feeder links for the broadcasting-satellite service.
5.484A	The use of the frequency bands 10.95-11.2 GHz (space-to-Earth), 11.45-11.7 GHz (space-to-Earth), 11.7-12.2 GHz (space-to-Earth) in Region 2, 12.2-12.75 GHz (space-to-Earth) in Region 3, 12.5-12.75 GHz (space-to-Earth) in Region 1, 13.75-14.5 GHz (Earth-to-space), 17.3-17.7 GHz (space-to-Earth) in Region 2, 17.8-18.6 GHz (space-to-Earth), 19.7-20.2 GHz (space-to-Earth), 27.5-28.6 GHz (Earth-to-space), 29.5-30 GHz (Earth-to-space) by a non-geostationary-satellite system in the fixed-satellite service is subject to application of the provisions of No. 9.12 for coordination with other non-geostationary-satellite systems in the fixed-satellite service. Non-geostationary-satellite systems in the fixed-satellite service shall not claim protection from geostationary-satellite networks in the fixed-satellite service operating in accordance with the Radio Regulations, irrespective of the dates of receipt by the Bureau of the complete coordination or notification information, as appropriate, for the non-geostationary-satellite systems in the fixed-satellite service and of the complete coordination or notification information, as appropriate, for the geostationary-satellite networks, and No. 5.43A does not apply. Non-geostationary-satellite systems in the fixed-satellite service in the above bands shall be operated in such a way that any unacceptable interference that may occur during their operation shall be rapidly eliminated. In Region 2, No. 22.2 shall continue to apply in the frequency band 17.3-17.7 GHz. (WRC-23)

Footnote Number	Footnote Content
5.484B	Resolution 155 (WRC-15) shall apply. (WRC-15)
5.487	In the band 11.7-12.5 GHz in Regions 1 and 3, the fixed, fixed-satellite, mobile, except aeronautical mobile, and broadcasting services, in accordance with their respective allocations, shall not cause harmful interference to, or claim protection from, broadcasting-satellite stations operating in accordance with the Regions 1 and 3 Plan in Appendix 30. (WRC-03)
5.487A	Additional allocation: in Region 1, the band 11.7-12.5 GHz, in Region 2, the band 12.2-12.7 GHz and, in Region 3, the band 11.7-12.2 GHz, are also allocated to the fixed-satellite service (space-to-Earth) on a primary basis, limited to non geostationary systems and subject to application of the provisions of No. 9.12 for coordination with other non-geostationary-satellite systems in the fixed-satellite service. Non-geostationary-satellite systems in the fixed satellite service shall not claim protection from geostationary-satellite networks in the broadcasting-satellite service operating in accordance with the Radio Regulations, irrespective of the dates of receipt by the Bureau of the complete coordination or notification information, as appropriate, for the non-geostationary-satellite systems in the fixed-satellite service and of the complete coordination or notification information, as appropriate, for the geostationary-satellite networks, and No. 5.43A does not apply. Non-geostationary-satellite systems in the fixed-satellite service in the above bands shall be operated in such a way that any unacceptable interference that may occur during their operation shall be rapidly eliminated. (WRC-03)
5.492	Assignments to stations of the broadcasting-satellite service which are in conformity with the appropriate regional Plan or included in the Regions 1 and 3 List in Appendix 30 may also be used for transmissions in the fixed-satellite service (space-to-Earth), provided that such transmissions do not cause more interference, or require more protection from interference, than the broadcasting-satellite service transmissions operating in conformity with the Plan or the List, as appropriate. (WRC-2000)
5.494	Additional allocation: in Algeria, Saudi Arabia, Bahrain, Cameroon, the Central African Rep., Congo (Rep. of the), Ivory Coast, Djibouti, Egypt, the United Arab Emirates, Eritrea, Ethiopia, Gabon, Ghana, Guinea, Iraq, Israel, Jordan, Kuwait, Lebanon, Libya, Madagascar, Mali, Morocco, Mongolia, Nigeria, Oman, Palestine*, Qatar, the Syrian Arab Republic, the Dem. Rep. of the Congo, Somalia, Sudan, South Sudan, Chad, Togo and Yemen, the frequency band 12.5-12.75 GHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference, and taking into account the Israeli Palestinian Interim Agreement of 28 September 1995.
5.495	Additional allocation: in Greece, Monaco, Montenegro, Uganda and Tunisia, the frequency band 12.5- 12.75 GHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a secondary basis. (WRC-19)
5.496	Additional allocation: in Austria, Azerbaijan, Kyrgyzstan and Turkmenistan, the band 12.5-12.75 GHz is also allocated to the fixed service and the mobile, except aeronautical mobile, service on a primary basis. However, stations in these services shall not cause harmful interference to fixed-satellite service earth stations of countries in Region 1 other than those listed in this footnote. Coordination of these earth stations is not required with stations of the fixed and mobile services of the countries listed in this footnote. The power flux-density limit at the Earth's surface given in Table 21-4 of Article 21, for the fixed-satellite service shall apply on the territory of the countries listed in this footnote. (WRC-2000)
5.496A	The frequency band 12.75-13.25 GHz (Earth-to-space) may be used by earth stations in motion, limited to earth stations on aircraft and vessels, communicating with geostationary space stations in the fixed-satellite service. Resolution 121 (WRC-23) shall apply. (WRC-23)
5.497	The use of the band 13.25-13.4 GHz by the aeronautical radionavigation service is limited to Doppler navigation aids.
5.498A	The Earth exploration-satellite (active) and space research (active) services operating in the band 13.25-13.4 GHz shall not cause harmful interference to, or constrain the use and development of, the aeronautical radionavigation service. (WRC-97)
5.499	Additional allocation: in Bangladesh and India, the band 13.25-14 GHz is also allocated to the fixed service on a primary basis. In Pakistan, the band 13.25-13.75 GHz is allocated to the fixed service on a primary basis. (WRC-12)
5.499A	The use of the frequency band 13.4-13.65 GHz by the fixed-satellite service (space-to-Earth) is limited to geostationary-satellite systems and is subject to agreement obtained under No. 9.21 with respect to satellite systems operating in the space research service (space-to-space) to relay data from space stations in the geostationary-satellite orbit to associated space stations in non-geostationary satellite orbits for which advance publication information has been received by the Bureau by 27 November 2015. (WRC-15)
5.499B	Administrations shall not preclude the deployment and operation of transmitting earth stations in the standard frequency and time signal-satellite service (Earth-to-space) allocated on a secondary basis in the frequency band 13.4-13.65 GHz due to the primary allocation to FSS (space-to-Earth). (WRC-15)
5.499C	The allocation of the frequency band 13.4-13.65 GHz to the space research service on a primary basis is limited to: - satellite systems operating in the space research service (space-to-space) to relay data from space stations in the geostationary-satellite orbit to associated space stations in non-geostationary satellite orbits for which advance publication information has been received by the Bureau by 27 November 2015, - active spaceborne sensors, - satellite systems operating in the space research service (space-to-Earth) to relay data from space stations in the geostationary-satellite orbit to associated earth stations. Other uses of the band by the space research service are on a secondary basis. (WRC-15)

Footnote Number	Footnote Content
5.499D	In the frequency band 13.4-13.65 GHz, satellite systems in the space research service (space-to-Earth) and/or the space research service (space-to-space) shall not cause harmful interference to, nor claim protection from, stations in the fixed, mobile, radiolocation and Earth exploration-satellite (active) services. (WRC-15)
5.499E	In the frequency band 13.4-13.65 GHz, geostationary-satellite networks in the fixed-satellite service (space-to-Earth) shall not claim protection from space stations in the Earth exploration-satellite service (active) operating in accordance with these Regulations, and No. 5.43A does not apply. The provisions of No. 22.2 do not apply to the Earth exploration-satellite service (active) with respect to the fixed-satellite service (space-to-Earth) in this band. (WRC-15)
5.500	Additional allocation: in Algeria, Saudi Arabia, Bahrain, Brunei Darussalam, Cameroon, Djibouti, Egypt, the United Arab Emirates, Gabon, Indonesia, Iran (Islamic Republic of), Iraq, Israel, Jordan, Kuwait, Lebanon, Madagascar, Malaysia, Mali, Morocco, Mauritania, Niger, Nigeria, Oman, Qatar, the Syrian Arab Republic, Singapore, Somalia, Sudan, South Sudan, Chad and Tunisia, the frequency band 13.4-14 GHz is also allocated to the fixed and mobile services on a primary basis. In Pakistan, the frequency band 13.4-13.75 GHz is also allocated to the fixed and mobile services on a primary basis. (WRC-23)
5.501	Additional allocation: in Hungary, Japan, Kyrgyzstan, Romania and Turkmenistan, the frequency band 13.4-14 GHz is also allocated to the radionavigation service on a primary basis. (WRC-23)
5.501A	The allocation of the frequency band 13.65-13.75 GHz to the space research service on a primary basis is limited to active spaceborne sensors. Other uses of the frequency band by the space research service are on a secondary basis. (WRC-15)
5.501B	In the band 13.4-13.75 GHz, the earth exploration-satellite (active) and space research (active) services shall not cause harmful interference to, or constrain the use and development of, the radiolocation service. (WRC-97)
5.502	In the band 13.75-14 GHz, an earth station of a geostationary fixed-satellite service network shall have a minimum antenna diameter of 1.2 m and an earth station of a non-geostationary fixed-satellite service system shall have a minimum antenna diameter of 4.5 m. In addition, the e.i.r.p., averaged over one second, radiated by a station in the radiolocation or radionavigation services shall not exceed 59 dBW for elevation angles above 2° and 65 dBW at lower angles. Before an administration brings into use an earth station in a geostationary-satellite network in the fixed-satellite service in this band with an antenna size smaller than 4.5 m, it shall ensure that the power flux-density produced by this earth station does not exceed: - 115 dB(W/(m ² · 10 MHz)) for more than 1% of the time produced at 36 m above sea level at the low water mark, as officially recognized by the coastal state; - 115 dB(W/(m ² · 10 MHz)) for more than 1% of the time produced 3 m above ground at the border of the territory of an administration deploying or planning to deploy land mobile radars in this band, unless prior agreement has been obtained. For earth stations within the fixed-satellite service having an antenna diameter greater than or equal to 4.5 m, the e.i.r.p. of any emission should be at least 68 dBW and should not exceed 85 dBW. (WRC-03)
5.503	In the band 13.75-14 GHz, geostationary space stations in the space research service for which information for advance publication has been received by the Bureau prior to 31 January 1992 shall operate on an equal basis with stations in the fixed-satellite service; after that date, new geostationary space stations in the space research service will operate on a secondary basis. Until those geostationary space stations in the space research service for which information for advance publication has been received by the Bureau prior to 31 January 1992 cease to operate in this band: - in the band 13.77-13.78 GHz, the e.i.r.p. density of emissions from any earth station in the fixed-satellite service operating with a space station in geostationary-satellite orbit shall not exceed: i) 4.7D + 28 dB(W/40 kHz), where D is the fixed-satellite service earth station antenna diameter (m) for antenna diameters equal to or greater than 1.2 m and less than 4.5 m; ii) 49.2 + 20 log(D/4.5) dB(W/40 kHz), where D is the fixed-satellite service earth station antenna diameter (m) for antenna diameters equal to or greater than 4.5 m and less than 31.9 m; iii) 66.2 dB(W/40 kHz) for any fixed-satellite service earth station for antenna diameters (m) equal to or greater than 31.9 m; iv) 56.2 dB(W/4 kHz) for narrow-band (less than 40 kHz of necessary bandwidth) fixed-satellite service earth station emissions from any fixed-satellite service earth station having an antenna diameter of 4.5 m or greater; - the e.i.r.p. density of emissions from any earth station in the fixed-satellite service operating with a space station in non-geostationary-satellite orbit shall not exceed 51 dBW in the 6 MHz band from 13.772 to 13.778 GHz. Automatic power control may be used to increase the e.i.r.p. density in these frequency ranges to compensate for rain attenuation, to the extent that the power flux-density at the fixed-satellite service space station does not exceed the value resulting from use by an earth station of an e.i.r.p. meeting the above limits in clear-sky conditions. (WRC-03)
5.504	The use of the band 14-14.3 GHz by the radionavigation service shall be such as to provide sufficient protection to space stations of the fixed-satellite service.
5.504A	In the band 14-14.5 GHz, aircraft earth stations in the secondary aeronautical mobile-satellite service may also communicate with space stations in the fixed-satellite service. The provisions of Nos. 5.29, 5.30 and 5.31 apply. (WRC-03)
5.504B	Aircraft earth stations operating in the aeronautical mobile-satellite service in the frequency band 14-14.5 GHz shall comply with the provisions of Annex 1, Part C of Recommendation ITU-R M.1643-0, with respect to any radio astronomy station performing observations in the 14.47-14.5 GHz frequency band located on the territory of Spain, France, India, Italy, the United Kingdom and South Africa. (WRC-15)

Footnote Number	Footnote Content
5.504C	In the frequency band 14-14.25 GHz, the power flux-density produced on the territory of the countries of Saudi Arabia, Bahrain, Botswana, Ivory Coast, Egypt, Guinea, India, Iran (Islamic Republic of), Kuwait, Nigeria, Oman, the Syrian Arab Republic and Tunisia by any aircraft earth station in the aeronautical mobile-satellite service shall not exceed the limits given in Annex 1, Part B of Recommendation ITU-R M.1643-0, unless otherwise specifically agreed by the affected administration(s). The provisions of this footnote in no way derogate the obligations of the aeronautical mobile-satellite service to operate as a secondary service in accordance with No. 5.29. (WRC-15)
5.505	Additional allocation: in Algeria, Saudi Arabia, Bahrain, Botswana, Brunei Darussalam, Cameroon, China, Congo (Rep. of the), Korea (Rep. of), Djibouti, Egypt, the United Arab Emirates, Eswatini, Gabon, Guinea, India, Indonesia, Iran (Islamic Republic of), Iraq, Israel, Japan, Jordan, Kuwait, Lebanon, Malaysia, Mali, Morocco, Mauritania, Oman, the Philippines, Qatar, the Syrian Arab Republic, the Dem. People's Rep. of Korea, Singapore, Somalia, Sudan, South Sudan, Chad, Viet Nam and Yemen, the frequency band 14-14.3 GHz is also allocated to the fixed service on a primary basis. (WRC-19)
5.506	The band 14-14.5 GHz may be used, within the fixed-satellite service (Earth-to-space), for feeder links for the broadcasting-satellite service, subject to coordination with other networks in the fixed-satellite service. Such use of feeder links is reserved for countries outside Europe.
5.506A	In the frequency band 14-14.5 GHz, ship earth stations with an equivalent isotropically radiated power (e.i.r.p.) greater than 21 dBW shall operate under the same conditions as earth stations located on board vessels, as provided in Resolution 902 (Rev.WRC-23). This footnote shall not apply to ship earth stations for which the complete Appendix 4 information has been received by the Bureau prior to 5 July 2003. (WRC-23)
5.506B	Earth stations located on board vessels communicating with space stations in the fixed-satellite service may operate in the frequency band 14-14.5 GHz without the need for prior agreement from Cyprus and Malta, within the minimum distance given in Resolution 902 (Rev.WRC-23) from these countries. (WRC-23)
5.508	Additional allocation: in Germany, Italy, Libya, North Macedonia and the United Kingdom, the frequency band 14.25-14.3 GHz is also allocated to the fixed service on a primary basis. (WRC-23)
5.508A	In the frequency band 14.25-14.3 GHz, the power flux-density produced on the territory of the countries of Saudi Arabia, Bahrain, Botswana, China, Ivory Coast, Egypt, Guinea, India, Iran (Islamic Republic of), Italy, Kuwait, Nigeria, Oman, the Syrian Arab Republic, the United Kingdom and Tunisia by any aircraft earth station in the aeronautical mobile-satellite service shall not exceed the limits given in Annex 1, Part B of Recommendation ITU-R M.1643-0, unless otherwise specifically agreed by the affected administration(s). The provisions of this footnote in no way derogate the obligations of the aeronautical mobile-satellite service to operate as a secondary service in accordance with No. 5.29. (WRC-23)
5.509A	In the frequency band 14.3-14.5 GHz, the power flux-density produced on the territory of the countries of Saudi Arabia, Bahrain, Botswana, Cameroon, China, Ivory Coast, Egypt, Gabon, Guinea, India, Iran (Islamic Republic of), Italy, Kuwait, Morocco, Nigeria, Oman, the Syrian Arab Republic, the United Kingdom, Sri Lanka, Tunisia and Viet Nam by any aircraft earth station in the aeronautical mobile-satellite service shall not exceed the limits given in Annex 1, Part B of Recommendation ITU-R M.1643-0, unless otherwise specifically agreed by the affected administration(s). The provisions of this footnote in no way derogate the obligations of the aeronautical mobile-satellite service to operate as a secondary service in accordance with No. 5.29. (WRC-23)
5.509B	The use of the frequency bands 14.5-14.75 GHz in countries listed in Resolution 163 (WRC-15) and 14.5-14.8 GHz in countries listed in Resolution 164 (WRC-15) by the fixed-satellite service (Earth-to-space) not for feeder links for the broadcasting-satellite service is limited to geostationary-satellites. (WRC-15)
5.509C	For the use of the frequency bands 14.5-14.75 GHz in countries listed in Resolution 163 (WRC-15) and 14.5-14.8 GHz in countries listed in Resolution 164 (WRC-15) by the fixed-satellite service (Earth-to-space) not for feeder links for the broadcasting-satellite service, the fixed-satellite service earth stations shall have a minimum antenna diameter of 6 m and a maximum power spectral density of -44.5 dBW/Hz at the input of the antenna. The earth stations shall be notified at known locations on land. (WRC-15)
5.509D	Before an administration brings into use an earth station in the fixed-satellite service (Earth-to-space) not for feeder links for the broadcasting-satellite service in the frequency bands 14.5-14.75 GHz (in countries listed in Resolution 163 (WRC-15)) and 14.5-14.8 GHz (in countries listed in Resolution 164 (WRC-15)), it shall ensure that the power flux-density produced by this earth station does not exceed -151.5 dB(W/(m ² · 4 kHz)) produced at all altitudes from 0 m to 19 000 m above sea level at 22 km seaward from all coasts, defined as the low-water mark, as officially recognized by each coastal State. (WRC-15)
5.509E	In the frequency bands 14.50-14.75 GHz in countries listed in Resolution 163 (WRC-15) and 14.50-14.8 GHz in countries listed in Resolution 164 (WRC-15), the location of earth stations in the fixed-satellite service (Earth-to-space) not for feeder links for the broadcasting-satellite service shall maintain a separation distance of at least 500 km from the border(s) of other countries unless shorter distances are explicitly agreed by those administrations. No. 9.17 does not apply. When applying this provision, administrations should consider the relevant parts of these Regulations and the latest relevant ITU-R Recommendations. (WRC-15)

Footnote Number	Footnote Content
5.509F	In the frequency bands 14.50-14.75 GHz in countries listed in Resolution 163 (WRC-15) and 14.50-14.8 GHz in countries listed in Resolution 164 (WRC-15), earth stations in the fixed-satellite service (Earth-to-space) not for feeder links for the broadcasting-satellite service shall not constrain the future deployment of the fixed and mobile services. (WRC-15)
5.509G	The frequency band 14.5-14.8 GHz is also allocated to the space research service on a primary basis. However, such use is limited to the satellite systems operating in the space research service (Earth-to-space) to relay data to space stations in the geostationary-satellite orbit from associated earth stations. Stations in the space research service shall not cause harmful interference to, or claim protection from, stations in the fixed and mobile services and in the fixed-satellite service limited to feeder links for the broadcasting-satellite service and associated space operations functions using the guardbands under Appendix 30A and feeder links for the broadcasting-satellite service in Region 2. Other uses of this frequency band by the space research service are on a secondary basis. (WRC-15)
5.510	Except for use in accordance with Resolution 163 (WRC-15) and Resolution 164 (WRC-15), the use of the frequency band 14.5-14.8 GHz by the fixed-satellite service (Earth-to-space) is limited to feeder links for the broadcasting-satellite service. This use is reserved for countries outside Europe. Uses other than feeder links for the broadcasting-satellite service are not authorized in Regions 1 and 2 in the frequency band 14.75-14.8 GHz. (WRC-15)
5.510A	The allocation of the frequency band 14.8-15.35 GHz to the space research service on a primary basis is limited to satellite systems operating in the space-to-space, space-to-Earth and Earth-to-space directions at distances from the Earth of less than 2×10^6 km in accordance with Resolution 678 (WRC-23). Other uses of the frequency band by the space research service are on a secondary basis. The use of the frequency band 14.8-15.35 GHz by the space research service (space-to-Earth) (Earth-to-space) is on a secondary basis with respect to the terrestrial services in Algeria, Saudi Arabia, Bahrain, Korea (Rep. of), Egypt, the United Arab Emirates, the United States, India, Iraq, Japan, Kuwait, Libya, Morocco, Mauritania, Oman, Qatar, the Syrian Arab Republic, Tunisia and Yemen. (WRC-23)
5.511	Additional allocation: in Saudi Arabia, Bahrain, Cameroon, Djibouti, Egypt, the United Arab Emirates, Guinea, Iran (Islamic Republic of), Iraq, Israel, Kuwait, Lebanon, Oman, Pakistan, Qatar, the Syrian Arab Republic and Somalia, the frequency band 15.35-15.4 GHz is also allocated to the fixed and mobile services on a secondary basis. (WRC-23)
5.511A	Use of the band 15.43-15.63 GHz by the fixed-satellite service (Earth-to-space) is limited to feeder links of non-geostationary systems in the mobile-satellite service, subject to coordination under No. 9.11A. (WRC-15)
5.511C	Stations operating in the aeronautical radionavigation service shall limit the effective e.i.r.p. in accordance with Recommendation ITU-R S.1340-0. The minimum coordination distance required to protect the aeronautical radionavigation stations (No. 4.10 applies) from harmful interference from feeder link earth stations and the maximum e.i.r.p. transmitted towards the local horizontal plane by a feeder link earth station shall be in accordance with Recommendation ITU-R S.1340-0. (WRC-15)
5.511E	In the frequency band 15.4-15.7 GHz, stations operating in the radiolocation service shall not cause harmful interference to, or claim protection from, stations operating in the aeronautical radionavigation service. (WRC-12)
5.511F	In order to protect the radio astronomy service in the frequency band 15.35-15.4 GHz, radiolocation stations operating in the frequency band 15.4-15.7 GHz shall not exceed the power flux-density level of -156 dB(W/m ²) in a 50 MHz bandwidth in the frequency band 15.35-15.4 GHz, at any radio astronomy observatory site for more than 2 per cent of the time. (WRC-12)
5.511G	Stations in the aeronautical mobile (OR) service operating in the frequency band 15.41-15.7 GHz shall not cause harmful interference to the radio astronomy service operating in the frequency band 15.35-15.4 GHz. The aggregate power flux-density (pfd) received from stations in the aeronautical mobile (OR) service operating in the frequency band 15.41-15.7 GHz at any radio astronomy station operating in the frequency band 15.35-15.4 GHz shall be in compliance with the protection criteria provided in Recommendations ITU-R RA.769-2 and ITU-R RA.1513-2, unless specifically agreed by the affected administration(s). (WRC-23)
5.512	Additional allocation: in Algeria, Saudi Arabia, Austria, Bahrain, Bangladesh, Brunei Darussalam, Cameroon, Congo (Rep. of the), Egypt, El Salvador, the United Arab Emirates, Eritrea, Finland, Guatemala, India, Indonesia, Iran (Islamic Republic of), Jordan, Kenya, Kuwait, Lebanon, Libya, Malaysia, Mali, Morocco, Mauritania, Montenegro, Nepal, Nicaragua, Niger, Oman, Pakistan, Qatar, Syrian Arab Republic, the Dem. Rep. of the Congo, Singapore, Somalia, Sudan, South Sudan, Chad, Togo and Yemen, the frequency band 15.7-17.3 GHz is also allocated to the fixed and mobile services on a primary basis. (WRC-15)
5.513	Additional allocation: in Israel, the band 15.7-17.3 GHz is also allocated to the fixed and mobile services on a primary basis. These services shall not claim protection from or cause harmful interference to services operating in accordance with the Table in countries other than those included in No. 5512.
5.513A	Spaceborne active sensors operating in the band 17.2-17.3 GHz shall not cause harmful interference to, or constrain the development of, the radiolocation and other services allocated on a primary basis. (WRC-97)

Footnote Number	Footnote Content
5.514	Additional allocation: in Algeria, Saudi Arabia, Bahrain, Bangladesh, Cameroon, Djibouti, El Salvador, the United Arab Emirates, Guatemala, India, Iran (Islamic Republic of), Iraq, Israel, Italy, Japan, Jordan, Kuwait, Libya, Lithuania, Nepal, Nicaragua, Nigeria, Oman, Uzbekistan, Pakistan, Qatar, Kyrgyzstan, Somalia, Sudan and South Sudan, the frequency band 17.3-17.7 GHz is also allocated to the fixed and mobile services on a secondary basis. The power limits given in Nos. 21.3 and 21.5 shall apply. (WRC-23)
5.516	The use of the band 17.3-18.1 GHz by geostationary-satellite systems in the fixed-satellite service (Earth-to-space) is limited to feeder links for the broadcasting-satellite service. The use of the band 17.3-17.8 GHz in Region 2 by systems in the fixed-satellite service (Earth-to-space) is limited to geostationary satellites. For the use of the band 17.3-17.8 GHz in Region 2 by feeder links for the broadcasting satellite service in the band 12.2-12.7 GHz, see Article 11. The use of the bands 17.3-18.1 GHz (Earth-to-space) in Regions 1 and 3 and 17.8-18.1 GHz (Earth-to-space) in Region 2 by non geostationary-satellite systems in the fixed-satellite service is subject to application of the provisions of No. 9.12 for coordination with other non-geostationary-satellite systems in the fixed-satellite service. Non geostationary-satellite systems in the fixed satellite service shall not claim protection from geostationary-satellite networks in the fixed-satellite service operating in accordance with the Radio Regulations, irrespective of the dates of receipt by the Bureau of the complete coordination or notification information, as appropriate, for the non-geostationary-satellite systems in the fixed-satellite service and of the complete coordination or notification information, as appropriate, for the geostationary-satellite networks, and No. 5.43A does not apply. Non-geostationary-satellite systems in the fixed-satellite service in the above bands shall be operated in such a way that any unacceptable interference that may occur during their operation shall be rapidly eliminated. (WRC-2000)
5.516A	In the band 17.3-17.7 GHz, earth stations of the fixed-satellite service (space-to-Earth) in Region 1 shall not claim protection from the broadcasting-satellite service feeder-link earth stations operating under Appendix 30A, nor put any limitations or restrictions on the locations of the broadcasting-satellite service feeder-link earth stations anywhere within the service area of the feeder link. (WRC-03)
5.516B	The following bands are identified for use by high-density applications in the fixed-satellite service: 17.3-17.7 GHz (space-to-Earth) in Region 1, 18.3-19.3 GHz (space-to-Earth) in Region 2, 19.7-20.2 GHz (space-to-Earth) in all Regions, 39.5-40 GHz (space-to-Earth) in Region 1, 40-40.5 GHz (space-to-Earth) in all Regions, 40.5-42 GHz (space-to-Earth) in Region 2, 47.5-47.9 GHz (space-to-Earth) in Region 1, 48.2-48.54 GHz (space-to-Earth) in Region 1, 49.44-50.2 GHz (space-to-Earth) in Region 1, and 27.5-27.82 GHz (Earth-to-space) in Region 1, 28.35-28.45 GHz (Earth-to-space) in Region 2, 28.45-28.94 GHz (Earth-to-space) in all Regions, 28.94-29.1 GHz (Earth-to-space) in Region 2 and 3, 29.25-29.46 GHz (Earth-to-space) in Region 2, 29.46-30 GHz (Earth-to-space) in all Regions, 48.2-50.2 GHz (Earth-to-space) in Region 2. ART5 – 37 – This identification does not preclude the use of these frequency bands by other fixed-satellite service applications or by other services to which these frequency bands are allocated on a co-primary basis and does not establish priority in these Radio Regulations among users of the frequency bands. Administrations should take this into account when considering regulatory provisions in relation to these frequency bands. See Resolution 143 (Rev.WRC-19). (WRC-19)
5.517A	The operation of earth stations in motion communicating with geostationary fixed-satellite service space stations within the frequency bands 17.7-19.7 GHz (space-to-Earth) and 27.5-29.5 GHz (Earth-to-space) shall be subject to the application of Resolution 169 (Rev.WRC-23). (WRC-23)
5.517B	The operation of aeronautical and maritime earth stations in motion communicating with non-geostationary space stations in the fixed-satellite service in the frequency bands 17.7-18.6 GHz, 18.8-19.3 GHz and 19.7-20.2 GHz (space-to-Earth) and 27.5-29.1 GHz and 29.5-30 GHz (Earth-to-space) shall be subject to the application of Resolution 123 (WRC-23). (WRC-23)
5.519	Additional allocation: the bands 18.0-18.3 GHz in Region 2 and 18.1-18.4 GHz in Regions 1 and 3 are also allocated to the meteorological-satellite service (space-to-Earth) on a primary basis. Their use is limited to geostationary satellites. (WRC-07)
5.520	The use of the band 18.1-18.4 GHz by the fixed-satellite service (Earth-to-space) is limited to feeder links of geostationary-satellite systems in the broadcasting-satellite service. (WRC-2000)
5.521	Alternative allocation: in the United Arab Emirates, the frequency band 18.1-18.4 GHz is allocated to the fixed, fixed-satellite (space-to-Earth) and mobile services on a primary basis (see No. 5.33). The provisions of No. 5.519 also apply. (WRC-23)
5.521A	For use of the frequency bands 18.1-18.6 GHz, 18.8-20.2 GHz and 27.5-30 GHz, or parts thereof, by space stations in the inter-satellite service, Resolution 679 (WRC-23) shall apply. Such use is limited to space research, space operation and/or Earth exploration-satellite applications, and also transmissions of data originating from industrial and medical activities in space. When using these frequencies, administrations shall ensure that this inter-satellite service is used only for the aforementioned purposes and is not subject to coordination under No. 9.11A. For use of the frequency bands 18.1-18.6 GHz, 18.8-20.2 GHz, 27.5-29.1 GHz and 29.5-30 GHz by space stations, the allocation is limited to inter-satellite links between non-geostationary satellites or between non-geostationary satellites and geostationary satellites. For use of the frequency band 29.1-29.5 GHz by space stations, the allocation is limited to inter-satellite links between non-geostationary satellites and geostationary satellites. No. 4.10 does not apply. (WRC-23)
5.522A	The emissions of the fixed service and the fixed-satellite service in the band 18.6-18.8 GHz are limited to the values given in Nos. 21.5A and 21.16.2, respectively. (WRC-2000)

Footnote Number	Footnote Content
5.522B	The use of the band 18.6-18.8 GHz by the fixed-satellite service is limited to geostationary systems and systems with an orbit of apogee greater than 20 000 km. (WRC-2000)
5.522C	In the band 18.6-18.8 GHz, in Algeria, Saudi Arabia, Bahrain, Egypt, the United Arab Emirates, the Libyan Arab Jamahiriya, Jordan, Lebanon, Morocco, Oman, Qatar, the Syrian Arab Republic, Tunisia and Yemen, fixed-service systems in operation at the date of entry into force of the Final Acts of WRC-2000 are not subject to the limits of No. 21.5A. (WRC-2000)
5.523A	The use of the bands 18.8-19.3 GHz (space-to-Earth) and 28.6-29.1 GHz (Earth-to-space) by geostationary and non-geostationary fixed-satellite service networks is subject to the application of the provisions of No. 9.11A and No. 22.2 does not apply. Administrations having geostationary-satellite networks under coordination prior to 18 November 1995 shall cooperate to the maximum extent possible to coordinate pursuant to No. 9.11A with non-geostationary-satellite networks for which notification information has been received by the Bureau prior to that date, with a view to reaching results acceptable to all the parties concerned. Non-geostationary-satellite networks shall not cause unacceptable interference to geostationary fixed-satellite service networks for which complete Appendix 4 notification information is considered as having been received by the Bureau prior to 18 November 1995. (WRC-97)
5.523B	The use of the band 19.3-19.6 GHz (Earth-to-space) by the Fixed-satellite service is limited to feeder links for non-geostationary-satellite systems in the mobile-satellite service. Such use is subject to the application of the provisions of No. 9.11A, and No. 22.2 does not apply.
5.523C	No. 22.2 of the Radio Regulations shall continue to apply in the bands 19.3-19.6 GHz and 29.1-29.4 GHz, between feeder links of non-geostationary mobile-satellite service networks and those fixed-satellite service networks for which complete Appendix 4 coordination information, or notification information, is considered as having been received by the Bureau prior to 18 November 1995. (WRC-97)
5.523D	The use of the band 19.3-19.7 GHz (space-to-Earth) by geostationary fixed-satellite service systems and by feeder links for non-geostationary-satellite systems in the mobile-satellite service is subject to the application of the provisions of No. 9.11A, but not subject to the provisions of No. 22.2. The use of this band for other non-geostationary fixed-satellite service systems, or for the cases indicated in Nos. 5.523C and 5.523E, is not subject to the provisions of No. 9.11A and shall continue to be subject to Articles 9 (except No. 9.11A) and 11 procedures, and to the provisions of No. 22.2. (WRC-97)
5.523DA	In order to protect feeder links of non-geostationary networks in the mobile-satellite service in the frequency band 19.3-19.7 GHz, the power flux-density values produced at the surface of the Earth for all angles of arrival by a space station in the inter-satellite service operating in this band in accordance with Resolution 679 (WRC-23) shall not exceed -140 dB(W/m ²) in any 1 MHz within 150 km of any of the above feeder-link earth stations recorded in the Master International Frequency Register. (WRC-23)
5.523E	No. 22.2 shall continue to apply in the bands 19.6-19.7 GHz and 29.4-29.5 GHz, between feeder links of non-geostationary mobile-satellite service networks and those fixed-satellite service networks for which complete Appendix 4 coordination information, or notification information, is considered as having been received by the Bureau by 21 November 1997. (WRC-97)
5.524	Additional allocation: in Afghanistan, Algeria, Saudi Arabia, Bahrain, Brunei Darussalam, Cameroon, China, Congo (Rep. of the), Costa Rica, Djibouti, Egypt, the United Arab Emirates, Gabon, Guatemala, Guinea, India, Iran (Islamic Republic of), Iraq, Israel, Japan, Jordan, Kuwait, Lebanon, Malaysia, Mali, Morocco, Mauritania, Nepal, Nigeria, Oman, Pakistan, Palestine*, the Philippines, Qatar, the Syrian Arab Republic, the Dem. Rep. of the Congo, the Dem. People's Rep. of Korea, Singapore, Somalia, Sudan, South Sudan, Chad, Togo and Tunisia, the frequency band 19.7-21.2 GHz is also allocated to the fixed and mobile services on a primary basis. This additional use shall not impose any limitation on the power flux-density of space stations in the fixed-satellite service in the frequency band 19.7-21.2 GHz and of space stations in the mobile-satellite service in the frequency band 19.7-20.2 GHz where the allocation to the mobile-satellite service is on a primary basis in the latter frequency band. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995
5.525	In order to facilitate interregional coordination between networks in the mobile-satellite and fixed-satellite services, carriers in the mobile-satellite service that are most susceptible to interference shall, to the extent practicable, be located in the higher parts of the bands 19.7-20.2 GHz and 29.5-30 GHz.
5.526	In the bands 19.7-20.2 GHz and 29.5-30 GHz in Region 2, and in the bands 20.1-20.2 GHz and 29.9-30 GHz in Regions 1 and 3, networks which are both in the fixed-satellite service and in the mobile-satellite service may include links between earth stations at specified or unspecified points or while in motion, through one or more satellites for point-to-point and point-to-multipoint communications.
5.527	In the bands 19.7-20.2 GHz and 29.5-30 GHz, the provisions of No 4.10 do not apply with respect to the mobile-satellite service.
5.527A	The operation of earth stations in motion communicating with the FSS is subject to Resolution 156 (Rev.WRC-23). (WRC-23)
5.528	The allocation to the mobile-satellite service is intended for use by networks which use narrow spot-beam antennas and other advanced technology at the space stations. Administrations operating systems in the mobile-satellite service in the band 19.7 - 20.1 GHz in Region 2 and in the band 20.1 - 20.2 GHz shall take all practicable steps to ensure the continued availability of these bands for administrations operating fixed and mobile systems in accordance with the provisions of No. 5.524.

Footnote Number	Footnote Content
5.529A	In the frequency bands 20.2-21.2 GHz and 30-31 GHz, non-geostationary-satellite systems for which complete coordination or notification information, according to the case, is received by the Bureau as of 1 January 2025 shall not cause unacceptable interference to and shall not claim protection from geostationary-satellite networks in the mobile-satellite service operating in accordance with these Regulations. No. 5.43A does not apply. (WRC-23)
5.530A	Unless otherwise agreed between the administrations concerned, any station in the fixed or mobile services of an administration shall not produce a power flux-density in excess of $-120.4 \text{ dB(W/(m}^2 \cdot \text{MHz))}$ at 3 m above the ground of any point of the territory of any other administration in Regions 1 and 3 for more than 20% of the time. In conducting the calculations, administrations should use the most recent version of Recommendation ITU-R P.452 (see also the most recent version of Recommendation ITU-R BO.1898). (WRC-15)
5.530B	In the band 21.4-22 GHz, in order to facilitate the development of the broadcasting-satellite service, administrations in Regions 1 and 3 are encouraged not to deploy stations in the mobile service and are encouraged to limit the deployment of stations in the fixed service to point-to-point links. (WRC-12)
5.531A	The use of the aeronautical mobile (OR) service in the frequency band 22-22.2 GHz is limited to non-safety applications. (WRC-23)
5.531B	Aircraft stations in the aeronautical mobile (OR) service operating in the frequency band 22-22.2 GHz are subject to agreement obtained under No. 9.21 with respect to the fixed service and shall not cause harmful interference to, nor claim protection from, the fixed service. The following power flux-density values shall be used as a threshold for coordination under No. 9.21: $-110 \text{ dB(W/(m}^2 \cdot \text{MHz))}$ for $0^\circ \leq \theta \leq 12.6^\circ$ $2.86 \theta - 146 \text{ dB(W/(m}^2 \cdot \text{MHz))}$ for $12.6^\circ < \theta \leq 15^\circ$ $0.87 \theta - 116 \text{ dB(W/(m}^2 \cdot \text{MHz))}$ for $15^\circ < \theta \leq 30^\circ$ $0.067 \theta - 92 \text{ dB(W/(m}^2 \cdot \text{MHz))}$ for $30^\circ < \theta \leq 90^\circ$ where θ is the angle of arrival of the incident wave above the horizontal plane, in degrees. This criterion should be applied at the border of the territory of another administration for any aircraft station located at an altitude of up to 15 km above the ground. In conducting the calculations, the most recent version of Recommendation ITU-R P.525 should be used. (WRC-23)
5.531C	Stations in the aeronautical mobile (OR) service operating in the frequency band 22-22.2 GHz shall not cause harmful interference to the radio astronomy service operating in the frequency band 22.21-22.5 GHz. The aggregate power flux-density (pfd) received from these stations at any radio astronomy station operating in the frequency band 22.21-22.5 GHz shall be in compliance with the protection criteria provided in Recommendations ITU-R RA.769-2 and ITU-R RA.1513-2, unless specifically agreed by the affected administration(s). (WRC-23)
5.531D	The use of the aeronautical mobile (OR) service in the frequency band 22-22.2 GHz outside national boundaries shall not cause harmful interference to, or claim protection from, services in other countries operating in accordance with the Table of Frequency Allocations. (WRC-23)
5.531F	In order to protect stations of the Earth exploration-satellite service (passive) operating in the frequency band 22.21-22.5 GHz, the unwanted equivalent isotropically radiated power (e.i.r.p.) of stations operating in the aeronautical mobile (OR) service shall not exceed -23 dBW in any 100 MHz band in the frequency band 22.21-22.5 GHz. (WRC-23)
5.532	The use of the band 22.21-22.5 GHz by the earth exploration-satellite (passive) and space research (passive) services shall not impose constraints upon the fixed and mobile, except aeronautical mobile, services.
5.532A	The location of earth stations in the space research service shall maintain a separation distance of at least 54 km from the respective border(s) of neighbouring countries to protect the existing and future deployment of fixed and mobile services unless a shorter distance is otherwise agreed between the corresponding administrations. Nos. 9.17 and 9.18 do not apply. (WRC-12)
5.532AB	The frequency band 24.25-27.5 GHz is identified for use by administrations wishing to implement the terrestrial component of International Mobile Telecommunications (IMT). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. Resolution 242 (Rev.WRC-23) applies. (WRC-23)
5.532B	Use of the band 24.65-25.25 GHz in Region 1 and the band 24.65-24.75 GHz in Region 3 by the fixed-satellite service (Earth-to-space) is limited to earth stations using a minimum antenna diameter of 4.5 m. (WRC-12)
5.534A	The allocation to the fixed service in the frequency band 25.25-27.5 GHz is identified in Region 2 for use by high-altitude platform stations (HAPS) in accordance with the provisions of Resolution 166 (Rev.WRC-23). Such use of the fixed-service allocation by HAPS shall be limited to the ground-to-HAPS direction in the frequency band 25.25-27.0 GHz and to the HAPS-to-ground direction in the frequency band 27.0-27.5 GHz. Furthermore, the use of the frequency band 25.5-27.0 GHz by HAPS shall be limited to gateway links. This identification does not preclude the use of this frequency band by other fixed-service applications or by other services to which this band is allocated on a co-primary basis, and does not establish priority in the Radio Regulations. (WRC-23)
5.535A	The use of the band 29.1-29.5 GHz (Earth-to-space) by the fixed-satellite service is limited to geostationary-satellite systems and feeder links to non-geostationary-satellite systems in the mobile-satellite service. Such use is subject to the application of the provisions of No. 9.11A, but not subject to the provisions of No. 22.2, except as indicated in Nos. 5.523C and 5.523E where such use is not subject to the provisions of No. 9.11A and shall continue to be subject to Articles 9 (except No. 9.11A) and 11 procedures, and to the provisions of No. 22.2. (WRC-97)

Footnote Number	Footnote Content
5.536	Use of the 25.25-27.5 GHz band by the inter-satellite service is limited to space research and Earth exploration-satellite applications, and also transmissions of data originating from industrial and medical activities in space.
5.536A	Administrations operating earth stations in the Earth exploration-satellite service or the space research service shall not claim protection from stations in the fixed and mobile services operated by other administrations. In addition, earth stations in the Earth exploration-satellite service or in the space research service should be operated taking into account the most recent version of Recommendation ITU-R SA.1862. Resolution 242 (Rev.WRC-23) applies. (WRC-23)
5.536B	In Algeria, Saudi Arabia, Austria, Bahrain, Belgium, Brazil, China, Korea (Rep. of), Denmark, Egypt, United Arab Emirates, Estonia, Finland, Hungary, India, Iran (Islamic Republic of), Iraq, Ireland, Israel, Italy, Jordan, Kenya, Kuwait, Lebanon, Libya, Lithuania, Moldova, Norway, Oman, Uganda, Pakistan, the Philippines, Poland, Portugal, Qatar, the Syrian Arab Republic, Turkiye, Dem. People's Rep. of Korea, Slovakia, the Czech Rep., Romania, the United Kingdom, Singapore, Slovenia, Somalia, Sudan, Sweden, Tanzania, Viet Nam and Zimbabwe, earth stations operating in the Earth exploration-satellite service in the frequency band 25.5-27 GHz shall not claim protection from, or constrain the use and deployment of, stations of the fixed and mobile services. Resolution 242 (Rev.WRC-23) applies. (WRC-23)
5.536C	In Algeria, Saudi Arabia, Bahrain, Botswana, Brazil, Cameroon, Comoros, Cuba, Djibouti, Egypt, United Arab Emirates, Estonia, Finland, Iran (Islamic Rep. of), Israel, Jordan, Kenya, Kuwait, Lithuania, Malaysia, Morocco, Nigeria, Oman, Qatar, Syrian Arab Republic, Somalia, Sudan, South Sudan, Tanzania, Tunisia, Uruguay, Zambia and Zimbabwe, earth stations operating in the space research service in the band 25.5-27 GHz shall not claim protection from, or constrain the use and deployment of, stations of the fixed and mobile services. (WRC-12)
5.537A	In Bhutan, Cameroon, China, Korea (Rep. of), the Russian Federation, India, Indonesia, Iran (Islamic Republic of), Iraq, Japan, Kazakhstan, Malaysia, Maldives, Mongolia, Myanmar, Uzbekistan, Pakistan, the Philippines, Kyrgyzstan, the Dem. People's Rep. of Korea, Sudan, Sri Lanka, Thailand and Viet Nam, the allocation to the fixed service in the frequency band 27.9-28.2 GHz may also be used by high altitude platform stations (HAPS) within the territory of these countries. Such use of 300 MHz of the fixed-service allocation by HAPS in the above countries is further limited to operation in the HAPS-to-ground direction and shall not cause harmful interference to, nor claim protection from, other types of fixed-service systems or other co-primary services. Furthermore, the development of these other services shall not be constrained by HAPS. See Resolution 145 (Rev.WRC-19). (WRC-19)
5.538	Additional allocation: the bands 27.500-27.501 GHz and 29.999-30.000 GHz are also allocated to the fixed-satellite service (space to Earth) on a primary basis for the beacon transmissions intended for up-link power control. Such space-to-Earth transmissions shall not exceed an equivalent isotropically radiated power (e.i.r.p.) of +10 dBW in the direction of adjacent satellites on the geostationary-satellite orbit. (WRC-07)
5.539	The band 27.5-30 GHz may be used by the fixed-satellite service (Earth-to-space) for the provision of feeder links for the broadcasting-satellite service.
5.540	Additional allocation: the band 27.501-29.999 GHz is also allocated to the fixed-satellite service (space-to-Earth) on a secondary basis for beacon transmissions intended for up-link power control.
5.541	In the band 28.5-30 GHz, the earth exploration-satellite service is limited to the transfer of data between stations and not to the primary collection of information by means of active or passive sensors.
5.541A	Feeder links of non-geostationary networks in the mobile-satellite service and geostationary networks in the fixed-satellite service operating in the band 29.1-29.5 GHz (Earth-to-space) shall employ uplink adaptive power control or other methods of fade compensation, such that the earth station transmissions shall be conducted at the power level required to meet the desired link performance while reducing the level of mutual interference between both networks. These methods shall apply to networks for which Appendix 4 coordination information is considered as having been received by the Bureau after 17 May 1996 and until they are changed by a future competent world radiocommunication conference. Administrations submitting Appendix 4 information for coordination before this date are encouraged to utilize these techniques to the extent practicable. (WRC-2000)
5.542	Additional allocation: in Algeria, Saudi Arabia, Bahrain, Brunei Darussalam, Cameroon, China, Congo (Rep. of the), Djibouti, Egypt, the United Arab Emirates, Eritrea, Ethiopia, Guinea, India, Iran (Islamic Republic of), Iraq, Japan, Jordan, Kuwait, Lebanon, Malaysia, Mali, Morocco, Mauritania, Nepal, Oman, Pakistan, Palestine*, Philippines, Qatar, the Syrian Arab Republic, the Dem. People's Rep. of Korea, Somalia, Sudan, South Sudan, Sri Lanka and Chad, the frequency band 29.5-31 GHz is also allocated to the fixed and mobile services on a secondary basis. The power limits specified in Nos. 21.3 and 21.5 shall apply. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995.
5.543	The band 29.95-30 GHz may be used for space-to-space links in the earth exploration-satellite service for telemetry, tracking, and control purposes, on a secondary basis.
5.543A	In Bhutan, Cameroon, Korea (Rep. of), the Russian Federation, India, Indonesia, Iran (Islamic Republic of), Irac, Japan, Kazakhstan, Malaysia, Maldives, Mongolia, Myanmar, Uzbekistan, Pakistan, the Philippines, Kyrgyzstan, the Dem. People's Rep. of Korea, Sudan, Sri Lanka, Thailand and Viet Nam, the allocation to the fixed service in the frequency band 31-31.3 GHz may also be used by systems using high altitude platform stations (HAPS) in the ground-to-HAPS direction. The use of the frequency band 31-31.3 GHz by systems using HAPS is limited to the territory of the countries listed above and shall not cause harmful interference to, nor claim protection from, other types of fixed-service systems, systems in the mobile service and systems operated under No. 5.545. Furthermore, the development of these services shall not be constrained by HAPS. Systems using HAPS in the frequency band 31-31.3 GHz shall not cause harmful interference to the radio astronomy service having a primary allocation in the frequency band 31.3-31.8 GHz, taking into account the protection criterion as given in the most recent version of Recommendation ITU-R RA.769. In order to ensure the protection of satellite passive services, the level of unwanted power density into a HAPS ground station antenna in the frequency band 31.3-31.8 GHz shall be limited to -106 dB(W/MHz) under clear-sky conditions, and may be increased up to -100 dB(W/MHz) under rainy conditions to mitigate fading due to rain, provided the effective impact on the passive satellite does not exceed the impact under clear-sky conditions. See Resolution 145 (Rev.WRC-12). (WRC-15)

Footnote Number	Footnote Content
5.543B	The allocation to the fixed service in the frequency band 31-31.3 GHz is identified for worldwide use by high-altitude platform stations (HAPS). This identification does not preclude the use of this frequency band by other fixed-service applications or by other services to which this frequency band is allocated on a co-primary basis, and does not establish priority in the Radio Regulations. Such use of the fixed-service allocation by HAPS shall be in accordance with the provisions of Resolution 167 (Rev.WRC-23). (WRC-23)
5.544	In the band 31-31.3 GHz the power flux-density limits specified in Article 21, Table 21-4 shall apply to the space research service.
5.545	Different category of service: in Armenia, Georgia, Kyrgyzstan, Tajikistan and Turkmenistan, the allocation of the band 31-31.3 GHz to the space research service is on a primary basis (see No. 5.33). (WRC-12)
5.546	Different category of service: in Saudi Arabia, Armenia, Azerbaijan, Bahrain, Belarus, Djibouti, Egypt, the United Arab Emirates, Spain, Estonia, the Russian Federation, Georgia, Hungary, Iran (Islamic Republic of), Israel, Jordan, Lebanon, Moldova, Mongolia, Oman, Uzbekistan, Poland, the Syrian Arab Republic, Turkiye, Kyrgyzstan, Romania, the United Kingdom, Somalia, South Africa, Tajikistan and Turkmenistan, the allocation of the frequency band 31.5-31.8 GHz to the fixed and mobile, except aeronautical mobile, services is on a primary basis (see No. 5.33). (WRC-23)
5.547	The frequency bands 31.8-33.4 GHz, 37-40 GHz, 40.5-43.5 GHz, 51.4-52.6 GHz, 55.78-59 GHz and 64-66 GHz are available for high-density applications in the fixed service. Administrations should take this into account when considering regulatory provisions in relation to these bands. Because of the potential deployment of high-density applications in the fixed-satellite service in the frequency bands 39.5-40 GHz and 40.5-42 GHz (see No. 5.516B), administrations should further take into account potential constraints to high-density applications in the fixed service, as appropriate. (WRC-23)
5.547A	Administrations should take practical measures to minimize the potential interference between stations in the fixed service and airborne stations in the radionavigation service in the 31.8-33.4 GHz band, taking into account the operational needs of the airborne radar systems. (WRC-2000)
5.547B	Alternative allocation: in the United States, the band 31.8-32 GHz is allocated to the radionavigation and space research (deep space) (space-to-Earth) services on a primary basis. (WRC-97)
5.547C	Alternative allocation: in the United States, the band 32-32.3 GHz is allocated to the radionavigation and space research (deep space) (space-to-Earth) services on a primary basis. (WRC-03)
5.547D	Alternative allocation: in the United States, the band 32.3-33 GHz is allocated to the inter-satellite and radionavigation services on a primary basis. (WRC-97)
5.547E	Alternative allocation: in the United States, the band 33-33.4 GHz is allocated to the radionavigation service on a primary basis. (WRC-97)
5.548	In designing systems for the inter-satellite service in the frequency band 32.3-33 GHz, for the radionavigation service in the frequency band 32-33 GHz, and for the space research service (deep space) in the frequency band 31.8-32.3 GHz, administrations shall take all necessary measures to prevent harmful interference between these services, bearing in mind the safety aspects of the radionavigation service (see Recommendation 707 (Rev.WRC-23)). (WRC-23)
5.549	Additional allocation: in Saudi Arabia, Bahrain, Bangladesh, Egypt, the United Arab Emirates, Gabon, Indonesia, Iran (Islamic Republic of), Iraq, Israel, Jordan, Kuwait, Lebanon, Libya, Malaysia, Mali, Morocco, Mauritania, Nepal, Nigeria, Oman, Pakistan, the Philippines, Qatar, the Syrian Arab Republic, the Dem. Rep. of the Congo, Singapore, Somalia, Sudan, South Sudan, Sri Lanka, Togo, Tunisia and Yemen, the band 33.4-36 GHz is also allocated to the fixed and mobile services on a primary basis. (WRC-12)
5.549A	In the band 35.5-36.0 GHz, the mean power flux-density at the Earth's surface, generated by any spaceborne sensor in the Earth exploration-satellite service (active) or space research service (active), for any angle greater than 0.8° from the beam centre shall not exceed -73.3 dB(W/m²) in this band. (WRC-03)
5.550A	For sharing of the band 36-37 GHz between the Earth exploration-satellite (passive) service and the fixed and mobile services, Resolution 752 (WRC-07) shall apply. (WRC-07)
5.550B	The frequency band 37-43.5 GHz, or portions thereof, is identified for use by administrations wishing to implement the terrestrial component of International Mobile Telecommunications (IMT). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. Because of the potential deployment of FSS earth stations within the frequency range 37.5-42.5 GHz and high-density applications in the fixed-satellite service in the frequency bands 39.5-40 GHz in Region 1, 40-40.5 GHz in all Regions and 40.5-42 GHz in Region 2 (see No. 5.516B), administrations should further take into account potential constraints to IMT in these frequency bands, as appropriate. Resolution 243 (Rev.WRC-23) applies. (WRC-23)
5.550C	The use of the frequency bands 37.5-39.5 GHz (space-to-Earth), 39.5-42.5 GHz (space-to-Earth), 47.2- 50.2 GHz (Earth-to-space) and 50.4-51.4 GHz (Earth-to-space) by a non-geostationary-satellite system in the fixed-satellite service is subject to the application of the provisions of No. 9.12 for coordination with other non-geostationary-satellite systems in the fixed-satellite service but not with non-geostationary-satellite systems in other services. Resolution 770 (WRC-19) shall also apply, and No. 22.2 shall continue to apply. (WRC-19)

Footnote Number	Footnote Content
5.550CA	Non-geostationary-satellite systems in the fixed-satellite service operating with an apogee altitude above 407 km and below 2 000 km in the frequency band 37.5-38 GHz shall not exceed an unwanted emission e.i.r.p. density of -21 dB(W/100 MHz) per space station for angles greater than 65.0° from nadir relative to the space station in the fixed-satellite service in the frequency band 36-37 GHz in order to protect the Earth exploration-satellite service (passive) operating in the latter frequency band. (WRC-23)
5.550D	The allocation to the fixed service in the frequency band 38-39.5 GHz is identified for worldwide use by administrations wishing to implement high-altitude platform stations (HAPS). In the HAPS-to-ground direction, the HAPS ground station shall not claim protection from stations in the fixed, mobile and fixed-satellite services; and No. 5.43A does not apply. This identification does not preclude the use of this frequency band by other fixed-service applications or by other services to which this frequency band is allocated on a co-primary basis and does not establish priority in the Radio Regulations. Furthermore, the development of the fixed-satellite, fixed and mobile services shall not be unduly constrained by HAPS. Such use of the fixed-service allocation by HAPS shall be in accordance with the provisions of Resolution 168 (Rev.WRC-23). (WRC-23)
5.550E	The use of the frequency bands 39.5-40 GHz and 40-40.5 GHz by non-geostationary-satellite systems in the mobile-satellite service (space-to-Earth) and by non-geostationary-satellite systems in the fixed-satellite service (space-to-Earth) is subject to the application of the provisions of No. 9.12 for coordination with other non-geostationarysatellite systems in the fixed-satellite and mobile-satellite services but not with non-geostationary-satellite systems in other services. No. 22.2 shall continue to apply for non-geostationary-satellite-systems. (WRC-19)
5.551H	The equivalent power flux-density (epfd) produced in the frequency band 42.5-43.5 GHz by all space stations in any non-geostationary-satellite system in the fixed-satellite service, or in the broadcasting-satellite service operating in the frequency band 42-42.5 GHz, shall not exceed the following values at the site of any radio astronomy station for more than 2% of the time: -230 dB(W/m ²) in 1 GHz and -246 dB(W/m ²) in any 500 kHz of the frequency band 42.5-43.5 GHz at the site of any radio astronomy station registered as a single-dish telescope; and -209 dB(W/m ²) in any 500 kHz of the frequency band 42.5-43.5 GHz at the site of any radio astronomy station registered as a very long baseline interferometry station. These epfd values shall be evaluated using the methodology given in Recommendation ITU-R S.1586-1 and the reference antenna pattern and the maximum gain of an antenna in the radio astronomy service given in Recommendation ITU-R RA.1631-0 and shall apply over the whole sky and for elevation angles higher than the minimum operating angle θ_{min} of the radiotelescope (for which a default value of 5° should be adopted in the absence of notified information). These values shall apply at any radio astronomy station that either: - was in operation prior to 5 July 2003 and has been notified to the Bureau before 4 January 2004; or - was notified before the date of receipt of the complete Appendix 4 information for coordination or notification, as appropriate, for the space station to which the limits apply. Other radio astronomy stations notified after these dates may seek an agreement with administrations that have authorized the space stations. In Region 2, Resolution 743 (WRC-03) shall apply. The limits in this footnote may be exceeded at the site of a radio astronomy station of any country whose administration so agreed. (WRC-15)
5.551I	The power flux-density in the band 42.5-43.5 GHz produced by any geostationary space station in the fixed-satellite service (space-to-Earth), or the broadcasting-satellite service (space-to-Earth) operating in the 42-42.5 GHz band, shall not exceed the following values at the site of any radio astronomy station: -137 dB(W/m ²) in 1 GHz and -153 dB(W/m ²) in any 500 kHz of the 42.5-43.5 GHz band at the site of any radio astronomy station registered as a single-dish telescope; and -116 dB(W/m ²) in any 500 kHz of the 42.5-43.5 GHz band at the site of any radio astronomy station registered as a very long baseline interferometry station. These values shall apply at the site of any radio astronomy station that either: - was in operation prior to 5 July 2003 and has been notified to the Bureau before 4 January 2004; or - was notified before the date of receipt of the complete Appendix 4 information for coordination or notification, as appropriate, for the space station to which the limits apply. Other radio astronomy stations notified after these dates may seek an agreement with administrations that have authorized the space stations. In Region 2, Resolution 743 (WRC-03) shall apply. The limits in this footnote may be exceeded at the site of a radio astronomy station of any country whose administration so agreed. (WRC-03)
5.552	The allocation of the spectrum for the fixed-satellite service in the bands 42.5-43.5 GHz and 47.2-50.2 GHz for Earth-to-space transmission is greater than that in the band 37.5-39.5 GHz for space-to-Earth transmission in order to accommodate feeder links to broadcasting satellites. Administrations are urged to take all practicable steps to reserve the band 47.2-49.2 GHz for feeder links for the broadcasting-satellite service operating in the band 40.5-42.5 GHz.
5.552A	The allocation to the fixed service in the frequency bands 47.2-47.5 GHz and 47.9-48.2 GHz is identified for use by high-altitude platform stations (HAPS). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated on a co-primary basis, and does not establish priority in the Radio Regulations. Such use of the fixed-service allocation in the frequency bands 47.2-47.5 GHz and 47.9-48.2 GHz by HAPS shall be in accordance with the provisions of Resolution 122 (Rev.WRC-19). (WRC-19)
5.553	In the bands 43.5-47 GHz and 66-71 GHz, stations in the land mobile service may be operated subject to not causing harmful interference to the space radiocommunication services to which these bands are allocated (see No. 5.43). (WRC-2000)
5.553A	In Algeria, Angola, Bahrain, Belarus, Benin, Botswana, Brazil, Burkina Faso, Cabo Verde, Korea (Rep. of), Ivory Coast, Croatia, Djibouti, Egypt, United Arab Emirates, Estonia, Eswatini, Gabon, Gambia, Ghana, Greece, Guinea, Guinea-Bissau, Hungary, Iran (Islamic Republic of), Iraq, Jordan, Kuwait, Lesotho, Latvia, Liberia, Lithuania, Madagascar, Malawi, Mali, Morocco, Mauritius, Mauritania, Mozambique, Namibia, Niger, Nigeria, Oman, Qatar, Senegal, Seychelles, Sierra Leone, Slovenia, Somalia, Sudan, South Africa, Sweden, Tanzania, Togo, Tunisia, Zambia and Zimbabwe, the frequency band 45.5-47 GHz is identified for use by administrations wishing to implement the terrestrial component of International Mobile Telecommunications (IMT), taking into account No. 5.553. With respect to the aeronautical mobile service and radionavigation service, the use of this frequency band for the implementation of IMT is subject to agreement obtained under No. 9.21 with concerned administrations and shall not cause harmful interference to, or claim protection from these services. This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. Resolution 244 (Rev.WRC-23) applies. (WRC-23)

Footnote Number	Footnote Content
5.553B	In Region 2 and Algeria, Angola, Saudi Arabia, Australia, Bahrain, Benin, Botswana, Burkina Faso, Burundi, Cameroon, Central African Rep., Comoros, Congo (Rep. of the), Korea (Rep. of), Ivory Coast, Djibouti, Egypt, United Arab Emirates, Eswatini, Ethiopia, Gabon, Gambia, Ghana, Guinea, Guinea-Bissau, Equatorial Guinea, India, Iran (Islamic Republic of), Iraq, Japan, Jordan, Kenya, Kuwait, Lesotho, Liberia, Libya, Lithuania, Madagascar, Malaysia, Malawi, Mali, Morocco, Mauritius, Mauritania, Mozambique, Namibia, Niger, Nigeria, Oman, Uganda, Qatar, the Syrian Arab Republic, the Dem. Rep. of the Congo, Rwanda, Sao Tome and Principe, Senegal, Seychelles, Sierra Leone, Singapore, Slovenia, Somalia, Sudan, South Sudan, South Africa, Sweden, Tanzania, Chad, Togo, Tunisia, Zambia and Zimbabwe, the frequency band 47.2-48.2 GHz is identified for use by administrations wishing to implement International Mobile Telecommunications (IMT). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated, and does not establish any priority in the Radio Regulations. Resolution 243 (Rev.WRC-23) applies. (WRC-23)
5.554	In the bands 43.5-47 GHz, 66-71 GHz, 95-100 GHz, 123-130 GHz, 191.8-200 GHz and 252-265 GHz, satellite links connecting land stations at specified fixed points are also authorized when used in conjunction with the mobile-satellite service or the radionavigation-satellite service. (WRC-2000)
5.554A	The use of the bands 47.5-47.9 GHz, 48.2-48.54 GHz and 49.44-50.2 GHz by the fixed-satellite service (space-to-Earth) is limited to geostationary satellites. (WRC-03)
5.555	Additional allocation: the band 48.94-49.04 GHz is also allocated to the radio astronomy service on a primary basis. (WRC-2000)
5.555B	The power flux-density in the band 48.94-49.04 GHz produced by any geostationary space station in the fixed-satellite service (space-to-Earth) operating in the bands 48.2-48.54 GHz and 49.44-50.2 GHz shall not exceed -151.8 dB(W/m ²) in any 500 kHz band at the site of any radio astronomy station. (WRC-03)
5.555C	The use of the frequency band 51.4-52.4 GHz by the fixed-satellite service (Earth-to-space) is limited to geostationary-satellite networks. The earth stations shall be limited to gateway earth stations with a minimum antenna diameter of 2.4 metres. (WRC-19)
5.556	In the bands 51.4-54.25 GHz, 58.2-59 GHz and 64-65 GHz, radio astronomy observations may be carried out under national arrangements. (WRC-2000)
5.556A	Use of the bands 54.25-56.9 GHz, 57-58.2 GHz and 59-59.3 GHz by the inter-satellite service is limited to satellites in the geostationary-satellite orbit. The single-entry power flux-density at all altitudes from 0 km to 1 000 km above the Earth's surface produced by a station in the inter-satellite service, for all conditions and for all methods of modulation, shall not exceed -147 dB(W/m ² /100 MHz) for all angles of arrival. (WRC-97)
5.556B	Additional allocation: in Japan, the band 54.25-55.78 GHz is also allocated to the mobile service on a primary basis for low-density use. (WRC-97)
5.557	Additional allocation: in Japan, the band 55.78-58.2 GHz is also allocated to the radiolocation service on a primary basis. (WRC-97)
5.557A	In the band 55.78-56.26 GHz, in order to protect stations in the Earth exploration-satellite service (passive), the maximum power density delivered by a transmitter to the antenna of a fixed service station is limited to -26 dB(W/MHz). (WRC-2000)
5.558	In the bands 55.78-58.2 GHz, 59-64 GHz, 66-71 GHz, 122.25-123 GHz, 130-134 GHz, 167-174.8 GHz and 191.8-200 GHz, stations in the aeronautical mobile service may be operated subject to not causing harmful interference to the inter-satellite service (see No. 5.43). (WRC-2000)
5.558A	Use of the band 56.9-57 GHz by inter-satellite systems is limited to links between satellites in geostationary-satellite orbit and to transmissions from non-geostationary satellites in high-Earth orbit to those in low Earth orbit. For links between satellites in the geostationary-satellite orbit, the single entry power flux-density at all altitudes from 0 km to 1 000 km above the Earth's surface, for all conditions and for all methods of modulation, shall not exceed -147 dB(W/(m ² □ 100 MHz)) for all angles of arrival. (WRC-97)
5.559	In the band 59-64 GHz, airborne radars in the radiolocation service may be operated subject to not causing harmful interference to the inter-satellite service (see No. 5.43). (WRC-2000)
5.559AA	The frequency band 66-71 GHz is identified for use by administrations wishing to implement the terrestrial component of International Mobile Telecommunications (IMT). This identification does not preclude the use of this frequency band by any application of the services to which this frequency band is allocated and does not establish priority in the Radio Regulations. Resolution 241 (Rev.WRC-23) applies. (WRC-23)
5.559B	The use of the frequency band 77.5-78 GHz by the radiolocation service shall be limited to short-range radar for ground-based applications, including automotive radars. The technical characteristics of these radars are provided in the most recent version of Recommendation ITU-R.M.2057. The provisions of No. 4.10 do not apply. (WRC-15)
5.560	In the band 78-79 GHz radars located on space stations may be operated on a primary basis in the earth exploration-satellite service and in the space research service.
5.561	In the band 74-76 GHz, stations in the fixed, mobile and broadcasting services shall not cause harmful interference to stations of the fixed-satellite service or stations of the broadcasting-satellite service operating in accordance with the decisions of the appropriate frequency assignment planning conference for the broadcasting-satellite service. (WRC-2000)

Footnote Number	Footnote Content
5.561A	The 81-81.5 GHz band is also allocated to the amateur and amateur-satellite services on a secondary basis. (WRC-2000)
5.562	The use of the band 94-94.1 GHz by the Earth exploration-satellite (active) and space research (active) services is limited to spaceborne cloud radars. (WRC-97)
5.562A	In the bands 94-94.1 GHz and 130-134 GHz, transmissions from space stations of the Earth exploration-satellite service (active) that are directed into the main beam of a radio astronomy antenna have the potential to damage some radio astronomy receivers. Space agencies operating the transmitters and the radio astronomy stations concerned should mutually plan their operations so as to avoid such occurrences to the maximum extent possible. (WRC-2000)
5.562B	In the frequency bands 105-109.5 GHz, 111.8-114.25 GHz and 217-226 GHz, the use of this allocation is limited to space-based radio astronomy only. (WRC-19)
5.562C	Use of the band 116-122.25 GHz by the inter-satellite service is limited to satellites in the geostationary-satellite orbit. The single-entry power flux-density produced by a station in the inter-satellite service, for all conditions and for all methods of modulation, at all altitudes from 0 km to 1 000 km above the Earth's surface and in the vicinity of all geostationary orbital positions occupied by passive sensors, shall not exceed -148 dB(W/(m ² · MHz)) for all angles of arrival. (WRC-2000)
5.562E	The allocation to the Earth exploration-satellite service (active) is limited to the band 133.5-134 GHz. (WRC-2000)
5.562H	Use of the bands 174.8-182 GHz and 185-190 GHz by the inter-satellite service is limited to satellites in the geostationary-satellite orbit. The single-entry power flux-density produced by a station in the inter-satellite service, for all conditions and for all methods of modulation, at all altitudes from 0 km to 1000 km above the Earth's surface and in the vicinity of all geostationary orbital positions occupied by passive sensors, shall not exceed -144 dB(W/(m ² · MHz)) for all angles of arrival. (WRC-2000)
5.563A	In the bands 200-209 GHz, 235-238 GHz, 250-252 GHz and 265-275 GHz, ground-based passive atmospheric sensing is carried out to monitor atmospheric constituents. (WRC-2000)
5.563AA	In the frequency band 235-238 GHz, stations in the Earth exploration-satellite service (passive) shall not claim protection from stations in the fixed and mobile services. (WRC-23)
5.563B	The band 237.9-238 GHz is also allocated to the Earth exploration-satellite service (active) and the space research service (active) for spaceborne cloud radars only. (WRC-2000)
5.564A	For the operation of fixed and land mobile service applications in frequency bands in the range 275-450 GHz: The frequency bands 275-296 GHz, 306-313 GHz, 318-333 GHz and 356-450 GHz are identified for use by administrations for the implementation of land mobile and fixed service applications where no specific conditions are necessary to protect Earth exploration-satellite service (passive) applications. The frequency bands 296-306 GHz, 313-318 GHz and 333-356 GHz may only be used by fixed and land mobile service applications when specific conditions to ensure the protection of Earth exploration-satellite service (passive) applications are determined in accordance with Resolution 731 (Rev.WRC-23). In those portions of the frequency range 275-450 GHz where radio astronomy applications are used, specific conditions (e.g. minimum separation distances and/or avoidance angles) may be necessary to ensure protection of radio astronomy sites from land mobile and/or fixed service applications, on a case-by-case basis, in accordance with Resolution 731 (Rev.WRC-23). The use of the above-mentioned frequency bands by land mobile and fixed service applications does not preclude use by, and does not establish priority over, any other applications of radio services in the range of 275-450 GHz. (WRC-23)
5.565	The following frequency bands in the range 275-1000 GHz are identified for use by administrations for passive services applications: - radio astronomy service: 275-323 GHz, 327-371 GHz, 388-424 GHz, 426-442 GHz, 453-510 GHz, 623-711 GHz, 795-909 GHz and 926-945 GHz; - Earth exploration-satellite service (passive) and space research service (passive): 275-286 GHz, 296-306 GHz, 313-356 GHz, 361-365 GHz, 369-392 GHz, 397-399 GHz, 409-411 GHz, 416-434 GHz, 439-467 GHz, 477-502 GHz, 523-527 GHz, 538-581 GHz, 611-630 GHz, 634-654 GHz, 657-692 GHz, 713-718 GHz, 729-733 GHz, 750-754 GHz, 771-776 GHz, 823-846 GHz, 850-854 GHz, 857-862 GHz, 866-882 GHz, 905-928 GHz, 951-956 GHz, 968-973 GHz and 985-990 GHz. The use of the range 275-1000 GHz by the passive services does not preclude use of this range by active services. Administrations wishing to make frequencies in the 275-1000 GHz range available for active service applications are urged to take all practicable steps to protect these passive services from harmful interference until the date when the Table of Frequency Allocations is established in the above-mentioned 275-1000 GHz frequency range. All frequencies in the range 1000-3000 GHz may be used by both active and passive services. (WRC-12)

CEPT Deliverables

Document	Description
ECC/DEC/(25)01	Designation of the frequency bands used by the Global Navigation Satellite System, Galileo, and Technical and operational measures for the use of the frequency band 1258-1300 MHz by the amateur and amateur-satellite services in order to protect the radionavigation-satellite service (space-to-Earth)
ECC/DEC/(24)01	Harmonised technical conditions for the shared use of the 3800-4200 MHz frequency band by low/medium power terrestrial wireless broadband systems (WBB LMP) providing local-area network connectivity
ECC/DEC/(23)01	On the use of the band 40.5-42.5 GHz by earth stations in the fixed-satellite service (space-to-Earth) and broadcasting-satellite service and on the use of the band 42.5-43.5 GHz by earth stations in the fixed-satellite service (Earth-to-space)
ECC/DEC/(22)07	Harmonised technical conditions for the usage of aerial UE for communications based on LTE and 5G NR in the bands 703-733 MHz, 832-862 MHz, 880-915 MHz, 1710-1785 MHz, 1920-1980 MHz, 2500-2570 MHz and 2570-2620 MHz harmonised for MFCN
ECC/DEC/(22)06	Harmonised technical conditions for Mobile/Fixed Communications Networks (MFCN) in the band 40.5-43.5 GHz
ECC/DEC/(22)03	Technical characteristics, exemption from individual licensing and free circulation and use of specific radiodetermination applications in the frequency range 116-260 GHz
ECC/DEC/(22)02	Regulation to operate Autonomous Maritime Radio Devices (AMRD) in CEPT
ECC/DEC/(22)01	Free circulation and use of Mobile/Fixed Communication Networks (MFCN) terminals operating under the control of terrestrial networks
ECC/DEC/(21)02	The harmonised frequency band 76-77 GHz, technical characteristics, exemption from individual licensing and free circulation and use of High Definition Ground Based Synthetic Aperture Radar (HD-GBSAR)
ECC/DEC/(21)01	The use of the bands 47.2-50.2 GHz and 50.4-52.4 GHz by the fixed-satellite service (Earth-to-space)
ECC/DEC/(20)02	Harmonised use of the paired frequency bands 874.4-880.0 MHz and 919.4-925.0 MHz and of the unpaired frequency band 1900-1910 MHz for Railway Mobile Radio (RMR)
ECC/DEC/(20)01	On the harmonised use of the frequency band 5945-6425 MHz for Wireless Access Systems including Radio Local Area Networks (WAS/RLAN)
ECC/DEC/(19)04	The harmonised use of spectrum, free circulation and use of earth stations on-board aircraft operating with GSO FSS networks and NGSO FSS systems in the frequency bands 12.75-13.25 GHz (Earth-to-space) and 10.7-12.75 GHz (space-to-Earth)
ECC/DEC/(19)03	Harmonised usage of the channels of the Radio Regulations Appendix 18 (Table of transmitting frequencies in the VHF maritime mobile band)
ECC/DEC/(19)02	Land mobile systems in the frequency ranges 68-87.5 MHz, 146-174 MHz, 406.1-410 MHz, 410-430 MHz, 440-450 MHz and 450-470 MHz
ECC/DEC/(18)06	The harmonised technical conditions for Mobile/Fixed Communications Networks (MFCN) in the band 24.25-27.5 GHz
ECC/DEC/(18)05	The harmonised use, exemption from individual licensing and free circulation and use of Earth Stations In-Motion (ESIM) operating with NGSO FSS satellite systems in the frequency bands 10.7-12.75 GHz and 14.0-14.5 GHz
ECC/DEC/(18)04	The harmonised use, exemption from individual licensing and free circulation and use of land based Earth Stations In-Motion (ESIM) operating with GSO FSS satellite systems in the frequency bands 10.7-12.75 GHz and 14.0-14.5 GHz
ECC/DEC/(17)06	The harmonised use of the frequency bands 1427-1452 MHz and 1492-1518 MHz for Mobile/Fixed Communications Networks Supplemental Downlink (MFCN SDL)
ECC/DEC/(17)04	The harmonised use and exemption from individual licensing of fixed earth stations operating with NGSO FSS satellite systems in the frequency bands 10.7-12.75 GHz and 14.0-14.5 GHz
ECC/DEC/(16)02	Harmonised technical conditions and frequency bands for the implementation of Broadband Public Protection and Disaster Relief (BB-PPDR) systems
ECC/DEC/(16)01	The harmonised frequency band 76-77 GHz, technical characteristics, exemption from individual licensing and free carriage and use of obstacle detection radars for rotorcraft use
ECC/DEC/(15)05	The harmonised frequency range 446.0-446.2 MHz, technical characteristics, exemption from individual licensing and free carriage and use of analogue and digital PMR 446 applications

Document	Description
ECC/DEC/(15)04	The harmonised use, free circulation and exemption from individual licensing of Land, Maritime and Aeronautical Earth Stations On Mobile Platforms (ESOMPs) operating with NGSO FSS satellite systems in the frequency ranges 17.3-20.2 GHz, 27.5-29.1 GHz and 29.5-30.0 GHz
ECC/DEC/(15)01	The harmonised technical conditions for mobile/fixed communications networks (MFCN) in the band 694-790 MHz including a paired frequency arrangement (Frequency Division Duplex 2x30 MHz) and an optional unpaired frequency arrangement (Supplemental Downlink)
ECC/DEC/(14)02	The harmonised technical and regulatory conditions for the use of the band 2300-2400 MHz for Mobile/Fixed Communications Networks (MFCN)
ECC/DEC/(13)03	The harmonised use of the frequency band 1452-1492 MHz for Mobile/Fixed Communications Networks Supplemental Downlink (MFCN SDL)
ECC/DEC/(13)01	The use, free circulation, and exemption from individual licensing of Earth stations on mobile platforms (ESOMPs) in the frequency bands available for use by uncoordinated FSS Earth stations within the ranges 17.3-20.2 GHz and 27.5-30.0 GHz
ECC/DEC/(12)03	The harmonised conditions for UWB applications onboard aircraft
ECC/DEC/(12)01	Exemption from individual licensing and free circulation and use of satellite mobile terminals operating under the control of networks in the range 1 to 3 GHz
ECC/DEC/(11)06	The harmonised frequency arrangements and Least Restrictive Technical Conditions (LRTCs) for Mobile/Fixed Communications Networks (MFCN) operating in the band 3400-3800 MHz
ECC/DEC/(11)03	The harmonised use of frequencies for Citizen' Band (CB) radio equipment
ECC/DEC/(11)02	Industrial Level Probing Radars (LPR) operating in frequency bands 6 - 8.5 GHz, 24.05 - 26.5 GHz, 57 - 64 GHz and 75 - 85 GHz
ECC/DEC/(11)01	The protection of the Earth exploration satellite service (passive) in the 1400-1427 MHz band
ECC/DEC/(10)02	Compatibility between the fixed satellite service in the 30-31 GHz band and the Earth exploration satellite service (passive) in the 31.3-31.5 GHz band
ECC/DEC/(10)01	Sharing conditions in the 10.6-10.68 GHz band between the fixed service, mobile service and Earth exploration satellite service (passive)
ECC/DEC/(09)04	Exemption from individual licensing and the free circulation and use of transmit-only mobile satellite terminals operating in the Mobile-Satellite Service allocations in the 1613.8-1626.5 MHz band
ECC/DEC/(09)03	Harmonised conditions for Mobile/Fixed Communications Networks (MFCN) operating in the band 791-862 MHz
ECC/DEC/(09)02	The harmonisation of the bands 1610-1626.5 MHz and 2483.5-2500 MHz for use by systems in the Mobile-Satellite Service
ECC/DEC/(09)01	Harmonised use of the 63.72-65.88 GHz frequency band for Intelligent Transport Systems (ITS)
ECC/DEC/(08)08	The harmonised use of GSM systems in the 900 MHz and 1800 MHz bands, UMTS systems in the 2 GHz band and LTE and 5G NR non-AAS systems in the 1800 MHz and 2.6 GHz (FDD) bands on board vessels
ECC/DEC/(08)05	The harmonisation of frequency bands for the implementation of digital Public Protection and Disaster Relief (PPDR) radio applications in bands within the 380-470 MHz range
ECC/DEC/(08)01	The harmonised use of Safety-Related Intelligent Transport Systems (ITS) in the 5875-5935 MHz frequency band
ECC/DEC/(07)01	The harmonised use, exemption from individual licensing and free circulation of Material Sensing Devices using Ultra-Wideband (UWB) technology
ECC/DEC/(06)13	Harmonised technical conditions for mobile/fixed communications networks (MFCN) including terrestrial IMT systems, other than GSM and EC-GSM IoT, in the bands 880-915/925-960 MHz and 1710-1785/1805-1880 MHz
ECC/DEC/(06)10	Transition of terrestrial service operations from the Bands 1980-2010 MHz and 2170-2200 MHz in order to facilitate the Harmonised Introduction and Development of Systems in the mobile-satellite service including those supplemented by a Complementary Ground Component
ECC/DEC/(06)09	The designation of the bands 1980-2010 MHz and 2170-2200 MHz for use by systems in the Mobile-Satellite Service including those supplemented by a Complementary Ground Component (CGC)
ECC/DEC/(06)07	The harmonised use of airborne GSM and LTE systems in the frequency bands 1710-1785 and 1805-1880 MHz, and airborne UMTS systems in the frequency bands 1920-1980 MHz and 2110-2170 MHz
ECC/DEC/(06)05	The harmonised frequency bands to be designated for Air-Ground-Air operation (AGA) of the Digital Land Mobile Systems for the Emergency Services

Document	Description
ECC/DEC/(06)04	The harmonised use, exemption from individual licensing and free circulation of devices using Ultra-Wideband (UWB) technology in bands below 10.6 GHz
ECC/DEC/(06)03	Exemption from Individual Licensing of high e.i.r.p. satellite terminals (HEST) operating with geostationary satellites and in the frequency bands 10.70-12.75 GHz or 19.70-20.20 GHz space-to-Earth and 14.00-14.25 GHz or 29.50-30.00 GHz Earth-to-space
ECC/DEC/(06)01	The harmonised utilisation of the bands 1920-1980 MHz and 2110-2170 MHz for mobile/fixed communications networks (MFCN) including terrestrial IMT systems
ECC/DEC/(05)11	The free circulation and use of Aircraft Earth Stations (AES) in the frequency bands 14-14.5 GHz (Earth-to-space), 10.7-11.7GHz (space-to-Earth) and 12.5-12.75 GHz (space-to-Earth)
ECC/DEC/(05)10	The free circulation and use of Earth Stations on board Vessels operating in fixed satellite service networks in the frequency bands 14-14.5 GHz (Earth-to-space), 10.7-11.7 GHz (space-to-Earth) and 12.5-12.75 GHz (space-to-Earth)
ECC/DEC/(05)09	The free circulation and use of Earth Stations on board Vessels operating in Fixed Satellite service networks in the frequency bands 5925-6425 MHz (Earth-to-space) and 3700-4200 MHz (space-to-Earth)
ECC/DEC/(05)08	The availability of frequency bands for high density applications in the Fixed-Satellite Service (space-to-Earth and Earth-to-space)
ECC/DEC/(05)05	Harmonised utilisation of spectrum for Mobile/Fixed Communications Networks (MFCN) operating within the band 2500-2690 MHz
ECC/DEC/(05)02	A harmonised frequency plan for the use of the band 169.4-169.8125 MHz
ECC/DEC/(05)01	The use of the band 27.5-29.5 GHz by the Fixed Service and uncoordinated Earth stations of the Fixed-Satellite Service (Earth-to-space)
ECC/DEC/(04)10	The frequency bands to be designated for the temporary introduction of Automotive Short Range Radars (SRR)
ECC/DEC/(04)09	Designation of the bands 1518-1525 MHz and 1670-1675 MHz for the Mobile Satellite Service
ECC/DEC/(04)08	The harmonised use of the 5 GHz frequency bands for Wireless Access Systems including Radio Local Area Networks (WAS/RLAN)
ECC/DEC/(04)03	The frequency band 77-81 GHz to be designated for the use of Automotive Short Range Radars
ECC/DEC/(03)04	The Exemption from Individual Licensing of Very Small Aperture Terminals (VSAT) operating in the frequency bands 14.25 - 14.50 GHz Earth-to-space and 10.70-11.70 GHz space-to-Earth
ERC/DEC/(99)06	The harmonised introduction of satellite personal communication systems operating in the bands below 1 GHz (S-PCS<1GHz)
ERC/DEC/(99)05	Free Circulation, Use and Exemption from Individual Licensing of Mobile Earth Stations.(S-PCS < 1GHz)
ERC/DEC/(98)22	Exemption from individual licensing and free circulation and use of DECT equipment
ERC/DEC/(97)02	The extended frequency bands to be used for the GSM Digital Pan-European Communications system
ERC/DEC/(95)03	The frequency bands to be designated for the introduction of DCS 1800
ERC/DEC/(94)03	The frequency band to be designated for the coordinated introduction of the Digital European Cordless Telecommunications system
ERC/DEC/(94)01	The frequency bands to be designated for the coordinated introduction of the GSM digital pan-European communications system
ERC/DEC/(01)19	Harmonised frequency bands to be designated for the Direct Mode Operation (DMO) of the Digital Land Mobile Systems for the Emergency Services
ERC/DEC/(01)17	Harmonised frequencies, technical characteristics and exemption from individual licensing of Ultra Low Power Active Medical Implant (ULP-AMI) communication systems operating in the frequency band 401 - 406 MHz on a secondary basis
ERC/DEC/(01)12	Harmonised frequencies, technical characteristics and exemption from individual licensing of Short Range Devices used for Model control operating in the frequencies 40.665, 40.675, 40.685 and 40.695 MHz
ERC/DEC/(01)11	Harmonised frequencies, technical characteristics and exemption from individual licensing of Short Range Devices used for Flying Model control operating in the frequency band 34.995 - 35.225 MHz

Document	Description
ERC/DEC/(00)08	The use of the band 10.7 - 12.5 GHz by the fixed service and Earth stations of the broadcasting-satellite and fixed-satellite Service (space-to-Earth)
ERC/DEC/(00)07	The shared use of the band 17.7 - 19.7 GHz by the fixed service and Earth stations of the fixed-satellite service (space-to-Earth)
ERC/DEC/(00)02	Use of the band 37.5-39.5 GHz by the fixed service and by earth stations of the fixed-satellite service (space-to-Earth) and use of the band 39.5-40.5 GHz by earth stations of the fixed-satellite service and the mobile-satellite service (space-to-Earth)
ECC/REC/(25)02	Cross-border coordination for Railway Mobile Radio (RMR) in the 900 MHz frequency band excluding GSM-R vs. GSM-R
ECC/REC/(24)03	Licensing of earth stations for space tracking, space telemetry and space telecommand in the bands 2025-2110 MHz (Earth-to-space) and 2200-2290 MHz (space-to-Earth)
ECC/REC/(24)02	Guidance for the use of governmental UAS operating within the frequency bands 1880-1900 MHz and 1910-1920 MHz
ECC/REC/(23)02	Cross-border coordination for Mobile/Fixed Communications Networks (MFCN) in the frequency band 24.25-27.5 GHz
ECC/REC/(23)01	Cross-border coordination for Railway Mobile Radio (RMR) in the 1900-1910 MHz TDD frequency band
ECC/REC/(22)02	Guidelines on measures to facilitate compatibility between MFCN operating in 40.5-43.5 GHz and FSS earth stations receiving in 39.5-40.5 GHz and to prevent and/or resolve interference issues
ECC/REC/(22)01	Guidelines to support the introduction of MFCN in 40.5-43.5 GHz while ensuring, in a proportionate way, the use of FSS receiving earth stations in the frequency band 40.5-42.5 GHz and the use of FSS transmitting earth stations in the frequency band 42.5-43.5 GHz and the possibility for future deployment of these earth stations
ECC/REC/(21)02	Guidance on the application of the least restrictive technical conditions (LRTC) in ECC Decision (11)06 (amended 26 October 2018) to ensure protection of the military radiolocation systems operating below 3400 MHz from indoor non-AAS small cells operating in the band 3400-3800 MHz
ECC/REC/(20)03	Frame structures to facilitate cross-border coordination of TDD MFCN in the frequency band 3400-3800 MHz
ECC/REC/(20)01	Guidelines to support the introduction of 5G while ensuring, in a proportionate way, the use of existing and planned FSS transmitting earth stations in the frequency band 24.65-25.25 GHz and the possibility for future deployment of these earth stations
ECC/REC/(19)01	Technical toolkit to support the introduction of 5G while ensuring, in a proportionate way, the use of existing and planned EESS/SRS receiving earth stations in the 26 GHz band and the possibility for future deployment of these earth stations
ECC/REC/(18)02	Radio frequency channel/block arrangements for fixed service systems operating in the bands 92-94 GHz, 94.1-100 GHz, 102-109.5 GHz and 111.8-114.25 GHz
ECC/REC/(18)01	Radio frequency channel/block arrangements for Fixed Service systems operating in the bands 130 - 134 GHz, 141-148.5 GHz, 151.5-164 GHz and 167 - 174.8 GHz
ECC/REC/(17)03	Guidance for the harmonised use and coordination of Maritime Broadband Radio (MBR) systems on board ships and off-shore platforms operating within the frequency bands 5852-5872 MHz and 5880-5900 MHz
ECC/REC/(16)03	Cross-border coordination for Broadband Public Protection and Disaster Relief (BB-PPDR) systems in the frequency band 698 to 791 MHz
ECC/REC/(15)04	Guidance for the implementation of a sharing framework between MFCN and PMSE within 2300-2400 MHz
ECC/REC/(15)01	Cross-border coordination for Mobile/Fixed Communications Networks (MFCN) in the frequency bands: 694-790 MHz, 1427-1518 MHz and 3400-3800 MHz
ECC/REC/(14)06	Implementation of Fixed Service Point-to-Point narrow channels (3.5 MHz, 1.75 MHz, 0.5 MHz, 0.25 MHz, 0.025 MHz) in the guard bands and center gaps of the lower 6 GHz (5925-6425 MHz) and upper 6 GHz (6425-7125 MHz) bands
ECC/REC/(14)04	Cross-border coordination for mobile/fixed communications networks (MFCN) and between MFCN and other systems in the frequency band 2300-2400 MHz
ECC/REC/(14)01	Radio frequency channel arrangements for fixed service systems operating in the band 92-95 GHz
ECC/REC/(11)10	Location Tracking Application for emergency and disaster situations
ECC/REC/(11)09	UWB Location Tracking Systems Type 2 (LT2)

Document	Description
ECC/REC/(11)08	Framework for authorisation regime of indoor global navigation satellite system (GNSS) pseudolites in the band 1559-1610 MHz
ECC/REC/(11)05	Cross-border Coordination for Mobile/Fixed Communications Networks (MFCN) in the frequency band 2500-2690 MHz
ECC/REC/(11)04	Cross-border Coordination for Mobile/Fixed Communications Networks (MFCN) in the frequency band 791-862 MHz
ECC/REC/(11)01	Guidelines for assignment of frequency blocks for Fixed Wireless Systems in the bands 24.5-26.5 GHz, 27.5-29.5 GHz and 31.8-33.4 GHz
ECC/REC/(10)02	A framework for authorisation regime of Global Navigation Satellite System (GNSS) repeaters
ECC/REC/(10)01	Guidelines for compatibility between Complementary Ground Components (CGC) operating in the band 2170-2200 MHz and EESS/SOS/SRS earth stations operating in the band 2200-2290 MHz
ECC/REC/(08)04	The identification of frequency bands for the implementation of Broad Band Disaster Relief (BBDR) radio applications in the 5 GHz frequency range
ECC/REC/(08)02	Frequency planning and frequency coordination for GSM / UMTS / LTE / WiMAX Land Mobile systems operating within the 900 and 1800 MHz bands
ECC/REC/(08)01	Use of the band 5855-5875 MHz for non-safety related Road Intelligent Transport Systems (ITS)
ECC/REC/(06)04	Use of the band 5725-5875 MHz for Broadband Fixed Wireless Access (BFWA)
ECC/REC/(05)08	Frequency planning and cross-border coordination between GSM Land Mobile Systems (GSM 900, GSM 1800 and GSM-R)
ECC/REC/(05)07	Radio frequency channel arrangements for Fixed Service Systems operating in the bands 71-76 GHz and 81-86 GHz
ECC/REC/(02)09	Protection of Aeronautical Radio Navigation Service in the band 2700-2900 MHz from interference caused by the operation of Digital Cordless Cameras
ECC/REC/(02)06	Preferred channel arrangements for digital Fixed Service Systems operating in the frequency range 7125-8500 MHz
ECC/REC/(02)02	Preferred channel arrangements for fixed service systems (point-to-point and point-to-multipoint) operating in the frequency band 31.0-31.3 GHz
ECC/REC/(01)04	Radio frequency channel arrangements for point-to-point (P-P) fixed wireless systems in the frequency band 40.5 - 43.5 GHz
ERC/REC/(01)02	Preferred channel arrangement for digital FS systems operating in the band 31.8-33.4 GHz
ERC/REC/(00)04	Harmonised frequencies and free circulation and use for meteor scatter applications
ERC/REC 70-03	Relating to the Use of Short Range Devices (SRD)
ERC/REC 62-02	Harmonised frequency band for civil and military airborne telemetry applications
ERC/REC 25-10	Frequency ranges for the use of terrestrial audio and video Programme Making and Special Events (PMSE) applications
ERC/REC 14-02	Radio-frequency channel arrangements for high, medium and low capacity digital Fixed Service systems operating in the band 6425-7125 MHz
ERC/REC 14-01	Radio-frequency channel arrangements for high capacity analogue and digital radio-relay systems operating in the band 5925 to 6425 MHz
ERC/REC 13-03	The use of the band 14.0 - 14.5 GHz for Very Small Aperture Terminals (VSAT) and Satellite News Gathering (SNG)
ERC/REC 12-12	Radio frequency channel arrangement for fixed service systems operating in the band 55.78-57.0 GHz (as amended in 2015)
ERC/REC 12-11	Radio frequency channel arrangements for Fixed Service systems operating in the bands 48.5-50.2 / 50.9-52.6 GHz
ERC/REC 12-08	Harmonised radio frequency channel arrangements and block allocations for low, medium and high capacity systems in the band 3600 MHz to 4200 MHz
ERC/REC 12-07	Harmonised radio frequency channel arrangements for digital terrestrial fixed systems operating in the band 14.5 - 14.62 GHz paired with 15.23 - 15.35 GHz
ERC/REC 12-06	Preferred channel arrangements for fixed service systems operating in the frequency band 10.7-11.7 GHz
ERC/REC 12-05	Harmonised radio frequency channel arrangements for digital terrestrial fixed systems operating in the band 10.0 - 10.68 GHz
ERC/REC 12-03	Harmonised radio frequency channel arrangements for digital terrestrial fixed systems operating in the band 17.7 GHz to 19.7 GHz

Document	Description
ERC/REC 12-02	Harmonised radio frequency channel arrangements for analogue and digital terrestrial fixed systems operating in the band 12.75 GHz to 13.25 GHz
ERC/REC 01-01	Cross-border coordination for mobile/fixed communications networks (MFCN) in the frequency bands: 1920-1980 MHz and 2110-2170 MHz
T/R 25-08	Planning criteria and cross-border coordination of frequencies for land mobile systems in the range 29.7-470 MHz
T/R 13-02	Preferred channel arrangements for fixed service systems in the frequency range 22.0-29.5 GHz
T/R 13-01	Preferred channel arrangements for fixed service systems operating in the frequency range 1-2-3 GHz
T/R 12-01	Harmonised radio frequency channel arrangements for analogue/digital terrestrial FS operating in 37-39.5 GHz

European Standards

Document	Description
EN 300 065	Narrow-band direct-printing telegraph equipment for receiving meteorological or navigational information (NAVTEX)
EN 300 066	Float-free maritime satellite Emergency Position Indicating Radio Beacons (EPIRBs) operating in the 406,0 MHz to 406,1 MHz frequency band
EN 300 086	Land Mobile Service; Radio equipment with an internal or external RF connector intended primarily for analogue speech
EN 300 113	Land Mobile Service; Radio equipment intended for the transmission of data (and speech) and having an antenna connector
EN 300 152	Maritime Emergency Position Indicating Radio Beacons (EPIRBs) intended for use on the frequency 121.5 MHz or the frequencies 121.5 MHz and 243 MHz for homing purposes only
EN 300 162	Radiotelephone transmitters and receivers for the maritime mobile service operating in VHF bands
EN 300 219	Land Mobile Service; Radio equipment transmitting signals to initiate a specific response in the receiver
EN 300 220	Short Range Devices (SRD) operating in the frequency range 25 MHz to 1 000 MHz
EN 300 224	On-site paging service
EN 300 296	Land Mobile Service; Radio equipment using integral antennas intended primarily for analogue speech
EN 300 328	Wideband Transmission systems; Data transmission equipment operating in the 2.4 GHz ISM band and using spread spectrum modulation techniques
EN 300 330	SRD; Radio equipment in the frequency range 9 kHz to 25 MHz and inductive loop systems in the frequency range 9 kHz to 30 MHz
EN 300 338	Radio equipment for generation, transmission and reception of Digital Selective Calling (DSC) in the maritime MF, MF/HF and/or VHF mobile service
EN 300 341	Land Mobile Service; Radio equipment using an integral antenna transmitting signals to initiate a specific response in the receiver
EN 300 390	Land Mobile Service; Radio equipment intended for the transmission of data (and speech) and using an integral antenna
EN 300 422	Wireless Microphones; Audio PMSE up to 3 GHz; Audio PMSE Equipment up to 3 GHz; part 1 and 4
EN 300 433	Citizens' Band (CB) radio equipment
EN 300 440	Radio equipment to be used in the 1 to 40 GHz frequency range
EN 300 454	Wide band audio links
EN 300 471	Rules for Access and the Sharing of common used channels by equipment complying with EN 300 113
EN 300 674	Transport and Traffic Telematics (TTT); Dedicated Short Range Communication (DSRC) transmission equipment (500 kbit/s / 250 kbit/s) operating in the 5 795 MHz to 5 815 MHz frequency band; Part 2: Harmonised Standard for access to radio spectrum; Sub-part 2: On-Board Units (OBU)
EN 300 676	Ground-based VHF hand-held, mobile and fixed radio transmitters, receivers and transceivers for the VHF aeronautical mobile service using amplitude modulation
EN 300 698	Radio telephone transmitters and receivers for the maritime mobile service operating in the VHF bands used on inland waterways
EN 300 718	Avalanche Beacons; Transmitter-receiver systems
EN 300 720	Ultra-High Frequency (UHF) on-board vessels communications systems and equipment
EN 301 025	VHF radiotelephone equipment for general communications and associated equipment for Class "D" Digital Selective Calling (DSC)
EN 301 091	Radar equipment operating in the 76 GHz to 77 GHz range
EN 301 166	Land Mobile Service; Radio equipment for analogue and/or digital communication (speech and/or data) and operating on narrow band channels and having an antenna connector

Document	Description
EN 301 178	Portable Very High Frequency (VHF) radiotelephone equipment for the maritime mobile service operating in the VHF bands (for non-GMDSS applications only)
EN 301 357	Cordless audio devices in the range 25 MHz to 2000 MHz
EN 301 406	Digital Enhanced Cordless Telecommunications (DECT)
EN 301 426	Low data rate Land Mobile satellite Earth Stations (LMES) and Maritime Mobile satellite Earth Stations (MMES) not intended for distress and safety communications operating in the 1.5/1.6 GHz frequency bands
EN 301 427	Low data rate Mobile satellite Earth Stations (MESs) except aeronautical mobile satellite earth stations, operating in the 11/12/14 GHz frequency bands
EN 301 428	Transmit-only, transmit/receive or receive-only satellite earth stations operating in the 11/12/14 GHz frequency bands
EN 301 430	Satellite News Gathering Transportable Earth Stations (SNG TES) operating in the 11-12/13-14 GHz frequency bands
EN 301 441	Handheld earth stations, for Satellite Personal Communications Networks (S-PCN) in the 1.6/2,4 GHz bands under the Mobile Satellite Service (MSS)
EN 301 442	Handheld earth stations, for Satellite Personal Communications Networks (S-PCN) in the 2.0 GHz bands under the Mobile Satellite Service (MSS)
EN 301 443	Harmonised Standard for Very Small Aperture Terminal (VSAT); Transmit-only, transmit-and-receive, receive-only satellite earth stations operating in the 4 GHz and 6 GHz frequency bands
EN 301 444	LMES operating in the 1.5 GHz and 1.6 GHz bands providing voice and/or data communications
EN 301 447	Harmonised Standard for satellite Earth Stations on board Vessels (ESVs) operating in the 4/6 GHz frequency bands allocated to the Fixed Satellite Service (FSS) covering the essential requirements of article 3.2 of the Directive 2014/53/EU
EN 301 449	Electromagnetic compatibility and Radio spectrum Matters (ERM); Harmonized EN for CDMA spread spectrum base stations operating in the 450 MHz cellular band (CDMA 450) and 410, 450 and 870 MHz PAMR bands (CDMA-PAMR) covering essential requirements of article 3.2 of the R&TTE; Directive
EN 301 459	SIT and SUT transmitting towards satellites in geostationary orbit in the 29.5 to 30.0 GHz frequency bands
EN 301 473	Aircraft Earth Stations (AES) operating below 3 GHz under the Aeronautical Mobile Satellite Service (AMSS)/Mobile Satellite Service (MSS) and/or the Aeronautical Mobile Satellite on Route Service (AMS(R)S)/Mobile Satellite Service (MSS)
EN 301 502	Global System for Mobile communications (GSM); Base Station and Repeater equipment
EN 301 511	Mobile stations in the GSM 900 and GSM 1800 bands
EN 301 526	Harmonized EN for CDMA spread spectrum mobile stations operating in the 450 MHz cellular band (CDMA 450) and 410, 450 and 870 MHz PAMR bands (CDMA-PAMR) covering essential requirements of article 3.2 of the R&TTE; Directive
EN 301 559	Low Power Active Medical Implants (LP-AMI) and associated Peripherals (LP-AMI-P) operating in the frequency range 2 483,5 MHz to 2 500 MHz; Harmonised Standard covering the essential requirements of article 3.2 of the Directive 2014/53/EU
EN 301 681	Geostationary mobile satellite systems, including handheld earth stations, for Satellite Personal Communications Networks (S-PCN) in the 1.5/1.6 GHz bands under the Mobile Satellite Service (MSS)
EN 301 721	Providing Low Bit Rate Data Communications (LBRDC) using Low Earth Orbiting (LEO) satellites operating below 1 GHz
EN 301 783	Land Mobile Service; Commercially available amateur radio equipment
EN 301 839	Ultra Low Power Active Medical Implants (ULP-AMI) and Peripherals (ULP-AMI-P) operating in the frequency range 402 MHz to 405 MHz
EN 301 841	VHF air-ground Digital Link (VDL) Mode 2; Technical characteristics and methods of measurement for ground-based equipment; Part 3
EN 301 842	VHF air-ground Digital Link (VDL) Mode 4 radio equipment; Technical characteristics and methods of measurement for ground-based equipment; Part 5
EN 301 893	5 GHz WAS/RLAN Harmonised Standard for access to radio spectrum

Document	Description
EN 301 908	IMT cellular networks
EN 301 929	VHF transmitters and receivers as Coast Stations for GMDSS and other appls in the maritime mobile service
EN 302 017	Transmitting equipment for the Amplitude Modulated (AM) sound broadcasting service
EN 302 018	Transmitting equipment for the Frequency Modulated (FM) sound broadcasting service
EN 302 054	Meteorological Aids (Met Aids); Radiosondes to be used in the 400.15 to 406 MHz frequency range with power levels ranging up to 200 mW
EN 302 064	Wireless Digital Video Links operating in the 1,3 GHz to 50 GHz frequency band
EN 302 065	Ultra Wide Band (UWB) technologies (multiple parts)
EN 302 077	Transmitting equipment for the Terrestrial - Digital Audio Broadcasting (T-DAB) service
EN 302 152	Satellite Personal Locator Beacons (PLBs) operating in the 406.0 MHz to 406.1 MHz frequency band
EN 302 186	Satellite mobile Aircraft Earth Stations (AESs) operating in the 11/12/14 GHz frequency bands
EN 302 194	Electromagnetic compatibility and Radio spectrum Matters (ERM); Navigation radar used on inland waterways
EN 302 195	Radio equipment in the frequency range 9 kHz to 315 kHz for ULP-AMI and accessories
EN 302 208	Radio Frequency Identification Equipment operating in the band 865 to 868 MHz with power levels up to 2 W and in the band 915 MHz to 921 MHz with power levels up to 4 W
EN 302 217	Characteristics and requirements for point-to-point equipment and antennas
EN 302 245	Transmitting equipment for the Digital Radio Mondiale (DRM) broadcasting service
EN 302 248	Navigation radar for use on non-SOLAS vessels
EN 302 264	Short Range Radar equipment operating in the 77 GHz to 81 GHz band
EN 302 288	Short range radar equipment operating in the 24 GHz range
EN 302 296	Transmitting equipment for the digital television broadcast service, Terrestrial (DVB-T)
EN 302 326	Multipoint Equipment and Antennas
EN 302 340	Harmonised Standard for satellite Earth Stations on board Vessels (ESVs) operating in the 11/12/14 GHz frequency bands allocated to the Fixed Satellite Service (FSS) covering the essential requirements of article 3.2 of the Directive 2014/53/EU
EN 302 372	Short Range Devices (SRD); Tank Level Probing Radar (TLPR) equipment operating in the frequency ranges 4,5 GHz to 7 GHz, 8,5 GHz to 10,6 GHz, 24,05 GHz to 27 GHz, 57 GHz to 64 GHz, 75 GHz to 85 GHz; Harmonised Standard covering the essential requirements of article 3.2 of the Directive 2014/53/EU
EN 302 426	Harmonized EN for CDMA spread spectrum Repeaters operating in the 450 MHz cellular band (CDMA450) and the 410 MHz, 450 MHz and 870 MHz PAMR bands (CDMA-PAMR) covering essential requirements of article 3.2 of the R&TTE; Directive
EN 302 448	Earth Stations on Trains (ESTs) operating in the 14/12 GHz frequency bands
EN 302 454	Radiosondes to be used in the 1 668.4 MHz to 1 690 MHz frequency range
EN 302 480	GSM onboard aircraft system
EN 302 502	Broadband Radio Access Networks (BRAN); 5800 MHz fixed broadband data transmitting systems
EN 302 510	Radio equipment in the range 30-37.5 MHz for Ultra Low Power Active Medical Membrane Implants and Accessories

Document	Description
EN 302 536	Radio equipment operating in the frequency range 315 kHz to 600 kHz for Ultra Low Power Animal Implantable Devices (ULP-AID) and associated peripherals
EN 302 537	Ultra Low Power Medical Data Service Systems operating in the frequency range 401-402 MHz and 405-406 MHz
EN 302 561	Radio equipment using constant or non-constant envelope modulation operating in a channel bandwidth of 25 kHz, 50 kHz, 100 kHz or 150 kHz
EN 302 567	60 GHz Multiple-Gigabit WAS/RLAN Systems
EN 302 571	Intelligent Transport Systems (ITS); Radiocommunications equipment operating in the 5855 MHz to 5925 MHz frequency band
EN 302 574	Satellite earth station for MSS operating in 1980-2010 MHz (E/s) and 2170-2200 MHz (s/E) frequency bands
EN 302 608	Radio equipment for Eurobalise railway systems
EN 302 609	Radio equipment for Euroloop communication systems
EN 302 617	Ground-based UHF radio transmitters, receivers and transceivers for the UHF aeronautical mobile service using amplitude modulation
EN 302 625	5 GHz BroadBand Disaster Relief applications (BBDR)
EN 302 645	Global Navigation Satellite Systems (GNSS) Repeaters
EN 302 686	Intelligent Transport Systems (ITS); Radiocommunications equipment operating in the 63 GHz to 64 GHz frequency band
EN 302 729	LPR equipment operating in the frequency ranges 6.0 GHz to 8.5 GHz, 24.05 GHz to 26.5 GHz, 57 GHz to 64 GHz, 75 GHz to 85 GHz
EN 302 752	Active Radar Target Enhancers
EN 302 858	Automotive radar equipment operating in the 24.05 GHz up to 24.25 GHz or 24.50 GHz frequency range
EN 302 885	VHF radiotelephone equipment for the maritime mobile service
EN 302 961	Maritime Personal Homing Beacon for search and rescue purposes intended for use on the frequency 121.5 MHz for search and rescue purposes only
EN 302 977	Vehicle-Mounted Earth stations (VMES) operating 14/12 GHz frequency bands
EN 303 039	Land Mobile Service; Multichannel transmitter specification for the PMR Service
EN 303 064	Primary Surveillance Radar (PSR);
EN 303 084	Technical characteristics and methods of measurement for ground-based equipment
EN 303 098	Maritime low power personal locating devices employing AIS
EN 303 132	Maritime low power VHF personal locating beacons employing Digital Selective Calling (DSC)
EN 303 135	Coastal Surveillance, Vessel Traffic Systems and Harbour Radars (CS/VTS/HR)
EN 303 203	Medical Body Area Network Systems (MBANS) operating in the 2483.5 MHz to 2500 MHz range
EN 303 204	Fixed Short Range Devices (SRD) in data networks; Radio equipment to be used in the 870 MHz to 876 MHz frequency range with power levels ranging up to 500 mW e.r.p.; Harmonised Standard for access to the radio spectrum
EN 303 213	Advanced Surface Movement Guidance and Control System (A-SMGCS)
EN 303 258	Wireless Industrial Applications (WIA); Equipment operating in the 5 725 MHz to 5 875 MHz frequency range with power levels ranging up to 400 mW
EN 303 276	Maritime Broadband Radio (MBR) links for ships and fixed installations engaged in off-shore activities
EN 303 340	Digital Terrestrial TV Broadcast Receivers; Harmonised Standard for access to radio spectrum

Document	Description
EN 303 345	Broadcast Sound Receivers; Part 3: FM broadcast sound service; Harmonised Standard for access to radio spectrum
EN 303 347	Meteorological Radars; Harmonised Standard for access to radio spectrum; Part 1: Meteorological Radar Sensor operating in the frequency band 2 700 MHz to 2 900 MHz (S band)
EN 303 360	Transport and Traffic Telematics (TTT); for heliborne obstacle detection radars operating in the 76-77 GHz range
EN 303 402	Maritime mobile transmitters and receivers for use in the MF and HF bands
EN 303 405	Analogue and Digital PMR446 Equipment
EN 303 406	Social Alarms Equipment operating in the frequency range 25 MHz to 1000 MHz;
EN 303 413	Global Navigation Satellite System (GNSS) receivers; Radio equipment operating in the 1 164 MHz to 1 300 MHz and 1 559 MHz to 1 610 MHz frequency bands
EN 303 447	Short Range Devices (SRD); Inductive loop systems for robotic mowers in the frequency range 0 Hz to 148,5 kHz
EN 303 454	Short Range Devices (SRD); Metal and object detection sensors in the frequency range 1 kHz to 148,5 kHz
EN 303 520	Ultra Low Power (ULP) wireless medical capsule endoscopy devices operating in the band 430 MHz to 440 MHz
EN 303 609	GSM Repeaters
EN 303 659	Short Range Devices (SRD) in Data Networks; Radio equipment to be used in the frequency ranges 865 MHz to 868 MHz and 915 MHz to 919,4 MHz; Harmonised Standard for access to radio spectrum
EN 303 661	Short Range Devices (SRD); Ground Based Synthetic Aperture Radar (GBSAR) in the frequency range 17,1 GHz to 17,3 GHz and High Definition Ground Based Synthetic Aperture Radar (HD-GBSAR) in the frequency range 76 GHz to 77 GHz; Harmonised Standard for access to radio spectrum
EN 303 687	6 GHz WAS/RLAN Harmonised Standard for access to radio spectrum
EN 303 699	Satellite Earth Stations and Systems (SES); Fixed earth stations communicating with non-geostationary satellite systems in the 20 GHz and 30 GHz FSS bands; Harmonised Standard for access to radio spectrum
EN 303 722	Wideband Data Transmission Systems (WDTS) for Fixed Network Radio Equipment operating in the 57 GHz to 71 GHz band
EN 303 753	Wideband Data Transmission Systems (WDTS) for Mobile and Fixed Radio Equipment operating in the 57 - 71 GHz band
EN 303 978	Earth Stations on Mobile Platforms ESOMP transmitting towards satellites in geostationary orbit in the 27.5-30.0 GHz frequency bands
EN 303 979	Fixed Earth Stations and Earth Stations on Mobile Platforms (ESOMPs) transmitting towards satellites in non-geostationary orbit in the 27.5 GHz to 29.1 GHz and 29.5 GHz to 30.0 GHz bands
EN 303 980	Fixed and in-motion Earth Stations communicating with non-geostationary satellite systems in the 11 GHz to 14 GHz frequency bands
EN 303 981	Satellite Earth Stations and Systems (SES); Fixed and in-motion Wide Band Earth Stations communicating with non-geostationary satellite systems (WBES) in the 11 GHz to 14 GHz frequency bands; Harmonised Standard for access to radio spectrum
EN 304 220	Wideband data transmission SRD; Harmonised Standard for access to radio spectrum; Part 1: Wideband data transmission devices: network access points operating in the frequency bands 863 MHz to 868 MHz and 915,8 MHz to 919,4 MHz
EN 305 550	Short Range Devices (SRD); Radio equipment to be used in the 40 GHz to 246 GHz frequency range

European Standards for Receive-Only Equipment

Document	Description
EN 300 487	Satellite Earth Stations and Systems (SES); Harmonised Standard for Receive-Only Mobile Earth Stations (ROMES) providing data communications operating in the 1,5 GHz frequency band
EN 303 372	Satellite Earth Stations and Systems (SES); Satellite broadcast reception equipment. Part 1: Outdoor unit receiving in the 10,7 GHz to 12,75 GHz frequency band
EN 303 345	Broadcast Sound Receivers

Abbreviations

Abbreviation	Description
(OR)	Off-Route
(R)	Route
1800	Global System for Mobile Communications using 1800 MHz band
ADS	Automatic Dependant Surveillance (Aeronautical)
AES	Aircraft Earth Stations
AGA	Air Ground Air
AIS	Automatic Identification System
ALS	Assistive Listening Systems
AM	Amplitude Modulation
AMRD	Autonomous Maritime Radio Device
AMS(R)S	Aeronautical Mobile Satellite (Route) Services
APP	Appendix of the ITU Radio Regulations
AVI	Automatic Vehicle Identification
BBDR	Broad Band Disaster Relief
BFWA	Broadband Fixed Wireless Access
BMA	Building Material Analysis
BSS	Broadcasting Satellite Service
CB	Citizen Band
CEPT	European Conference of Postal and Telecommunications Administrations
CGC	Complementary Ground Component
CRS	Central Radio Station
CT	Cordless Telephone
DEC	Decision
DECT	Digital Enhanced Cordless Telecommunication
D-GPS	Differential Global Positioning System
DME	Distance Measuring Equipment
DMO	Direct Mode Operation
DRM	Digital Radio Mondiale
DSC	Digital Selective Calling

Abbreviation	Description
DSI	Detailed Spectrum Investigation
DVB-T	Terrestrial Digital Video Broadcasting
E/s	Earth-to-space direction
ECA	European Common Allocation
ECC	Electronic Communications Committee
ECM	Electronic Countermeasures
ECP	European Common Proposal
EESS	Earth Exploration-Satellite Service
EFIS	European Frequency Information System
EGSM	Extended GSM
EISCAT	European Incoherent SCATter facility
ELT	Emergency locator transmitter
ENG	Electronic News Gathering
EPIRB	Emergency Position-Indicating Radiobeacon
ERC	European Radiocommunications Committee
ERO	European Radiocommunications Office
ESIM	Earth Stations In Motion
ESOMPs	Earth Stations On Mobile Platforms
EST	Earth Stations on Trains
ESV	Earth Stations on-board Vessels
EU	European footnote
FDD	Frequency Division Duplex
FM	Frequency Modulation
FSS	Fixed-Satellite Service
FWA	Fixed Wireless Access
GALILEO	European Global Navigation Satellite System
GBAS	Ground Based Augmentation System
GBSAR	Ground Based Synthetic Aperture Radar
GE06	Geneva 2006 Agreement
GE75	Geneva 1975 Agreement
GE85	Geneva 1985 Agreement

Abbreviation	Description
GMDSS	Global Maritime Distress and Safety System
GNSS	Global Navigation Satellite System
GPR/WPR	Ground Probing Radar / Wall Probing Radar
GPS	Global Positioning System
GSM	Global System for Mobile Communications
GSM 1800	Global System for Mobile Communications using 1800 MHz band
GSM-R	GSM for Railways
GSO	GeoStationary Orbit
HAPS	High Altitude Platform Systems
HDFS	High Density Fixed Service
HDFSS	High Density Fixed-Satellite Service
HDTV	High Definition Television
HEST	High E.i.r.p. Satellite Terminals
HF	High Frequency
HIPERLAN	High Performance Radio Local Area Network
IALA	International Association of Lighthouse Authorities
IBCN	Integrated Broadband Communications Network
IFF	Identification Friend or Foe
ILS	Instrument Landing System
IMO	International Maritime Organisation
IMT	International Mobile Telecommunications
IMT-2000	International Mobile Telecommunications-2000
IMT-Advanced	Systems beyond IMT-2000
IoT	Internet of Things
ISM	Industrial, Scientific and Medical
ITS	Intelligent Transport Systems
ITU	International Telecommunication Union
JTIDS	Joint Tactical Information Distribution System
LAES	Location Application for Emergency Services
LANs	Local Area Networks
LDC	Low Duty Cycle

Abbreviation	Description
LP-AMI	Low Power Active Medical Implants
LPD-S	Low Power Devices communicating with Satellites
LPR	Level Probing Radar
LT2	Location Tracking Type 2
MBANS	Medical Body Area Network Systems
MBR	Maritime Broadband Radio Links
MCA	Mobile Communications Services on Board Aircraft
MCV	Mobile Communication Services on Board Vessels
MES	Mobile Earth Stations
MFCN	Mobile/Fixed Communications Networks
MIDS	Multifunctional Information Distribution System
MLS	Microwave Landing System
MSI	Maritime Safety Information
MSS	Mobile-Satellite Service
MWS	Multimedia Wireless System
NATO	North Atlantic Treaty Organisation
NAVTEX	Narrow-band direct-printing telegraphy system for transmission of navigational and meteorological warnings and urgent information to ships
NDB	Non-Directional Beacon
NGSO	Non-GeoStationary Orbit
NJFA	NATO Joint Civil/Military Frequency Agreement
NMR	Nuclear Magnetic Resonance
NTFA	National Table of Frequency Allocation
OB	Outside Broadcasting
PAMR	Public Access Mobile Radio
PKO	Peace Keeping Operations
PLB	Personal Locator Beacons
PMR	Professional Mobile Radio, Private Mobile Radio
PMSE	Programme Making and Special Events
POCSAG	Post Office Code Standards Advisory Group
PPDR	Public Protection and Disaster Relief
PWAP	Private Wide Area Paging

Abbreviation	Description
RA	Radio Astronomy
REC	Recommendation
RFID	Radio Frequency Identification
RLAN	Radio Local Area Network System
RR	ITU Radio Regulations
RTE	Radar Target Enhancer
RTTT	Road Transport & Traffic Telematics
s/E	space-to-Earth direction
SAB	Services Ancillary to Broadcasting
SAP	Services Ancillary to Programming
SAR(communciations)	Search and Rescue
SIT	Satellite Interactive Terminal
SNG	Satellite News Gathering
S-PCS	Satellite Personal Communication System
SRD	Short Range Device
SRR	Short Range Radar
SRS	Space Research Service
SSR	Secondary Surveillance Radar
SUT	Satellite User Terminal
TACAN	Tactical Air Navigation
T-DAB	Terrestrial Digital Audio Broadcasting
TDD	Time Division Duplex
TETRA	Terrestrial Trunked Radio
TLPR	Tank Level Probing Radar
TRR	Tactical Radio Relays
TS	Terminal Station
TTT	Transport and Traffic Telematics
TV	Television
UIC	International Union for Railways
ULP-AID	Ultra Low Power Animal Implants Devices
ULP-AMI	Ultra Low Power Active Medical Implants

Abbreviation	Description
ULP-MMI	Ultra Low Power Medical Membrane Implants
ULP-WMCE	Ultra-Low Power Wireless Medical Capsule Endoscopy
UMTS	Universal Mobile Telecommunications System
UWB	Ultra – Wideband
VDB	VHF ground-air Data Broadcast
VLBI	Very Long Baseline Interferometry (Radio Astronomy)
VOR	VHF Omni-directional Range
VSAT	Very Small Aperture Terminal
VTs	Vessel Traffic System (radar)
WAIC	Wireless Avionics Intra-Communication systems
WARC	World Administrative Radio Conference
WAS	Wireless Access System
WIA	Wireless Industrial Applications